



TRANSMITTAL

October 18, 2024

PO Number: **20234972**
Sales Order#: **22367602**
Attention to: **mgarner@lincolnnassoc.com**

Equipment Description: **HCD-2250-EA-SFS**

Attached please find:

Operating and Maintenance Data for Record.

Please note that this information is for record only and not subject to approval.

If you have any questions, please feel free to contact the undersigned.

Sincerely,

A handwritten signature in black ink that reads 'John Savoy' in a cursive script.

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Munters

OPERATING AND MAINTENANCE MANUAL

**MODEL:
HCD-2250-EA-SFS**

**PREPARED FOR:
WAYNE FARMS, LLC**

**Customer Purchase Order Number:
20234972**

**Munters Sales Order Number:
22367602**

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PROTECTING YOUR INVESTMENT

**As with all machinery,
continuation of long term,
satisfactory operation can
be best assured by regular
servicing.
To help achieve this,
Munters Service has
developed a range of
service options.**

REPLACEMENT PARTS

Munters prides itself on being able to quickly respond to customer requests for spare parts, whether the part is for a unit one year old or twenty years old. All Munters manufactured parts carry a full one-year warranty, guaranteeing that they will be free from defects and provide continued performance that is expected from the dehumidifier. All vendor-supplied parts are manufactured to Munters' exacting specifications.

Days: 1-888-DHWheel 8am - 5 pm EST - ASK FOR PARTS
(1-888-349-4335)

SERVICING PROGRAMS

Regular preventive maintenance servicing is vital to maximizing equipment efficiency and longevity, and reducing unscheduled downtime. In-house maintenance personnel may not be familiar with some of the technical aspects required to ensure peak efficiency, such as performance monitoring and adjustment, and psychometric analysis. Periodic visits by an experienced Munters technician to ensure optimum performance may be the answer. A preventive maintenance service program is available on all Munters units - new and used. Munters' service programs are customized to customers' requirements with one to twelve visits annually. Parts or additional service needed for equipment under a service agreement are provided for at a discounted rate. Munters is willing to design service programs to specifically match customers' requirements.

For more information contact: 1-888-DHWheel (1-888-349-4335)

TECHNICAL SUPPORT

At any time should there be a mechanical or operational problem, whether or not the equipment is still in warranty, telephone support can be obtained from Munters' technical staff to assist in understanding or troubleshooting the problem, 24 hours a day.

Days: 1-888-DHWheel 8am - 5pm EST - ASK FOR TROUBLESHOOTING
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Downtime is costly Munters can help minimize unscheduled downtime and reduce production losses by providing quick response to service requirements from one of fourteen regional offices located throughout the United States and Canada. Munters factory-trained Field Service Technicians are ready to respond.

Days: 1-888-DHWheel 8am - 5 pm EST
Nights: 1-978-372-9782 5pm - 8am EST

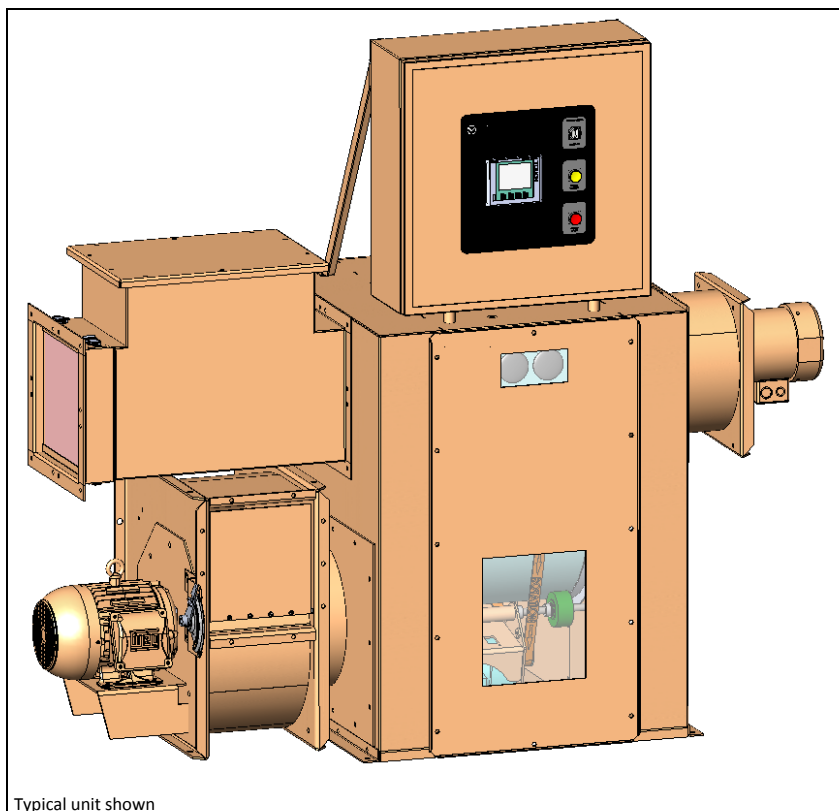
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I27092



Typical unit shown

MODEL HCD-SERIES DEHUMIDIFIERS

with Electric Reactivation

**Units with Siemens
Programmable
Logic Controller**

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1 - INTRODUCTION

The engineers at Munters have designed the HoneyCombe® dehumidifier using state-of-the-art technology. The unit is durable, simple to operate, and needs very little maintenance.

The HCD unit can provide years of trouble-free service when the recommendations listed in this manual are followed. It is strongly recommended that this entire manual is read prior to operating the equipment. The manual describes how the dehumidifier works, and how to get the best service from the unit. While a Munters dehumidifier is easy to operate, it is a bit different from most air handling equipment.

The HCD unit includes a heater which is used to remove moisture from the unit. (The heater is used to “reactivate” the material in the HCD unit which dries the air.) The HCD units described in this manual have electric reactivation. Figure 1-1 shows a typical unit.

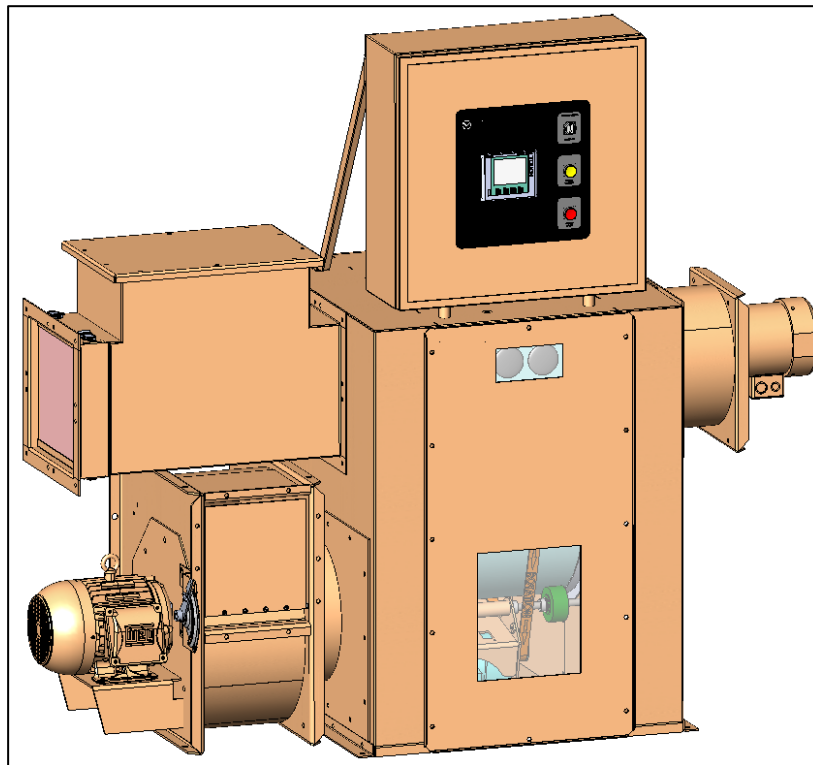
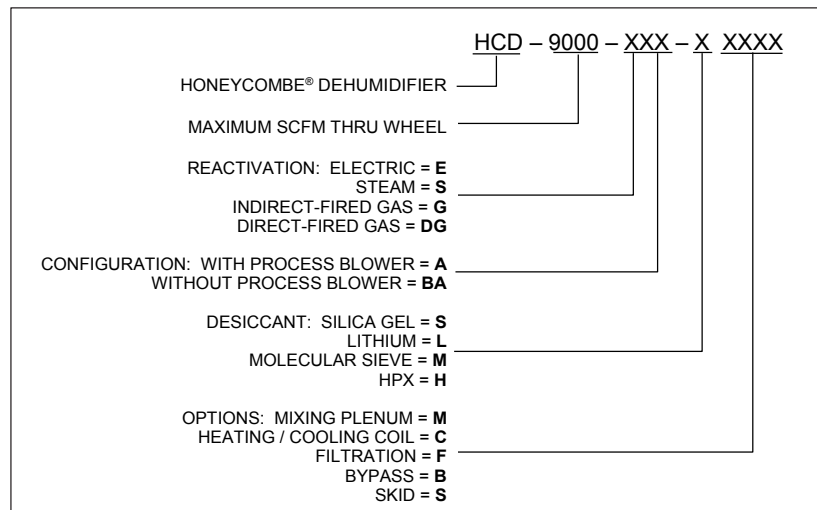


Figure 1-1
TYPICAL HCD UNIT WITH ELECTRIC REACTIVATION

The table below explains the identification system used for the model number:



If any part of this manual is not understood, or there is a question concerning the dehumidifier, call the Service Operations Department of Munters and speak with a Service Operation Technicians at (800) 843-5360.

2 - SAFETY NOTES

Munters is concerned about the safety of anyone who uses or services the HCD unit. Some of the parts inside the HCD unit can be dangerous if an untrained person tries to service the unit. Throughout this manual, hazards are pointed out which may occur in the use of the HCD unit. Precautions are also listed which should be taken to avoid these hazards.

In this manual, different kinds of messages are used to warn of possible hazards problems:



Immediate hazard which **will** result in severe personal injury or death.



Hazard or unsafe practice which **may** result in severe personal injury or death.



Hazard or unsafe practice which could result in minor personal injury or property damage.



Hazard or unsafe practice which will not result in personal injury, but could result in property damage.

Please keep these points in mind while using or servicing the unit:



The HCD unit may be wired for up to 575V AC. The unit can produce enough voltage and current to cause serious injury or death. Always turn off and lock out the power before working inside the unit. Only trained electricians should perform work with the electrical parts.



The two blowers inside the HCD unit spin at a high rate of speed. Hands may be badly hurt if put inside the blower while it is turning. Keep hands away from the blowers while the unit is turned on. Before working with the machine, allow enough time for the blowers to come to a complete stop.

The wheel used in a larger HCD unit can weigh several hundred pounds. Never attempt to lift the wheel without help from a second person. It is usually easier to roll the wheel. When working near the wheel when it is inside the unit, and the wheel is propped up, use blocks to prevent the wheel from moving accidentally.

Units designed for outdoor use are not waterproof and weatherproof unless all covers are secured in place.



During normal operation, do not run the unit with the access panels removed. This can allow moist air to reach parts of the machine which should be dry.

 WARNING

Do not use any kind of filter coating on the reactivation inlet filter. This filter is located just upstream of the electric heating elements. Some filter coatings give off volatile chemicals, which could be pulled into the heating elements. This could create a fire hazard. (If a filter is installed in the process air duct, it is OK to use filter coating on this filter. The process air is not pulled through the heating elements.)

The electric heaters and the solid state power controllers are mounted in an electrical control enclosure. These parts may carry high voltages from 208 to 575V AC. Only trained electricians should perform work on these parts.

Only for units with HoneyCombe® wheels containing lithium chloride (wheel type listed on tech data sheet, “GFR” engraved on wheel) –

NOTICE

Do not allow air to move through the HoneyCombe® wheel unless the wheel is turning and the reactivation heater is on. If the machine will be turned off for a long period of time, certain precautions must be taken to protect the HoneyCombe® wheel. Even when the machine is shut down, moist air can continue to move through the wheel. This is especially likely if a fan or airmoving device is still operating in the system. Eventually, a large amount of moisture can collect in the wheel. This can damage the wheel. For instructions on long-term shutdown, see Section 6 on “Maintenance.”

Do not wash the wheel with water or any solvent. This will permanently damage the wheel.

Do not use a lithium chloride wheel in a dehumidifier which uses direct-fired gas reactivation. The wheel may be damaged.

PLEASE READ ALL OF THIS MANUAL. PLEASE FOLLOW THE INSTRUCTIONS CAREFULLY AND COMPLETELY. PLEASE PAY PARTICULAR ATTENTION TO THE SAFETY INSTRUCTIONS AND PRECAUTIONS.

3 - OPERATING INSTRUCTIONS

3.1 CONTROLS AND INDICATORS

Figure 3-1 shows the controls on the main panel of the machine:

Auto/Off/Manual switch –

The center of the Auto/Off/Manual switch will light whenever the unit is receiving power and is ready to operate. When this switch is set to the Manual position, the unit will run continuously. When the switch is set to the Auto position, the unit will run under the control of a remote sensor, such as a humidistat. Turn the switch to the Off position to stop the unit. If the switch is set to either Manual or Auto, the unit will stop if one of the safety sensors detects a fault.

Operator display -

This panel shows the status of the machine, and also presents any fault or warning messages. Use this display to check the reactivation inlet and outlet temperatures and the machine run time. The operating set-points for the machine can also be set using this panel, but some of these functions require a password. (For more information on fault or warning messages, see Section 7 - TROUBLESHOOTING) Note that the operator display just reports information from the machine - it is not a controller. If the display is not working correctly for some reason, the rest of the dehumidifier may continue to operate normally. If there is an alarm, it must be cancelled on the machine itself and the alarm indication must also be cancelled on the display.

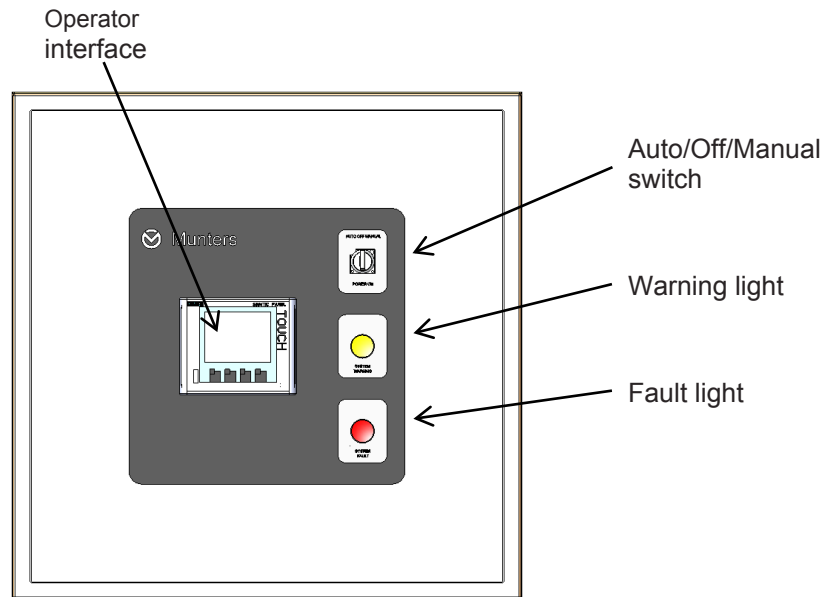


Figure 3-1
MAIN CONTROL PANEL

Fault and warning indicators –

The unit has several safety circuits. If one of these circuits detects a problem, the unit may indicate this in several ways:

Warning - The machine will continue to run, and a warning message will appear on the operator interface. The Warning indicator will light.

Fault - The machine will stop, and a fault message will appear on the operator interface. The Fault indicator will light.

For more information, see Section 7 - TROUBLESHOOTING. To test the Fault indicator, press the control knob.

Emergency stop switch -

Some machines are equipped with an optional E-Stop switch. Press in on this switch to stop the machine quickly in the event of an emergency. (Do not use this switch to stop the machine routinely - this may damage the heating elements.) To reset the E-Stop switch, turn the top so that it pops out.

Indicators on PLC –

This unit is controlled by a small computer called a “programmable logic controller” or “PLC.” The PLC has several LED indicators which may be helpful during troubleshooting. Normally, the operator does not have to use these indicators. The indicators are described in Section 7 - TROUBLESHOOTING.

3.2 STARTING THE UNIT

1. Some units have a disconnect switch on the front panel. Be sure this switch is turned on.
2. Check to see that the center of the Auto/Off/Manual switch is lighted. This means the unit has power. The operator interface will show that the machine is ready to run. See Figure 3-2.

React	This refers to just the dehumidifier itself (heater, reactivation fan, drive motor, etc.).
-------	--

Status	This refers to the whole line of equipment, including the dehumidifier and any Munters-supplied equipment mounted upstream or downstream of the dehumidifier (pre-heater, post-cooler, etc.).
--------	---

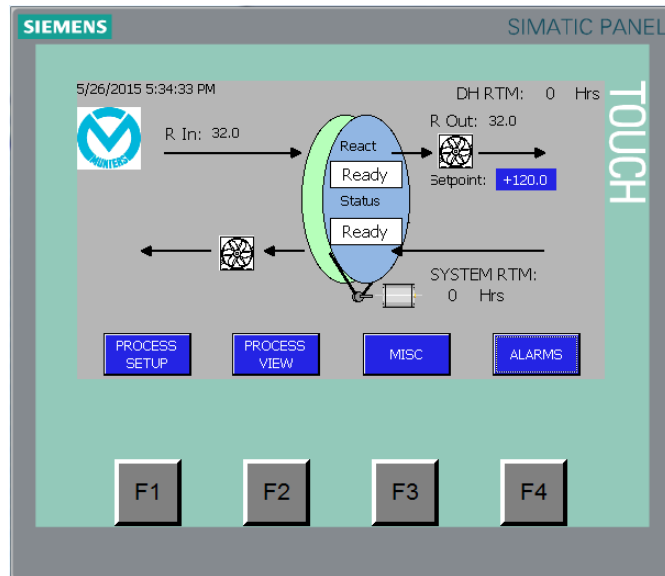


Figure 3-2
MACHINE READY TO RUN

3. Start the machine -

If the unit has a remote sensor such as a humidistat, or can be turned on by a remote control system, set the Auto/ Off/ Manual switch to Auto. The unit will begin to run, and the operator interface will show that the machine is running. See Figure 3-3. (If the unit does not run, check the setpoint on the humidistat.)

If the unit does not have a remote sensor, turn the Auto/ Off/ Manual switch to Manual. The unit will begin to run, and the operator interface will show that the machine is running. See Figure 3-3.

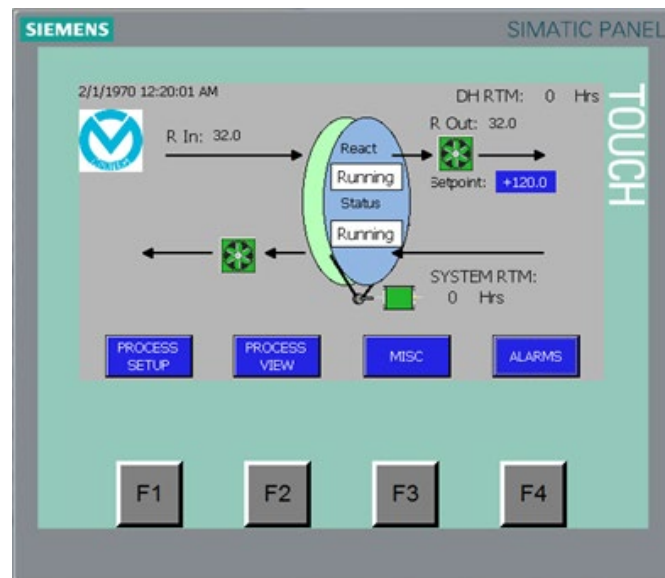


Figure 3-3
MACHINE RUNNING

4. After the machine is switched on, there will be a normal delay of 10 to 30 seconds before the fans start. As the machine is warming to normal operating temperature (first 60 minutes of machine operation), the unit will not display the Reactivation Low Temperature warning.

5. Notice the two lines at the top of the screen. The upper line (React) refers to just the dehumidifier itself. The lower line (Status) refers to the whole line of equipment, including the dehumidifier and any Munters-supplied equipment mounted upstream or downstream of the dehumidifier (pre-heater, post-cooler, etc.).

On the line for React, there are five possible indications:

READY	The machine has power. There are no alarms or warnings, and the system is ready to run whenever it receives a "start" signal.
RUNNING	The dehumidifier is actually running.
PURGING	When the selector switch is turned off, or the system stops calling for dehumidification, the dehumidification wheel must continue to turn so it can be "dried out" before it stops. Normally the purge cycle runs for three minutes.
STANDBY	The system is running in AUTO mode and the desired humidity condition is being met without the dehumidifier running (full bypass on Face and Bypass or Constant Process Fan)
FAULTED	The dehumidifier is faulted.

On the line for Status, there are four indications which may appear:

READY	The system has power. There are no alarms or warnings, and the system is ready to run whenever it receives a "start" signal.
RUNNING	The system is actually running.
PURGING	When the selector switch is turned off, or the system stops calling for dehumidification, the dehumidification wheel must continue to turn so it can be "dried out" before it stops. Normally the purge cycle runs for three minutes.
FAULTED	The system is faulted.

3.3 CHECKING THE REACTIVATION TEMPERATURES

1. The present value (PV) reactivation inlet and outlet temperatures can be viewed on the Machine Running screen (See Figure 3-3). To view details (CV, SP, and a PV chart) associated with the reactivation inlet and outlet temperatures, press the PROCESS VIEW button. See Figure 3-4.

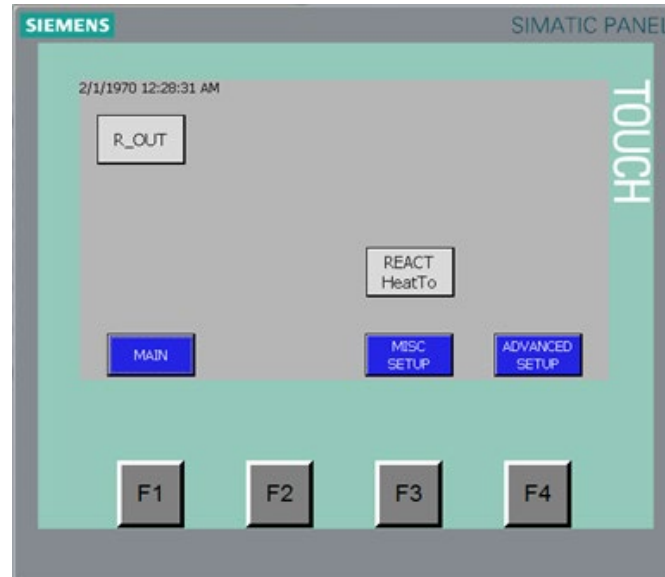


Figure 3-4
PROCESS VIEW

(The screen shown here would appear on a machine without a humidistat or other features. In more advanced installations, the screen on the machine may be slightly different.)

2. To see the reactivation inlet and outlet temperatures, press the REACT HeatTo or R_OUT buttons respectively. The following screen will show the detailed reactivation temperature info. See Figure 3-5.

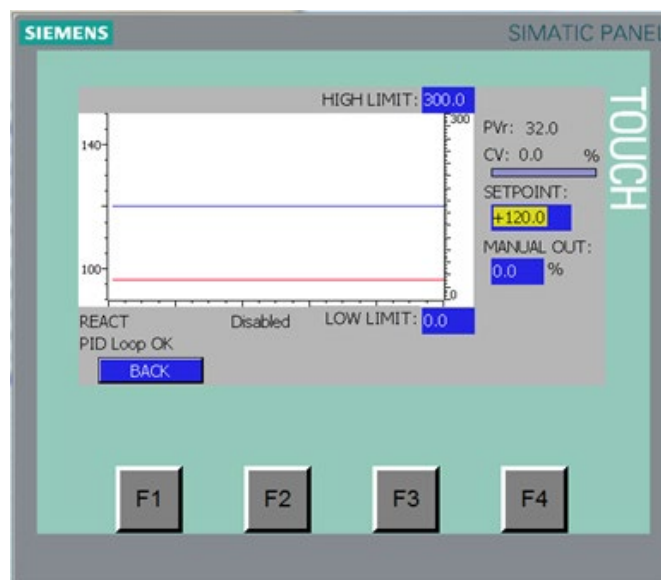


Figure 3-5
DETAILED REACTIVATION TEMPERATURE INFO

Here are the items on this screen:

SETPOINT Setpoint for reactivation outlet temperature

PVr Current reactivation inlet or outlet temperature

CV "Control Variable" - percentage of full reactivation heat output.

Although the temperatures may be displayed in °F or °C, the factory default is °F.

To return to the Machine Running display, press the BACK button, followed by the MAIN button.

3.4 CHECKING THE SYSTEM AND DH RUN TIME

The System and DH run time are displayed on the Machine Running screen. See Figure 3-6.

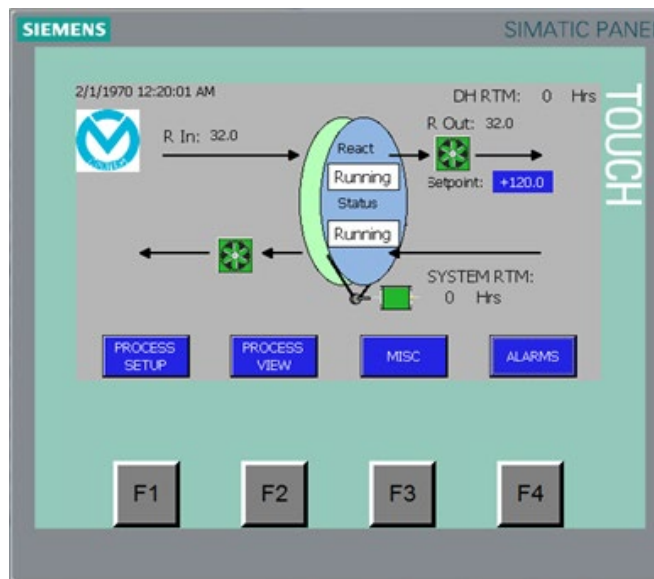


Figure 3-6
CHECKING SYSTEM AND DH RUN TIME

Here are the items on this screen:

SYSTEM RTM Total run time for whole system, including the dehumidifier and any additional equipment mounted upstream or downstream

DH RTM Total run time for dehumidifier only

3.5 LOGIN – ENTERING A USER AND PASSWORD

Some of the functions on the machine are protected so they can only be changed by certain personnel. Attempting to change one of these functions or settings will require a password.

1. When trying to change one of these protected functions or settings, the system will ask for a password. See Figure 3-7.

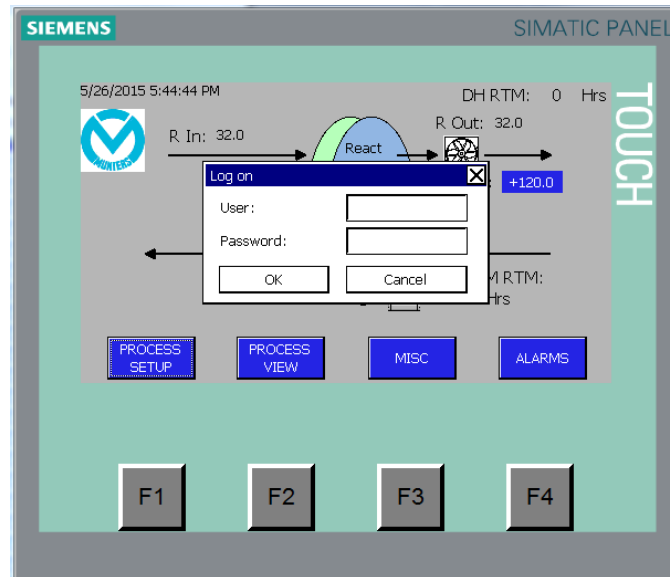


Figure 3-7
ENTERING PASSWORD

2. Start by entering an user name:
 - Press the text box next to User: to highlight the User box.
 - Use the pop-up keyboard to input the username.
 - Press Enter to accept the new entry in this box.
3. Next, enter a password:
 - Press the text box next to Password: to highlight the Password box.
 - Use the pop-up keyboard to input the password.
 - Press Enter to accept the new entry in this box.
4. Press OK to send the user name and password to the controller.

If the password is correct, the display will go to the screen that was protected by the password.

If the password is not correct, the display will continue to ask for the password.

3.6 CHECKING AND CHANGING THE DATE AND TIME

1. To check or change the date and time, start at the Machine Running screen (Figure 3-3). Press the MISC button followed by the Set PLC Date and Time to go to the Date and Time screen (Figure 3-8).

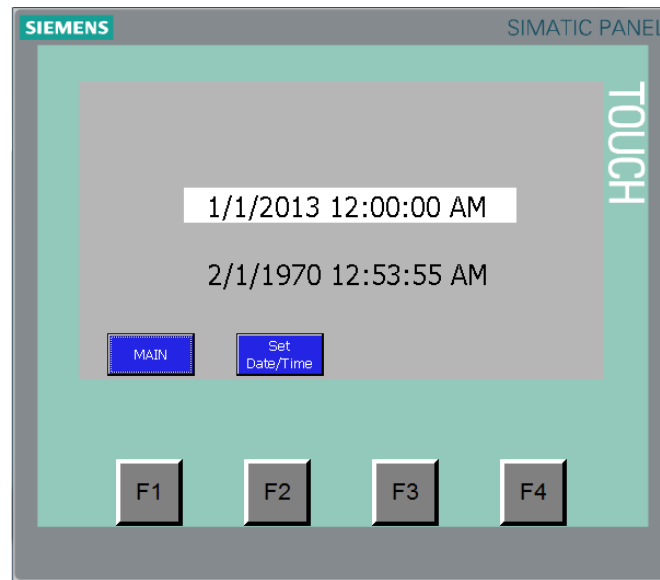


Figure 3-8
DATE AND TIME

2. Press the date and time in the white text box in order to change the date and time setting. The system will ask for a password. See Section 3.5 for instructions.
3. After the password has been entered, the machine will present a pop-up keyboard for setting the fields for year, month, day, hours, minutes, and seconds.
 - Use the pop-up keyboard to input the values.
 - Press Enter to accept the new entry in this box.

3.7 CHANGING THE REACTIVATION OUTLET TEMPERATURE AND PID SETTINGS

Normally, these settings should not need to be changed. **These settings should only be changed by Munters personnel or a trained technician.**

1. To check the reactivation outlet temperature, from the Machine Running screen (Figure 3-3) press the PROCESS SETUP button followed by the R_OUT button to go to the Reactivation PID screen. See Figure 3-9.

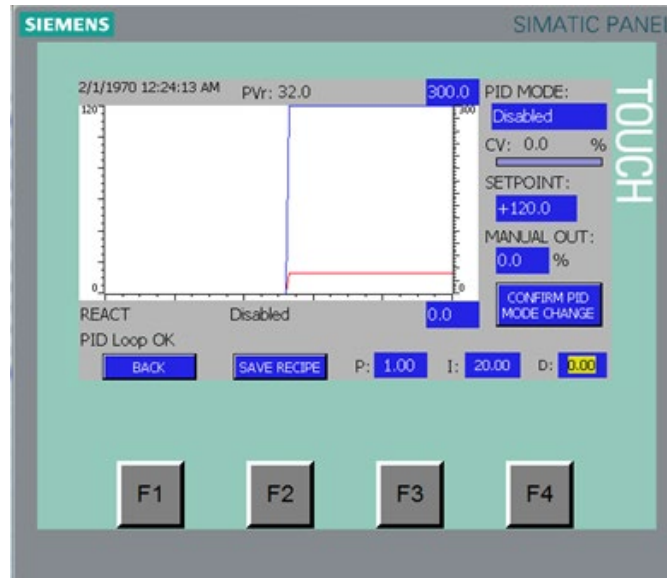


Figure 3-9
REACTIVATION SETPOINT AND PID

2. Press the SETPOINT text box to highlight the field for the setpoint. The system will ask for a password. See Section 3.5 for instructions.
3. Once the display returns to the Reactivation Setpoint and PID screen, use the pop-up keyboard to select any characters to change. Press Enter again to save the changes and return to the Reactivation Setpoint and PID screen.
4. From the Reactivation Setpoint and PID screen (Figure 3-9), press the P, I, or D text box to highlight the corresponding field for the control values. The system will present a screen which sets the PID parameters for the reactivation control system. Use the pop-up keyboard to select any characters to change. Press Enter again to save the changes and return to the Reactivation Setpoint and PID screen.
5. From the Reactivation Setpoint and PID screen (Figure 3-9), press the min or max text box to highlight the corresponding field for chart scale. The system will present a screen which sets the min or max chart scale values. Use the pop-up keyboard to select any characters to change. Press Enter again to save the changes and return to the Reactivation Setpoint and PID screen.

6. From the Reactivation Setpoint and PID screen (Figure 3-9), press the PID MODE text box to highlight the field for the PID mode. The system will present the following options for PID modes:
 - Disabled: Output value of the PID loop is always zero.
 - Pretuning: A PID tuning mode in which the process response is measured in response to a step change in the output value of the PID loop. The calculations are based on the maximum rate of rise and dead time of the controlled system. Successful pretuning will result in a switch to Auto mode upon completion.
 - Fine Tuning: Fine tuning should follow pretuning. This process will minimize the oscillation of the process value based upon the amplitude and frequency of the oscillation. Successful fine tuning will result in a switch to Auto mode upon completion.
 - Auto: The PLC corrects the output value based on the specified parameters of the PID loop.
 - Manual: The user specifies the output value of the PID loop.
 - ChangeReady: The prior mode change has been implemented and a subsequent mode change can be made.

Select the desired PID mode and press Enter. Once back in the Reactivation Setpoint and PID screen, press CONFIRM PID MODE CHANGE to save the mode change.

Additional limits (min / max Reactivation Heat To temperature, Process Fan Delay, etc) can be set from the Advanced Setup menu. To access the Advanced Setup menu from the Reactivation Setpoint and PID screen, press BACK followed by ADVANCED SETUP. In order to change these values, select the field and make the changes as described above.

Note: If Save Recipe is not pressed before exiting, the changed parameters will be lost when the power is turned off.

7. From the Reactivation Setpoint and PID screen (Figure 3-9), press the BACK button, followed by MISC SETUP. The system will present a screen which sets the maximum and minimum values for the system sensors. In order to change these values, select the field and make the changes as described above.

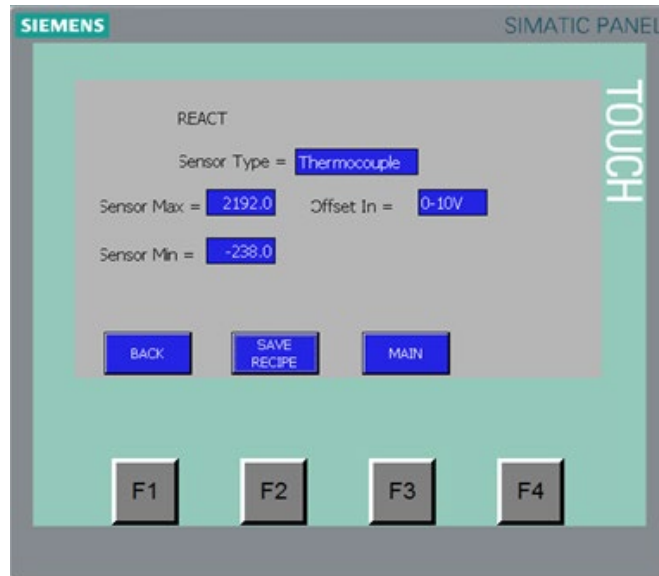


Figure 3-10
TEMPERATURE SENSOR MIN/MAX SETTINGS

3.8 CHANGING THE HUMIDITY SETPOINT AND PID SETTINGS

Some systems are wired so they react to an input from a humidity transmitter. The control system allows checking of the humidity setpoint and changing the values in the PID control loop. Normally, these settings should not need to be changed. **These settings should only be changed by Munters personnel or a trained technician.**

1. To check the humidity setpoint, from the Machine Running screen (Figure 3-3) press the PROCESS SETUP button followed by the F&B button to access the Humidity Setpoint and PID screen. See Figure 3-11.

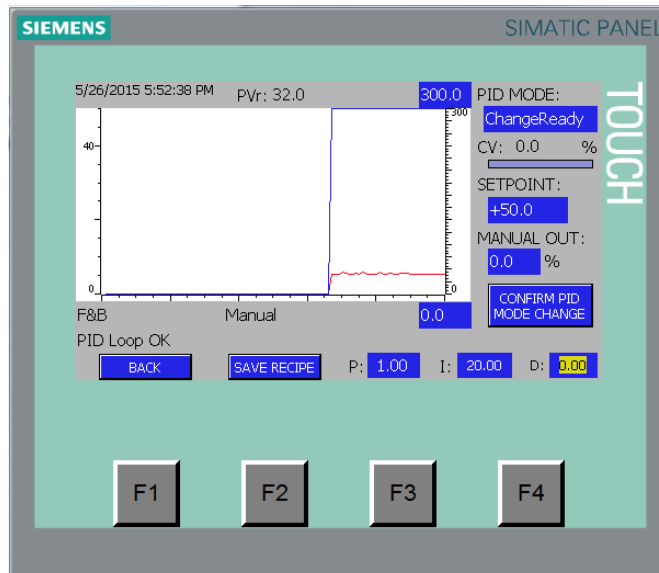


Figure 3-11
HUMIDITY SETPOINT AND PID SCREEN

2. Press the SETPOINT text box to highlight the field for the setpoint. The system will ask for a password. See Section 3.5 for instructions.
3. Once the display returns to the Humidity Setpoint and PID screen, use the pop-up keyboard to select any characters to change. Press Enter again to save the

changes and return to the Humidity Setpoint and PID screen.

4. From the Humidity Setpoint and PID screen (Figure 3-11), press the P, I, or D text box to highlight the corresponding field for the control values. The system will present a screen which sets the PID parameters for the reactivation control system. Use the pop-up keyboard to select any characters to change. Press Enter again to save the changes and return to the Reactivation Setpoint and PID screen.
5. From the Humidity Setpoint and PID screen (Figure 3-11), press the min or max text box to highlight the corresponding field for chart scale. The system will present a screen which sets the min or max chart scale values. Use the pop-up keyboard to select any characters to change. Press Enter again to save the changes and return to the Reactivation Setpoint and PID screen.
6. From the Humidity Setpoint and PID screen (Figure 3-11), press the PID MODE text box to highlight the field for the PID mode. The system will present the following options for PID modes:
 - Disabled: Output value of the PID loop is always zero.
 - Pretuning: A PID tuning mode in which the process response is measured in response to a step change in the output value of the PID loop. The calculations are based on the maximum rate of rise and dead time of the controlled system. Successful pretuning will result in a switch to Auto mode upon completion.
 - Fine Tuning: Fine tuning should follow pretuning. This process will minimize the oscillation of the process value based upon the amplitude and frequency of the oscillation. Successful fine tuning will result in a switch to Auto mode upon completion.
 - Auto: The PLC corrects the output value based on the specified parameters of the PID loop.
 - Manual: The user specifies the output value of the PID loop.
 - ChangeReady: The prior mode change has been implemented and a subsequent mode change can be made.

Select the desired PID mode and press Enter. Once back in the Humidity Setpoint and PID screen, press CONFIRM PID MODE CHANGE to save the mode change.

Note: If Save Recipe is not pressed before exiting, the changed parameters will be lost when the power is turned off.

3.9 CHANGING THE SETPOINTS AND PID SETTINGS FOR OTHER CONTROL LOOPS

Dehumidification systems may be set up in many different ways, and include different types of equipment installed both upstream and downstream from the dehumidifier. If the system includes some of these components, the control system for the unit will include controls for them.

When the machine is operating normally, none of the settings within these controls will need to be checked or changed. **These settings should only be changed by a Munters technician or other qualified personnel.**

1. From the Machine Running screen (Figure 3-3) press the PROCESS SETUP button to go to the control setup screen. From this screen, the following control functions can be accessed:

F&B ----- Face and bypass
HUM ----- Humidifier PID control
PAF ----- Process outflow PID control
PC1 ----- Precool 1 PID control
PC2 ----- Precool 2 PID control
PH1 ----- Preheat 1 heater PID control
PH2 ----- Preheat 2 heater PID control
POC ----- Postcool PID control
POH ----- Postheat PID control
R_OUT ----- Reactivation out PID control
REACT HeatTo ----- Reactivation heated to PID control
RPC ----- Reactivation precool PID control

2. To make changes to the PID variables, follow the procedure previously described for the reactivation heater control loop (Section 3.7).

3.10 REACTING TO ALARMS

Note - The operator interface display reports on operating conditions inside the dehumidifier, but they are really two separate units. If an alarm is triggered it must be corrected in two places - on the dehumidifier itself, and also on the operator display.

1. If the unit detects a fault condition, the ALARMS button on the display will blink red (See Figure 3-12) and the Fault Light on the control panel will light. For a warning, the ALARMS button on the display will blink yellow and the Warning Light on the control panel will light.

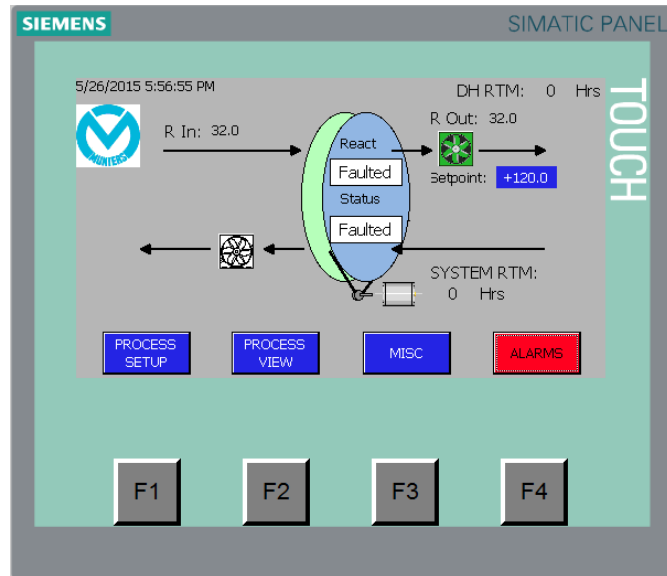


Figure 3-12
TYPICAL FAULT DISPLAY

2. To see the details on the fault or warning, press the ALARMS button. Figure 3-13 shows the type of screen that the display might present.

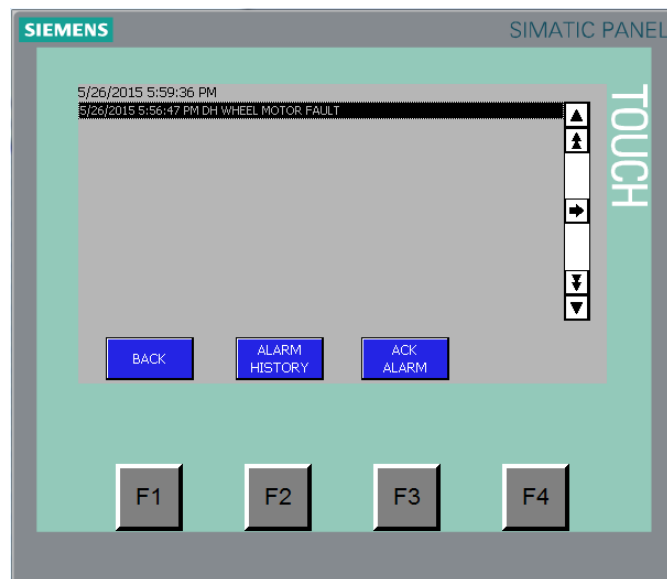


Figure 3-13
INFORMATION ON FAULT CONDITION

3. Recovering from a fault is a three-step process. The fault must be cleared on **both** the dehumidifier and the operator interface display.
 - On the dehumidifier, locate the cause of the problem that caused the fault, and correct it.
 - On the dehumidifier, reset the selector switch (Auto/Off/Manual) by moving the switch to the Off position, then returning it to Auto or Manual.
 - On the operator interface display, press the Acknowledge button (ACK ALARM) to clear the message and alarm symbol.
 - After a fault message has been cleared, return to the Main screen by pressing

BACK. The display will indicate "Ready." For more information, see Section 7 - TROUBLESHOOTING.

4. To see a history of past alarm conditions, press the ALARMS button followed by ALARM HISTORY. See Figure 3-14

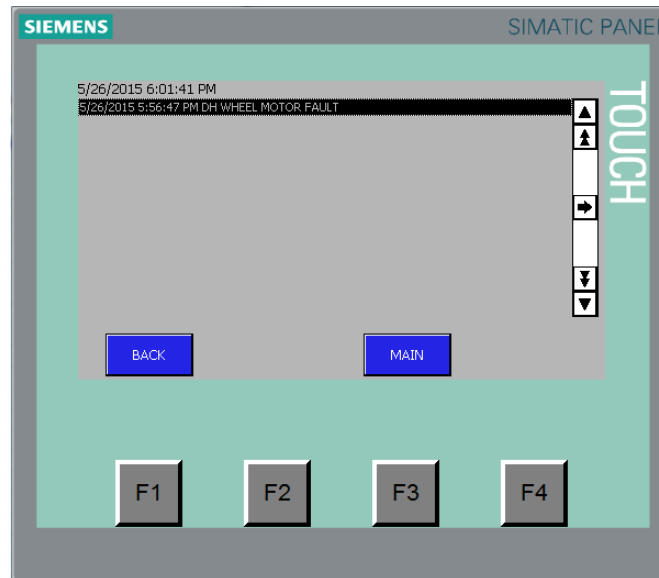


Figure 3-14
ALARM HISTORY DISPLAY

3.11 STOPPING THE UNIT

1. Turn the Auto/Off/Manual switch to Off.
2. After the switch has been set to Off, the unit will continue to run for three minutes. This allows time to cool the reactivation heater. As the machine does this, both the React and Status lines in the operator display will indicate Purging. After three minutes, the unit will stop, and the display will indicate Ready.

NOTICE

Other than emergency situations, do not shut down the system by de-energizing the main power disconnect. De-energizing the main power disconnect will cause all components to immediately shut down and power to be isolated. A hard shut down will damage the system, as the components will not be cooled down in a controlled manner.

3. Some units have a disconnect switch on the front panel. On a unit of this type, turn off the disconnect switch.

3.12 LONG-TERM SHUTDOWN

If the machine will be turned off for a long period of time, some special precautions must be taken to protect the HoneyCombe® wheel. Even when the machine is shut down, moist air can continue to move through the wheel. This is especially likely if a fan or airmoving device is still operating in the system. Eventually, a large amount of moisture can collect in the wheel. This can damage the wheel. For instructions on long-term shutdown, see Section 6 - MAINTENANCE.

4 - PRINCIPLES OF OPERATION

4.1 DEHUMIDIFIER OPERATING PRINCIPLE

Figure 4-1 shows how the HCD unit removes moisture from the air. The “heart” of the system is the HoneyCombe® wheel. The detail in Figure 4-1 shows how this wheel is made. The wheel has a series of air passages or channels arranged in a honeycomb pattern. Air can be forced through these channels in either direction.

The passages inside the wheel are impregnated with a special substance called a “desiccant.” When this substance contacts damp air, it soaks up moisture. When the desiccant is heated, it releases the moisture. Figure 4-2 shows how this principle is used in the HCD unit. An example is using the HCD unit to dry the air in a storage room. Damp “process” air is pulled into the unit from the storage room. This air is drawn through the channels in the HoneyCombe® wheel, as shown in the detail. The desiccant picks up most of the moisture in the air. Once it has been “dried out,” the process air is ducted back into the storage room. At this point, the moisture has been taken out of the process air, and “stored” in the HoneyCombe® wheel.

The next job is to remove this moisture from the wheel. As previously mentioned, the desiccant will give up moisture when it is heated. Hot air can hold a large amount of moisture. When hot air reaches the desiccant, and the moisture is released and the desiccant is “reactivated.” In the HCD unit, a stream of “reactivation” air is taken from the outdoors and heated. The normal temperature range of the reactivation air is from 190° to 300°F. Heat input is provided by electric heating elements (“electric reactivation”).

The heated air is pulled through the channels in the HoneyCombe® wheel. The desiccant releases the moisture into the heated air stream. Finally, the damp reactivation air is vented outside. At this point, the moisture has been moved from the storage room to the wheel, then from the wheel into the outside air. The process continues until the HCD unit is turned off.

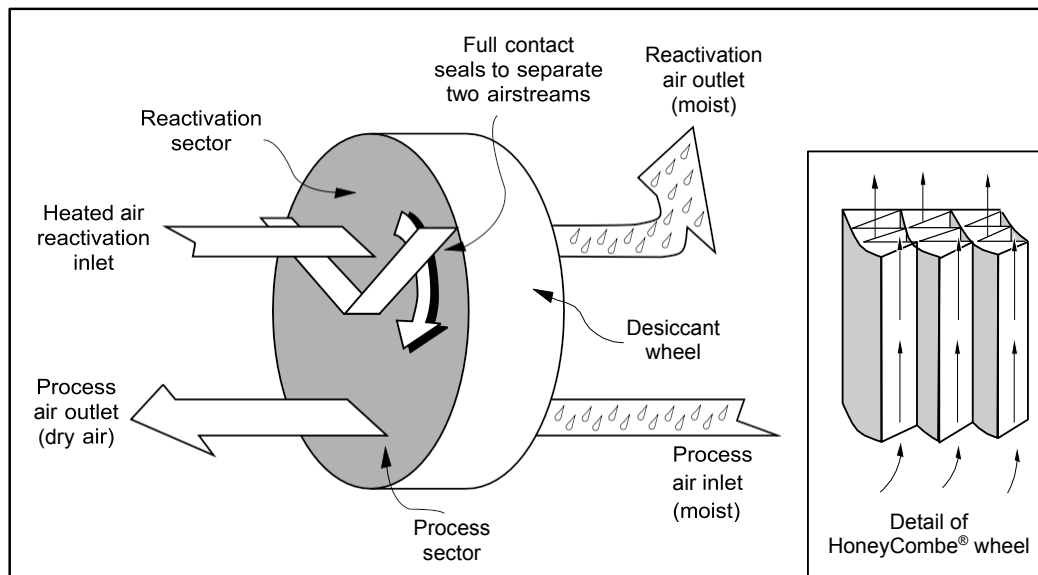


Figure 4-1
DESICCANT DEHUMIDIFIER OPERATING PRINCIPLE

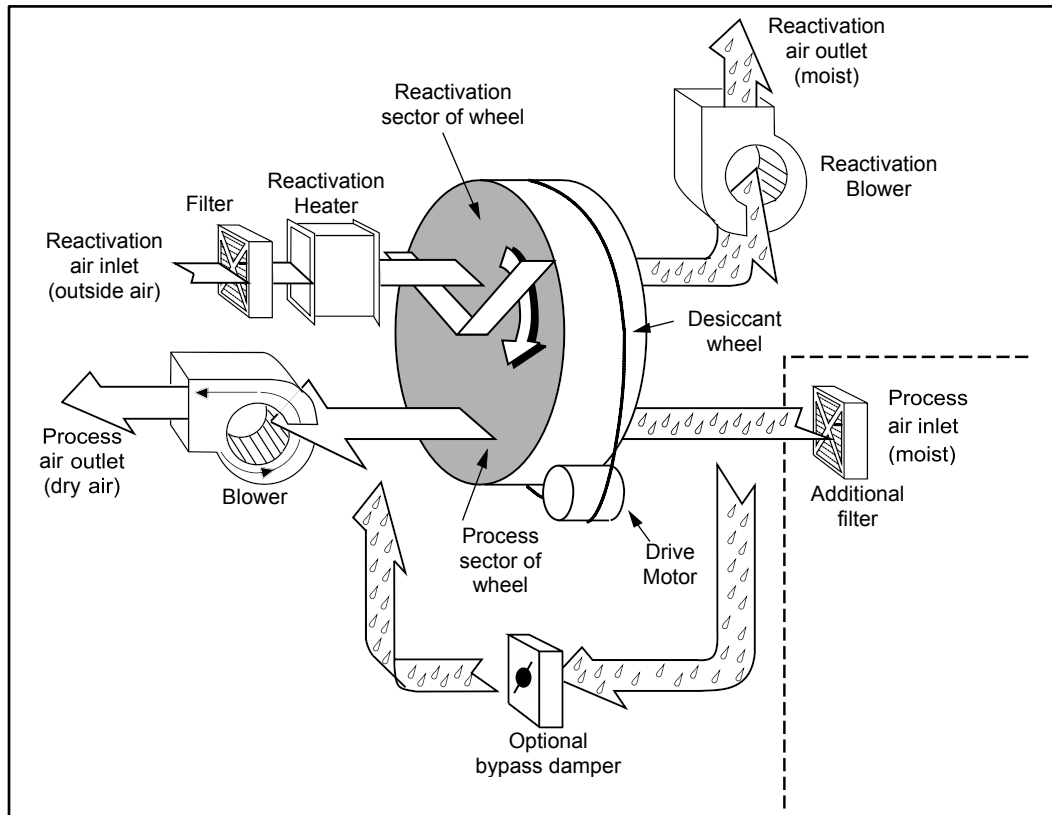


Figure 4-2
COMPLETE OPERATING UNIT

Figure 4-1 and Figure 4-2 show that, at one moment, the wheel is being used to pick up moisture, and a moment later, the wheel is being heated to drive off the moisture. In the HCD unit, both actions are happening at the same time, on different sections of the wheel. All sections of the wheel go through the following cyclic process:

- At the beginning of the process, this section of the wheel is exposed to the damp “process” air.
- As the wheel turns, this section of the wheel moves under the stream of heated “reactivation” air. This part of the wheel is then dried out by the heated air.
- Finally, the dried part of the wheel turns into the “process” area again so it can pick up more moisture.

During this cycle, the wheel does not stop. One side of the wheel is picking up moisture while the other side is being dried out. The wheel rotates quite slowly – 6 to 12 revolutions per hour.

The HoneyCombe® wheel has been carefully designed. The honeycomb material is non-metallic and does not react to bacteria. Three different types of desiccant material may

be used in this model. See the following table:

<i>Desiccant Type</i>	<i>Color</i>	<i>Marking</i>
Lithium chloride	Gray or black	Li2
Titanium enhanced silica gel	Red	Tigel
High performance	Silver/gray	GTR3
Low dew point	Green	HPX

Each type of wheel has a different way of collecting moisture from the air. A lithium chloride wheel uses a principle called “**absorption**.” The lithium chloride goes through a physical change as it picks up water. If enough water collects, the lithium chloride will dissolve. At this point, the structure of the wheel may be damaged by the large amount of moisture.

A wheel using silica gel or molecular sieves depends on the “**adsorption**” principle. As the moisture collects, the chemical does not go through a physical change. Each particle of chemical can bond with a large number of water molecules. Each water molecule is attracted to many sites on the outside of the silica gel molecule. A molecular sieve wheel uses a similar principle. In this type of wheel, the water molecules are attracted to sites within a crystalline structure.

In each type of wheel, the process can be reversed by adding heat. As the desiccant is heated, it releases water to the air. Several other parts must be added to make up a working machine. Two blowers may be provided: one for the process air stream and another for the reactivation air. A drive motor is included to drive the wheel.

4.2 ABOUT THE HCD UNITS

This is a simplified explanation of the operating principle. Further explanation illustrates how this principle works in the HCD unit. Figure 4-2 shows most of the important parts of the machine: the HoneyCombe® wheel, process air stream and reactivation air stream. However, additional parts are required for the unit to function. For example, the HCD unit has two seals, mounted on the wheel plates on each side of the HoneyCombe® wheel. These seals separate the two streams of air – the damp process air, and the heated reactivation air. Notice that the two air streams move through the HCD unit in opposite directions (“counter flow”). The HoneyCombe® wheel is turned by a small drive motor and a belt. A spring-type tensioner automatically adjusts the belt tension.

The process air is pulled through the wheel by a blower. (Notice that the blower is on the downstream side of the wheel – it “pulls” rather than “pushes.”) A damper is included downstream of the blower. By adjusting this damper, the amount of process air moving through the unit can be changed. This adjustment is important, and it will be discussed again in the section on “Installation.”

The reactivation side of the system has some of the same features. The air is pulled through the wheel by a second blower. Again, a damper is included to adjust the reactivation air flow. A filter is included to keep dirt from the reactivation air from plugging the wheel. Just upstream of the wheel, the reactivation system includes an energy source. The heat energy is provided by electric heating elements. These heaters provide the heat needed to “reactivate” the desiccant.

4.3 OPTIONAL BYPASS

This unit may be equipped with an optional bypass duct. See Figure 4-3. The HoneyCombe® wheel is sized to handle a certain volume of process air. When the bypass option is added, the unit can pass more process air than the wheel can handle

directly. (Refer to the “Technical Data Sheet” provided with this manual for the process and bypass air volumes.)

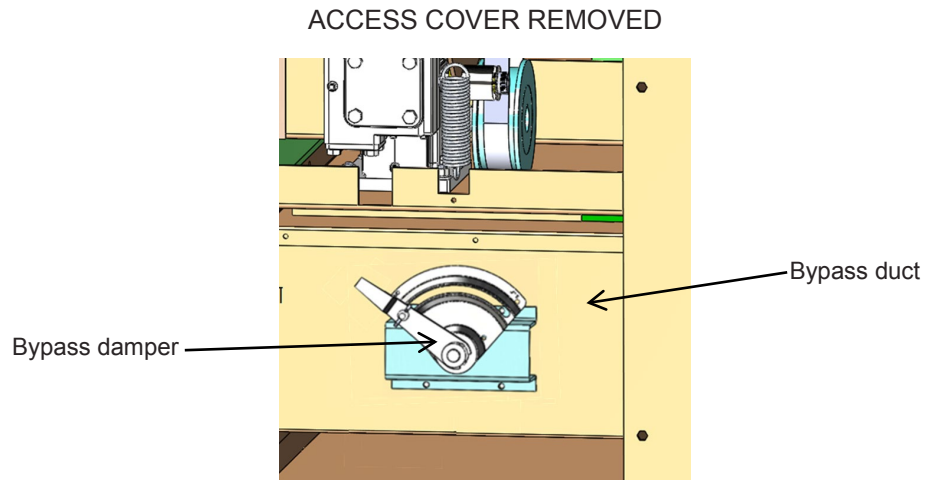


Figure 4-3
OPTIONAL BYPASS DUCT

4.4 CONTROL SYSTEM AND POWER DISTRIBUTION

Figure 4-4 shows the control system used on a unit with electric reactivation. The HCD unit is controlled by a “programmable logic controller” or “PLC.” The PLC is a small computer which executes a simple set of commands or “program.” The program is organized as a loop which repeats continuously. LEDs on the front of the PLC show the status of each of the inputs and outputs. For more information on these LEDs, see the section on “Troubleshooting.”

Different variations of the unit operate on various voltages of three-phase AC. The blowers use this voltage. Motor starters are provided for each of the blower motors. On a unit with electric reactivation, the PLC changes or “modulates” the power going to one zone of heating elements. Larger machines have additional heating elements, and additional contactors are included for these. A transformer provides 115V AC, which is used by the controls and indicators in the unit.

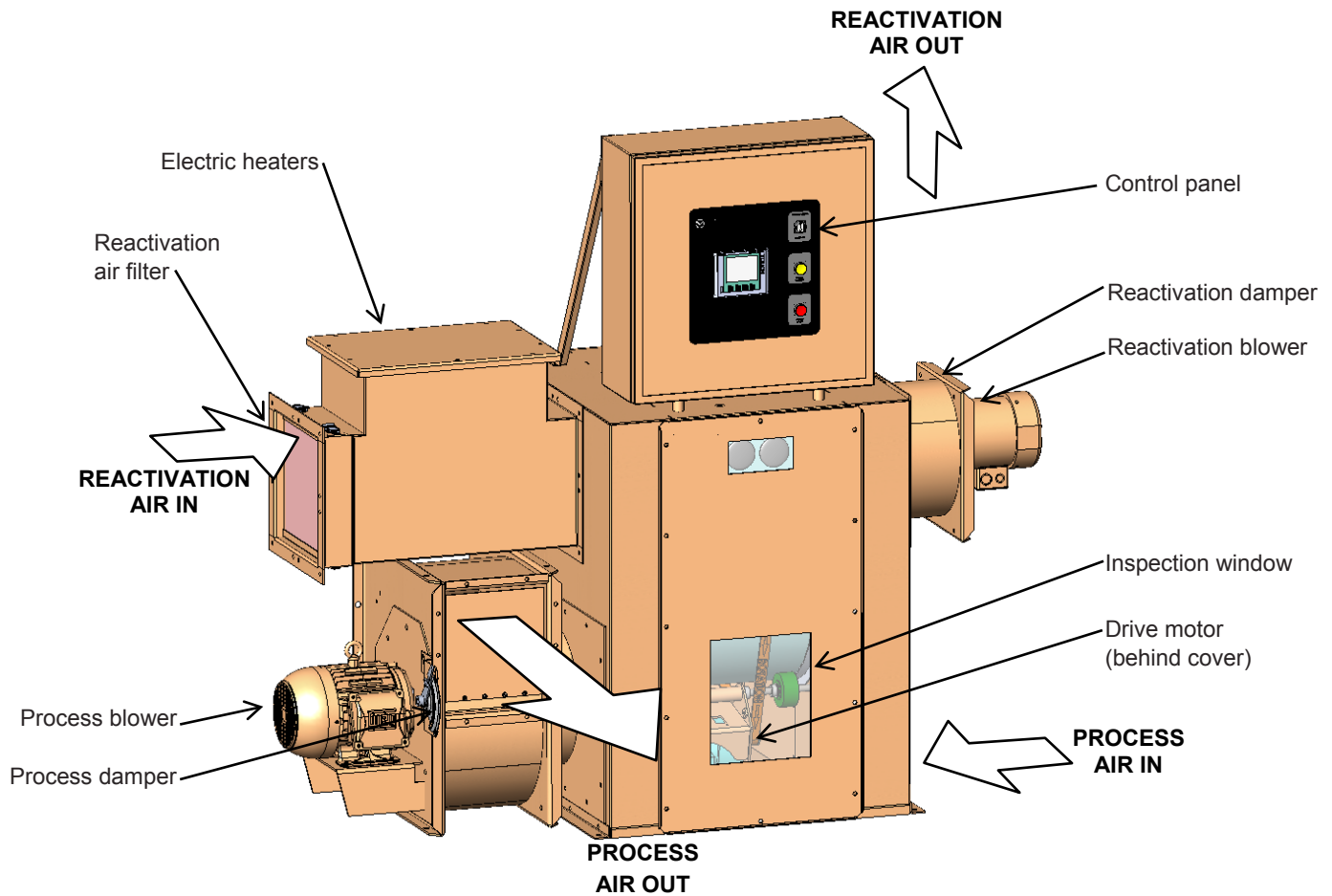


Figure 4-4
UNIT WITH ELECTRIC REACTIVATION - TYPICAL

If a blower motor is jammed, it will start to draw a large amount of electrical current. Each of the motor starters can detect this condition, and shut down the faulty motor automatically. This will also shut down the rest of the unit. Each motor starter can be reset by pressing the gray or red "Reset" button on the starter.

4.5 ENERGY MODULATION

The reactivation heaters are sized to provide enough heat to remove the maximum amount of moisture the unit was designed to handle. The HCD unit is efficient because the heat input can be changed or "modulated." The heaters will always supply enough heat to keep the wheel in "equilibrium." This means that the wheel releases moisture at the same rate it absorbs it. Because the heat input can be modulated, less energy is used when the moisture load drops.

As the reactivation air passes through the wheel, the air temperature drops. (the moisture "cools off" the reactivation air as it evaporates.) Tests have shown that when the wheel is in equilibrium, the ideal temperature at the reactivation outlet is 120°F*.

* - Another value may be listed on the Tech. Data sheet.

There are four factors which affect the temperature at the reactivation outlet:

- Changes in temperature at the reactivation inlet
- Changes in the flow of reactivation air
- Changes in moisture loads in the process air stream
- Changes in the amount of heat provided by the electric heaters

If the temperature at the reactivation outlet is a bit higher than 120°F*, this does not

necessarily indicate a problem. A low outlet temperature indicates a potential problem.

* - Another value may be listed on the Tech. Data sheet.

There are four possible causes for a low outlet temperature:

- Large drop in temperature at the reactivation inlet
- Large drop in flow of reactivation air
- Large increase in moisture load in the process air stream
- Reduced output from the electric heaters

The controller will not allow the machine to operate for over 30 minutes if the temperature drops below 110°F*.

* - Another value may be listed on the Tech. Data sheet.

The reactivation heaters are designed to produce enough heat to handle the full design moisture load. If the moisture load is less than maximum, the amount of energy input may be reduced. The amount of energy used is controlled or “modulated” to maintain a constant 120°F* at the output of the reactivation air stream. Less moisture absorbed by the wheel means less evaporative cooling taking place in the reactivation air stream. This means that less heat is required to maintain 120°F* at the reactivation outlet. Accurate energy modulation saves energy whenever the unit is handling less than the full load. The reactivation air entering the wheel must never rise above 325°F*. High temperatures can damage the wheel.

* - Another value may be listed on the Tech. Data sheet.

4.6 FACE AND BYPASS OPTION

This option helps to provide precise humidity control. (It is also used for other purposes, but this section will concentrate on humidity control.) The optional equipment includes two dampers: the face damper, which controls airflow through the desiccant wheel, and the bypass damper, which controls airflow through a bypass duct. The bypass duct provides a path for airflow around the desiccant wheel. Usually it is sized to handle the same amount of air that flows through the desiccant wheel. See Figure 4-5.

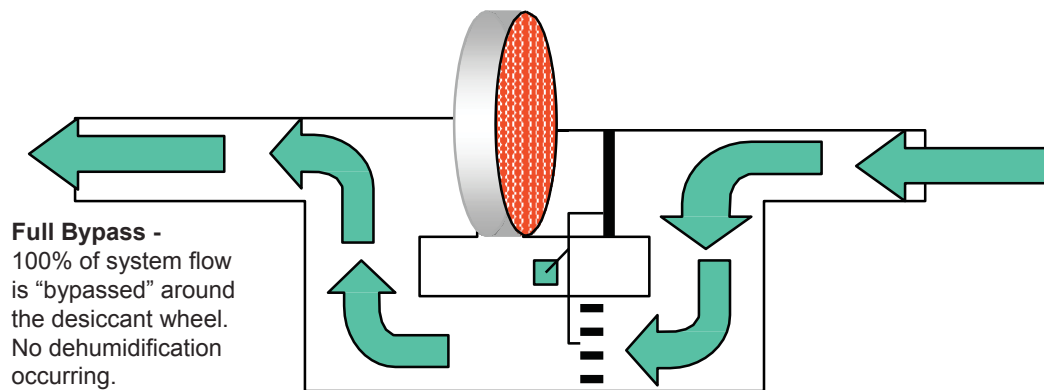
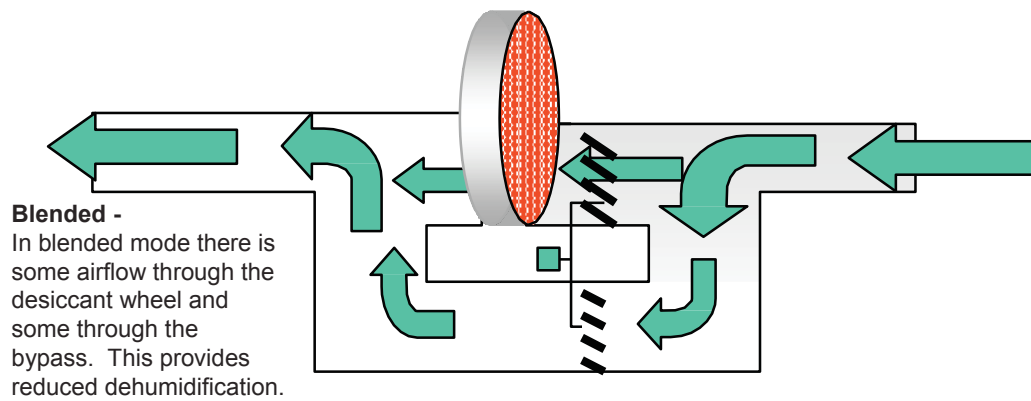
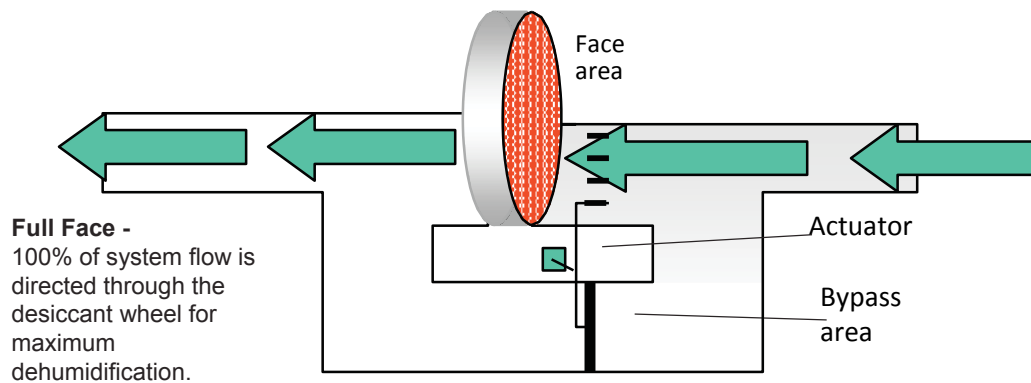


Figure 4-5
OPERATION OF FACE AND BYPASS EQUIPMENT

5 - INSTALLATION

5.1 INSPECTION

1. When the unit arrives, check it immediately for signs of shipping damage. If there is any noticeable damage, report it to the trucking company right away.
2. Check the outside of the unit. Open the damper for the reactivation blower. Try to turn the blower by hand. It should turn easily. Do the same with the process blower.
3. Remove the access cover on the front of the unit. Check the following items:
 - Check to be sure the HoneyCombe® wheel is in position. The wheel should be evenly supported on four rollers.
 - Check the seals around both faces of the HoneyCombe® wheel for any obvious signs of damage.
 - Check to be sure the drive belt is in place. The drive belt and rollers should be free of grease.
 - Remove the drive belt from the drive pulley. Try to turn the HoneyCombe® wheel by hand. It should be able to turn it with some resistance. The wheel will turn in one direction more easily than the other. Replace the drive belt on the drive pulley.
 - Check to be sure the reactivation air filter is in place.
4. Check the electrical panels and controls:
 - Check for any signs of damage.
 - Check the tightness of all electrical connections. Carefully check the power wiring terminals.
 - Be sure all control relays are firmly seated in their sockets.

5.2 POSITIONING THE UNIT

1. Place the unit on a flat, level surface. A dimension drawing is included in this manual. Several clearances must be allowed around the unit:
 - Allow enough space for removal and replacement of the HoneyCombe® wheel. Allow the space specified on the General Arrangement drawing (within this manual) around the machine.
 - If the process air intake is operated without ductwork, allow a clearance of at least 12" in front of the intake. Also allow 12" for the reactivation openings.
 - Allow a space of at least 15" on one of the access sides of the filter at the reactivation inlet. The filter is removed by lifting it upward.
 - If other filters are added to the machine, allow space to remove these filters.
2. Figure 5-1, Figure 5-2, Figure 5-3, and Figure 5-4 show four different ways of installing the HCD unit.

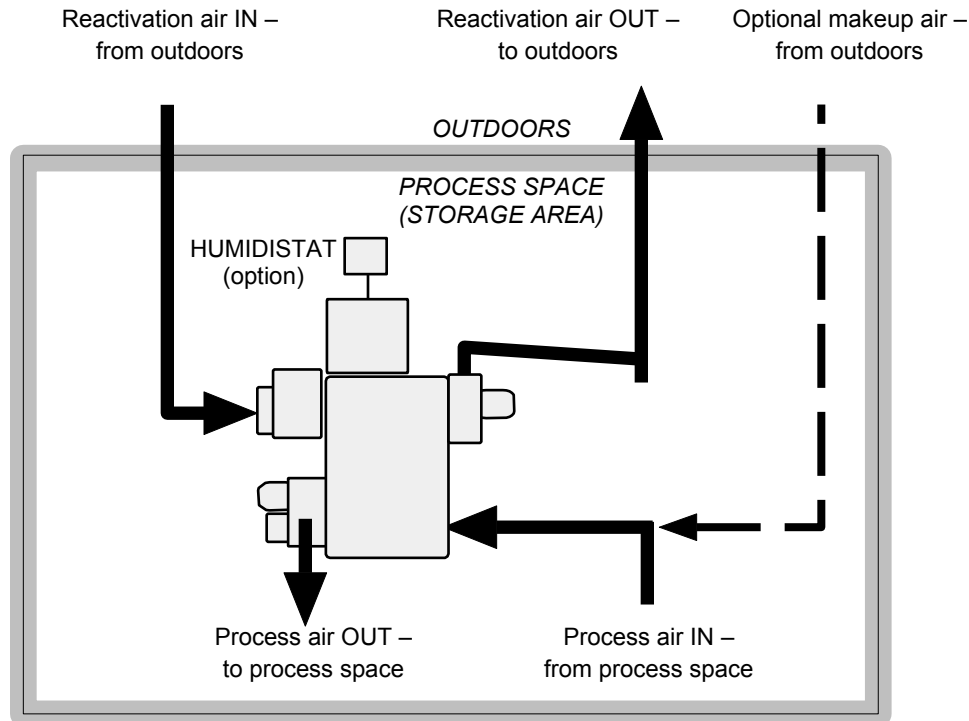


Figure 5-1
INSTALLATION IN PROCESS SPACE

3. To minimize condensation on the outside of the unit when it is not running, the dehumidifier can be placed inside a cool space, rather than outdoors. This will reduce the amount of condensation forming on the outside of the unit. See Figure 5-1 thru Figure 5-4.
4. The HCD unit handles two separate streams of air:
 - The process air, which comes from the space which is to be dehumidified
 - The reactivation air, which comes from outside of the space to be dehumidified. This air is used to remove the moisture from the HoneyCombe® wheel inside the unit.

The process air intake should come from the conditioned space, and the process air outlet should be vented to the same space. The process air intake can be quite close to the machine. In fact, it can be operated without any ductwork on the process intake. (The other openings will still need ductwork, however.) If the two process air ducts have different lengths, and one of the ducts must run a long way, the long duct should handle the process output air.

The reactivation air should come from outside of the conditioned space. There is no point using dehumidified air for this purpose. The air at the reactivation outlet is very damp and should be vented to a space where this is not a problem. Even though this air is heated, it is too damp to use for most space heating purposes. Normally, the reactivation air is taken directly from the outdoors, and the damp air is returned outside. The reactivation air can also be taken from and returned to an indoor space where the temperature and humidity levels are not important.

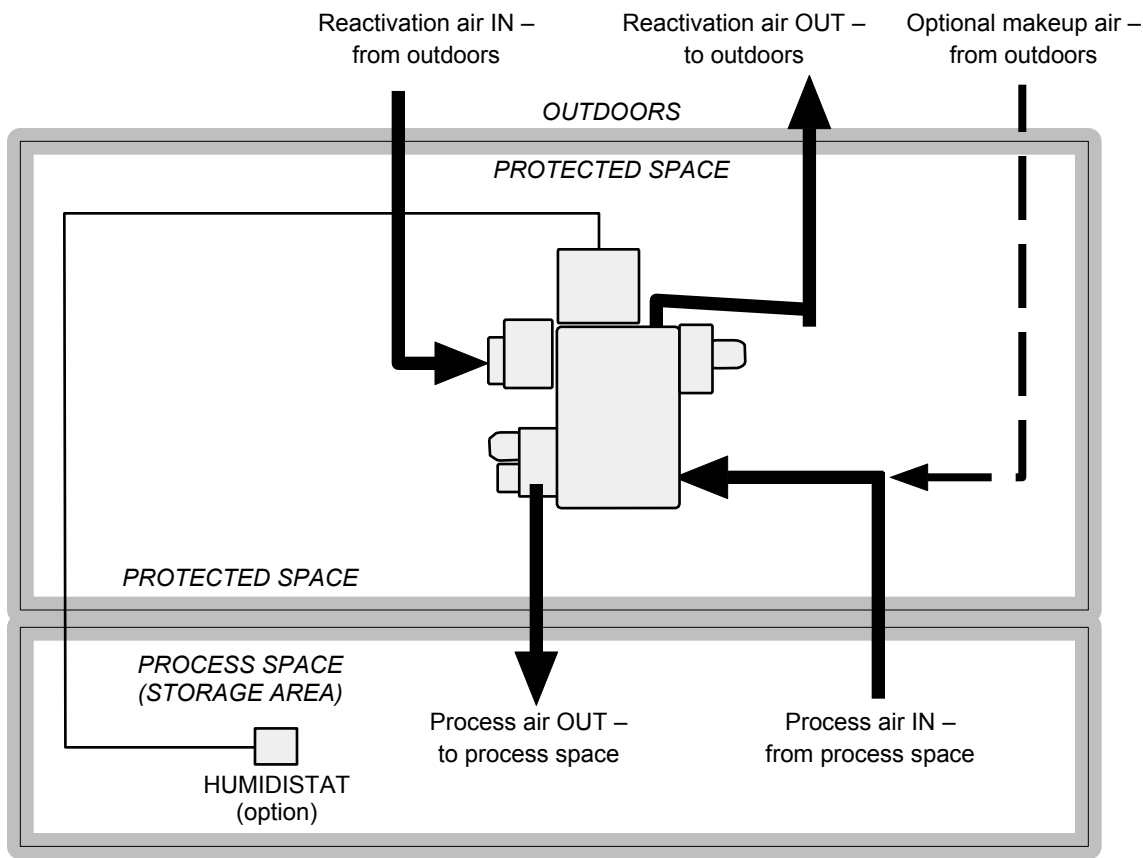


Figure 5-2
INSTALLATION IN PROTECTED SPACE

5.3 CONNECTING THE DUCTWORK

1. Do not try to operate the unit without ductwork. The ducts shield the blowers from possible access by workers' hands. The ducts also separate the reactivation and process air streams.
2. The general arrangement drawing shows the details of the duct connections. This drawing is included at the back of this manual.

<i>Duct Connection</i>	<i>Note</i>
Process intake	If no ducting – allow at least 12" clearance
Process outlet	If no ducting – allow at least 12" clearance
Reactivation intake	Allow at least 12" clearance
Reactivation outlet	Allow at least 12" clearance

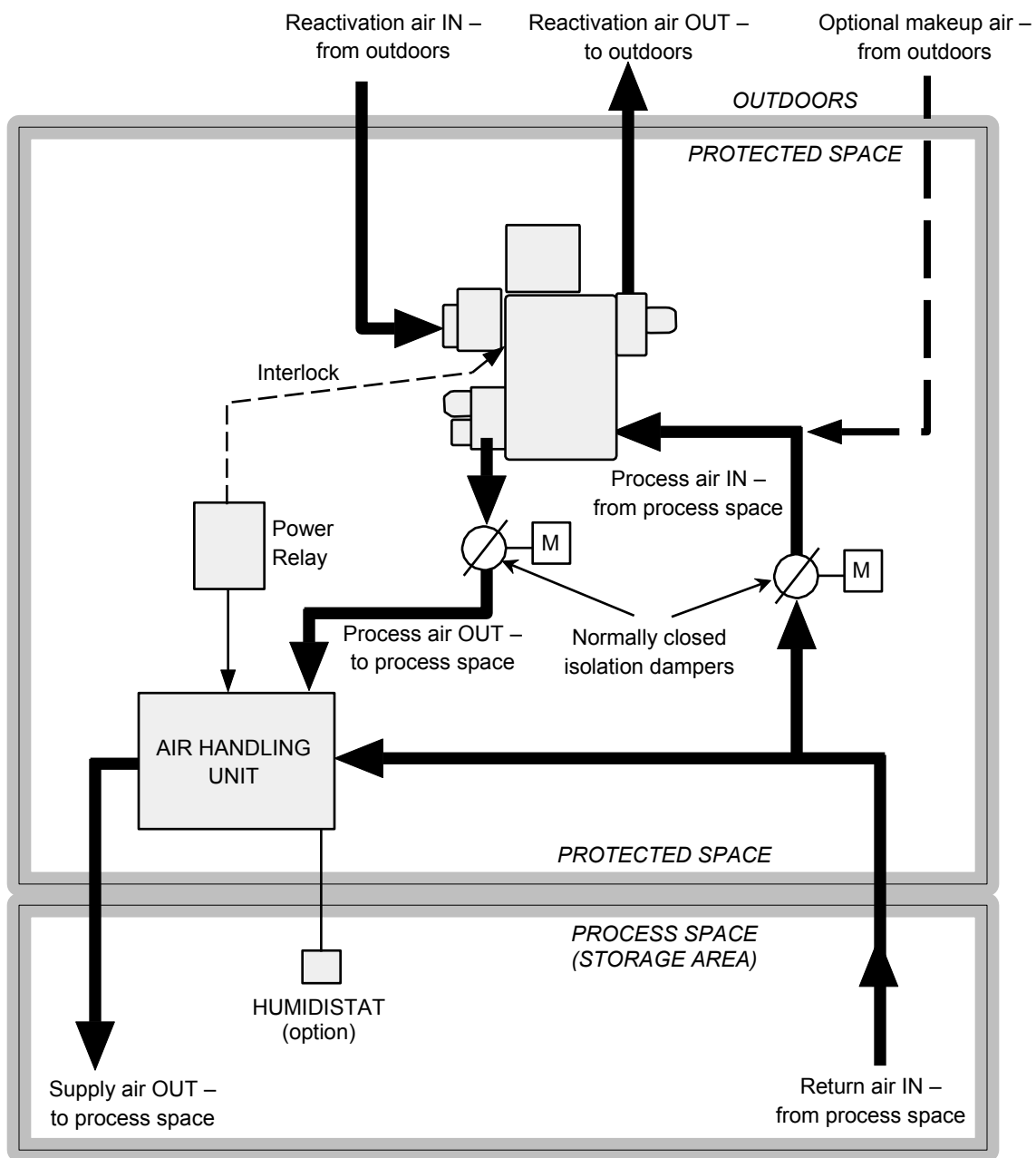


Figure 5-3
INSTALLATION WITH EXISTING AIR HANDLING UNIT

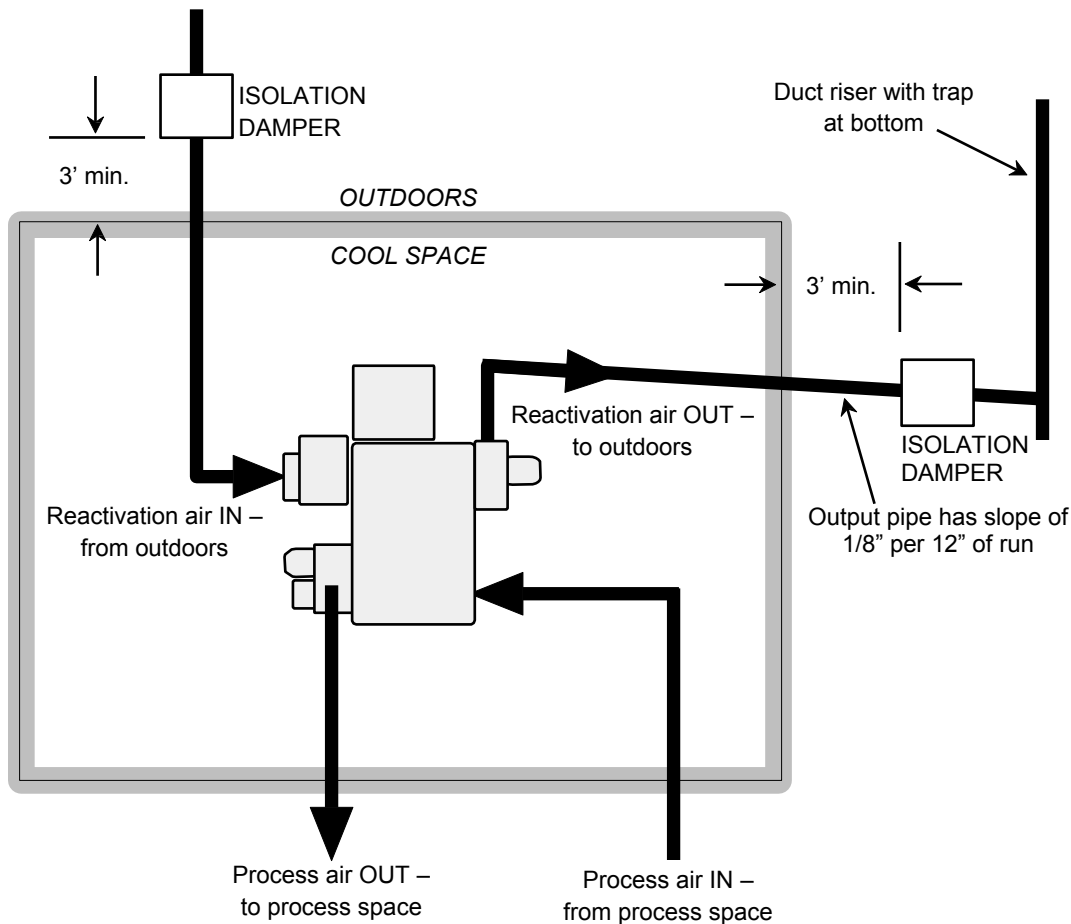


Figure 5-4
INSTALLATION INSIDE COOL SPACE

3. By planning and installing the ductwork carefully, problems with condensation can be avoided inside the ducts. Always place any horizontal runs of ductwork so they slope away from the unit. Allow a slope of 1/8" per 12" of run. Before any vertical run, create a trap which can be drained.

NOTICE

Never allow condensation from the ducts to run back into the HoneyCombe® wheel. This will prevent the unit from operating as it should.

4. If the unit is installed in a cold space, air from the reactivation air stream must not be allowed to flow into the HCD unit when the unit is turned off. This moist air would condense inside the unit. To prevent this, install isolation dampers at two points in the reactivation air stream – one damper in the intake and the other in the exhaust. The dampers should be installed outside of the conditioned space.

BAD ARRANGEMENT

Do not set up the dehumidifier so it bypasses the air-handling unit. The air-handling unit can create back-pressure. This can cause the air to flow backwards through the dehumidifier when it is off.

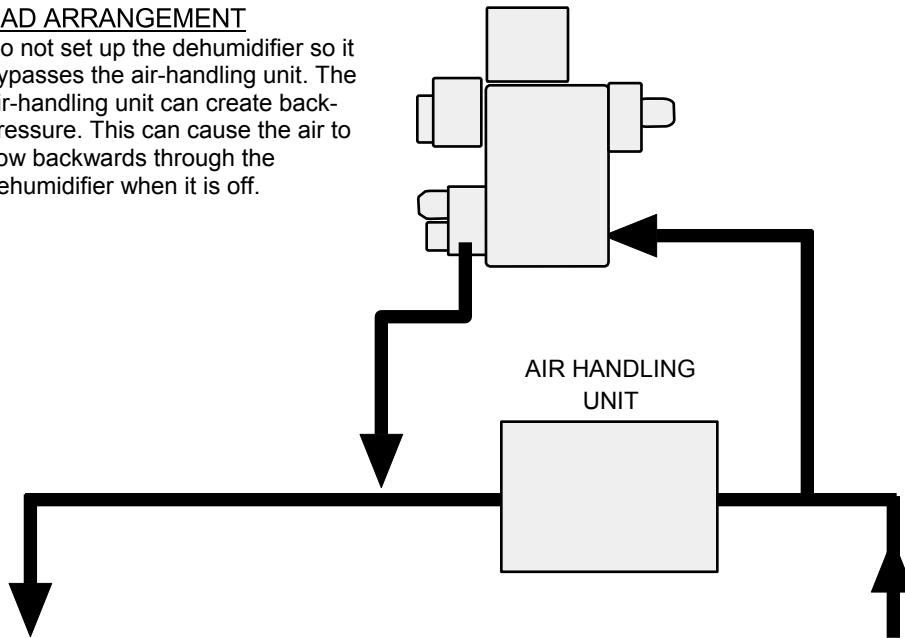


Figure 5-5
INCORRECT INSTALLATION

BAD ARRANGEMENT

Do not set up the dehumidifier on the downstream side of the air-handling unit. When set up this way, the air-handling unit cannot condition the air after it leaves the dehumidifier.

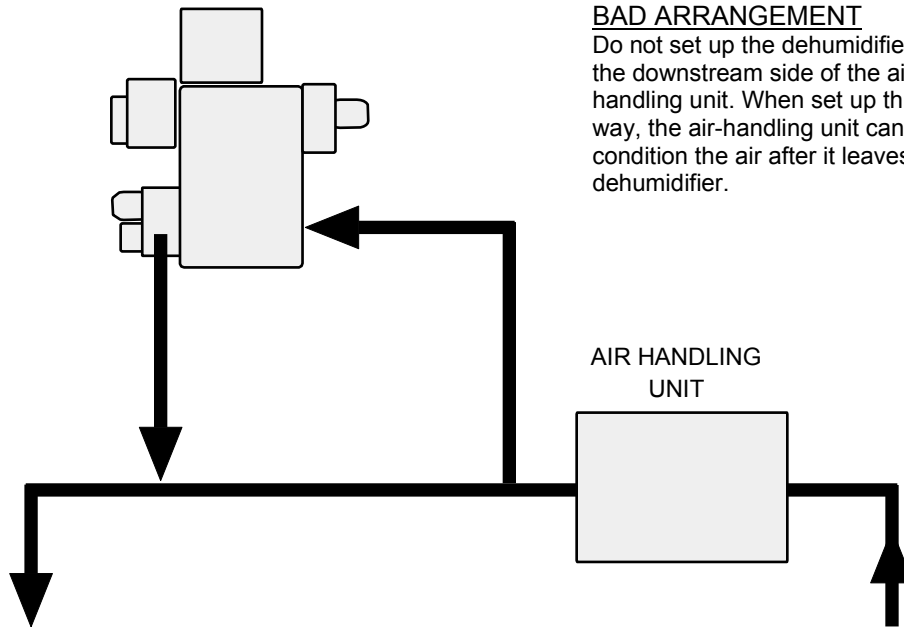


Figure 5-6
INCORRECT INSTALLATION

5. It is very important not to allow any leaks on the output side of the process air stream. Leaks can allow the moisture from the process space to move into the process airstream, even though the moisture must move “upstream” against the air flow. It is best to weld the joints in this ductwork.

It is also important that the hot, moist reactivation air not be allowed to leak back into the conditioned space. All of the joints for the reactivation ducts must be completely sealed.

6. After the reactivation air passes through the unit, it is very hot and can put a heat load on the conditioned space. If any ductwork for the reactivation output air runs through the conditioned space, insulate these sections. This will help prevent condensation from forming inside the duct. It is also helpful to insulate the ducts for the process output air. Seal the joints on these ducts before adding the insulation.
7. Wherever the intake or outlet ducts open outdoors, protect them from the elements. Install weather hoods and bird screens.
8. Do not locate the intake and outlet for the process air too close together. This would allow the dried process outlet air to be pulled into the process inlet. The machine would be trying to dry air which has just been dehumidified (“short-cycling”).

Also allow some distance between the intake and outlet for the reactivation air. The reactivation outlet air is very damp – do not allow it to be pulled into the reactivation inlet. Either condition can make the unit work harder than necessary. The unit handles large volumes of air, so try to maintain a good distance between the inlet and outlet ducts.

9. Figure 5-3 shows a possible setup if the HCD is being installed in a system with an existing air-handling unit. Notice that both sides of the HCD unit are connected “upstream” of the air-handling unit. Do not connect the HCD unit so that it bypasses the air-handling unit. See Figure 5-5. If the ductwork is connected this way, some of the air from the air-handling unit may be forced back through the HCD unit when the dehumidifier is turned off.

Both sides of the HCD unit may be connected “downstream” of the air-handling unit, as shown in Figure 5-6. The arrangement shown in Figure 5-3 is better, however. This setup allows the air-handling unit to heat or cool the processed air after it leaves the HCD unit.

Install a control interlock between the HCD unit and the air-handling unit, as shown by the dotted line in Figure 5-3. The control system must follow three rules:

- The dehumidifier may operate whenever the air handling unit is operating.
- The air handling unit must shut down if the dehumidifier faults.
- When the air handling unit stops, the dehumidifier must shut down after being allowed to complete its three minute cool down purge. This allows the heaters and wheel to cool to safe levels.

For the control system to operate this way, the air handling unit must be able to sense whether the dehumidifier is running. The wiring diagram included in this manual shows the contacts which may be used to turn on the air handling unit.

Do not set up the power wiring so that the HCD unit is turned on and off directly by the air handling unit. The HCD unit must be able to switch itself on and off when

necessary, independent of the power switching for the air handling unit.

10. On some installations, “makeup” air is taken from outside the process space, and added to the process air stream. Unconditioned makeup air can add a moisture load to the HCD unit, and this can overload the unit. For recommendations, consult the facility engineer or the Service Operations Division at Munters.

5.4 ELECTRICAL CONNECTIONS

1. The HCD unit is designed to operate on three-phase AC at up to 575V. The required voltage is listed on the nameplate on the machine. Be sure the line voltage matches the voltage required by the machine. A wiring diagram for this unit is included in this manual.



The HCD unit may be wired for up to 575V AC. The unit can produce enough voltage and current to cause serious injury or death. Always turn off the power before working inside the unit. Follow standard lock-out tag-out procedures. Only trained electricians should perform work with the electrical parts.

Be sure the unit is connected to a source with the correct line voltage. If a higher voltage is connected than the unit can accept, a shock hazard may be created or damage may be done to the machine. The correct line voltage is listed on the nameplate on the machine.



Do not try to control the humidity by switching the power to the dehumidifier on and off. Use a humidistat to control the unit. The wiring diagram included in this manual shows the contacts for the humidistat.

2. Make an opening for the power leads in the control box located on top of the unit. The control cabinet is weatherproof, with a NEMA4 rating. If the unit is mounted outdoors, make the opening in the bottom or side of the cabinet, so that the cabinet continues to be weather proof.
3. Make the connections to the “line” side of the terminal block in the upper right-hand corner of the cabinet, where the label says, “Connect power leads here.”
4. Be sure the chassis of the HCD unit is connected to a good earth ground.

5.5 CONNECTING A REMOTE HUMIDISTAT

1. In some installations, the HCD unit operates in the manual mode. In this type of installation, the HCD unit operates whenever the power switch is turned on. Other units are set up for automatic cycling. In an installation of this type, the HCD unit is controlled by a device called a “humidistat.” The humidistat works much like the thermostat in a home cooling system. When the humidity rises above a preset point, the humidistat turns on the HCD unit. (On a unit with a “face and bypass” arrangement, a “humidity transmitter” performs a similar function.)
2. The humidistat is wired to a power source separate from the HCD unit (115V AC). Use a “close on rise” humidistat, with contacts which are “normally open.” (The contacts should be open when the humidity is below the preset limit, and closed when the humidity is too high.) The humidistat contacts should be rated for at least 2 amps.

3. Mount the humidistat in the space to be dehumidified. For best results, place the humidistat near the process air return duct (the duct returning process air to the dehumidifier). This will provide the most accurate sensing of the relative humidity in the process space. If possible, mount the humidistat away from the floor and ceiling, and do not mount it near any doors and windows. Do not mount the humidistat near the supply duct bringing the process air from the HCD unit.
4. Make the wiring connections between the humidistat and the HCD unit. Use 18 AWG wire. See the wiring diagram included in this manual. (If the optional 24V humidistat is being wired, see the wiring diagram supplied with that unit.)

5.6 CONNECTING THE AUXILIARY INTERLOCKS

1. The unit includes contacts for two kinds of auxiliary inputs. One auxiliary input can stop the machine immediately (a "hard stop"). This can be connected to an Emergency Stop switch or a smoke detector. A second input can act as a remote start/ stop switch, assuming that the Auto/Off/Manual switch is set to Auto. If this input is opened, the machine will stop normally. See the wiring diagram included in this manual.
2. The machine includes four auxiliary outputs which report on various conditions inside the unit: fault, warning, system run and dehumidification run. All of these are normally-open dry contacts.

5.7 START-UP TESTS

1. Turn on power to the unit. For a moment, set the Auto/Off/Manual switch to Manual, then back to Off. Check the direction of rotation of one of the blower fans. Each fan should turn in the direction shown on the attached label. If a fan is moving the other way, turn off the HCD unit and disconnect the power. Reverse any two of the three leads to the main power terminal block.

NOTICE

Do not operate the unit if the fans are turning backwards. The fans have much lower output when they operate backwards.

Always use the Auto/Off/Manual switch to turn off the machine. Other than emergency situations, do not shut down the system by de-energizing the main power disconnect. De-energizing the main power disconnect will cause all components to immediately shut down and power to be isolated. A hard shut down will damage the system, as the components will not be cooled down in a controlled manner.

2. Before operating the unit normally, be sure to complete these steps:

Function	Section
Adjusting the process and reactivation dampers.....	5.8
Adjusting the bypass damper	5.9
Setup procedure for face and bypass damper	5.10
Final checks	5.11

5.8 ADJUSTING THE PROCESS AND REACTIVATION DAMPERS

1. It is very important to provide the correct air flows for the process and reactivation air streams. Dampers are installed in both air streams.
2. Two pressure gauges are installed in the panel on the front of the machine. To

adjust the air flow, the pressure drop must be checked as the air moves through the HoneyCombe® wheel. The left-hand gauge measures the pressure drop as the process air moves through the desiccant wheel. The greater the air flow, the greater the pressure drop, and the higher the reading on the gauge.

3. Turn on the unit, using the circuit breaker. Set the Auto/Off/Manual switch to the Manual position.
4. The damper for the reactivation air is located on the reactivation blower. Adjust this damper to get the correct reading on the right-hand pressure gauge. The ideal reading is listed on the data sheet included in this manual.
5. The damper for the process air stream is located on the process blower. Adjust this damper to get the correct reading on the left-hand pressure gauge. The ideal reading is listed on the data sheet included in this manual.

5.9 ADJUSTING THE OPTIONAL BYPASS DAMPER

1. For an explanation of the purpose of the bypass duct, see Section 4.3.
2. Set the damper for the reactivation air. For instructions, see the last section.
3. Remove the rear access panel so the bypass damper can be reached.
4. The setting for the bypass damper depends on the total amount of process air moving through the machine. The data sheet includes a table which lists damper settings for different air volumes. Use this table to get an initial setting for the damper.
5. Adjust the damper for the process air. For instructions, see the last section.
6. Operate the unit, and check the total volume of process air on the output side of the unit. The easiest way of doing this is to measure the pressure drop across a downstream device, such as a cooling coil. See Figure 5-6.

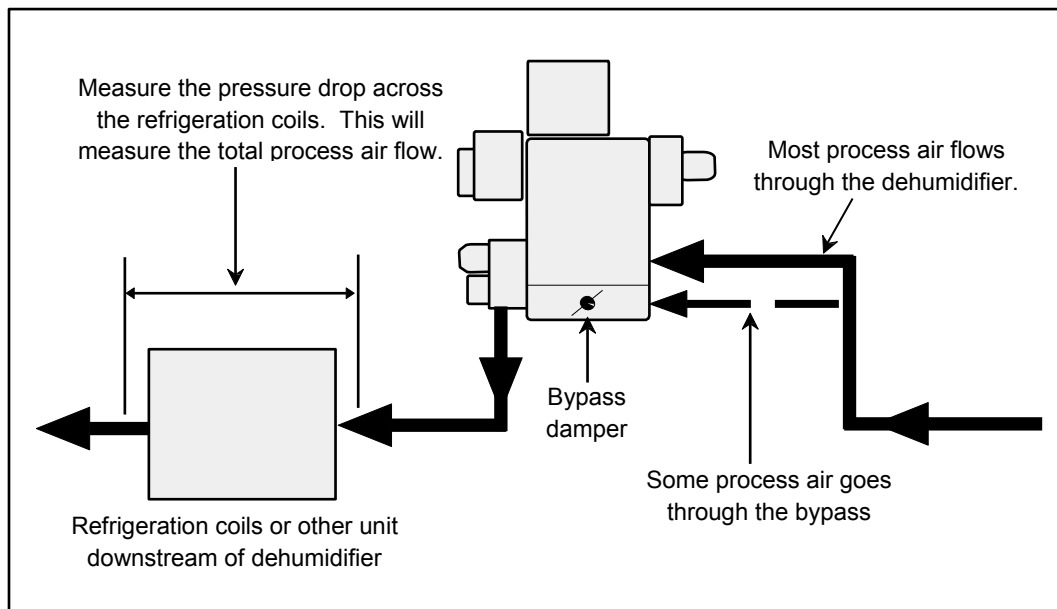


Figure 5-7
MEASURING TOTAL PROCESS AIR FLOW
(UNITS WITH OPTIONAL BYPASS ONLY)

7. Now that the actual total air flow is known, refer to the table again. Readjust the bypass damper based on the table.
8. Check the setting on the process damper again. Continue steps 5 through 7 until the settings do not change anymore.

5.10 SETUP PROCEDURE FOR OPTIONAL FACE AND BYPASS

For an explanation of the operation of this optional equipment, see Section 4.6. Regardless of the positions of the face and bypass dampers, the volume of air through the whole system must remain constant. This equipment does not change the airflow through the system. Instead, it just redirects the air through the wheel or around it, based on the humidity load.

Keep these points in mind while setting up the equipment:

- The face damper is always set up with a full 90° of damper travel.
- A measurement reference point must be selected for the system air volume. Usually it is best to measure the pressure drop across a postcooling and/or postheating coil. Caution - Do not use a precool coil for this purpose. Changes in the condensate load on the coil face can affect the pressure drops.
- The travel of the bypass damper may need to be set in the field. The volume of air through the system must remain constant, regardless of the positions of the dampers.
- A restrictor plate is installed in the outlet of the bypass. This simulates the pressure loss provided by the desiccant wheel when the air is not flowing through the bypass.

The following is the setup procedure:

1. Establish a monitoring point to check for the total volume of air moving through the system. The pressure drop can be checked across the postcool and/or postheat

coils. Monitor the pressure drop using a manometer or slope gauge. (As previously stated, do not use a precool coil for this purpose. Changes in the condensate load on the coil face can affect the pressure drops.) See Figure 5-7.

2. Use a manual balancing damper to adjust the volume of air moving through the system.
3. Start the system and place the face and bypass dampers in the "full face" positions. See Figure 5-8.
4. Refer to the Tech. Data Sheet and find the correct pressure drop for the process air stream. Using this as a reference, adjust the manual balancing damper until the correct pressure drop across the desiccant wheel is reached (measured using a manometer or the minihelic gauge). Make a note of the reading for the pressure drop across the postcool and/or postheat coils.

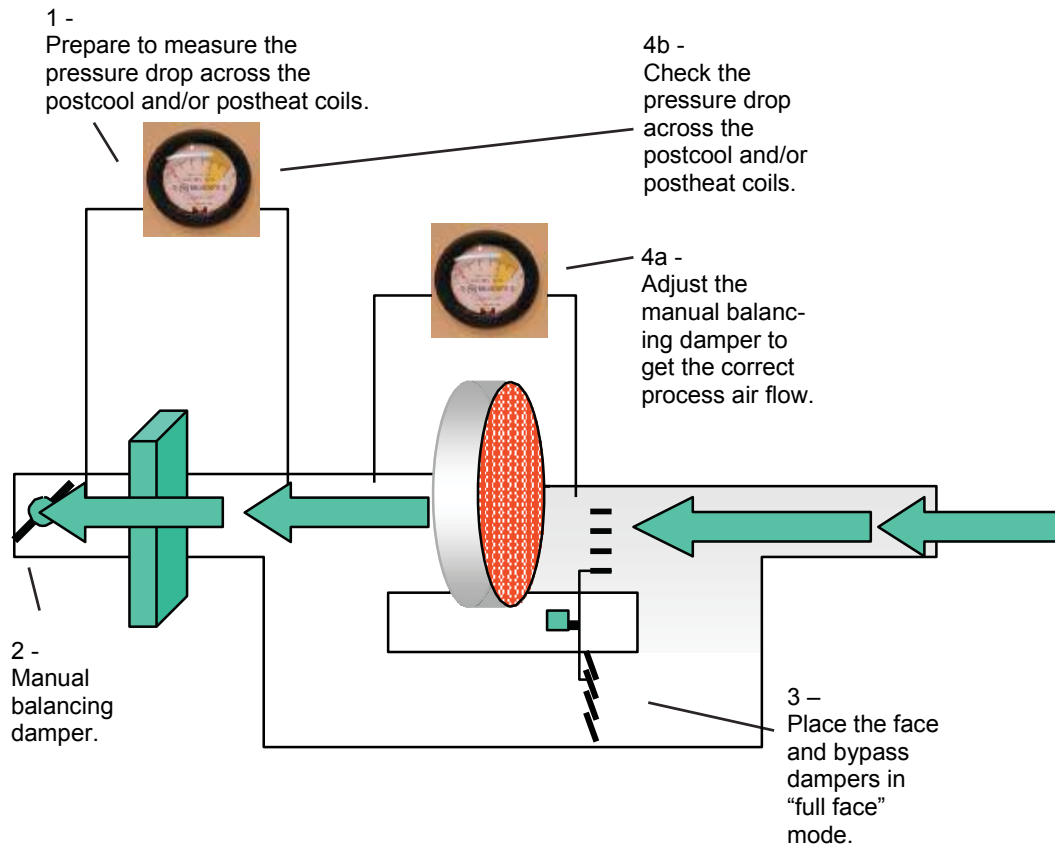


Figure 5-8
CHECKING PROCESS AIR FLOW - FULL FACE MODE
(UNITS WITH OPTIONAL FACE AND BYPASS ONLY)

5. Place the face and bypass dampers in the "full bypass" positions. See Figure 5-9. Again, check the pressure drop across the postcool and/or postheat coils. It should match the reading taken in the last step. If it does not, the travel of the bypass damper must be adjusted.

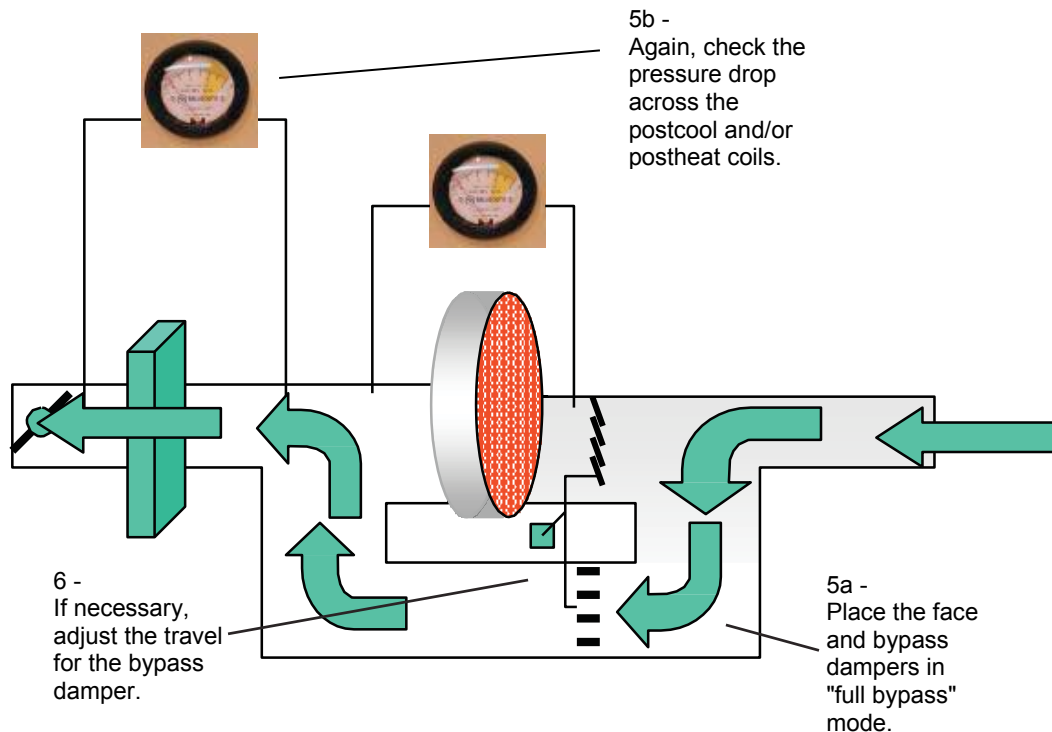


Figure 5-9
CHECKING PROCESS AIR FLOW - FULL BYPASS MODE
(UNITS WITH OPTIONAL FACE AND BYPASS ONLY)

6. If necessary, correct the position of the bypass damper.
 - If the reading at the postcool and/or postheat coils increased in the bypass mode, the travel of the bypass damper must be reduced.
 - If the reading at the postcool and/or postheat coils decreased in the bypass mode, the travel of the bypass damper must be increased.

Adjust the travel of the bypass damper until the same reading is achieved at the postcool and/or postheat coils that was recorded when the machine was in the "full face" mode.

7. Check the setting. Cycle the face and bypass through several complete cycles, from "full face" to "full bypass." As this is done, watch the pressure drop across the postcool and/or postheat coils. There should be very little change - $\pm 10\%$ is the maximum allowable difference. Once the pressure drop remains constant, the setting is correct.

5.11 FINAL CHECKS

Do not allow the unit to operate unattended until the following points have been checked:

- Secure electrical connections, including a good earth ground.
- Blower motors turning in correct direction.
- HoneyCombe® wheel rotating 6-12 rph.
- Ductwork connected and sealed. Reactivation outlet ductwork insulated.
- Reactivation damper and process damper set to achieve correct pressure drops across wheel.
- Optional bypass damper or face and bypass equipment set as described in text.
- After unit has stabilized, reactivation outlet temperature is near 120°F ±5°F, and always above 110°F.

6 - MAINTENANCE

This section includes some suggested maintenance requirements for this unit. This information is offered as a guide and some systems may require more frequent maintenance. For example, a unit used in a very dusty atmosphere should have the air filters changed more often. Use these suggested schedules as a starting point, and modify them to suit the specific installation.

Munters dehumidifiers are designed to be very rugged and reliable. In fact, they have the lightest maintenance requirements in the industry. However, this does not mean the maintenance can be skipped completely. Poor maintenance can result in poor performance, increased downtime and additional operating costs.

Munters strongly recommends that a comprehensive maintenance program be followed. This program should include not only the dehumidifier, but any support systems involved in the dehumidification system.

At some installations, downtime can be very expensive. If downtime of a few days is not possible, then it is important to maintain an on-site inventory of spare parts. A listing of recommended spare parts is included in this manual.

6.1 QUICK MAINTENANCE CHECK

As the unit operates, make routine checks of these three basic functions:

- The process air flow and reactivation air flow should be checked to be sure they remain at design levels. If the air flows change, this can change the operation of the machine. A change in air flow may call for maintenance or troubleshooting. If the application changes and this will result in different air flows, contact the Service Operations Department at Munters.
- Check the temperature at the reactivation outlet to be sure it remains near 120°F ±5°F, and always above 110°F. If the temperature drops below this level, check the section on Troubleshooting.
- The wheel must be rotating whenever the unit is operating.

6.2 SUGGESTED MAINTENANCE SCHEDULE

The following table lists the recommended maintenance schedule for this unit. For detailed instructions on each procedure, see the sections which follow.

EVERY 30 DAYS (See Figure 6-1)

<i>Check this</i>	<i>Location</i>
Clean the air filters. (See Sect. 6.5)	Inlet for reactivation air Inlet for process air (if equipped)
Inspect the seals around the HoneyCombe® wheel. (See Sect. 6.11)	Both faces of the HoneyCombe® wheel
Inspect the HoneyCombe® wheel. (See Sect. 6.7)	Behind access panel, center of unit
Check the temperature at the reactivation outlet (See Sect. 6.3)	Outlet for reactivation air
Check all ductwork. (See Sect. 6.4)	Inlet and outlet for process air Inlet and outlet for reactivation air
Large units only - Check the drive belt tension on the process air blower	Behind long cover on process blower
Check the reactivation air volume	Readings on reactivation airflow gauge behind front access panel should match tech. data sheet.

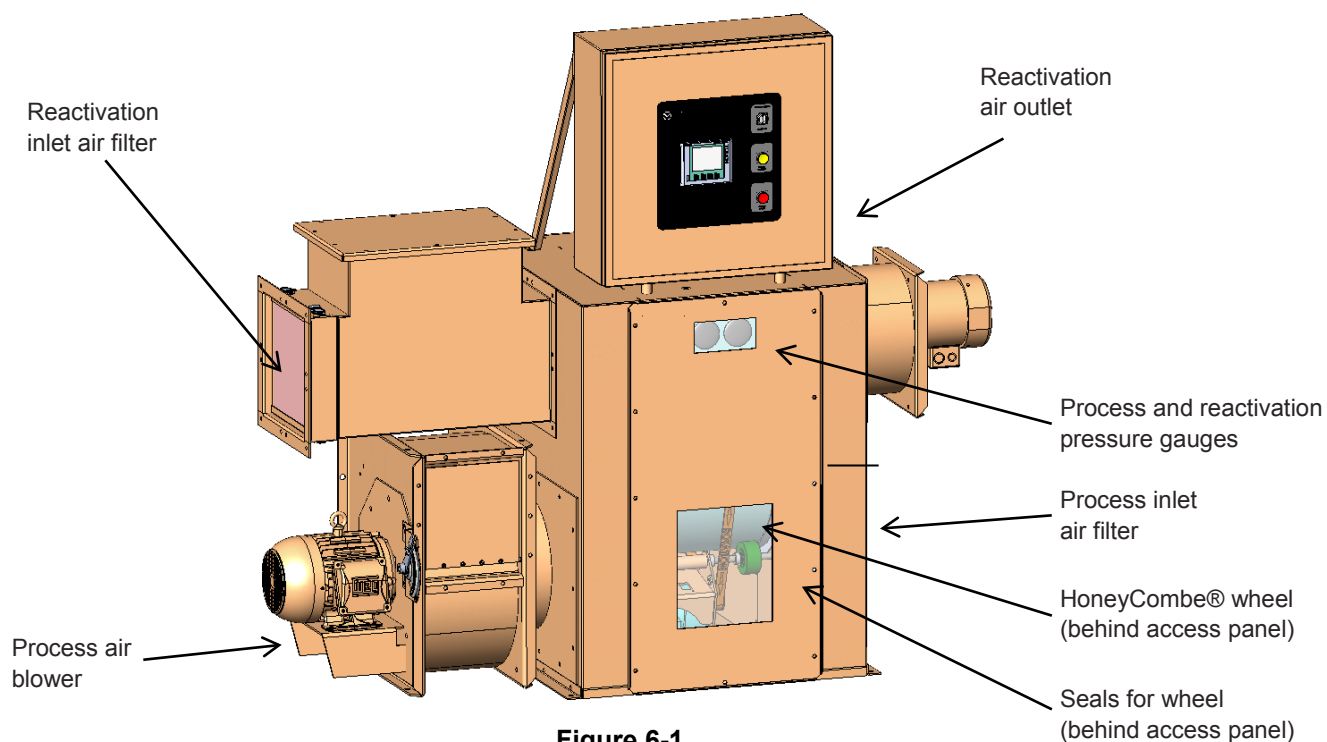


Figure 6-1
MAINTENANCE POINTS – EVERY 30 DAYS

ADDITIONAL MAINTENANCE, EVERY 60 DAYS (See Figure 6-2)

<i>Check this</i>	<i>Location</i>
Inspect the blower fans. (See Sect. 6.13)	Blower for process air Blower for reactivation air
Inspect the electronic controls. (See Sect. 6.15)	Cabinet behind control panel Units with electric heaters only – Reactivation chamber to left of HoneyCombe® wheel
Inspect the wheel drive motor and belt. (See Sect. 6.16)	Behind front access panel

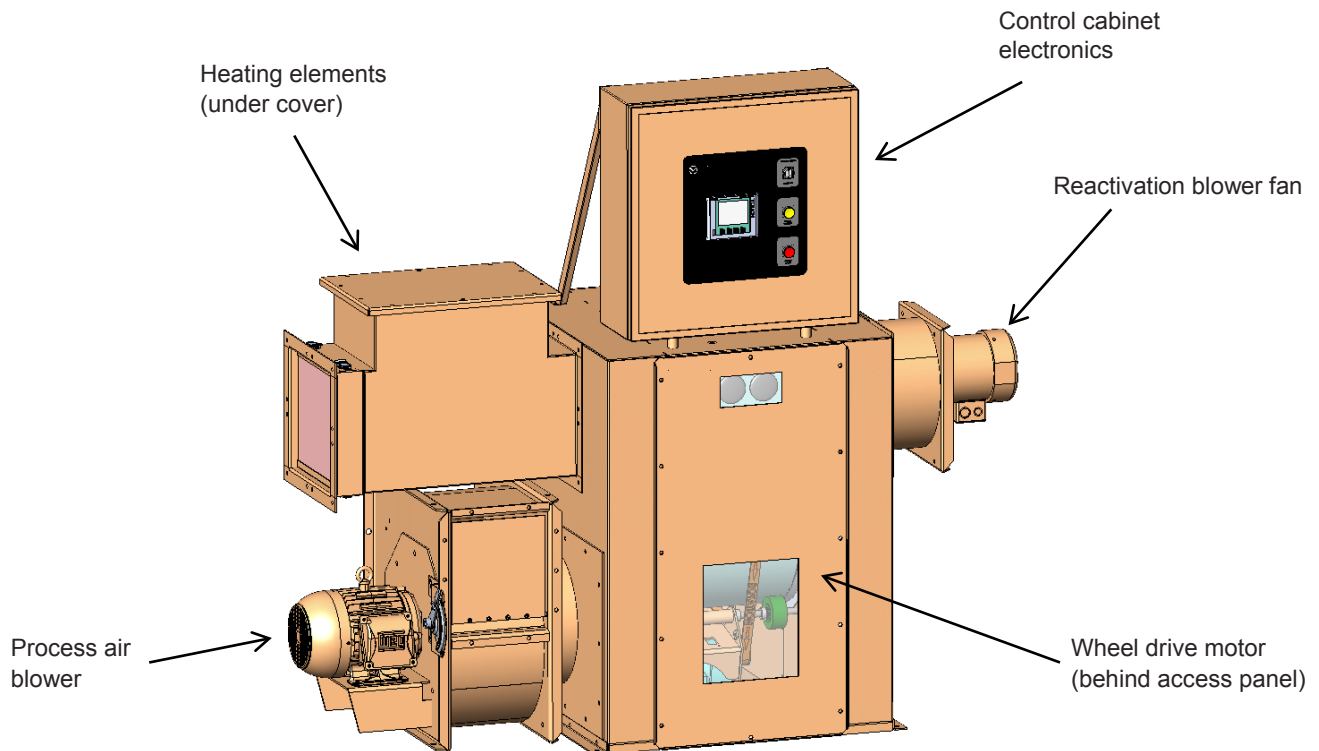


Figure 6-2
MAINTENANCE POINTS – EVERY 60 DAYS

ADDITIONAL MAINTENANCE, EVERY 6 MONTHS

<i>Check this</i>	<i>Location</i>
Inspect the roller wheels. (See Sect. 6.14)	Behind access panel, center of unit
Grease the bearings on the process air blower. (Texaco Molytex #2 E.P.)	Blower for process air

ADDITIONAL MAINTENANCE, EVERY 12 MONTHS

<i>Check this</i>	<i>Location</i>
Larger units only - Change the oil in the bevel gear drive for the wheel. Use Mobil Glygoyle 460 (Munters P/N 94602-01)	Drive unit, behind access panel

6.3 CHECKING THE REACTIVATION OUTLET TEMPERATURE

After the unit has been operating for 30 minutes, the temperature at the outlet of the reactivation air stream should be about 120°F. Check this outlet temperature with a thermometer. It should be within $\pm 5^\circ\text{F}$. If the outlet temperature falls outside this range, see Section 7 - TROUBLESHOOTING.

6.4 INSPECTING THE DUCTWORK

Check the ductwork for signs of air leaks or blockage. When the ducts run outdoors, rain hoods and bird screens should have been installed. Be sure the screens are not blocked.

Even small leaks in the ductwork can hurt the efficiency of the unit. Leaks in the reactivation outlet duct can be important, since the reactivation air is very damp. If this damp air leaks back into the conditioned space, this makes the unit work harder. If there are leaks in the process output duct, moisture from the process space can get into the process air stream. This can happen even though the moisture must move “upstream” against the flow of process air. Either condition can make the unit work harder than necessary.

Condensation may collect in the ductwork for the reactivation air stream. If condensation traps have been installed, be sure they are draining correctly.

6.5 CLEANING THE AIR FILTERS

This unit has an air filter at the intake for the reactivation air. See Figure 6-1. (An additional filter should be added at the intake for the process air.) It is important to keep all of the air filters clean. If the filters become dirty, the efficiency of the unit will drop off. One symptom of this may be a decreased temperature reading at the outlet for the reactivation air stream. This condition can also cause an overheating fault.

1. Turn off the machine and allow it to cool before changing the filter.
2. The filter at the intake for the reactivation air can be cleaned and reused. The filter is held in place by two screws. Remove the screws and lift the filter upward.
3. Wash the filter using mild soap and water. Allow the filter to air-dry, or use compressed air. When the filter is dry, reinstall it.

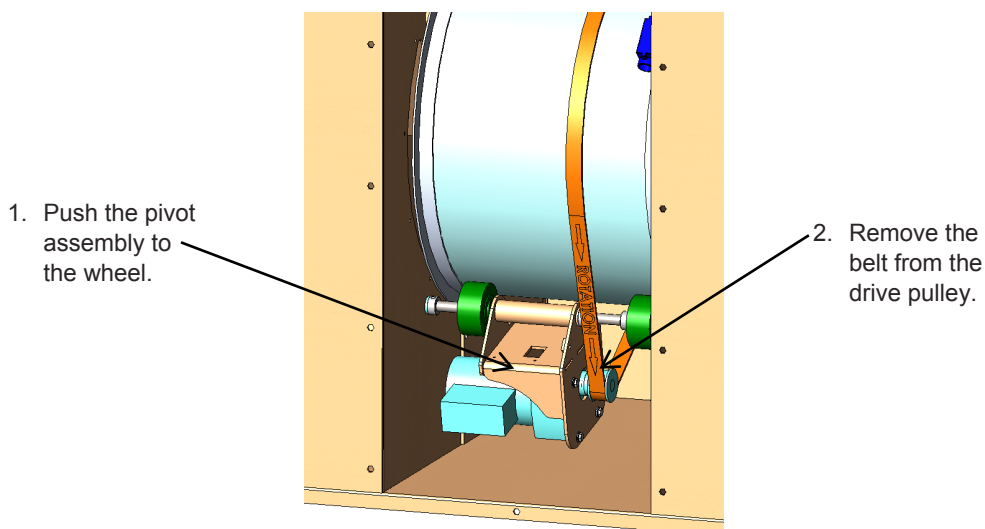


Figure 6-3
DRIVE BELT MECHANISM FOR SMALLER UNITS

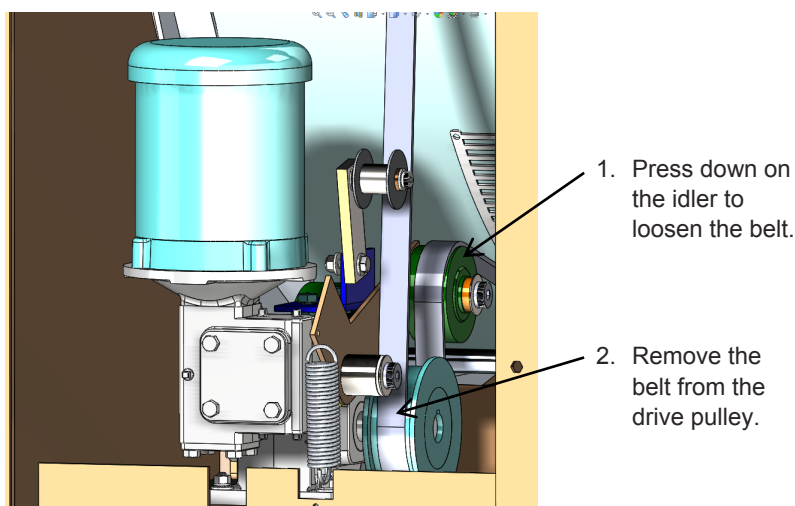


Figure 6-4
DRIVE BELT MECHANISM FOR LARGER UNITS

⚠ CAUTION

Do not use any kind of filter coating on the reactivation inlet filter. This filter is located just upstream of the electric heating elements. Some filter coatings give off volatile chemicals, which could be pulled into the heating elements. This could create a fire hazard. (It is OK to use filter coating on the filter for the process air, since this air is not pulled through the heating elements.)

6.6 REMOVING AND REPLACING THE HONEYCOMBE® WHEEL

NOTICE

It is important to work carefully when removing and replacing the wheel. The HoneyCombe® section in the center of the wheel can be damaged if it is handled roughly.

1. Turn off the unit. Switch off the power supply to the unit and lock it out so that nobody else can turn it on by mistake.

2. Remove the access panel on the front of the unit. If there is space behind the rear panel, remove this panel also.
3. At the front of the machine, press down on the idler roller. See Figure 6-3 and Figure 6-4. This will release the tension on the drive belt. Remove the drive belt from the drive pulley.
4. Carefully rock the wheel back, so it is supported by the rear roller. Support the wheel with a block of wood, so that it cannot roll forward.



The wheel can be heavy, especially on a larger unit. Allowing the wheel to roll forward suddenly could result in personal injury.

5. If possible, work from the rear of the unit. Remove the three braces which run across the wheel opening. Find a strong board which can be used as a ramp for the wheel. Set this in place at the rear of the machine.



Be sure the board is strong enough to support the wheel. Be sure the board is fitted firmly in place.

6. Now it is possible to remove the HoneyCombe® wheel. Roll the wheel out of the rear of the machine. The drive belt can be used as a handle.



Be careful to avoid damage to the seals. Until the wheel is clear of the seals, pull the wheel straight out of the machine. If there is room, place a helper on the opposite side of the machine to guide the wheel and push the seals back. Be especially careful when replacing the wheel.



The wheel assembly is heavy, particularly if it is wet. Use a helper when lifting the HoneyCombe® wheel. It is safe to roll the wheel.

7. To replace the wheel, reverse the steps listed above. It is easiest to work from the rear of the unit. Roll the new wheel carefully into position in the center of the machine.

The wheel includes a metal cam. When viewed from the rear of the machine, this cam should be closest to the left side (process inlet) of the unit. When viewed from the front of the machine, the cam should be closest to the right side of the machine.

Work carefully to not damage the seals. Do not seat the wheel against the front roller yet.

8. Go to the front of the machine. Slip the end of the drive belt over the end of the shaft for the front wheels. Replace the right end of the shaft in the supporting hole. Move the left-hand collar into position and tighten the set screw. Allow a small amount of side-to-side play on the shaft.
9. Lift the front of the motor again, and place the drive belt over the drive pulley. Allow the drive motor to drop.



Be sure the drive belt is centered on the drive pulley. If the drive belt is off center, it will not provide full power to the HoneyCombe® wheel. This can cause the wheel to stall.

10. Replace the access panels. The unit is now ready to operate.

11. Make a quick check of the airflows and operating temperatures as the machine runs.
12. The new wheel may behave slightly differently from the old one.

6.7 INSPECTING THE HONEYCOMBE® WHEEL

The HoneyCombe® wheel contains a lightweight framework which supports the desiccant chemical. This framework is manufactured in a shape which looks like the honeycomb in a bee hive. The honeycomb pattern includes many small air passages or “flutes” which run through the wheel. These passages are designed to allow air to flow across the desiccant chemical with the least amount of resistance.

Near the seam in the wheel housing there is some engraved text. The text includes the wheel serial number, the date of manufacture, and the manufacturing process used. The following table will help to identify the type of desiccant chemical used in the wheel.

<i>Wheel Chemical</i>	<i>Color</i>	<i>Marking</i>
Lithium chloride	Gray or black	GFR
Titanium enhanced silica gel	Red	Tigel
Molecular sieve	Blue	MS
High performance	Green	HPX

The HoneyCombe® wheel will generally not foul or plug up unless one of the air streams has not been properly filtered. Plugging is most likely to happen when a unit must operate in a very dusty environment. Problems can be minimized by adding prefilters.

If the wheel becomes blocked, this will increase the pressure drop across the wheel for both the process and the reactivation air streams. Checking the pressure drop is described in Section 5 - INSTALLATION on “Installation.”

To check for moisture damage and plugging, do the following:

1. Shut down the dehumidifier. Turn off and lock out the power to the unit to prevent someone else from starting it by mistake.
2. Remove the wheel as described in Section 6.6.
3. Check the faces of the HoneyCombe® wheel for signs of softening. Place the palm of your hand flat against the face of the wheel and apply moderate pressure. Run your hand over the entire face of the wheel on both sides. If end of the flutes are flattened, then the wheel structure is sound. If some of the flutes deform, or the surface depresses under your palm, try the wheel drying procedure listed in Section 6.8. If, after the drying procedure the wheel is still soft, consult the Service Operations Department and/or the Parts Department at Munters.
4. Also check for damage to the ends of the flutes. This may indicate a problem with the alignment of the wheel. This type of problem is most likely if the support rollers have become worn or misadjusted. Be sure the ends of each roller shaft are pressed down completely in the supporting slots.

Small isolated areas of damage are usually caused by rough handling of the wheel. If the total area of damage is less than 10% of total wheel area, then no action is

required. However, if the damaged area is large and equipment performance has deteriorated, the wheel should be replaced.

5. To check for plugging, use a “drop light” with a bulb of at least 60 Watts or higher. Hold the light on the far side of the wheel, facing toward the wheel. The light should be four to six inches from the face of the wheel. By watching the near face of the wheel, look to see if the wheel is plugged. Because the passages are very small, it will not be possible to see the light directly. Instead, there should be a “glow” from the light. If any part of the wheel is plugged, it will appear as a dark area. Scan the entire area of the wheel. Light should be visible at all points on the wheel. If the light is not visible, or if some spots on the wheel are dark, the passages in the wheel are plugged. See the cleaning instructions in Section 6.9.

6.8 DRYING THE HONEYCOMBE® WHEEL

1. Turn off the power to the machine and open the cabinet behind the control panel. Remove the fuse for the process air blower. See the wiring diagram.
2. Operate the machine for 30 minutes. This will give the wheel a chance to get rid of moisture without receiving more moisture from the process air stream.
3. Replace the fuse and operate the machine normally.

6.9 CLEANING THE HONEYCOMBE® WHEEL

1. Remove the HoneyCombe® wheel by following the procedure listed earlier.
2. Vacuum both faces of the wheel. Use a “wet/dry” vacuum, and a “dusting brush” attachment with a soft bristle brush.
3. Vacuuming alone may not always remove the blockage. The force of the air can be increased by applying a stream of low-pressure compressed air to one side of the wheel while vacuuming the other. This must be done carefully as to not damage the wheel. Use dry, oil-free compressed air of no more than 10 PSIG. Do not allow the air jet to come closer than 12" to the face of the wheel. The compressed air will work with the vacuum to clear the blockage.



Do not use high-pressure air to clean the wheel. Do not hold the air jet near the surface of the wheel. The honeycomb structure inside the wheel may be damaged.

Do not use any other procedure to clean the HoneyCombe® wheel. Do not wash the wheel with water or any solvent. The wheel can be easily damaged and make it necessary to buy a replacement. Wheels using lithium chloride (gray or black color wheel, code “GFR”) may be cleaned and reimpregnated at the factory. Wheels using silica gel (orange or rust color wheel, code “SI”) may be cleaned.

4. After cleaning, recheck the wheel using the drop light. If the blockage has been removed, reinstall the wheel. If the blockage is still extensive, consult the factory.

6.10 SAMPLING WHEEL CORE MATERIAL

Units with Lithium Chloride Wheel Only – (CODE GFR)

A lithium chloride wheel has a gray or black color, and is marked with the code letters “GFR.” If the wheel has been inspected thoroughly as outlined above, and cause of poor performance

cannot be found, it is possible to analyze the desiccant content within the HoneyCombe® wheel. Take samples of the core material and send them to Munters. The samples can be analyzed to determine the condition of the desiccant material in the wheel. If the condition of the honeycomb structure is good, it may be possible to reimpregnate or recharge the wheel. For details of the sampling procedure, contact Munters Service Operations.

6.11 INSPECTING THE SEALS

1. The seals are positioned against each face of the HoneyCombe® wheel. One seal is mounted on each side of the wheel. The standard high pressure seal has an outer layer of black Rulon®, with a red supporting material. The seal is held in place by aluminum pop-rivets.
2. Remove the front and rear access covers. Leave the HoneyCombe® wheel in place.
3. Check the clearance between each seal and the face of the wheel. Slide a business card or a feeler gauge of 0.030" between the face of the wheel and the seal face. Check all of the seal area on both sides of the wheel. At each point, there should be a moderate drag or resistance as the card or feeler is slid along the seal.

If at any point on the seal there is little or no resistance, replace the seals with new ones. See Section 6.12.

If there is little or no drag on one side, and excessive drag on the other side, check the rollers which support the wheel. Look for wear on the roller bushings. See Section 6.15.

4. If the HoneyCombe® wheel is removed, the surface of each seal can be inspected. On the face of the seal (the portion that comes in contact with the HoneyCombe® wheel) is a black coating known as Rulon®, which is a Teflon® material. Inspect the upper surface of each seal for any signs of excess wear. If the red portion of the seal is visible through the black, the seal should be replaced.
5. If a seal is torn, a temporary repair can be made by using high temperature silicone caulking (such as GE RTV®). Bond the torn section back together. Avoid getting the silicone on the outer or wearing portion of the seal. This can cause excess drag. Plan to replace the damaged seal as soon as possible.

6.12 CHANGING THE SEALS

NOTICE

The HoneyCombe® seals are delicate, and require extra care during handling. Failure to handle them carefully could result in damage to the seals and/or equipment.

1. Remove the HoneyCombe® wheel as outlined earlier.
2. Each seal is held in place by a series of clips. Slide the old seal out of the clips.

NOTICE

Each HoneyCombe® seal must be installed so that the seal joints are not in contact with the seal clips. Failure to install the seal this way could result in damage to the seal and/or equipment.

3. Fit the new seal into the same clips.

6.13 INSPECTING THE BLOWER FANS

1. Turn off the unit before checking the blower fans. After the unit is turned off, wait three minutes to allow the unit to stop operating.



WARNING

The two blowers inside the HCD unit spin at a high rate of speed. Hands may be badly hurt if put inside the blower while it is turning. Keep hands away from the blowers while the unit is turned on. Turn off the power, using the disconnect or circuit breaker, before inspecting the fans. Follow standard lock-out tag-out procedures.

2. The blower for the process air is located in the lower left-hand part of the machine. See Figure 4-4. Open the damper beside the blower motor. (If ductwork is connected, it may have to be removed temporarily.) Reach through the damper opening and touch the fan itself. The fan should spin easily.
3. The blower for the reactivation air is located in the upper right-hand part of the machine. See Figure 4-4. Again, open the damper beside the blower motor to get access to the fan. (Some of the ductwork may have to be removed to do this.)
4. If one of the fans is difficult to turn, or there is a noticeable grinding or “gritty” feel, something is wrong. Check for anything which could be rubbing against the fan.
5. If this does not reveal the problem, remove the fan and motor. Be sure the power is off before doing this! The blower assembly for the reactivation air is mounted on a plate which is held in place by a number of bolts. Remove the plate, and then remove the assembly. Use an Allen wrench to loosen the fan, and then remove it. This will allow access to the bolts which attach the motor to the plate.
6. The blower assembly for the process air is mounted on a plate which is held in place by a number of bolts. Remove the plate, and then remove the assembly. Use an Allen wrench to loosen the fan wheel hub, and then remove it. This will allow access to the bolts which attach the motor to the plate.
7. Turn the motor shaft by hand to check the motor bearings. There should be no free play when the motor shaft is tried to move side-to-side. A motor with stiff or gritty bearings must be replaced.
8. Check the blower blades for excessive dirt or corrosion. If dirt is attached to the blades, this can reduce the blower's performance. If any blades are broken or missing, this will cause an imbalance, and can damage the motor bearings.

6.14 INSPECTING THE ROLLER WHEELS

1. The HoneyCombe® wheel is supported on four wheels. See Figure 6-3. Remove the front and rear access panels so the wheels can be accessed.
2. Each pair of wheels is supported on a shaft. Each shaft is held in place by two collars. There should be a small side-to-side play on the shaft.
3. If the wheels become worn, excessive play may develop. This can cause uneven support of the HoneyCombe® wheel. Inspect the wheels and check for signs of wear. Replace them if necessary.
4. The wheel assemblies use greaseless bushings made of Delrin®. These bushings do not require lubrication.

NOTICE

Do not lubricate the bushings on the support wheels. The lubricant may damage the plastic material on the outer layer of each wheel.

6.15 INSPECTING THE ELECTRONIC CONTROLS

1. Turn off the power to the machine and lock it out before opening any of the electrical cabinets.
2. Open the cabinet behind the main control panel. Check the contactors for signs of pitting.
3. Make a general check for possible problems – frayed wires, overheated parts, etc.
4. Remove the cover above the electric heating elements. See Figure 4-4. Some of the electronic controls are located under this cover. Make a visual check.

6.16 INSPECTING THE WHEEL DRIVE MOTOR AND BELT

1. The drive motor is located behind the front access panel.
2. Check the belt for signs of stiffness or cracking. Replace it if necessary. In order to do this, the wheel must be removed.
3. Figure 6-3 shows how to disconnect the belt from the drive pulley. When the belt is disconnected, try to turn the drive pulley by hand. The motor shaft should not be able to move side-to-side. There should not be any significant backlash between the drive gears. If either condition exists, replace the motor.
4. If the drive pulley is being replaced, mount the pulley as close as possible to the body of the motor. This will reduce the side load on the shaft and bearings.
5. When replacing the drive belt, be sure the belt is centered on the drive pulley. If the belt is off-center, the pulley may not be able to provide enough drive to turn the wheel. This can cause the wheel to turn more slowly than it should.

6.17 CHECKING THE HUMIDISTAT

1. Start by adjusting the humidistat to the high end of the scale (close to 100% relative humidity). At this point, the humidistat should not be calling for drying. The contacts inside the humidistat should be open, and 115V AC should be able to be measured across the contacts.



The contacts inside the humidistat carry 115V AC. This voltage and current can be dangerous. Only trained electricians should perform work with the electrical parts.

2. Next, adjust the humidistat to the low end of the scale (close to 10% or 20% relative humidity). Now the humidistat should be calling for drying, and the contacts inside the humidistat should be closed. 0V AC should be able to be measured across the contacts.
3. The wiring between the humidistat and the unit can also be checked. See the wiring diagram included in this manual. When the humidistat is not calling for dehumidification, full AC voltage should be present across the contacts. When the humidistat does call for dehumidification, the voltage across the terminals should drop to 0V. (A 24V humidistat is an option. See the separate wiring diagram supplied with the unit.)

6.18 STORAGE AND LONG-TERM SHUT-DOWN

Turn off all energy inputs to the machine. Turn off all electrical power. On a unit with steam reactivation, turn off the steam supply valves. On a unit with gas reactivation, turn off the gas supply.

The desiccant in the HoneyCombe® wheel will continue to absorb some moisture after the dehumidifier is shut down. If the wheel is allowed to absorb too much moisture over a period of time, the wheel may not be able to perform normally when the unit is turned on again. A wheel which uses lithium chloride (gray or black color) is more likely to have this problem.

Air may continue to move through the system, even if the HCD unit is turned off. This is a more important problem if the system includes a separate fan or other air-moving device. If the fan is turned on, and the wheel inside the HCD unit is stopped, a section of the wheel may be exposed to a large amount of moisture.

It is best to remove the wheel from the HCD unit. This is not always possible, however. If the rest of the air-handling system will continue to be used, it may not be possible to remove the wheel. With the wheel removed from the system, the back-pressure in the ductwork will change. Air from the process ducts will be able to escape to the reactivation ducts. Here are some possible ways of dealing with this situation:

If the rest of the system will not be used –

1. If possible, remove the HoneyCombe® wheel. Wrap the wheel with strong plastic sheeting, seal with duct tape, and store the wheel separately.
2. If this is not possible, seal the dehumidifier to prevent the open air from reaching the HoneyCombe® wheel. This can be done by fitting moisture barriers over the openings in the unit:
 - Process inlet
 - Process outlet
 - Reactivation inlet
 - Reactivation outlet

To make the moisture barriers, use strong plastic sheeting sealed with duct tape.

If the rest of the system will be used –

Remove the wheel. Plug up the openings for the ductwork for the reactivation inlet and reactivation outlet – the points where these ducts exit the building.

It is also possible to set up the unit so that it operates periodically. This provides an automatic way of purging the wheel. Consult the factory for details.

7 - TROUBLESHOOTING

The HCD-series dehumidifiers have a state-of-the-art design, with a sophisticated control system using solid-state electronics. These units have proven to be very reliable in a wide variety of installations.

There are two parts to this section of the manual. The first part lists each of the fault and warning indicators. In the second part, there are some additional detailed troubleshooting checks.

In order to check these trouble conditions, the HCD unit must be turned on and operating, or trying to operate.



The HCD unit can be wired for high-voltage AC. The unit can produce enough voltage and current to result in death or cause severe burning. Always turn off the power before working inside the unit. Follow standard lock-out tag-out procedures. Only trained electricians should perform work with the electrical parts.



The two blowers inside the HCD unit spin at a high rate of speed. Hands may be badly hurt if put inside the blower while it is turning. Keep hands away from the blowers while the unit is turned on.

If this documentation does not address a specific issue, call the Technical Support Department at Munters. Technicians are available to answer questions and assist with troubleshooting:

Tel: 1-978-241-1100

Fax: 1-978-241-1217

The control system on this machine is set up to react to problems in two possible ways:

- A "fault" is caused by a condition which might create a danger to personnel or damage to the machine. A fault condition lights a red indicator on the front panel and causes the machine to shut down. Before the machine can be restarted, the Auto/Off/Manual must be set to Off, then back to Auto or Manual.
- A "warning" is caused by a condition which might make the machine run less efficiently. A warning condition allows the machine to continue to run, but signals that corrective action is advised.
- Fault and warning messages are normally displayed on the operator interface panel to assist in troubleshooting.

7.1 TROUBLESHOOTING GUIDE

Problem	Possible Cause	Recommended Remedy
Power turned on, unit does not run, no indicator lights	No power to the unit.	Check power to the unit.
		Customer controlled power interlock is open or E-stop is pressed.
	Phases of supply power are out of sequence.	Check three phase power monitor and switch two of the incoming phases if necessary.
	Failed fuses	Determine which fuse failed and why. Replace fuses if necessary.
Power turned on, unit does not run, fault indicator continuously blinks	Customer-controlled power interlock is open or E-stop switch is pressed	Check system and close interlock or disable E-stop as necessary.
Unit runs in Manual position but does not run in Auto.	Humidistat (option) failed.	Check humidistat wiring, replace humidistat if necessary.
Humidity is high, dehumidifier performance is poor.	Controlled space is not completely sealed.	Seal all ductwork.
		Make process space vapor-tight.
	Unit is not operating at design conditions.	Consult technical data sheet for correct pressure drops across wheel. See "Maintenance"
		Reduce moisture load or temperature to dehumidifier in order to operate at design conditions.
		Check for leakage in air handler outside of dehumidifier.
	Wheel is clogged.	Monitor wheel pressure drop for multiple revolutions, looking for changes in pressure drop readings.
		Visually inspect wheel. See "Maintenance"
	Wheel is not rotating at proper speed (6-12 rph; consult technical data sheet).	Check wheel drive motor and belt.
Reactivation over temperature warning	Reactivation air flow is low.	Inspect reactivation blower for blockage.
		Inspect reactivation blower damper position.
		Inspect reactivation filters for clogging.
Blower and / or wheel drive motor is not running	Motor overload has tripped.	Inspect blowers and motors for obstructions, remove obstruction, and reset with button on starter.

Problem	Possible Cause	Recommended Remedy
		Inspect unit operation for overblowing or closed dampers. Adjust and reset with button on starter.
Low reactivation outlet temperature warning	Reactivation air flow is high.	Inspect reactivation blower damper position.
	Unit is not operating at design conditions.	Consult technical data sheet for correct pressure drops across wheel. See "Maintenance"
		Reduce moisture load or temperature to dehumidifier in order to operate at design conditions.
		Check for leakage in air handler outside of dehumidifier.
	Wheel drive belt is slipping	Inspect drive belt and motor. Consult factory if belt is slipping on wheel.
	Reactivation energy is not operating.	Inspect fuses. Determine cause of failure and replace.
		Control components have failed. Inspect and replace as necessary.
		Control signal is not being sent from PLC. Inspect wiring and control module. Inspect and replace as necessary.
		Inspect for open heating element. Replace as necessary.
	Reactivation blower is not operating.	See "Blower and / or wheel drive motor is not running."
DH wheel fault	Reactivation duct and / or filter is blocked.	Inspect duct and filters for clogging. Repair / replace as necessary.
	Temperature sensor has failed.	Inspect sensors and wiring. Replace as necessary.
Airflow pressure fault	Wheel drive belt is slipping	Inspect drive belt and motor. Consult factory if belt is slipping on wheel.
	Cam is not making contact with rotation detector switch (LCS01)	Inspect location of cam and correct as necessary.
Airflow pressure fault	Reactivation blower is not operating.	See "Blower and / or wheel drive motor is not running."
	Reactivation duct and / or filter is blocked.	Inspect duct and filters for clogging. Repair / replace as necessary.

7.2 FAULT ALARMS

1. If the unit detects a fault condition, the ALARMS button on the display will blink red (See Figure 7-1) and the Fault Light on the control panel will light.

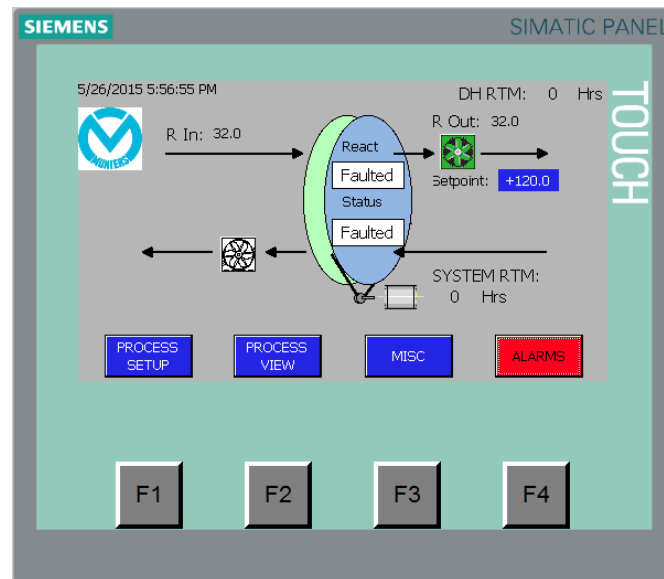


Figure 7-1
ALARM INDICATION ON OPERATOR DISPLAY

2. To see the details on the fault condition, press the ALARMS button. Figure 7-2 shows the type of screen that the display might present.

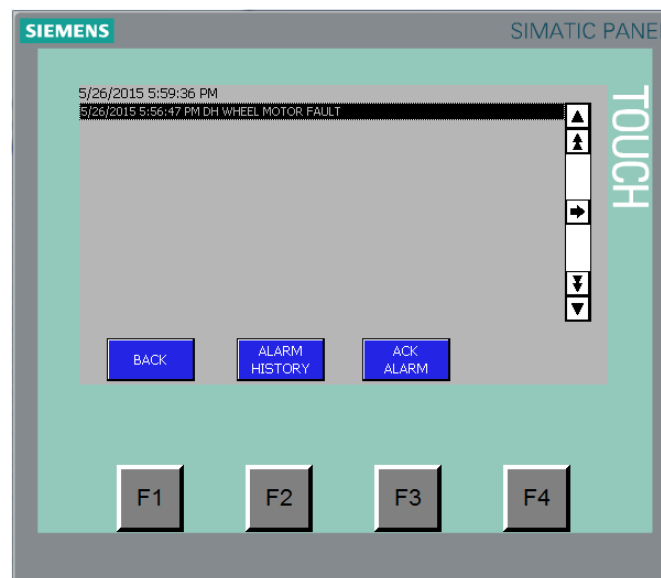


Figure 7-2
ALARM DISPLAY

3. Recovering from a fault is a three-step process. The fault must be cleared on **both** the dehumidifier and the operator interface display.
 - On the dehumidifier, locate the cause of the problem that caused the fault, and correct it.
 - On the dehumidifier, reset the selector switch (Auto/Off/Manual) by moving the

switch to the Off position, then returning it to Auto or Manual.

- On the operator interface display, press the Acknowledge button (ACK ALARM) to clear the message and alarm symbol.
 - After a fault message has been cleared, return to the Main screen by pressing BACK. The display will indicate "Ready."
4. To see a history of past alarm conditions, press the ALARMS button followed by ALARM HISTORY. See Figure 7-3.

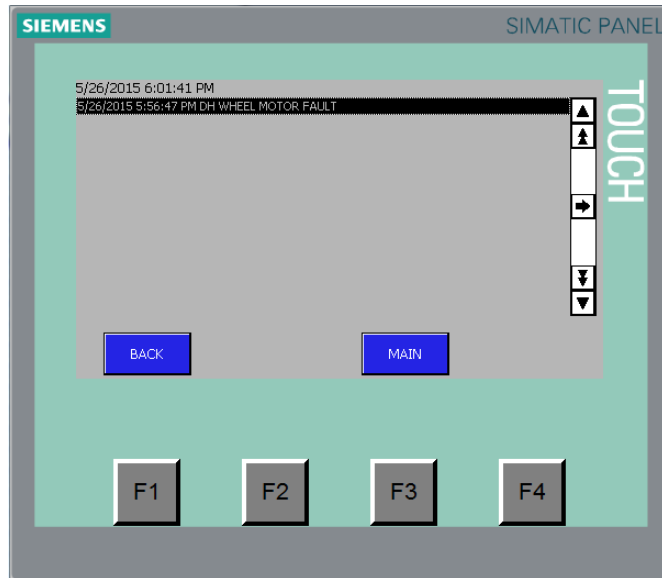


Figure 7-3
ALARM HISTORY DISPLAY

To troubleshoot beyond this point, see the Wiring Diagram and Section 7.7 Troubleshooting Using the PLC.

7.3 CHECKING THE POWER DISTRIBUTION SYSTEM

For details of the power distribution system, see the Wiring Diagram.

1. If supplied, the three-phase AC entering the unit is checked by a three-phase power monitor (PMO1). If one of the phases is missing, or if the rotation of the motors is reversed, a red LED on this unit will light. The power monitor is also tied to the E-Stop interlock circuit.
2. The contactor for each of the blower motors (OLR1 and OLR2) has integral short-circuit and overload protection. An overload will trigger a fault indication on the front panel (3 flashes). A short-circuit will trip the contactor. It can be reset by pressing a button on the front of the unit.
3. A thermocouple (TC01) senses the reactivation outlet temperature.

Machine with one zone of heat -

Based on the input from TC01, the PLC uses solid-state relays (SSR1) to control or modulate the output of the heating elements. Each SSR receives an analog input from the PLC, and switches one phase of the AC to the heating elements. When the SSR is receiving the analog input, and is supposed to be turned on, an amber LED on the SSR lights. See Figure 7-4. Whenever the amber LED is on, the SSR should be producing an output.

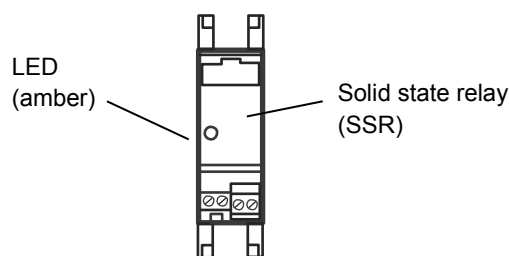


Figure 7-4
SOLID STATE RELAY

Machine with several zones of heat -

Based on the input from TC01, the PLC uses solid-state relays (SSR1) to control or modulate the output of the first set of heating elements. Each SSR receives an analog input from the PLC, and switches one phase of the AC to the heating elements. When the SSR is receiving the analog input, and is supposed to be turned on, an amber LED on the SSR lights. See Figure 7-4. Whenever the amber LED is on, the SSR should be producing an output.

Each additional heating zone can be switched on by the PLC via a contactor. As additional heating is required, the PLC turns on these additional zones. The total heat output is still "fine-tuned" by the PLC modulating the first zone via the SSR.

7.4 CHECKING THE HEATING ELEMENTS

1. The heating elements are located in the intake side of the reactivation air stream (upstream of the desiccant wheel). Usually they are mounted in a box or chamber.
2. Turn off the power to the unit!



Turn off the power to the HCD unit before working with the heating elements. These parts carry high voltage and current which can result in death or cause severe burning. Set the Auto/Off/Manual switch to the Off position. Follow standard lock-out tag-out procedures.

3. Remove the cover on the box or chamber.
4. To check the elements, remove the buss bars which connect the individual heating elements. Check the resistance across each of the heating elements using an Ohmmeter. Check inside the heater compartment, on the underside of the cover, to find the wiring diagram for the heating elements. If an element is operational, there should be a resistance of a few Ohms through the element. An open circuit or a high resistance indicates a failed element.

7.5 CHECKING THE HUMIDISTAT

Some HCD units are wired to a humidistat. Check the wiring diagram for the machine. For reliable operation, the humidistat must be accurate.

1. Start by adjusting the humidistat to the high end of the scale (close to 100% relative humidity). At this point, the humidistat should not be calling for drying. The contacts inside the humidistat should be open, and full AC voltage should be able to be measured across the contacts.



The contacts inside the humidistat carry 115V AC. This voltage and current can result in death or cause severe burning. Only trained electricians should perform work with the electrical parts.

2. Next, adjust the humidistat to the low end of the scale (close to 10% or 20% relative humidity). Now the humidistat should be calling for drying, and the contacts inside the humidistat should be closed. 0V should be seen across the contacts.
3. Also check the wiring between the humidistat and the HCD unit. When the humidistat is not calling for dehumidification, full AC voltage should be seen across the terminals. When the humidistat does call for dehumidification, the voltage across the terminals should drop to 0V.

7.6 CHECKING THE THERMOCOUPLES

HCDs have two thermocouples: TC01 and TC02. Because of the way these parts operate, they cannot be checked with a voltmeter. If a thermocouple fails, it usually does so in an “open-circuit” condition. It is possible to check continuity through the suspect unit with an Ohmmeter. Be sure to turn off the HCD unit before performing this check. Disconnect one of the leads, so other circuitry will not affect the reading.

7.7 TROUBLESHOOTING USING THE PLC

The HCD unit is controlled by a “programmable logic controller” or “PLC.” The PLC is a small computer which executes a simple set of commands or “program.” The program is organized as a loop which repeats continuously.

The PLC is mounted inside the control enclosure. Three LEDs on the front of the PLC show the status of the PLC itself. See Figure 7-5 and the following table.

The following table may help to interpret the three LEDs.

LED	Color	Normal Behavior
Error	Solid Red Flashing Red	Defective hardware PLC error
Run / Stop	Solid Yellow Solid Green Alternating Green & Yellow	PLC in Run mode (not program mode) PLC in Stop mode PLC is in Startup mode
Maintenance (Maint)	Yellow	Flashes whenever a memory card is inserted. The CPU then changes to STOP mode and one of the following functions needs to be performed to initiate the evaluation of the memory card: • Change the CPU to RUN mode • Perform a memory reset (MRES) • Power-cycle the CPU

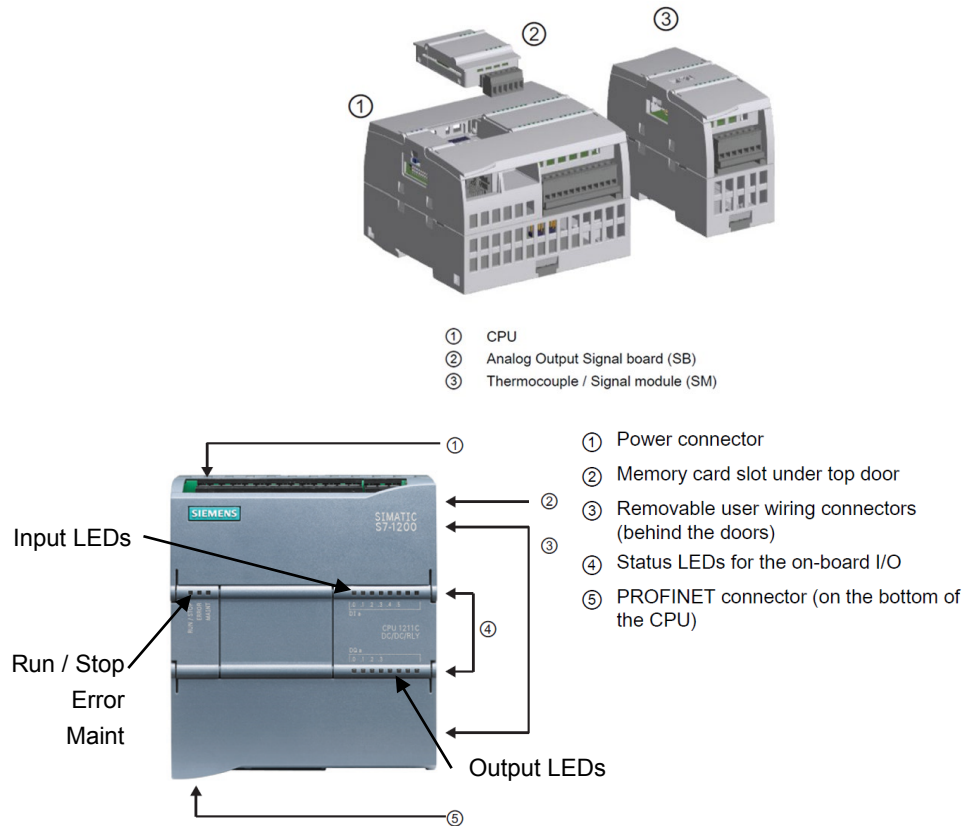


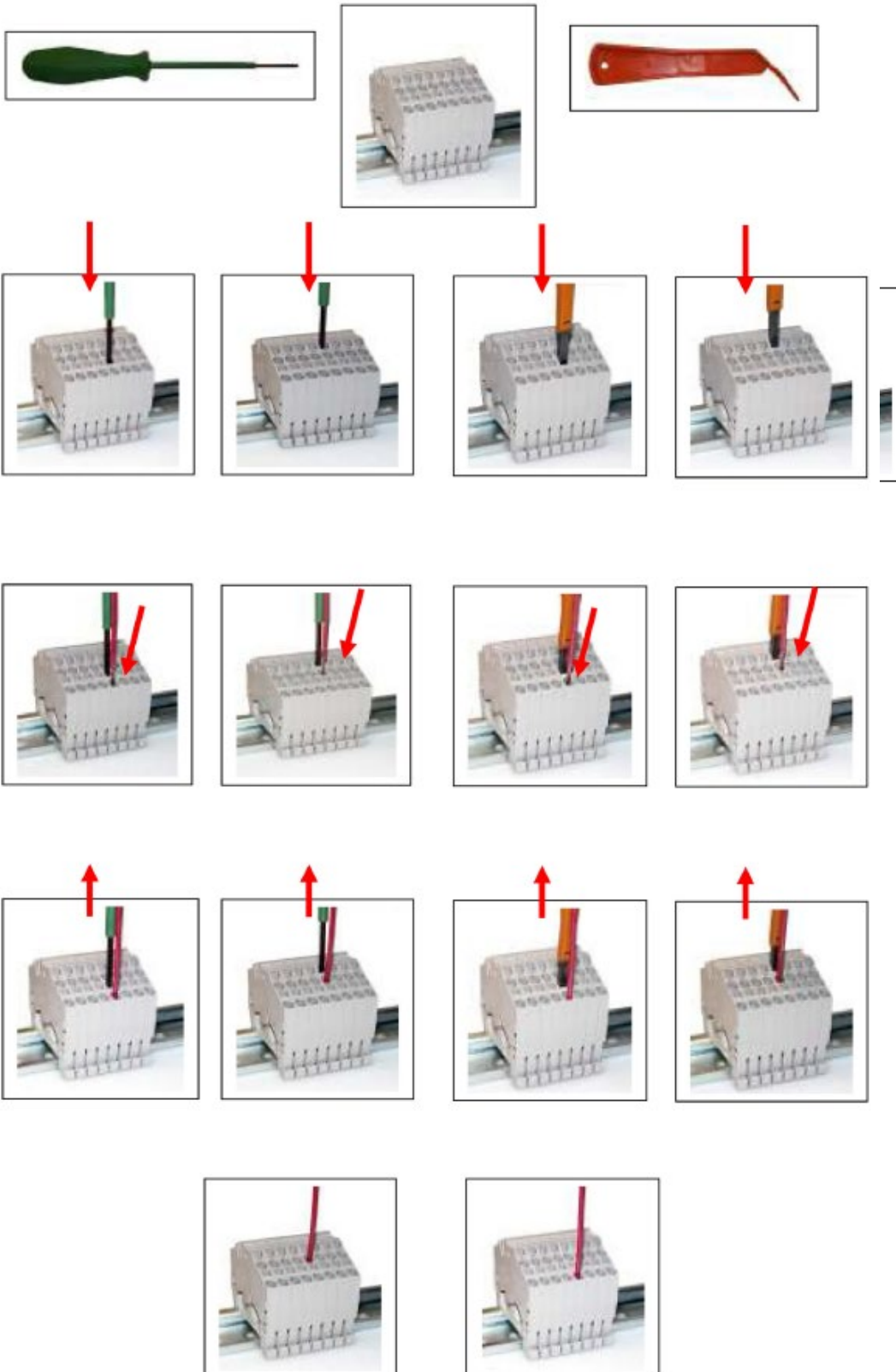
Figure 7-5
PROGRAMMABLE LOGIC CONTROLLER (PLC)

For more information, see the manual supplied by Siemens. This is available online:

<http://www.automation.siemens.com/>

The front panel of the PLC also includes LEDs which show the status of each of the inputs and outputs. These LEDs may be helpful in troubleshooting. For example, to check a switch on the machine, operate the switch and watch the PLC. See if the LED for that input lights up. These LEDs can be used to quickly check the condition of the whole machine. The Wiring Diagram gives more information on the functions indicated by the LEDs.

7.8 TERMINAL BLOCK WIRING INSTRUCTIONS



Munters Corporation
79 Monroe Street P.O. Box 540
Amesbury, MA 01913-0640
TEL (978) 241-1100 or 1-800-843-5360
FAX (978) 241- 1217



I27092

SCOPE & RESPONSIBILITY
22367602-01 WAYNE FARMS, LLC

NOTE: THIS UNIT IS NOT SUITABLE FOR OPEN ROOF CURB MOUNTING.

DESCRIPTION	RESPONSIBILITY	
	MUNTERS	WAYNE FARMS
ELECTRICS		
NEMA 4 CONTROLLER	X	
REACTIVATION ENERGY MODULATION	X	
460/3/60 POWER SUPPLY		X
CIRCUIT BREAKER DISCONNECT	X	

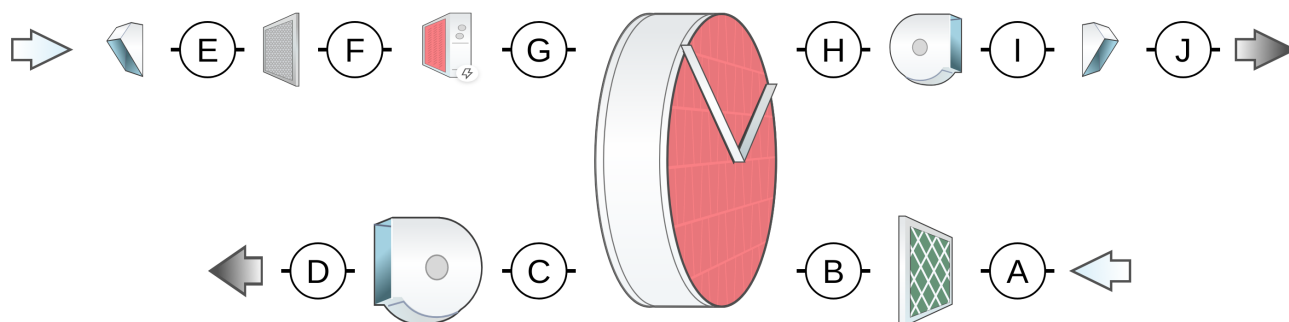
Please Note:

SUBSTITUTION OF PARTS AND COMPONENTS. In the event any delay in transportation, defect or delay in the performance of any supplier or subcontractor, obsolescence, or inability to obtain necessary materials or components results in the inability of Munters to procure or manufacture any part included in any Product, Munters shall have the right to procure, manufacture, and use any substantially similar part in replacement thereof.

Product Code: HCD-2250-EA-SFS
 Genesys No: G341624-1
 System Name: Wayne Farms Union Springs
 Sales Order No: 22367602-01
 Revision: 0

Prepared by: Brian Cull
 Date: 7/3/2024
 Prepared for: Wayne Farms, LLC
 Project Name: Wayne Farms – Union Springs

Flow diagram



Climate data location: Montgomery, Alabama, United States of America (203 ft / 14.6 psi)

Summer

	A	B	C	D	E	F	G	H	I	J
SCFM	2250	2250	2250	2250	562	562	562	562	562	562
°F	50	50	94	97	84	84	285	112	119	119
gr/lb	48.0	48.0	5.2	5.2	140.6	140.6	140.6	312.0	312.0	312.0
in WC	0.00	-0.50	-2.46	1.99	-0.35	-0.95	-2.77	-4.69	2.59	2.24

Product Code: HCD-2250-EA-SFS
Genesys No: G341624-1
System Name: Wayne Farms Union Springs
Sales Order No: 22367602-01
Revision: 0

Prepared by: Brian Cull
Date: 7/3/2024
Prepared for: Wayne Farms, LLC
Project Name: Wayne Farms – Union Springs

Summary

System

Product Code	HCD-2250-EA-SFS
Installed location	Outdoor
Electrical power source	460/3/60
Full load amps	See Wiring Diagram 22367602-01-WD
SCCR	5 KA RMS SYMMETRICAL, 600 VAC MAX
Plenum size	HCD-2250
Dehumidifier model	HCD-2250
Reactivation heater type	Electric
Gas type	N/A
Desiccant type	Tigel
Skid type	Formed steel
Adsorption dehumidification	62.0 lb/h
System total dehumidification	62.0 lb/h

Product Code: HCD-2250-EA-SFS
Genesys No: G341624-1
System Name: Wayne Farms Union Springs
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Revision: 0

Prepared by: Brian Cull
Date: 7/3/2024
Prepared for: Wayne Farms, LLC
Project Name: Wayne Farms – Union Springs

Technical data - Pre-Filter

Main selections

Filter efficiency bank 1

MERV 8

Results

Filter pressure drop design

0.50 in WC

Filter pressure drop clean

0.29 in WC

Filter pressure drop dirty

0.50 in WC

Face area

6.00 ft²

Face velocity

375 ft/min

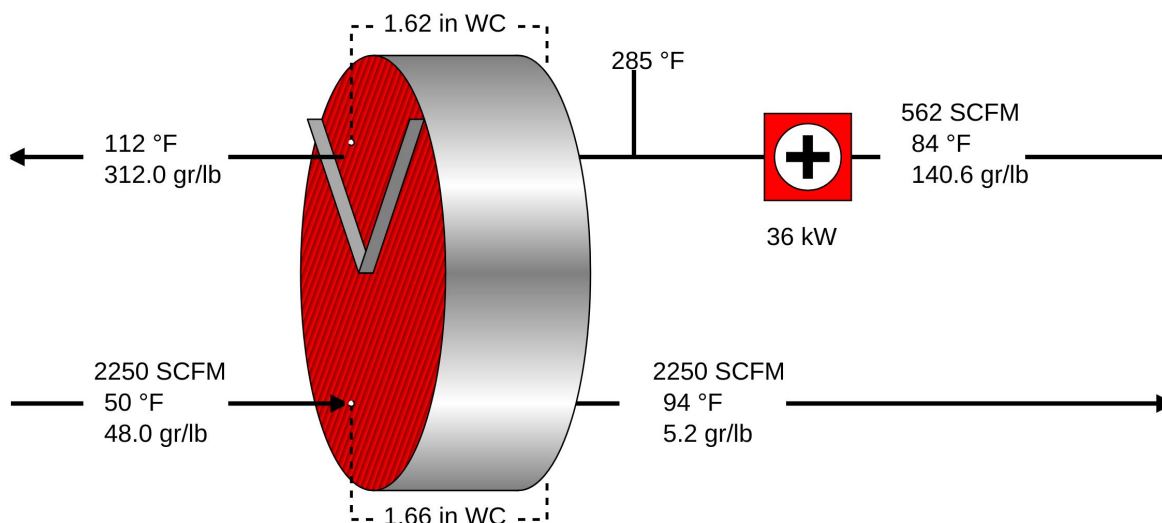
Filter dimensions

Qty 1 @ 12in x 24in & Qty 1 @ 24in x 24in

Product Code: HCD-2250-EA-SFS
 Genesys No: G341624-1
 System Name: Wayne Farms Union Springs
 Sales Order No: 22367602-01
 Revision: 0

Prepared by: Brian Cull
 Date: 7/3/2024
 Prepared for: Wayne Farms, LLC
 Project Name: Wayne Farms – Union Springs

Technical data - Dehumidifier



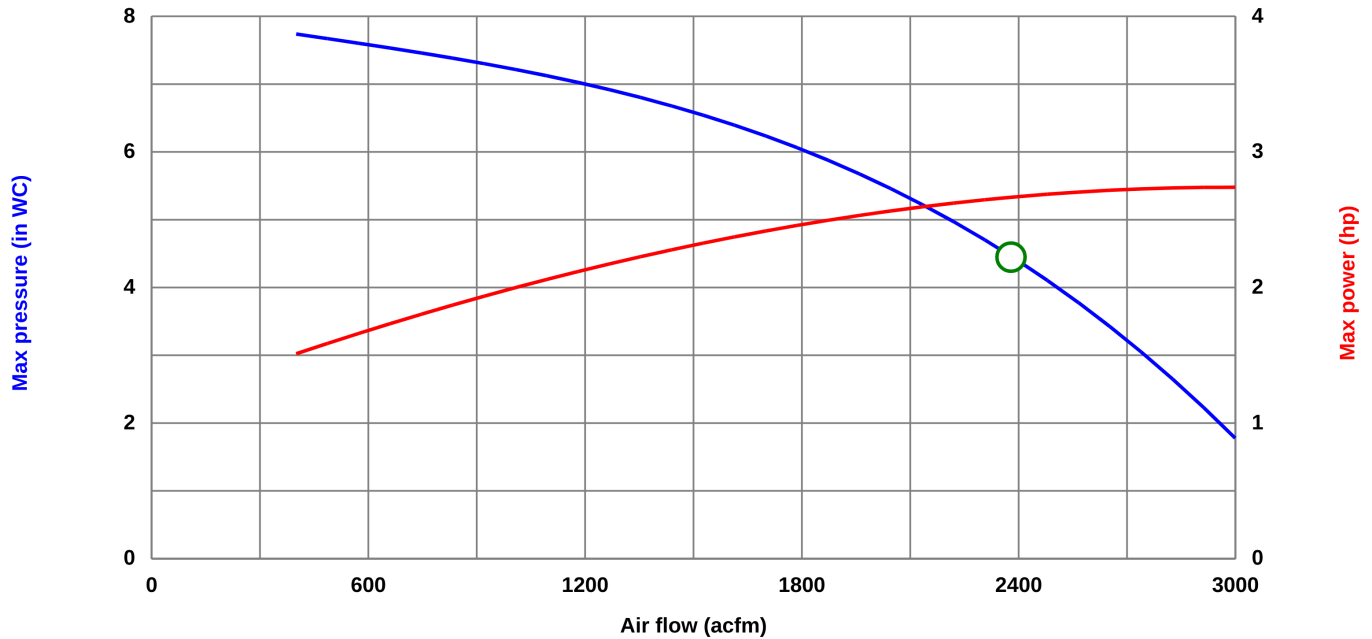
Main selections

Air flow direction	Right to left
Installed location	Outdoor
Electrical power source	460/3/60
Dehumidifier model	HCD-2250
Desiccant type	Tigel
Bypass control type	None
Reactivation heater type	Electric
Bypass section type	No
Electric heater capacity	42 kW
Rotor speed	8 rph
Adsorption dehumidification	62.0 lb/h

Product Code: HCD-2250-EA-SFS
 Genesys No: G341624-1
 System Name: Wayne Farms Union Springs
 Sales Order No: 22367602-01
 Revision: 0

Prepared by: Brian Cull
 Date: 7/3/2024
 Prepared for: Wayne Farms, LLC
 Project Name: Wayne Farms – Union Springs

Technical data - Process fan



Main selections

Fan size	12-1/4
Fan mounting location	End
Percent width	100 %
Drive type	Direct
Fan speed	3450 rpm
Fan discharge orientation	Bottom horizontal rear
Fan type	SWSI airfoil
Motor HP	3
Motor type	TEFC

Results

Inlet air temperature	94 °F
Leaving air temperature	97 °F
Air flow out	2379 acfm
Fan total static pressure	4.45 in WC
Fan Static Efficiency	59 %
Fan shaft power	2.7 hp
External Static Pressure	1.99 in WC

Product Code: HCD-2250-EA-SFS
Genesys No: G341624-1
System Name: Wayne Farms Union Springs
Sales Order No: 22367602-01
Revision: 0

Prepared by: Brian Cull
Date: 7/3/2024
Prepared for: Wayne Farms, LLC
Project Name: Wayne Farms – Union Springs

Technical data - Reactivation filter

Main selections

Filter efficiency bank 1
High volume metal mesh

Metal mesh
Yes

Results

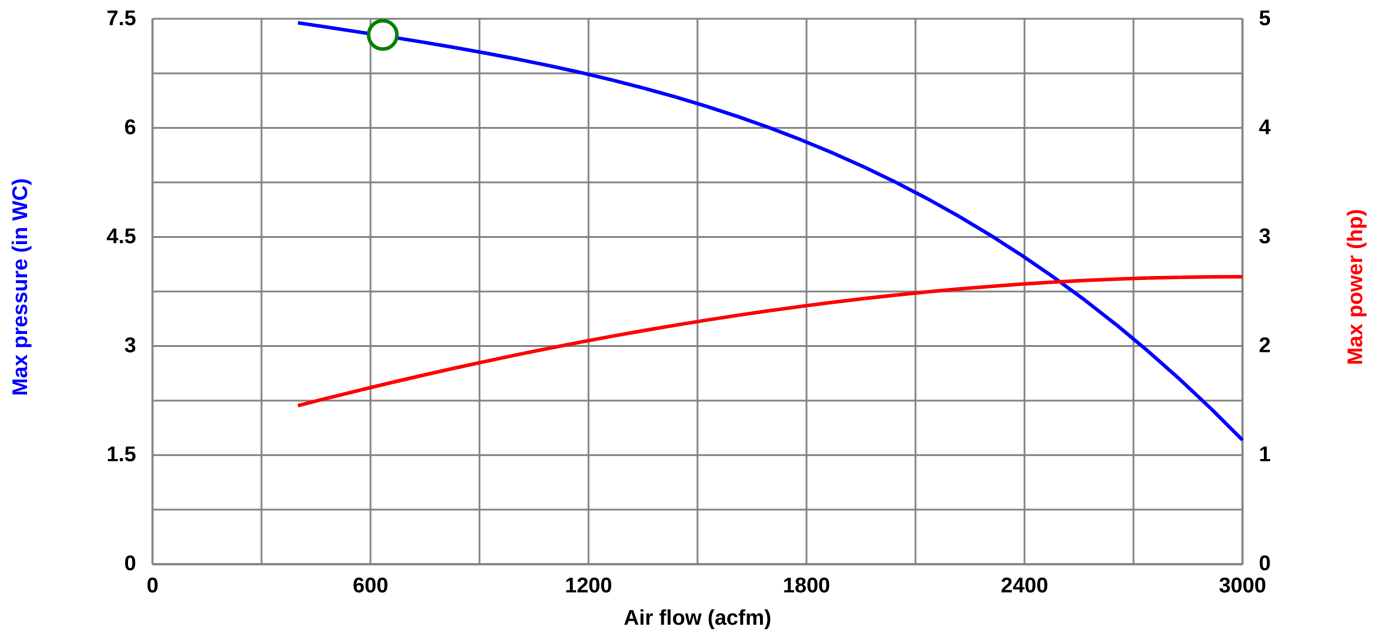
Filter pressure drop design
Filter pressure drop clean
Filter pressure drop dirty
Face area
Face velocity
Filter dimensions

0.60 in WC
0.11 in WC
0.60 in WC
2.25 ft²
250 ft/min
Qty 1 @ 18in x 18in

Product Code: HCD-2250-EA-SFS
 Genesys No: G341624-1
 System Name: Wayne Farms Union Springs
 Sales Order No: 22367602-01
 Revision: 0

Prepared by: Brian Cull
 Date: 7/3/2024
 Prepared for: Wayne Farms, LLC
 Project Name: Wayne Farms – Union Springs

Technical data - Reactivation fan



Main selections

Dehumidifier model	HCD-2250
Outlet options	Weatherhood
Fan size	12-1/4
Percent width	100 %
Drive type	Direct
Fan speed	3450 rpm
Fan discharge orientation	Top horizontal rear
Fan wheel rotation	CW
Fan type	SWSI airfoil
Motor HP	3
Motor type	TEFC

Results

Inlet air temperature	112 °F
Leaving air temperature	119 °F
Air flow out	634 acfm
Fan total static pressure	7.28 in WC
Fan Static Efficiency	39 %
Fan shaft power	1.6 hp
External Static Pressure	2.24 in WC

Product Code: HCD-2250-EA-SFS
Genesys No: G341624-1
System Name: Wayne Farms Union Springs
Sales Order No: 22367602-01
Revision: 0

Prepared by: Brian Cull
Date: 7/3/2024
Prepared for: Wayne Farms, LLC
Project Name: Wayne Farms – Union Springs

Technical data - Electrical options

Controller type	Siemens S7-1200 (Standard)
Electrical power source	460/3/60
Certification type	UL508A

Process fan enable

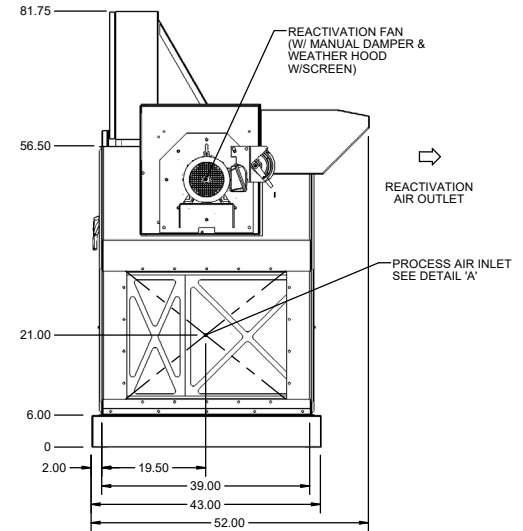
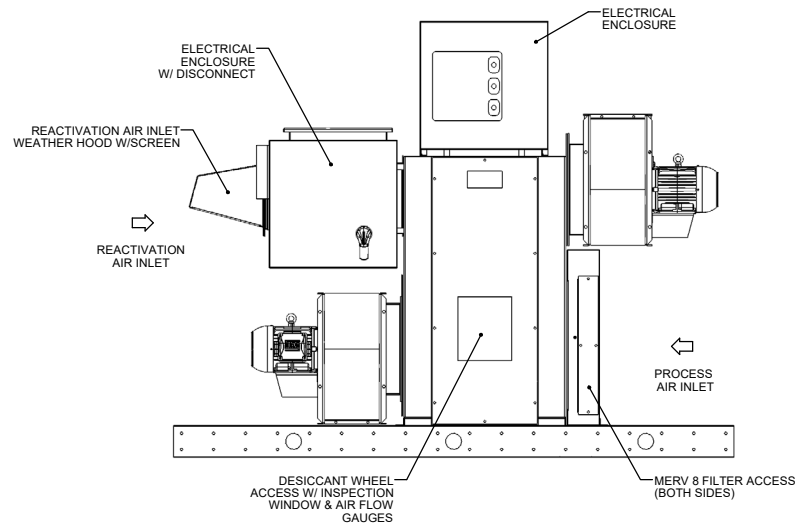
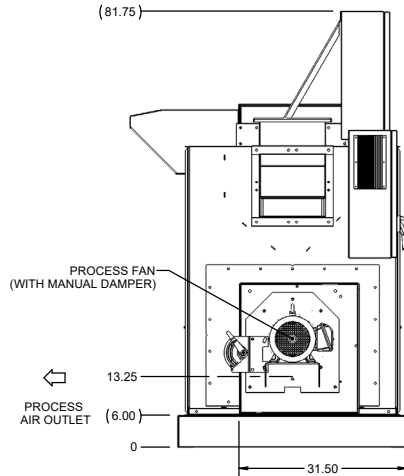
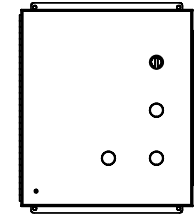
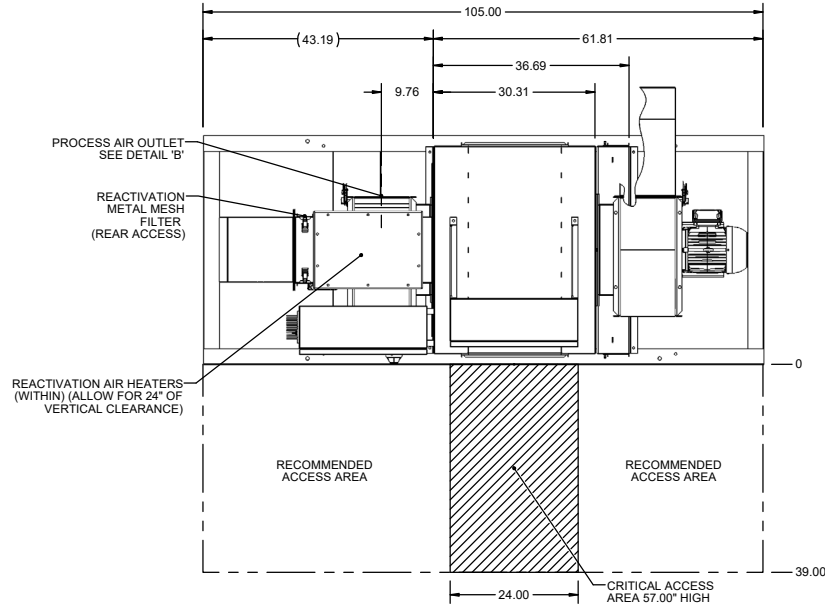
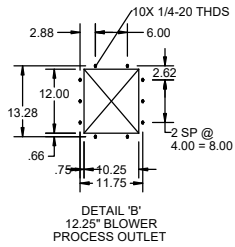
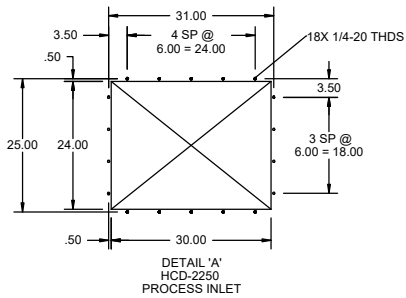
Process fan enabled when	System is enabled (constant process fan)
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1 x Remote start/stop station

Station type	with selector switch and pilot lights
--------------	---------------------------------------

1 x Electrical disconnect

Disconnect type	Circuit breaker
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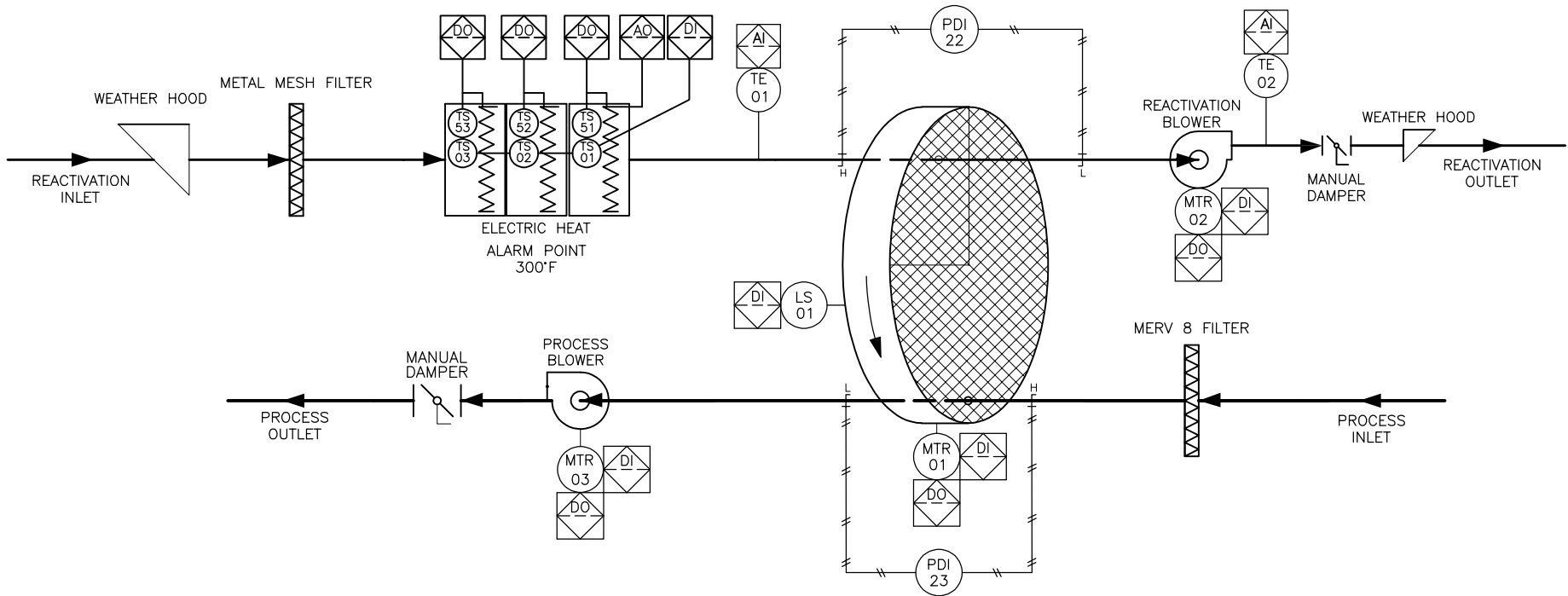
NOTES:

- CAPACITIES, ENERGY REQUIREMENTS AND FEATURES WILL BE ALTERED TO MATCH APPLICATIONS. CONSULT TECHNICAL DATA SHEET AND MANUAL BEFORE INSTALLING OR OPERATING UNIT.
 - BLOWERS ARE PROVIDED WITH DAMPER AND MANUAL LOCKING QUADRANT.
 - UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES.
 - ALL DIAMETERS ARE OUTSIDE DIAMETER U.O.S.
 - ALL PIPING CONNECTIONS $\pm 1/2$ ".
 - ANY ELECTRICAL AND/OR MECHANICAL REASSEMBLY IS THE RESPONSIBILITY OF OTHERS UNLESS PRIOR ARRANGEMENTS HAVE BEEN MADE WITH MUNTERS. THIS INCLUDES DISASSEMBLY THAT WAS NECESSARY TO FACILITATE SHIPPING.
 - MUNTERS REQUIRES THAT SKID MOUNTED UNITS SHALL BE INSTALLED LEVEL, FRONT TO BACK, AND SIDE TO SIDE.
 - THE UNIT MUST BE SUPPORTED ALONG THE FULL PERIMETER OF THE SKID.
 - IF FULL PERIMETER SUPPORT IS NOT POSSIBLE OR PRACTICAL, IT IS ACCEPTABLE TO SHIM THE UNIT TO LEVEL AT A MAXIMUM OF 1 FOOT INCREMENTS.
 - FOR TEMPORARY NON-OPERATIONAL STORAGE, UNIT SHALL BE SUPPORTED AND LEVELED UNDER EACH LIFTING LUG LOCATION.
 - THE ALLOWABLE TOLERANCE TO ACCOUNT FOR DEFLECTION OF THE SKID SHALL NOT EXCEED 1/4" OVER A 10 LINEAL FOOT SPAN.
 - SYSTEM IS NOT DESIGNED FOR OPEN ROOF CURB MOUNTING.
 - REFER TO SECTION 110.26 IN THE NEC CODE BOOK FOR CLEARANCES AND REQUIREMENTS TO ELECTRICAL PANELS. THESE REQUIREMENTS MAY IMPACT THE LOCATION OF THE SUPPLIED EQUIPMENT. IT IS THE CUSTOMER'S RESPONSIBILITY TO PLAN THE LOCATION OF THIS EQUIPMENT USING THE REQUIREMENTS SET FORTH IN NEC AND OTHER LOCAL CODES.
- APPROXIMATE WT. = 1400 LBS.

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REVISIONS					
REV.	ECR	CHANGE NOTES	DWN BY	DWN DATE	APPVD BY
					APPRVD DATE

CUSTOMER WAYNE FARMS, LLC				JOB NO. 22367602-01	
Description GENERAL ARRANGEMENT HCD-2250-EA-SFS					
Tolerances, if not indicated, according to:		See MUNTERS Doc.No. 160-027139-001			Sheet 1 / 1
					Revision 0
 Munters	DESIGN BY	DATE	PROJ ENG CRD	DATE	Drawing / Part No 22367602-01
	USBOSJOWH	7/3/2024	BDC	7/8/2024	
	ELECTRICAL CRD	DATE			
	DL	7/9/2024			



NOTES:

1. DASHED LINES INDICATE CONTROLS/EQUIPMENT BY OTHERS.
2. SOLID LINES INDICATE CONTROLS/EQUIPMENT BY MUNTERS.
3. SEE DRAWING 22602 FOR SYMBOLS & IDENTIFICATION.
4. SEE EQUIPMENT SCHEDULE/TECH DATA SHEET FOR UTILITY CONNECTION SIZES, RATINGS & SET POINTS.

Description		PROCESS FLOW & INSTRUMENTATION DIAGRAM HCD-2250-EA-SFS	
Customer		Job No.	
WAYNE FARMS, LLC		22367602-01	
Tolerances, if not indicated, according to:		Sheet	Revision
See MUNTERS Doc.No. 160-010266-001		1 / 1	0
Design by		Date	Drawing / Part No
BDC		7/3/24	22367602-01-PID
PM. ckd		Date	
BDC		7/3/24	
EE. ckd		Date	
DL		7/3/24	

STANDARD DEVICE AND COMPONENT IDENTIFICATION AND PIPING IDENTIFICATION								Revisions										
SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION	Rev	Description	Date	Dft	Chk	App					
AI	ANALOG INPUT	ESD	EMERGENCY STOP	MT	MOISTURE TRANSMITTER	RES	RESISTOR	1	ADDED SUPPLY AIR PRESSURE NUMBER	1/16/97	DMR	KPW	KPW					
AO	ANALOG OUTPUT	ETM	ELAPSED TIME METER	MTR	MOTOR	RG	REGULATOR	2	ADDED IDENTIFICATIONS, CAPILARY TUBE LINE SYMBOL, & 3 LETTER WIRE ABRV.	3/6/97	DMR	KPW	KPW					
AFS	AIR FLOW SWITCH	FB	FUSE BLOCK	NC	NORMALLY CLOSED	RSS	ROTARY SWITCH	3	ADDED PIPING IDENTIFICATION	8/25/97	DMR	KPW	KPW					
AH	ALARM HIGH SET POINT	FC	FAIL CLOSED	NO	NORMALLY OPEN	RTD	RESISTIVE TEMPERATURE DETECTOR	4	ADDED RFQ, PV-- & SSR--	5/15/97	CJT	KPW	KPW					
ATM	ATMOSPHERE	FIT	FLOW INDICATING TRANSMITTER	OLR	OVERLOAD RELAY	SD	SMOKE DETECTOR	5	ADDED- (AMB)-AMBER COLOR	5/15/97	RRM	KPW	KPW					
BU	BULB	FS	FLOW SWITCH	OOF	OPEN ON FALL	SEN	SENSOR	6	ADDED SYBOLS MIT THRU PDS	5/03/01	TJM	KPW	KPW					
C	CLOSED/CONTROL/COOL	FO	FAIL OPEN	OOR	OPEN ON RISE	SOL	SOLENOID	7	ADDED VIB SYMBOL	8/21/03	CAG	KPW	CAG					
CAP	CAPACITOR	FSC	FLAME SAFETY CONTROLLER	P/I	PNEUMATIC TO CURRENT TRANSDUCER	SP	SETPOINT VALUE	8	NEW SYMBOLS ON SHEET 2 ADDED	9/04/12	YD	KPL	CAG					
CB	CIRCUIT BREAKER	FU	FUSE	P/P	PNEUMATIC TO PNEUMATIC POSITIONER	SS	SELECTOR SWITCH	9	CHANGED FSV TO FSC	1/06/21	CLS	DSL	CLS					
CCR	CIRCULAR CHART RECORDER	GFCI	GROUND FAULT CIRCUIT INTERRUPTER	P/S	PRESSURE SWITCH	SSP	SOLID STATE POWER CONTROLLER	10	TAG NAME CHANGES AND UPDATE	09/16/22	TGR	DSL	TGR					
CD	CONDENSING UNIT	GMV	GAS MODULATING VALVE	PB	PUSHBUTTON	SV	SOLENOID VALVE	LINE SYMBOLS GENERAL MAIN PROCESS EQUIPMENT OUTLINE // // // PNEUMATIC SIGNAL OPTIONAL JUMPER PIPING × × × CAPILLARY TUBE CONTROLLER CONNECTION COLOR CODE ABBREVIATIONS BLK - BLACK BLU - BLUE BRN - BROWN GRY - GRAY RED - RED WHT - WHITE ORG - ORANGE CLR - CLEAR YLW - YELLOW VLT - VIOLET GRN - GREEN GENERAL SYMBOLS ⊥ ⊥ ⊥ GROUNDIED TERMINAL ⏏ GROUND CONNECTION CHASSIS OR FRAME ✱ WRAP AND HEATSHRINK CABLE SHIELD										
CF	CARTRIDGE FILTER	GND	GROUND	PCB	PRINTED CIRCUIT BOARD	SW	SWITCH											
CMDP	CHILLER MIRROR DEW POINT	GPS	GAS PROVING SYSTEM	PD	PRESSURE DIFFERENTIAL	TB	TERMINAL BLOCK											
CN	CONTACTOR	H	HIGH	PDI	PRESSURE DIFFERENTIAL INDICATOR	T/C	THERMOCOUPLE											
COR	CLOSE ON RISE	HGV	HOT GAS VALVE	PDIS	PRESSURE DIFFERENTIAL INDICATING SWITCH	TC	TEMPERATURE CONTROLLER											
CR	CONTROL RELAY	HGSV	HOT GAS VALVE & SOLENOID	PDIT	PRESS. DIFF. INDICATING TRANSMITTER	TE	TEMPERATURE ELEMENT											
CV	CONTROL VALVE	HOA	HAN OFF AUTO	PDITS	PRESS. DIFF. INDICATING TRANSMITTER & SWITCH	TG	TEMPERATURE GAUGE											
DH	DEHUMIDIFIER	HRN	HORN	PDS	PRESSURE DIFFERENTIAL SWITCH	TR	TIMING RELAY											
DA	DAMPER ACTUATOR	HT	HUMIDITY TRANSMITTER	PDT	PRESSURE DIFFERENTIAL TRANSMITTER	TS	TEMPERATURE SWITCH											
DFG	DIRECT FIRE GAS	HTR	ELECTRIC HEATER	PDTS	PRESS. DIFF. TRANSMITTER & SWITCH	TT	TEMPERATURE TRANSMITTER											
DI	DIGITAL INPUT	HUM	HUMIDIFIER	PG	PRESSURE GAUGE	T	TRANSFORMER											
DO	DIGITAL OUTPUT	I/P	CURRENT TO PNEUMATIC TRANSDUCER	PL	PLUG	VA	VALVE ACTUATOR											
DIS	FUSED DISCONNECT	IC	IGNITION CONTROLLER	PLC	PROGRAMMABLE LOGIC CONTROLLER	VFD	VARIABLE FREQUENCY DRIVE											
DP	DEWPOINT TRANSMITTER/DISPLAY	L	LOW	PM	PHASE LINE MONITOR	VIB	VIBRATION SENSOR/ACCELEROMETER											
DPG	DIFFERENTIAL PRESSURE GAUGE	LS	LIMIT SWITCH	PNL	PANEL	VM	VENT MOTOR											
DPS	DIFFERENTIAL PRESSURE SWITCH	LLSV	LIQUID LINE SOLENOID VALVE	PRS	PROXIMITY SWITCH	VT	VENT											
DPT	DEWPOINT TRANSMITTER	LT	LIGHT	PS	POWER SUPPLY	VVA	VARIABLE VANE ACTUATOR											
DS	DISCONNECT	MDW	MOTOR DH WHEEL	PSH	PRESSURE SWITCH HIGH	ZS	POSITION SWITCH											
DSP	DISPLAY	MPF	MOTOR PROCESS FAN	PSI	POUNDS PER SQUARE INCH	ZSC	POSITION SWITCH CLOSED											
E/P	ELECTRIC TO PNEUMATIC SOLENOID VALVE	MPPF	MOTOR POWER PURGE FAN	PSL	PRESSURE SWITCH LOW	ZSD	POSITION SWITCH OPEN											
ENC	ENCLOSURE	MRF	MOTOR REACT FAN	PT	PRESSURE TRANSMITTER	CONFIDENTIALITY STATEMENT THE TECHNICAL INFORMATION AND DESIGN DATA DISCLOSED HERIN CONSTITUTE PROPRIETARY INFORMATION OF MUNTERS CORPORATION - CARGOCAIRE DIVISION AND ARE TO BE MAINTAINED IN STRICT CONFIDENCE. THIS INFORMATION IS FOR THE SOLE USE OF OUR CUSTOMERS AND END USERS OF OUR EQUIPMENT.		Job No.	STD	Date	Munters Cargocaire 79 Monroe St. Amesbury, Ma 01913 TEL 508-388-0600 FAX 508-388-0292							
EPL	ELECTRICAL PARTS LIST	MGV	MAIN GAS VALVE	R/C	RESISTOR CAPACITOR			Customer										
EPR	EVAPORATOR PRESSURE REGULATING VALVE	MIT	MOISTURE INDICATOR TRANSMITTER	RC	RECEIVER CONTROLLER			Dr.By.	TAB	10/15/96				STANDARD WIRING, PROCESS & INSTRUMENTATION LEGENDS & SYMBOLS IDENTIFICATION				
EPS	ENCLOSURE PROTECTION SYSTEM	MS	MOTOR STARTER	RCP	RECEPTACLE			Chk.By.	KPW	10/16/96								
						Aprv.By.	KPW	10/16/96	SCALE	SHEET	SIZE	22602	REV					
								NTS	1 of 2	B			10					

SYMBOL IDENTIFICATION					
SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION
	LOCALLY MOUNTED		CHILLED WATER RETURN		PRESSURE RELIEF REGULATOR
	PANEL MOUNTED		CHILLED WATER SUPPLY		MOTOR ACTUATED VALVE
	INSIDE PANEL MOUNTED		GLYCOL RETURN		GLOBE VALVE
	ALARM		GLYCOL SUPPLY		BACK DRAFT DAMPER
	AMBER INDICATOR LIGHT		HOT WATER RETURN		ROUND OR RECTANGULAR DAMPER
	BLUE INDICATOR LIGHT		HOT WATER SUPPLY		ADJUSTABLE GAS COCK
	GREEN INDICATOR LIGHT		NATURAL GAS		MANUAL OPERATION
	RED INDICATOR LIGHT		REFRIGERATION HOT GAS LINE		STRAINER
	WHITE INDICATOR LIGHT		REFRIGERATION LIQUID LINE		LEAK TESTER
	SUPPLY AIR PRESSURE NUMBER		REFRIGERATION SUCTION LINE		FILTER/REGULATOR
	CONTROLS INTERLOCK		STEAM LINE PRESSURE		EXPANSION/FLEXIBLE PIPE
	PROGRAMMABLE LOGIC CONTROLLER OUTPUT		SOLID STATE CONTROL RELAY		FILTER
	ANALOG OR DIGITAL INPUT OR OUTPUT TO OR FROM INSIDE PANEL MOUNTED PLC		STEAM		FILTER, IN LINE
	INSTRUMENT AIR		1/2" NPT PORT		FLOW SWITCH
	COMPRESSED AIR		FLANGED CONNECTION		THERMAL EXPANSION VALVE
	WATER		EXPANSION/FLEXIBLE JOINT		HUMIDIFIER
	AMBIENT AIR		GENERAL VALVE		ELECTRICAL HEATER
	LIQUID PROPANE		MANUAL VALVE W/TEST PORT		COIL
	FUEL OIL		PRESSURE REGULATOR		INDIRECT FIRED GAS BURNER
	COMBUSTION AIR		PRESSURE REGULATOR (SELF CONTAINED)		DIRECT FIRED GAS HEATER
	CONDENSATION		3-WAY VALVE		WEATHER HOOD/LOUVER

CONFIDENTIALITY STATEMENT

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Job No. STD

Customer

Dr.By. TAB

Ch.By. KPW

Appr.By. KPW

Date

10/15/96

10/16/96

10/16/96

Munters Cargocaire
79 Monroe St. Amesbury, Ma 01913
TEL 508-388-0600 FAX 508-388-0292

STANDARD WIRING, PROCESS & INSTRUMENTATION LEGENDS & SYMBOLS IDENTIFICATION

SCALE NTS

SHEET 2 of 2

SIZE B

22602

REV 10

0		1		2		3		4		5		6		7		8		9			
WAYNE FARMS, LLC HCD-2250-EA-SFS 22367602-01												SHEET	DESCRIPTION								
												1	TITLE								
												2	P&ID								
												3	460V POWER								
												4	SPARE								
												5	120 VAC								
												6	SPARE								
												7	SPARE								
												8	SPARE								
												9	PLC INPUTS								
												10	SPARE								
												11	PLC OUTPUTS								
												12	SPARE								
												13	ANALOG IN/OUT								
												14	SPARE								
												15	SPARE								
												16	TC INPUT								
												17	SPARE								
												18	SPARE								
												19	PLC LAYOUT								
												20	ELECTRIC REACT								
												21	SPARE								
												22	SPARE								
												23	SPARE								
												24	SPARE								
												25	SPARE								
												26	SPARE								
												27	REMOTE								
												28	MAIN PANEL								
												29	ELECTRIC HEAT PANEL								
												30	NOTES								
 Munters 79 Monroe St. Amesbury, MA 01913 TELEPHONE: 1-888-349-4335 WEBSITE: MUNTERS.COM/HELP		PROJECT NUMBER: 22367602-01		MODEL: HCD-2250-EA-SFS		DESCRIPTION:		CONFIDENTIALITY STATEMENT				DRAWN DSL		APPROVED BW		SIZE B		DRAWING NUMBER: 22367602-01-WD			
		PROJECT NAME: WAYNE FARMS, LLC				SEQUENCE OF OPERATIONS: K71200		DRAWING NAME: WIRING DIAGRAM		THE TECHNICAL INFORMATION AND DESIGN DATA HEREIN CONSTITUTES PROPRIETARY INFORMATION OF THE MUNTERS CORPORATION AND ARE TO BE MAINTAINED IN STRICT CONFIDENCE. THIS INFORMATION IS FOR THE SOLE USE OF OUR CUSTOMERS AND END USERS OF OUR EQUIPMENT.				CHECKED KPW		DATE 7/15/2024		REVISION 0		SHEET 1 OF 30	

FOR P&ID, SEE DRAWING 22367602-01-PID



Munters

PROJECT NUMBER:
22367602-01

MODEL:
HCD-2250-EA-SFS

DESCRIPTION:

CONFIDENTIALITY STATEMENT

DRAWN	DSL
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APPROVED BW

SIZE	B
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DRAWING NUMBER:
22367602-01-WD

PROJECT NAME:
WAYNE FARMS, LLC

SEQUENCE OF OPERATIONS:
K71200

DRAWING NAME:
WIRING DIAGRAM

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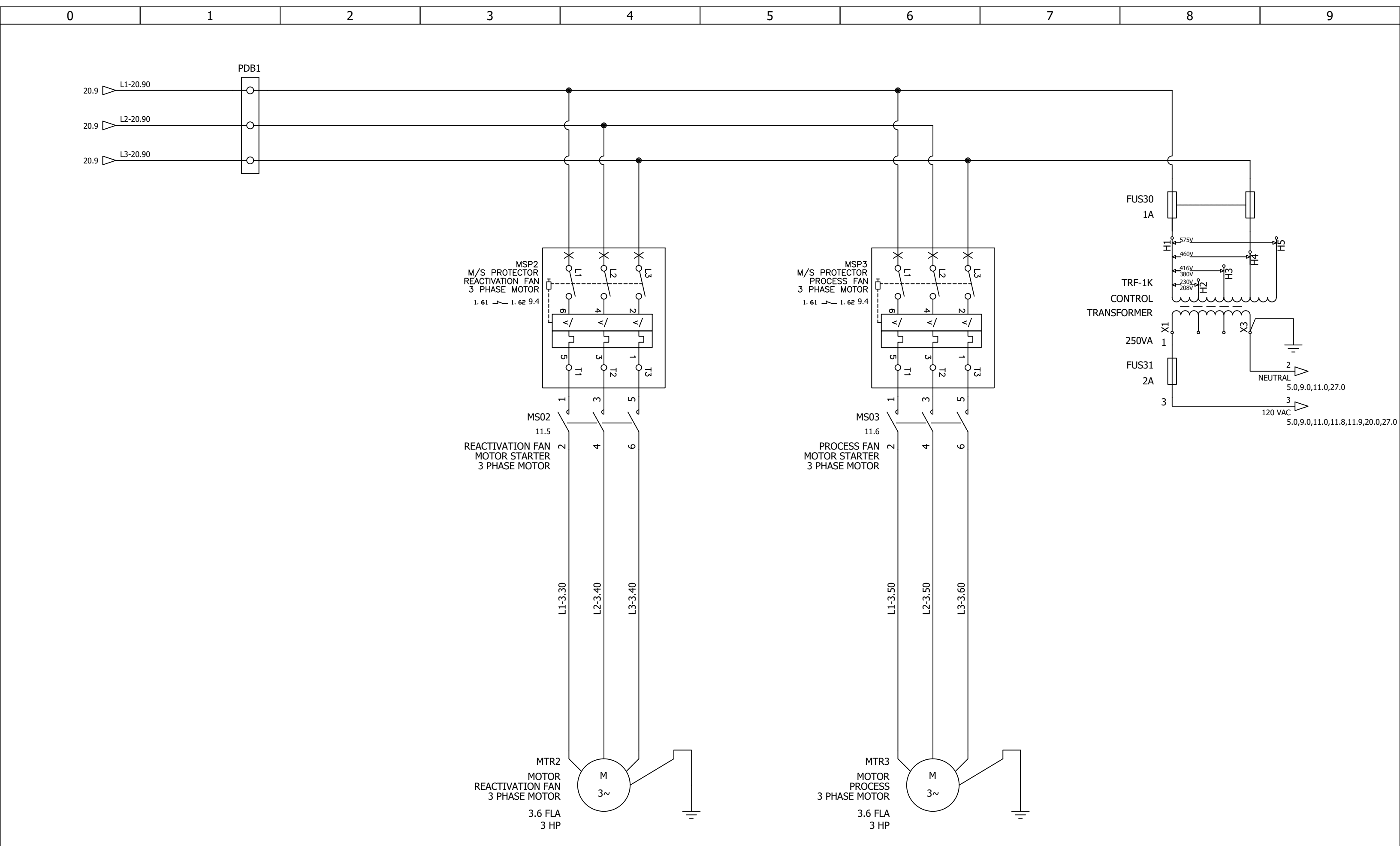
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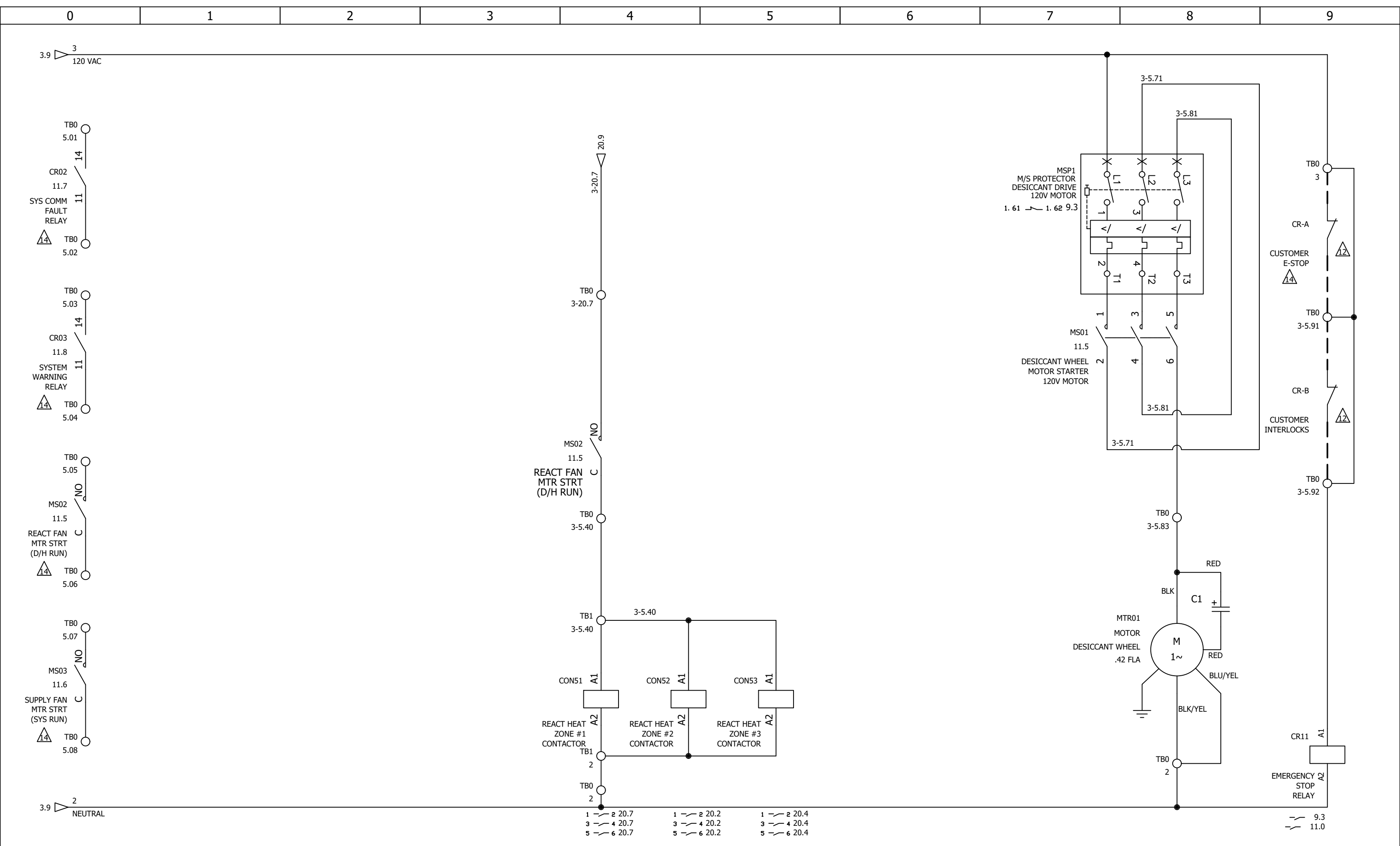
DATE	7/15/2024
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REVISION	0
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SHEET
2 OF 30

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SPARE SHEET



Munters

PROJECT NUMBER:
22367602-01

MODEL:
HCD-2250-EA-SFS

DESCRIPTION:

CONFIDENTIALITY STATEMENT

	DRAWN
	DSL

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BW

SIZE	B
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DRAWING NUMBER:
22367602-01-WD

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WEBSITE: MUNTERS.COM/HELP

PROJECT NAME:
WAYNE FARMS, LLC

SEQUENCE OF OPERATIONS:	K71200
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DRAWING NAME:
WIRING DIAGRAM

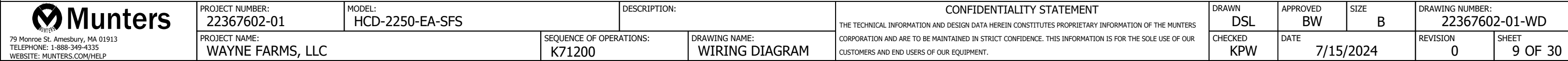
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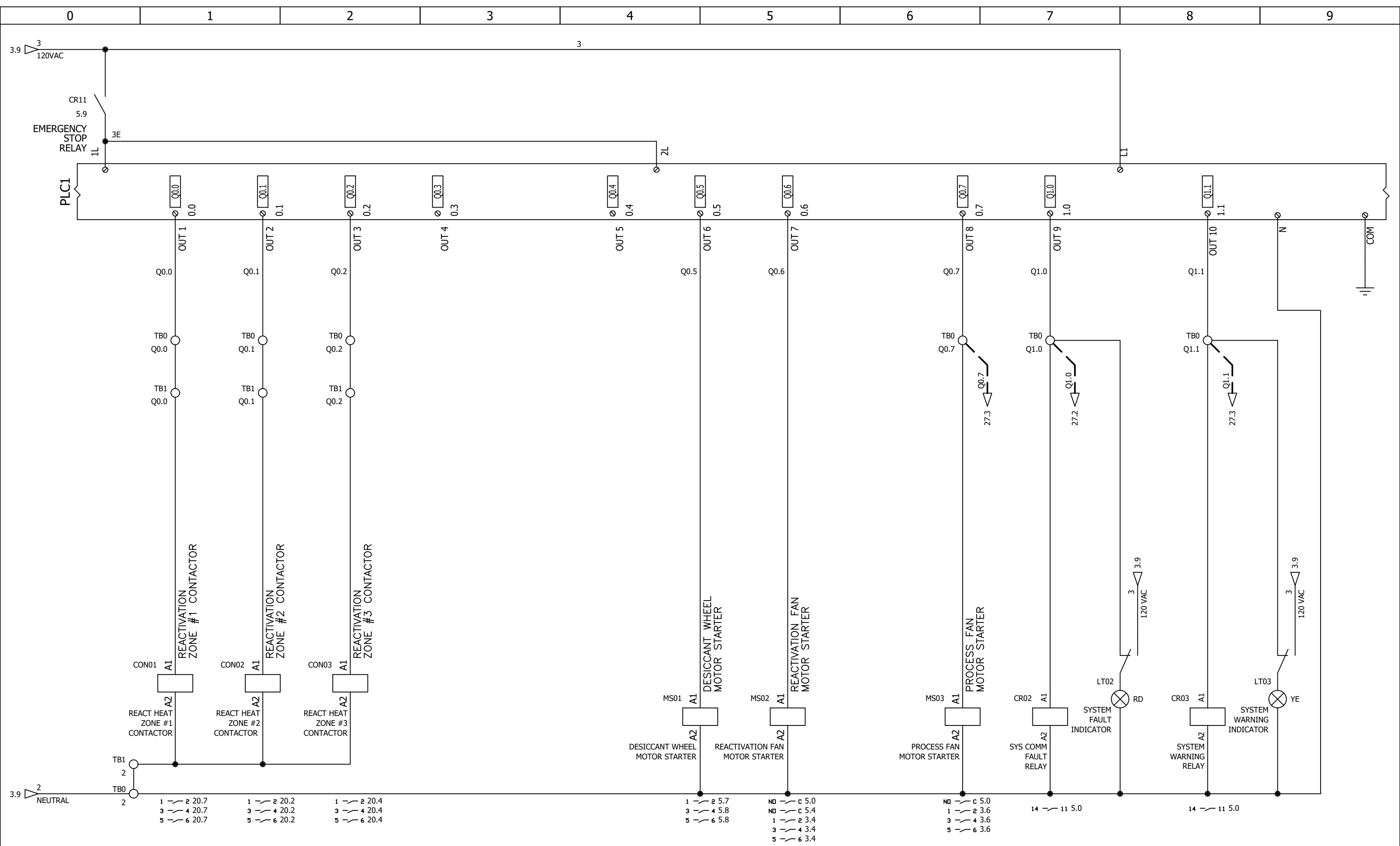
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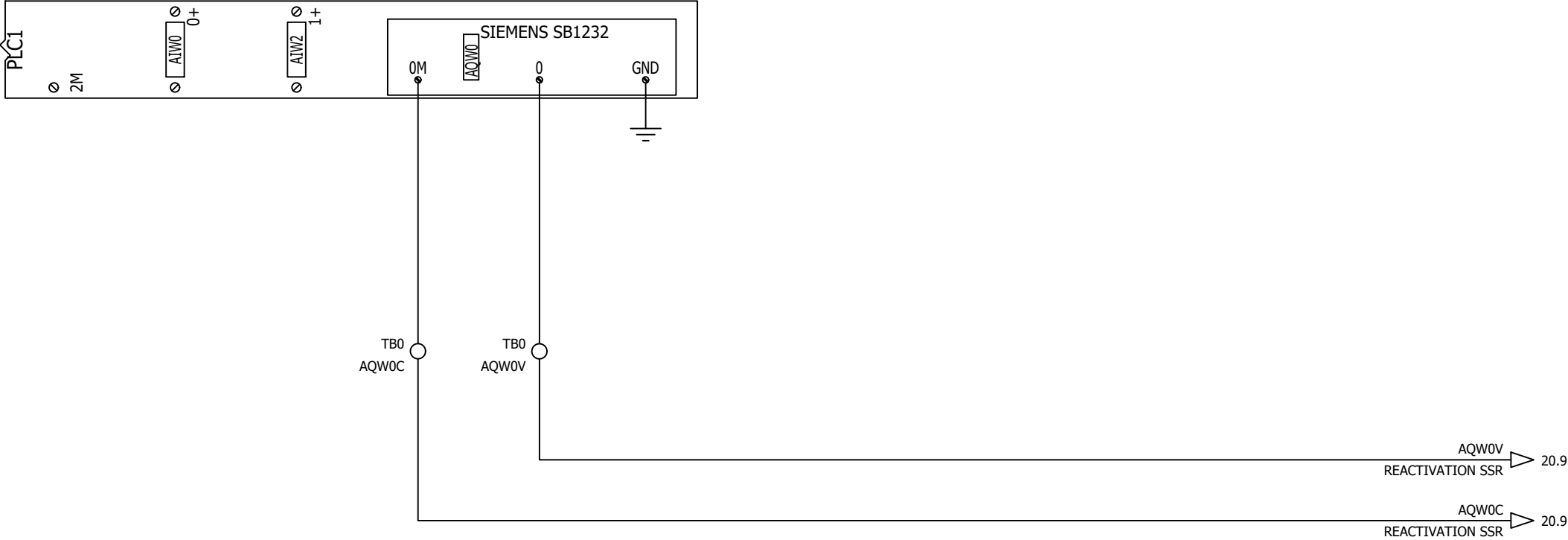
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REVISION	0
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SHEET
7 OF 30







SPARE SHEET



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DRAWN DSL

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BW

SIZE	B
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DRAWING NUMBER:
22367602-01-WD

PROJECT NAME:
WAYNE FARMS, LLC

SEQUENCE OF OPERATIONS:
K71200

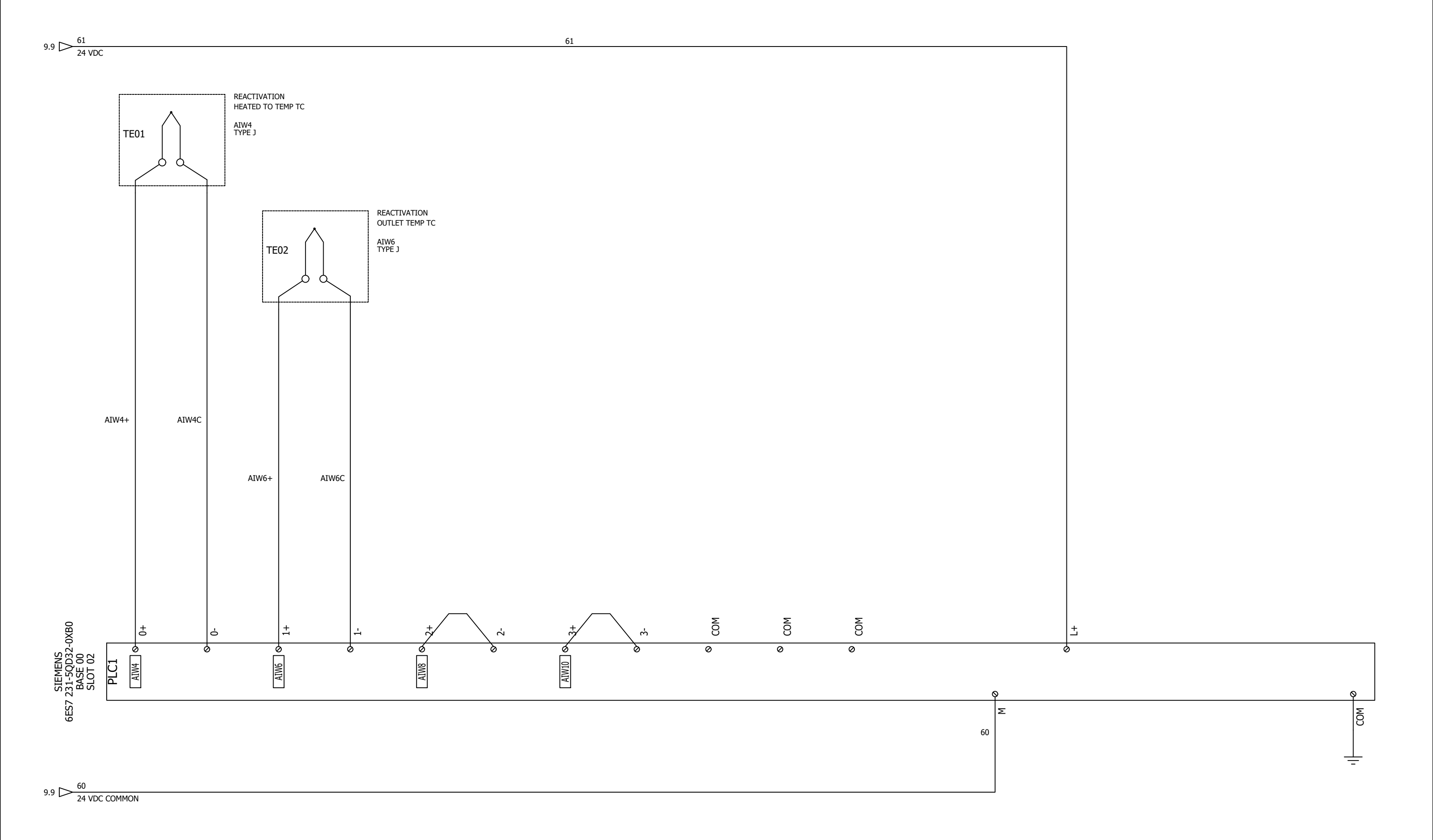
DRAWING NAME:
WIRING DIAGRAM

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KPW

DATE	7/15/2024
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REVISION	0
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SHEET
15 OF 30



SPARE SHEET



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DRAWN	DSL
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BW

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SEQUENCE OF OPERATIONS:
K71200

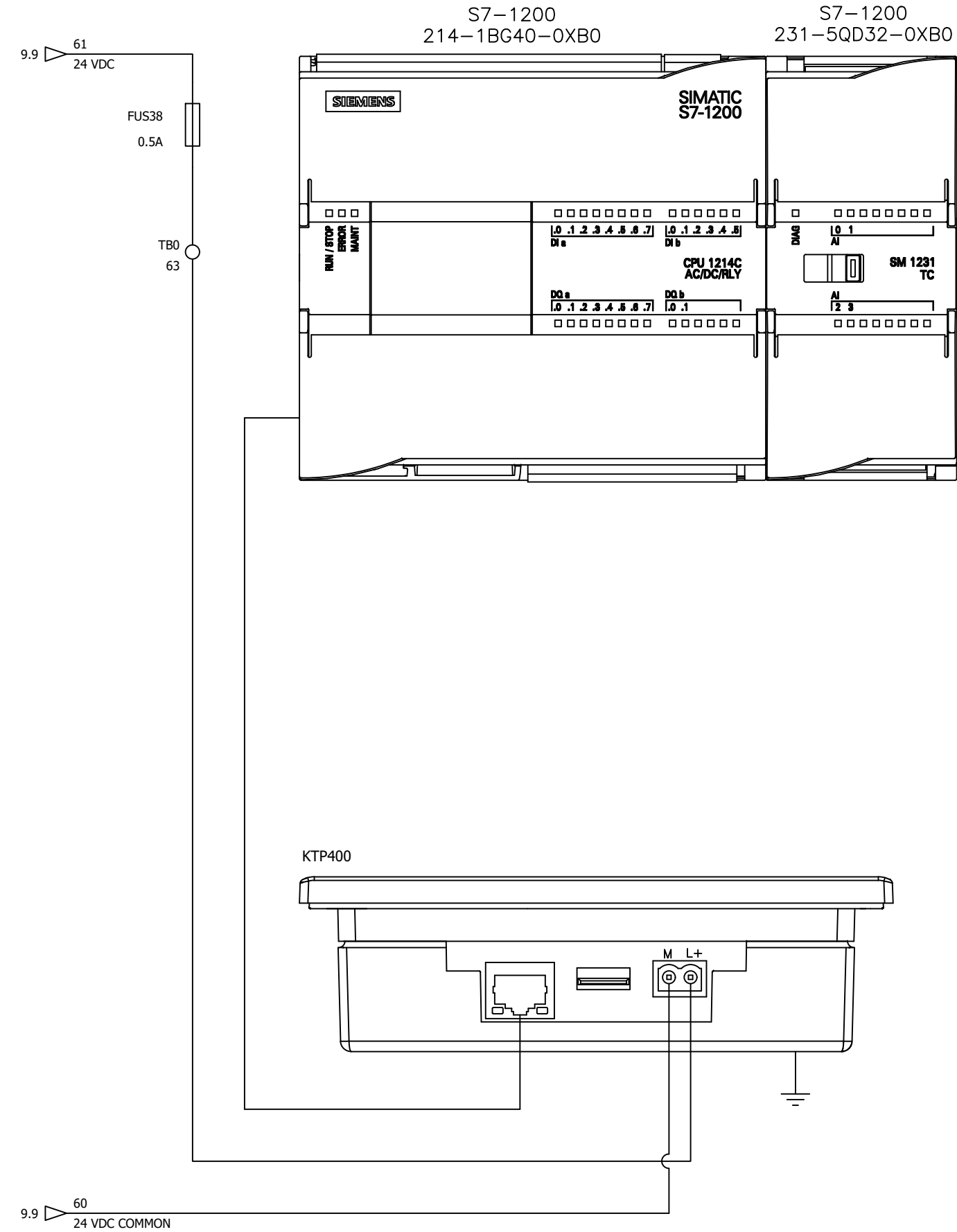
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WIRING DIAGRAM

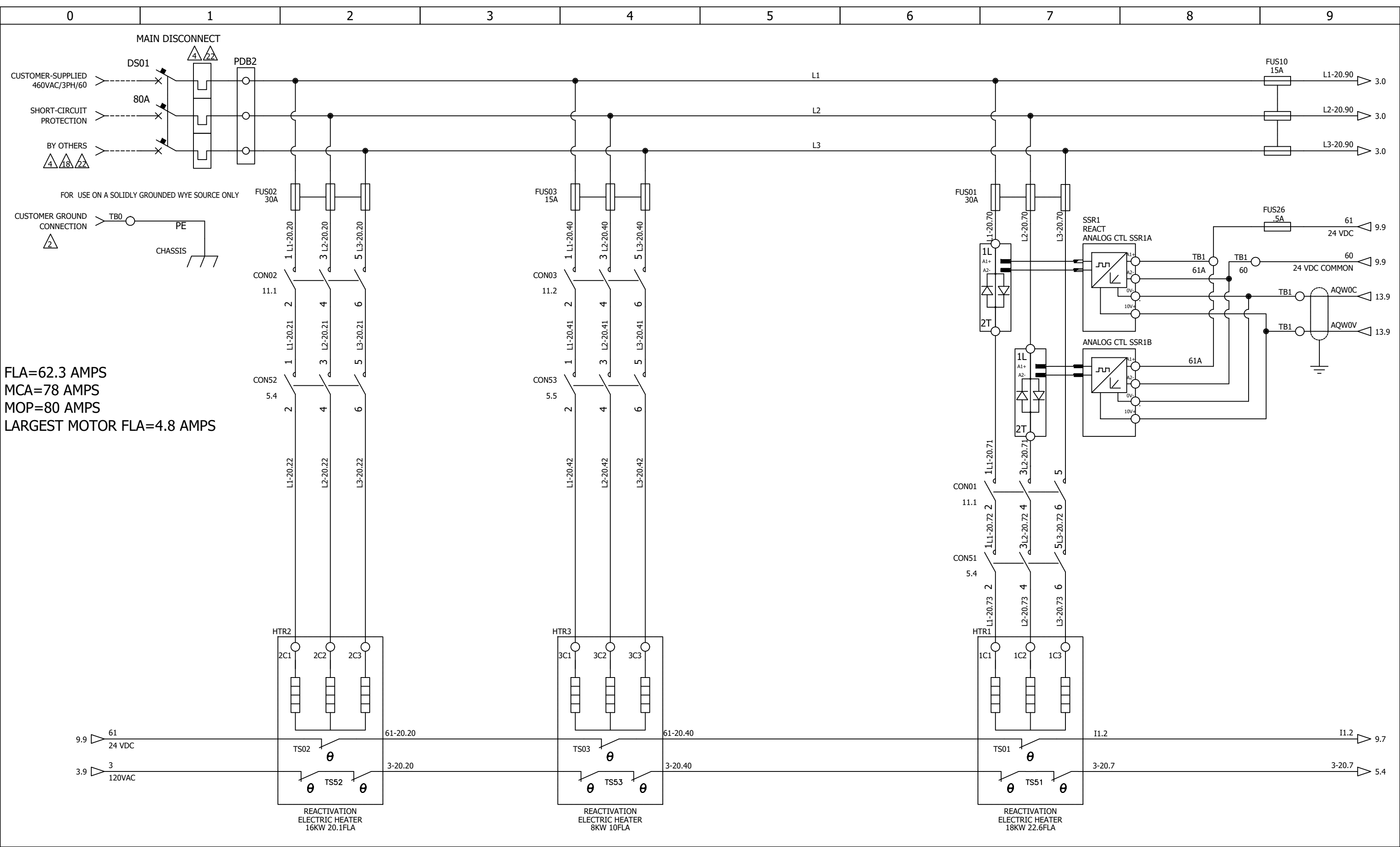
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KPW

DATE	7/15/2024
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REVISION	0
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SHEET
17 OF 30





FLA=62.3 AMPS
MCA=78 AMPS
MOP=80 AMPS
LARGEST MOTOR FLA=4.8 AMPS

SPARE SHEET



Munters

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DRAWN DSL

APPROVED
BW

SIZE	B
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22367602-01-WD

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KPW

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REVISION	0
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SHEET
21 OF 30

SPARE SHEET



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APPROVED
BW

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DRAWING NUMBER:
22367602-01-WD

CHECKED
KPW

DATE	7/15/2024
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REVISION	0
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SHEET
22 OF 30

SPARE SHEET



Munters

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22367602-01-WD

PROJECT NAME:
WAYNE FARMS, LLC

SEQUENCE OF OPERATIONS:
K71200

DRAWING NAME:
WIRING DIAGRAM

CHECKED
KPW

DATE	7/15/2024
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REVISION	0
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SHEET
24 OF 30

SPARE SHEET



Munters

79 Monroe St. Amesbury, MA 01913
TELEPHONE: 1-888-349-4335
WEBSITE: MUNTERS.COM/HELP

PROJECT NUMBER:
22367602-01

MODEL:
HCD-2250-EA-SFS

DESCRIPTION:

CONFIDENTIALITY STATEMENT

THE TECHNICAL INFORMATION AND DESIGN DATA HEREIN CONSTITUTES PROPRIETARY INFORMATION OF THE MUNTERS CORPORATION AND ARE TO BE MAINTAINED IN STRICT CONFIDENCE. THIS INFORMATION IS FOR THE SOLE USE OF OUR CUSTOMERS AND END USERS OF OUR EQUIPMENT.

DRAWN	DSL
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CHECKED
KPW

APPROVED
BW

7/15/2024

SIZE	B
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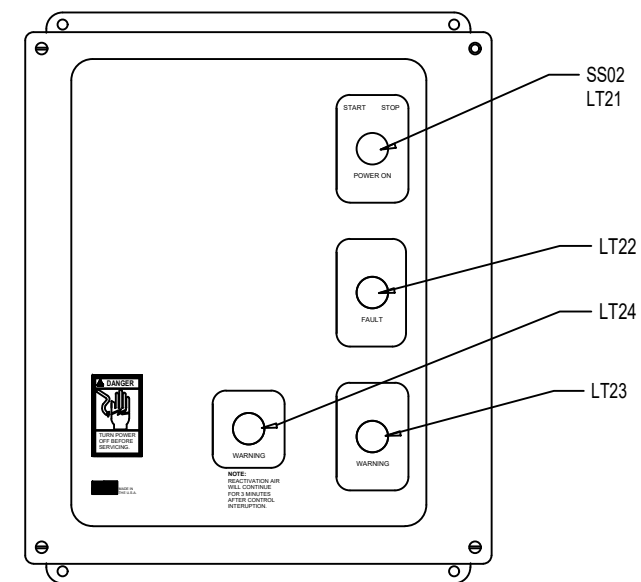
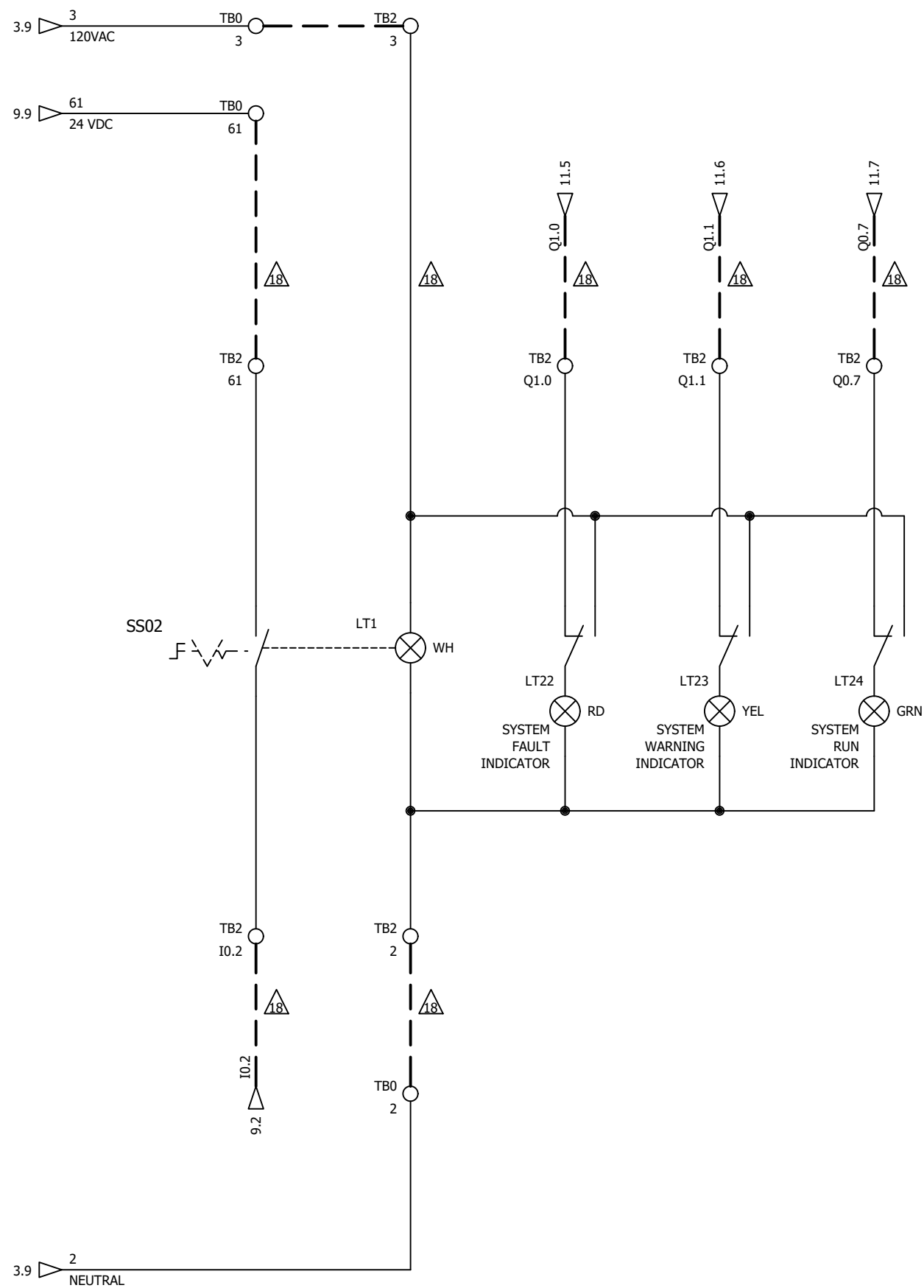
24	REVISION 0
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DRAWING NUMBER:
22367602-01-WD

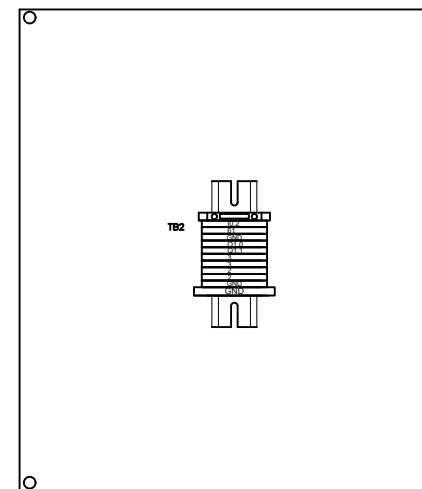
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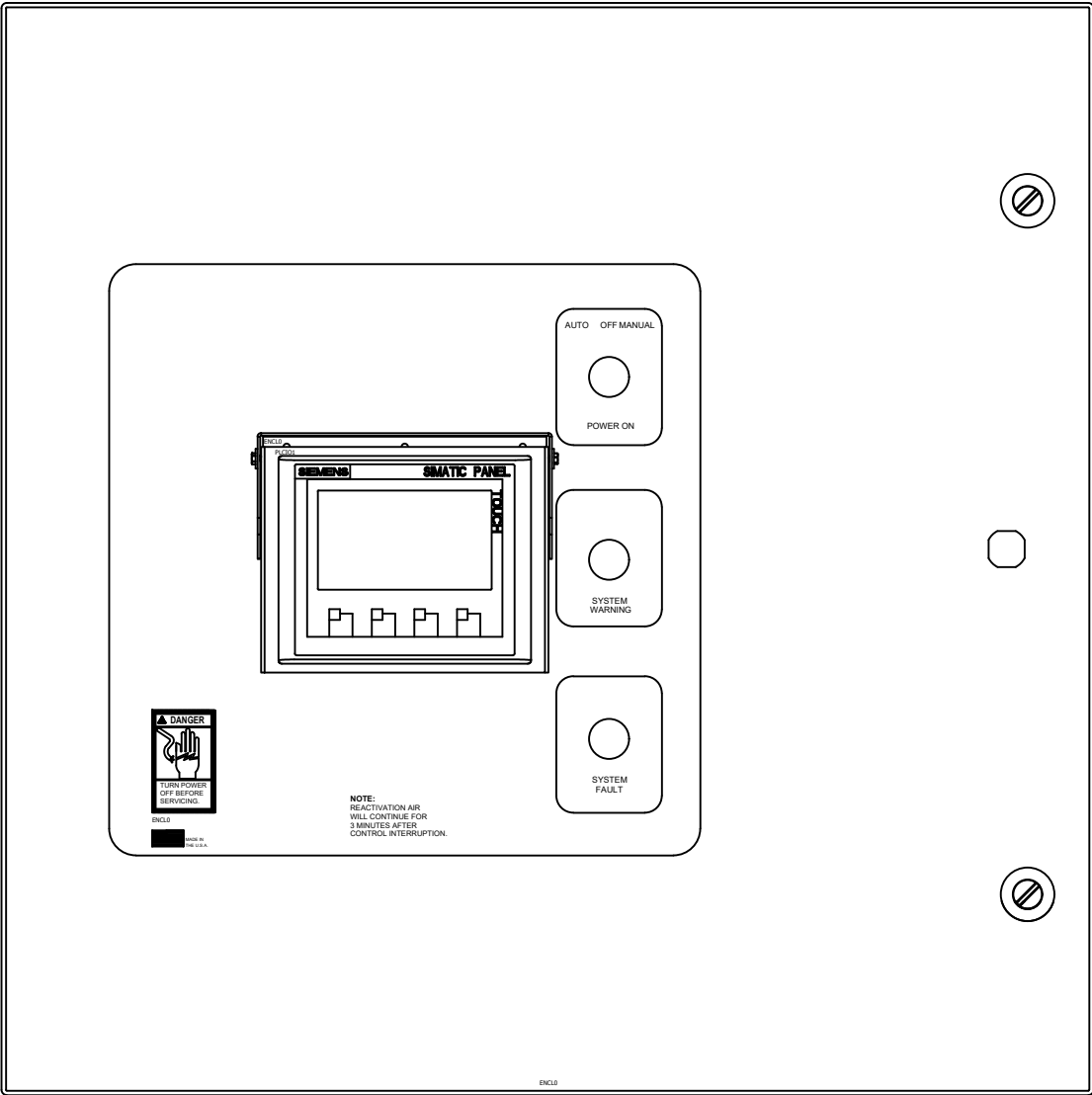
02-01-WD

SHEET
26 OF 30



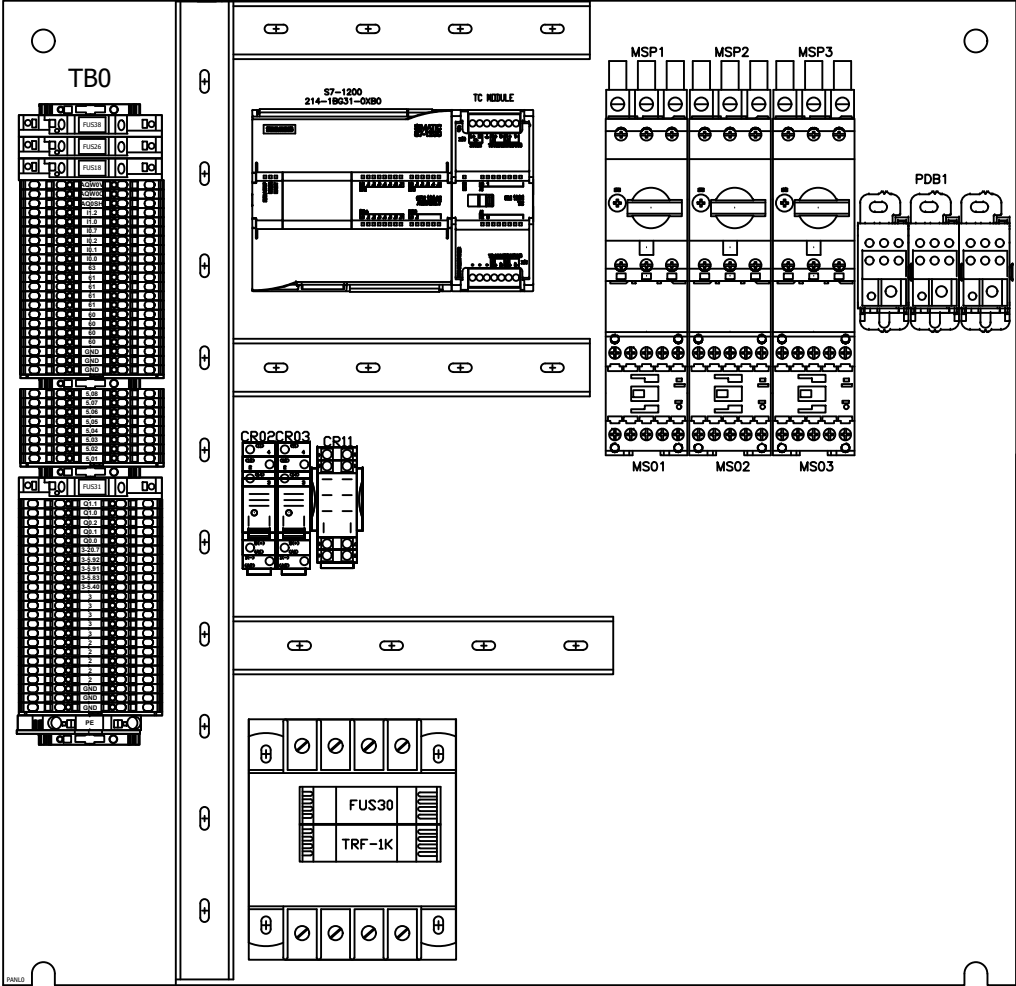
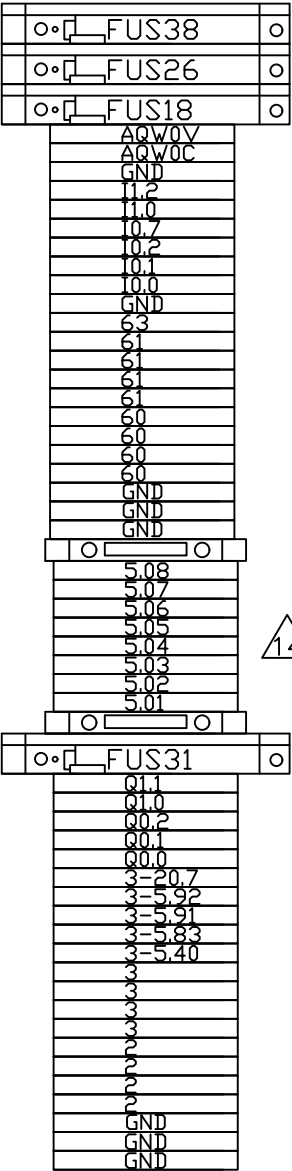
16 X 14 X 6 ENCLOSURE



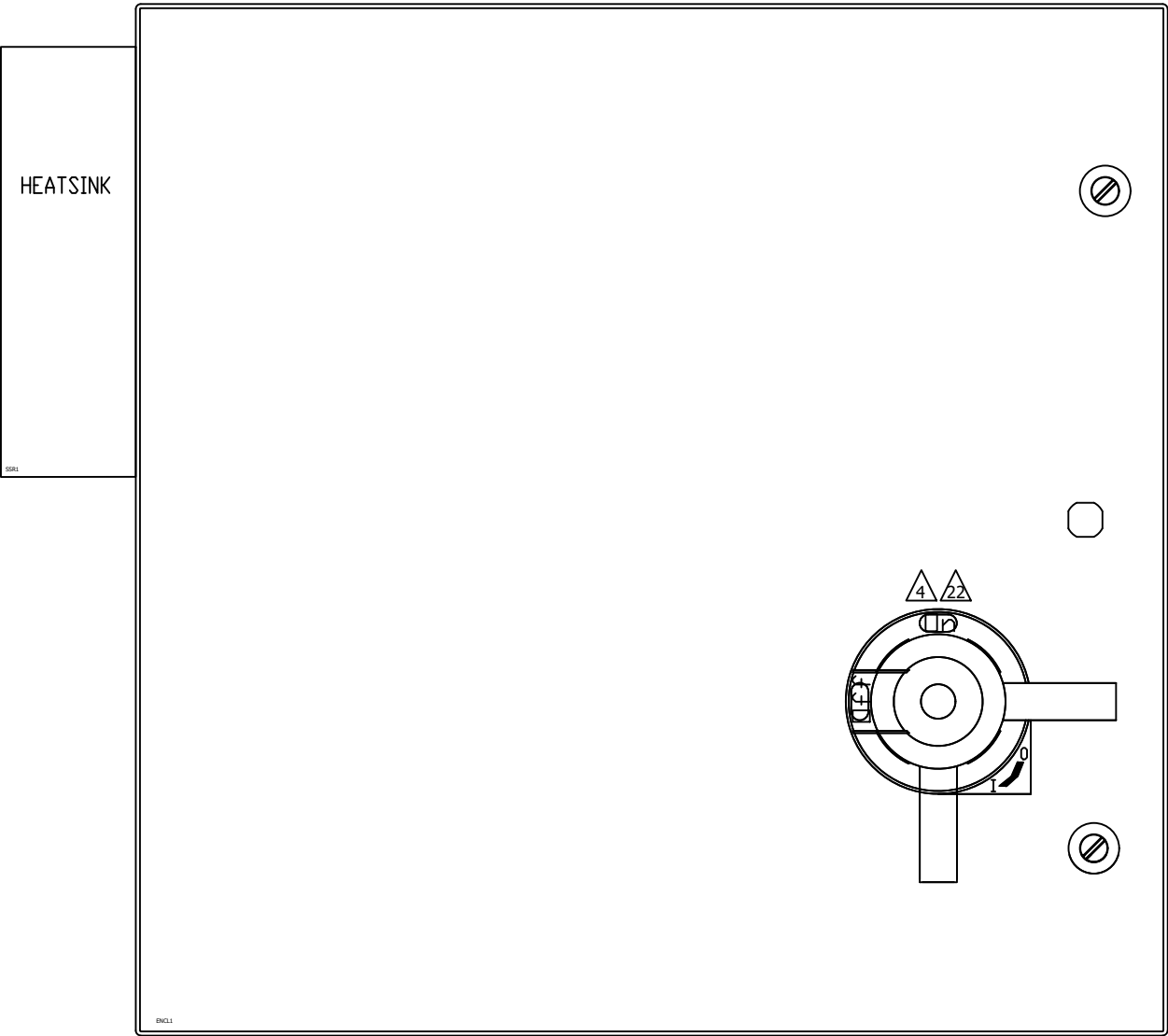


CONTROL ENCLOSURE 24 X 24 X 8 (NEMA 4)
(ENCLO)

TB0

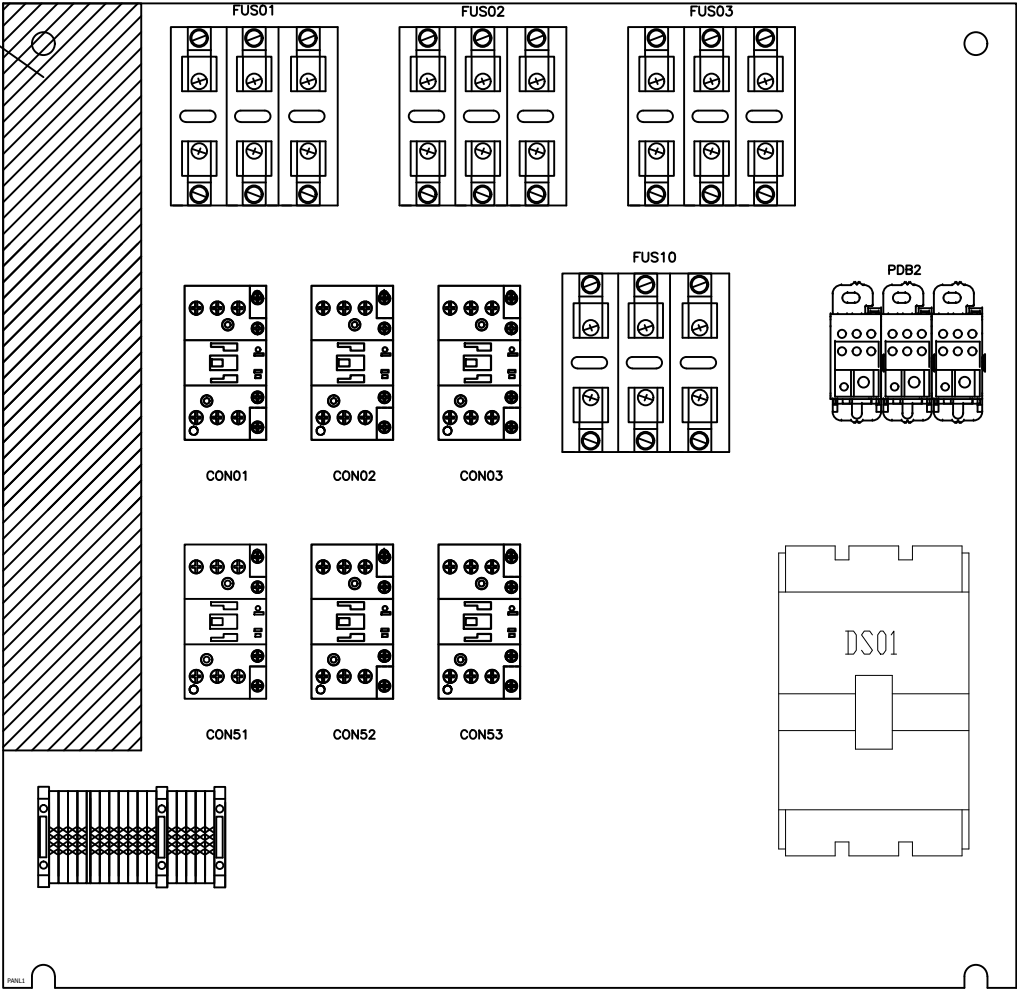


PANEL LAYOUT
(PANLO)

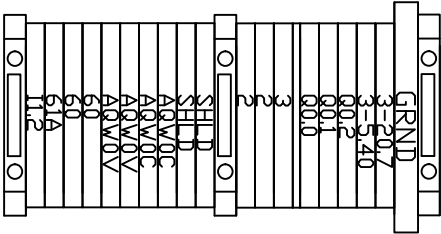


CONTROL ENCLOSURE 24 X 24 X 8 (NEMA 4)
(ENCL1)

THIS AREA REPRESENTS THE
LOCATION OF THE SOLID
STATE RELAYS MOUNTED ON
THE SIDE OF THE
ENCLOSURE



PANEL LAYOUT
(PANL1)



TB1

NOTES:

1. REVIEW OPERATING & MAINTENANCE MANUAL BEFORE ENERGIZING EQUIPMENT.
2. CUSTOMER-SUPPLIED EARTH GROUND TO BE TERMINATED AT KP LUG OR GROUNDING STRIP PROVIDED.
3. PROTECTIVE DEVICES HAVE BEEN PRESET AND/OR SELECTED AND ARE RECORDED AT FACTORY PER SPECIFICATION 22096/22576.
4. OPTIONAL EQUIPMENT.
5. N/A.
6. ALL WIRING SHOWN WITH SOLID LINES IS INSTALLED AT THE FACTORY.
7. SEE ELECTRICAL PARTS LIST FOR COMPONENT SPECIFICATION.
8. NO = NORMALLY OPEN, OOR = OPEN ON RISE,
NC = NORMALLY CLOSED, OOF = OPEN ON FALL.
10. INTERLOCK WIRING FOR AUTOMATIC START/STOP IS NOT BY MUNTERS.
11. USE SHIELDED CABLE, NOT LESS THAN #18 AWG.
12. REMOVE JUMPERS WHEN INTERLOCKING SAFETIES.
13. ELECTRICAL WIRE SIZING DATA: STANDARD: A72001/CANADIAN: A72011
14. AUXILIARY INTERLOCK IS PROVIDED FOR CUSTOMER INTERLOCKING.
15. SHIPPED REMOTE FROM FACTORY, INSTALLED BY OTHERS.
16. CUSTOMER CONTROLS.
17. N/A.
18. FIELD WIRED, BY INSTALLER.
19. TRANSFORMER VOLTAGES SHOWN COVER ALL STANDARD 50/60Hz APPLICATIONS.
THE ACTUAL TRANSFORMER USED IS DETERMINED BY THE POWER SOURCE.
20. DASHED LINES ARE CUSTOMER WIRED FOR INTERCONNECT.
21. INTERLOCK PROVIDED FOR OPTIONAL HUMIDISTAT OR LIMIT SWITCH ON.
22. MAIN POWER DISCONNECT DEVICE NOT SUPPLIED BY MUNTERS AS STANDARD AND MUST
BE PROVIDED BY CUSTOMER. FUSED DISCONNECT OR CIRCUIT BREAKER AVAILABLE AS
OPTIONAL EQUIPMENT.
23. ACTUAL MOTOR FLA RATING REFER TO UL TABLE.
24. MOTOR OVERLOAD PROTECTION RATING IS BASED ON ACTUAL MOTOR NAMEPLATE FLA.
25. OPTIONAL SAFETIES FOR "INDOOR" INDIRECT GAS REACTIVATION ONLY. REMOVE JUMPER WHEN INSTALLING
POWER VENTOR SAFETIES.
26. N/A.
27. N/A.
28. N/A.

SEQUENCE of OPERATION

HCD/ICA-EA

NOTE: Before servicing the dehumidifier, refer to the "OPERATION AND MAINTENANCE MANUAL" provided with the dehumidifier

1.0 CUSTOMER INTERFACE

1.1 Sequence is as follows:

- 1.1.1 "AUTO / OFF / MANUAL" selector switch is provided for starting the system in the Auto mode by interfacing with the customer supplied equipment. A contact closure via the "ON/OFF" selector switch between TB06#61 and TB06#10.1 will let the system start in auto mode. The MANUAL mode does not require the dry contact closure input to enable the system to start. The selector switch has an integral light and will stay illuminated at all times to indicate the equipment has power.

2.0 AUXILIARY DRY CONTACTS

2.1 Operational System States

- 2.1.1 SYSTEM FAULT: Auxiliary dry contacts of CR02 are provided for customer interlocking and shall close whenever the dehumidifier goes into any fault as described in Section 6.0.
- 2.1.2 SYSTEM RUN (MASTER CONTROL RELAY): Auxiliary dry contacts of MS03 (optional) are provided for customer interlocking and shall close whenever the process blower is in operation.
- 2.1.3 SYSTEM WARNING: Auxiliary dry contacts of CR03 are provided for customer interlocking and shall close whenever the dehumidifier goes into a warning condition.
- 2.1.4 D/H RUN: Auxiliary dry contacts of MS02 are provided for customer interlocking and shall close whenever the dehumidifier is in operation.

3.0 START-UP

3.1 Main Power Initialization Sequence:

- 3.1.1 Energize the main power to the equipment by supplying correct voltage to either a customer connection directly to the main TB01 terminal block or by using the system's main disconnect DS1. (The selector switch will illuminate indicating the equipment is energized)
- 3.1.2 The equipment's high voltage is supplied to all motor starters and the control transformer(s). The control voltage (115V/1Ø/60Hz) is now being supplied to the control circuits.
- 3.1.3 The HMI LCD, located on or in (visible through the door) the Main control enclosure is energized and will display the "System Screen" (Figure 1.0).
- 3.1.4 There is a 10 sec delay in the equipment startup of the system once power is activated via the selector switch.

3.2 Startup:

- 3.2.1 Normal startup operation shall occur once the main selector switch (SS01) is placed in the MANUAL or in AUTO position with a customer-supplied auxiliary interlock (dry-contact) closed between TB06#61 and TB06#10.2

3.3 SHUTDOWN

3.3.1 Conditions for Normal Shutdown

Normal shutdown shall occur with the following.

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3.3.1.1 Selector Switch (SS01) is maintained in the OFF position

3.3.1.2 Selector Switch (SS01) is maintained in Auto position with a contact opening across TB06 # 61 and TB06 # I0.2.

3.3.1.3 Selector Switch (SS01) is maintained in Auto position and BAS system sends stop command. The modbus control is enabled through operator interface (figure).

3.3.1.4 A fault has occurred.

3.3.2 Sequence of Normal Shutdown

3.3.2.1 The reactivation control Proportional Control Loop, and associated fault circuits are de-energized.

3.3.2.2 The process control loops are de-energized (section 6.0).

3.3.2.3 A purge timer in the (PLC1) is a delay time with a fixed on time of three (3) minutes. This allows the air to continue to cool the desiccant wheel. When the timer has completed its count cycle, the desiccant wheel drive motor (MTR1), the reactivation blower motor (MTR2), and supply blower motor (MTR3-optional) are all de-energized.

IMPORTANT: Do not shut down unit by de-energizing main power prior to completion of purge cycle. (Wait for "PURGING" to change to either "STANDBY" or "READY" before de-energizing main power.)

3.3.3 Sequence of Customer Interlock Shutdown

3.3.3.1 The customer interlock for the emergency stop is between TB03#3, TB03#3-3.91 and TB03-3.92 to energize CR11 that will enable the system to run. A contact open between TB06#61 and TB06#I0.2 will cause an immediate dehumidifier shutdown. The interlock will deactivate the system.

4.0 FAULTS AND WARNINGS

ESC or Help button may be pressed twice quickly to exit back to the system screen at any time during screen navigation.

4.1 Alarm Indication

4.1.1 Fault and warning indication is seen in two possible manners on the equipment's control enclosure. The HMI display and fault warning lights. The HMI will display the blinking triangle with exclamation point and the fault message on the Alarm screen (figure 1.5).

FAULTS:

4.1.1.1 A Red fault Light (LT02) on the enclosure door remains illuminated.

4.1.2 The system will shut down and annunciate on the first fault that occurs. Any subsequent fault that occurs after the fault that caused the shutdown will be ignored.

4.1.3 The system will go into the SHUTDOWN SEQUENCE on any FAULT. The FAULT relay (CR02) energizes. Fault contacts are available for customer use (TB09-3.01 & 3.02).

To restart the dehumidifier from a fault condition, perform the following steps:

4.1.3.1 Clear the Fault condition.

4.1.3.2 Press the acknowledge 'ACK' on the HMI.

4.1.3.3 Go to the ALARM screen and press the F4 key or ACK ALARM button on the touch screen key to reset the PLC.

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To restart the system, the selector switch (SS01) must be moved to the "OFF" position then moved back to the "AUTO" or "MANUAL" position:

4.1.4 WARNINGS:

4.1.4.1 A Yellow warning Light (LT03) on the enclosure door remains illuminated.

4.1.4.2 The system will go into a WARNING condition but continue to operate. The WARNING relay (CR03) energizes. Warning contacts are available for customer use (TB09-3.03 & 3.04).

4.1.4.3 The system WARNING will remain active until the condition is cleared.

4.2 List of System and DH Faults

4.2.1 *FAULT* - D/H WHEEL MOTOR FAULT

4.2.1.1 If the desiccant drive motor starter/protector (MSP1) trips the system is faulted.

4.2.1.2 The system will go into the shutdown sequence as described in Section 4.0, and the system auxiliary dry contacts close in sequence (described in Section 2.0), also process blower will purge.

4.2.1.3 The dehumidifier will be held in the 'Fault' condition by (PLC1). To restart the dehumidifier clear the alarm pressing the acknowledge ACK on the HMI, reset the overload, and go to the ALARM screen and press the F4 RESET key this resets the PLC. To restart the system, the selector switch (SS01) must be moved to the "OFF" position, then move (SS01) back to the "AUTO" or "MANUAL" position.

4.2.2 *FAULT* - REACTIVATION MOTOR FAULT

4.2.2.1 If the reactivation blower motor starter/protector (MSP2) trips the system is faulted.

4.2.2.2 The dehumidifier goes into the shutdown sequence, as described in Section 4.0, and the system auxiliary dry contacts sequence as described in Section 2.0 The dehumidifier will be held in the 'Fault' condition by (PLC1). To restart the dehumidifier clear the alarm pressing the acknowledge ACK on the HMI, reset the overload, and go to the ALARM screen and press the F4 RESET key this resets the PLC. To restart the system, the selector switch (SS01) must be moved to the "OFF" position, then move (SS01) back to the "AUTO" or "MANUAL" position.

4.2.3 *FAULT* - SUPPLY BLOWER MOTOR FAULT

4.2.3.1 If the supply blower motor starter/protector (MSP3) trips or the customer-supplied variable frequency drive (VFD) goes into a Fault, the system is faulted.

4.2.3.2 The system will go into the shutdown sequence (described in Section 4.0), and the system auxiliary dry contacts close in sequence (described in Section 2.0).

4.2.3.3 The dehumidifier will be held in the 'Fault' condition by (PLC1). To restart the dehumidifier clear the alarm pressing the acknowledge ACK on the HMI, reset the overload, and go to the ALARM screen and press the F4 RESET key this resets the PLC. To restart the system, the selector switch (SS01) must be moved to the "OFF" position, then move (SS01) back to the "AUTO" or "MANUAL" position.

4.2.4 *FAULT* - DH WHEEL ROTATION FAULT

4.2.4.1 Limit switch (LS01) is wired normally closed, and is located inside the dehumidifier wheel compartment next to desiccant wheel. As the wheel turns, a cam fastened to the wheel housing opens this limit switch.

4.2.4.2 If limit switch (LS01) is not opened once every 600 seconds, a timer in (PLC1) times out the system is faulted.

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4.2.5 FAULT- E-STOP/CUSTOMER SHUTDOWN OPENED

- 4.2.5.1 Auxiliary wire terminals are provided for the customer to interface with the supplied equipment. An open connection between TB03 #3 and TB03 3-3.91 or TB03 3-3.92 will de-energize the Emergency Stop relay. The system is faulted.

4.2.6 FAULT- REACTIVATION ELECTRIC HEATER OVERTEMP SWITCH

- 4.2.6.1 Temperature switch (TS01) is located in the reactivation Electric Heater assembly and is furnished to detect extreme temperature level or heat buildup. If the temperature switch input to the PLC is opened, the system is faulted.

4.3 List of System and DH Warnings

4.3.1 WARNING – REACTIVATION AIR OUTLET LOW TEMP

- 4.3.1.1 If the reactivation outlet temperature is lower than preset low temp set point for 30 minutes.
- 4.3.1.2 The typical low temp alarm point =110° F and the adjustment range is 100.0° F to 165.0° F.

5.0 PROPORTIONAL CONTROL LOOPS

5.1 PID – REACTIVATION - ELECTRIC HEAT ENERGY MODULATION CONTROL

- 5.1.1 The Reactivation electric heater solid state relays accept a 0-10V modulating signal from the logic controller to control Reactivation inlet air temperature, as sensed by the reactivation air inlet temperature transmitter (TC01), the reactivation inlet loop temperature set point will be varied based on the reactivation outlet PID loop to maintain maximum required reactivation inlet temperatures over a wide range of DH load conditions. The typical reactivation outlet temperature set point is 120°F.
- 5.1.2 When the reactivation outlet PID loop output is 0% the reactivation inlet set point will be 120°F and when the reactivation outlet PID loop output is 100% the reactivation inlet set point will be 325°F.

5.2 Dehumidification Humidistat Interlock (OPTIONAL)

- 5.2.1 The Humidistat is an ON/OFF Controller (HS01), and is furnished to interlock the Dehumidifier heel and associated controls, based on moisture removal demand.
- 5.2.2 Provided that the System is running, as the humidity level, as sensed by (HS01), exceeds the desired set point, HS01 will close contacts energizing relay (CR69) which, in turn, signals the PLC (PLC1) to start the Dehumidifier.
- 5.2.3 The PLC (PLC1) will enable the Desiccant Wheel Drive Motor Starter/ Relay (CR01), Reactivation Fan Motor Starter (MS02), and the associated Reactivation Energy modulation controls.
- 5.2.4 When the humidity level, as sensed by (HS01) decreases below the desired set point, the Dehumidifier will initial a 90 second purge/ shutdown sequence. Upon completion of the purge sequence, the PLC (PLC1) will de-energize the Desiccant Wheel Motor Starter/ Relay (CR01), Reactivation Fan Motor Starter (MS02) and associated Reactivation Energy Modulation controls. The Operator Interface (OIT1) will display Dehumidifier: STANDBY (if system is still running) or "READY" (if the system is stopped).

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6.0 HUMAN MACHINE INTERFACE CONTROL (KTP400 or KTP600 SIMATIC PANEL)

6.1 Screen descriptions, navigation and security:

* Indicates a level of security to adjust equipment parameters.

Note: Other visual options are password protected for FACTORY USE ONLY.

- 6.1.1 System Screen This screen is the first screen that comes up when power is applied to the equipment (figure 1.0). The System screen displays the equipment's operational status along with separate indication for the Reactivation air stream. The System screen also acts as the navigation portal for all of the screens configured in the system. All navigation references in this section assume the operator starts on the System screen.

Equipment Status Indications

"SYSTEM" displays the modes of operation of the process as listed below:

READY = The system is ready to run and there are no faults and the interlocks are made.

RUNNING = The system is running and process control loops are running if they are enabled.

FAULTED = The system has faulted.

PURGING = The system is going thru a shutdown sequence and is purging.

"REACT" displays the modes of operation of the dehumidifier as listed below:

READY = The DH is ready to run and there are no faults and the interlocks are made.

RUNNING = The DH is running and process control loops are running if they are enabled.

FAULTED = The DH has faulted.

PURGING = The DH is going thru a shutdown sequence and is purging.

STANDBY = The System is Running and the DH is stopped because there is no call for dehumidification (full bypass).

- 6.2 Reactivation Temperature The Reactivation Heat to Temperature and R_Out temperature screen displays the real time indication of the reactivation inlet and outlet temperatures (degrees Fahrenheit).
- 6.3 Date/Time The Date/Time screen displays the current date and time. This may be configured by the customer using the correct username and password. The display will change to the Username and Password screen prompted by pressing the button on the touch screen. Fig 1.6 and fig 1.7.
- 6.4 Run Time Meter the system and DH runtime meters are displayed on the main start screen. The Run Time Meter displays the amount of time in hours that the equipment has been in operation. There is an indication for DH displaying the amount of time dehumidification has been requested and in operation. These values, shown in hours, may vary dependent on the equipment's configuration.
- 6.5 Control Group Selection 1* Pressing PROCESS SETUP will navigate to this screen (figure 1.1). The Siemens KTP400 Display may prompt the operator for Username and Password, The Control Group Selection screen displays the various system loops available on the equipment.
- 6.6 Alarms Pressing F4 (ALM) will navigate to this screen (figure 1.4). The Alarm screen displays the Faults and Warnings the system encountered during operation. The current alarm message may be cleared without clearing the history by pressing the ACK ALARM button

The Alarm Screen displays and records information on triggered alarms. Each time an alarm is triggered, it is added to the Alarm History list, and the last alarm shall be displayed.

- 6.6.1 F3 ACK ALARM The operator must press the F3 button to reset a fault in the PLC; this will clear all system faults and enable the equipment to be restarted by cycling the selector switch. The selector switch may also be used as the system reset by turning it to the off position and return it to the auto or manual position. The system screen will display READY after a fault has been reset.

- 6.7 React PID Pressing F1 (REACT) while on the Process Set-up screen will navigate to this screen (figure 1.2). In the React PID screen the operator may adjust the reactivation temperature set-point (R_SP). To set the value Press ENTER. This temperature is in degrees Fahrenheit.

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Temperature Sensor Min/Max Pressing “Misc SETUP” and selecting the sensor you want to configure from the Process Setup screen will navigate to this screen (figure 2.1). Here the operator may adjust the sensor maximum and minimum. To set the values Touch the parameter you wish to change and enter the new value. Pressing “Save Recipe” will save these values to memory in the PLC allowing the values to remain in case of power failure or shutdown.

7.0 SIMATIC PANEL SCREEN SHOTS

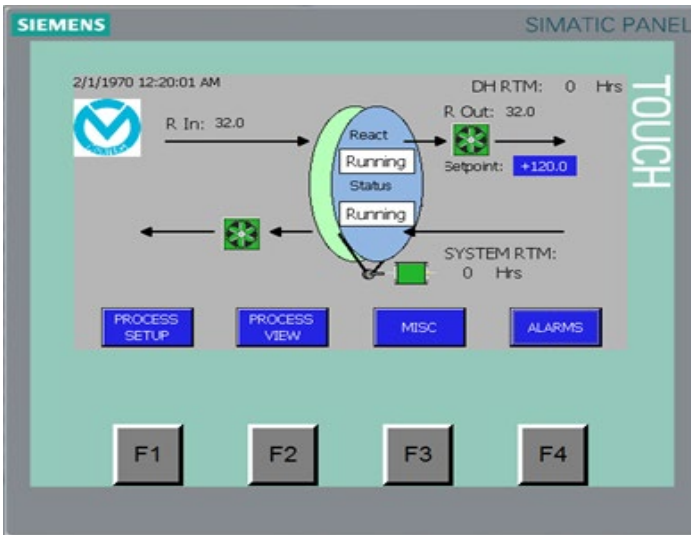


Figure 1.0

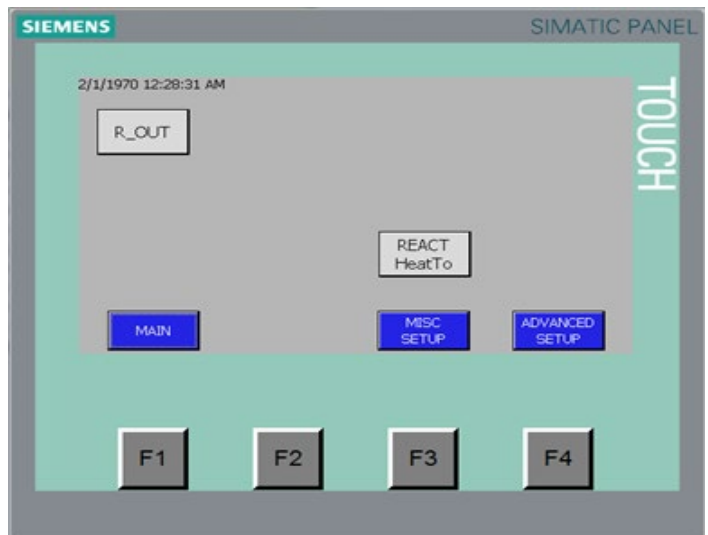


Figure 1.1

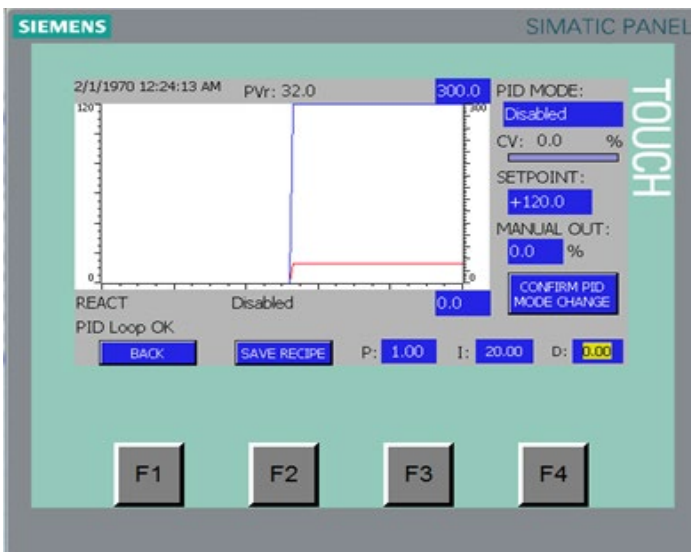


Figure 1.2

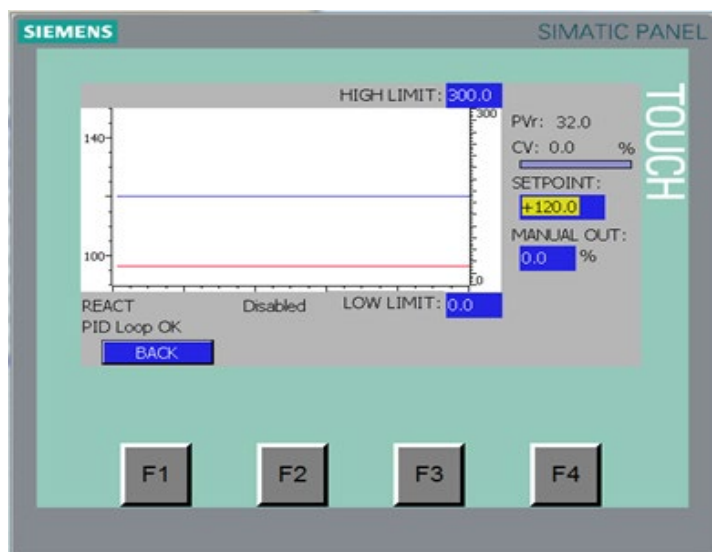


Figure 1.3

Munters Corporation

79 Monroe Street
Amesbury, MA 01913

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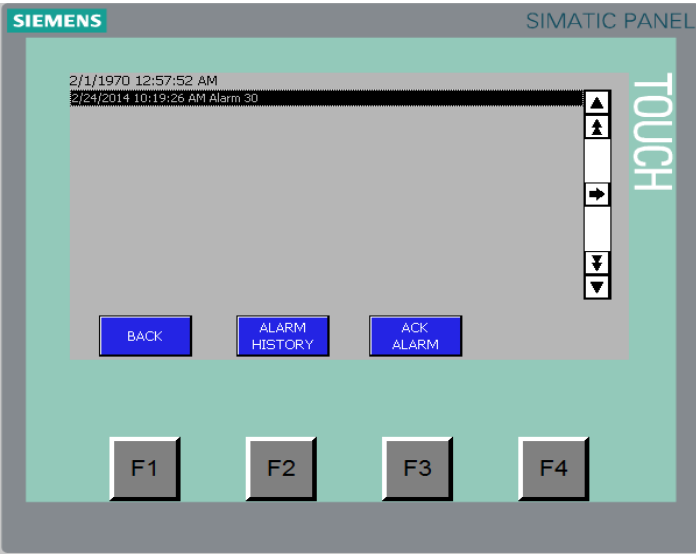


Figure 1.4

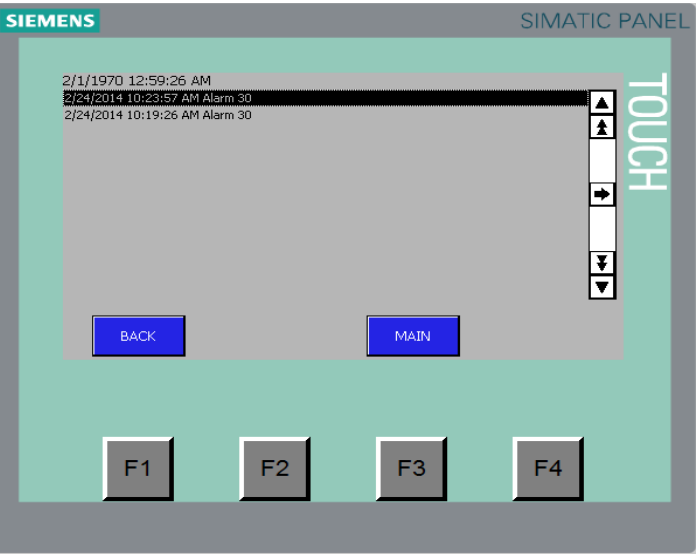


Figure 1.5

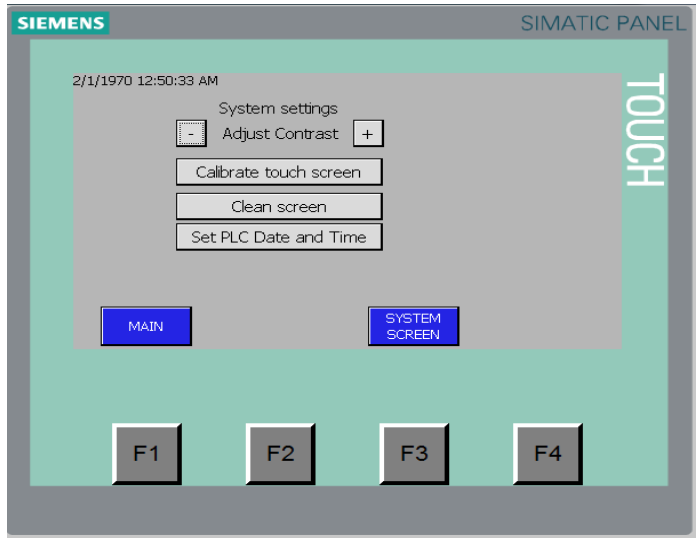


Figure 1.6

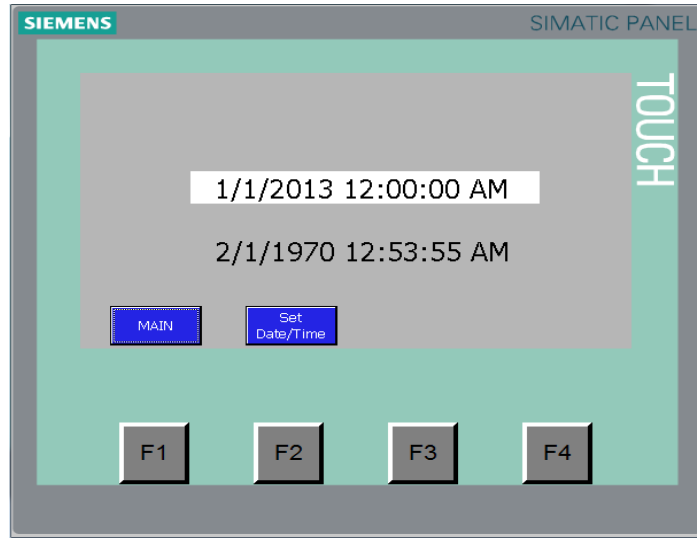


Figure 1.7

Munters Corporation 79 Monroe Street Amesbury, MA 01913	DFT RDS	DATE 03/20/2014	DWG NO. K71200	
	CHK HZ	3/20/2014	CUST NAME: HCD/ICA-EA STANDARD	
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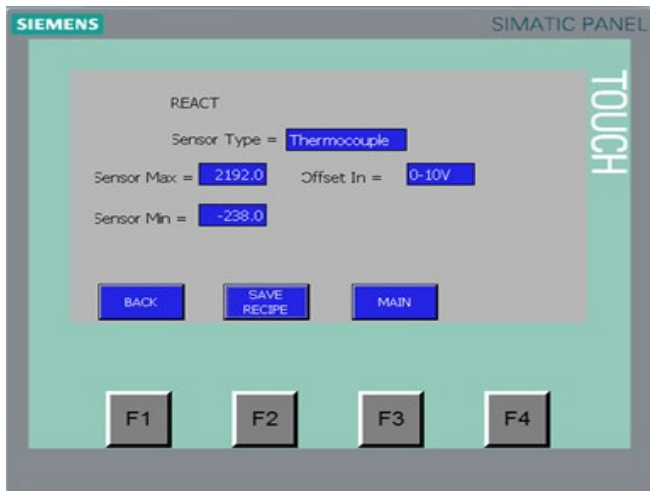


Figure 2.1

3	CORRECTED SECTION 6.7 SCREEN FIGURE FROM 1.1 TO 1.2	HZ	03/04/19	HZ
2	Removed Obsolete Login Data in Section 6	RGV	4/14/15	TRF
1	ADDED H-STAT INTERLOCK OPTION	RDS	10/20/14	RDS
0	RELEASE	RDS	03/20/14	RDS
REV	DESCRIPTION	DFT	DATE	APP
REVISIONS				

Munters Corporation 79 Monroe Street Amesbury, MA 01913	DFT RDS	DATE 03/20/2014	DWG NO. K71200	
	CHK HZ	3/20/2014	CUST NAME: HCD/ICA-EA STANDARD	
	APP RDS	3/20/2014	PAGE 8 OF 8	REV 3

MOTOR FUSE SIZING TABLE 1														
LPJ - FUSE STYLE			LPJ - FUSE STYLE			LPJ - FUSE STYLE			LPCC - FUSE STYLE			LPCC - FUSE STYLE		
MOTOR AMPS	FUSE SIZE	PART NUMBER	MOTOR AMPS	FUSE SIZE	PART NUMBER	MOTOR AMPS	FUSE SIZE	PART NUMBER	MOTOR AMPS	FUSE SIZE	PART NUMBER	MOTOR AMPS	FUSE SIZE	PART NUMBER
0.40 - 0.61	1	91981-01	3.50 - 3.99	7	91981-17	40.0 - 44.9	80	91981-32	0.40 - 0.49	1	92252-04	3.20 - 3.59	8	92252-25
0.62 - .0.79	1.25	91981-02	4.00 - 4.49	8	91981-18	45.0 - 49.9	90	91981-33	0.50 - 0.63	1.25	92252-06	3.60 - 3.99	9	92252-26
0.80 - 0.89	1.6	91981-03	4.50 - 4.99	9	91981-19	50.0 - 54.9	100	91981-34	0.64 - 0.71	1.6	92252-09	4.00 - 4.79	10	92252-27
0.90 - 0.99	1.8	91981-04	5.00 - 5.99	10	91981-20	55.0 - 62.4	110	91981-35	0.72 - 0.89	1.8	92252-10	4.80 - 5.99	12	92252-28
1.00 - 1.12	2	91981-05	6.00 - 7.49	12	91981-48	62.5 - 74.9	125	91981-36	0.90 - 0.99	2.25	92252-11	6.00 - 7.99	15	92252-29
1.13 - 1.24	2.25	91981-06	7.50 - 8.74	15	91981-21	75.0 - 87.4	150	91981-37	1.00 - 1.11	2.5	92252-12	8.00 - 9.99	20	92252-30
1.25 - 1.39	2.5	91981-07	8.75 - 9.99	17.5	91981-22	87.5 - 99.9	175	91981-38	1.12 - 1.19	2.8	92252-13	10.0 - 11.9	25	92252-31
1.40 - 1.49	2.8	91981-08	10.0 - 12.4	20	91981-23	100 - 112	200	91981-39	1.20 - 1.27	3	92252-14	12.0 - 15.0	30	92252-32
1.50 - 1.59	3	91981-09	12.5 - 14.9	25	91981-24	113 - 124	225	91981-40	1.28 - 1.39	3.2	92252-15			
1.60 - 1.74	3.2	91981-10	15.0 - 17.4	30	91981-25	125 - 149	250	91981-41	1.40 - 1.59	3.5	92252-16			
1.75 - 1.99	3.5	91981-11	17.5 - 20.0	35	91981-26	150 - 174	300	91981-42	1.60 - 1.79	4	92252-17			
2.00 - 2.24	4	91981-12	20.1 - 22.4	40	91981-27	175 - 199	350	91981-43	1.80 - 1.99	4.5	92252-18			
2.25 - 2.49	4.5	91981-13	22.5 - 24.9	45	91981-28	200 - 224	400	91981-44	2.00 - 2.23	5	92252-19			
2.50 - 2.79	5	91981-14	25.0 - 29.9	50	91981-29	225 - 249	450	91981-45	2.24 - 2.39	5.6	92252-20			
2.80 - 2.99	5.6	91981-15	30.0 - 34.9	60	91981-30	250 - 285	500	91981-46	2.40 - 2.79	6	92252-21			
3.00 - 3.49	6	91981-16	35.0 - 39.9	70	91981-31				2.80 - 3.19	7	92252-23			

Fuse sizes for all other amperes will be based on the following formulas: **LP-CC = 2.5*FLA** **LPJ = 2*FLA**.

TABLE 2A			
MOTOR OVERLOADS			
3 PH MOTORS		1,3 PH MOTORS	
F.L.A. RANGE	PART#	0.1 - 0.16	92902-13
0.1 - 0.32	92902-01	0.16 - 0.24	92902-14
0.32 - 1.0	92902-02	0.24 - 0.40	92902-15
1.0 - 2.9	92902-03	0.40 - 0.60	92902-16
1.6 - 5.0	92902-04	0.60 - 1.0	92902-17
3.7 - 12	92902-05	1 - 1.6	92902-18
12 - 32	92902-06	1.6 - 2.4	92902-19
12 - 37	92902-07	2.4 - 4	92902-20
14 - 45	92902-08	4 - 6	92902-21
26 - 85	92902-09	6 -10	92902-22
		10 -16	92902-23
		16 - 24	92902-24
		18 -30	92902-25
		30 - 45	92902-26
		18 -30	92902-27
		30 -45	92902-28
		45 - 60	92902-29
		60 - 75	92902-30
		70 - 90	92902-31

Munters Corporation 79 Monroe Street Amesbury, MA 01913	REV 3 - REVISED TABLE 2 2/91 G.G KWC KWC	DFT	DATE	SIZE	DESCRIPTION		
	REV 4 - REVISED TABLE 2 8/91 DMR KWC KWC	JSR	9/11/90				
	REV 5 - REVISED TABLE 2 8/91 JOS KWC KWC	CHK KWC	DATE	B	FUSE SELECTION AND OVERLOAD PROTECTION		
	REV 6 - REVISED TABLE 4-5 6/93 MH KWC KWC		2/91				
	REV 7 - REVISED TABLE 1 10/98 TTM KPW KPW						
	REV 8 - UPDATED MOTOR AMP 12/98 TTM KPW TTM	APP	DATE	SCALE	DRAWING NUMBER		REV
	REV 10 - TABLE ADDED 2A 8/00 EAC EAC EAC	KWC	2/91				
REV 11- UPDATED TABLE 4 AND 5 RDS CAG CAG			NTS				

TRANSFORMER PRIMARY TABLE 4											
VA	208V		230V		380V		480V		575V		
	AMPS	PART#	AMPS	PART#	AMPS	PART#	AMPS	PART#	AMPS	PART#	
200	1.60	91410-07	1.60	91410-07	1.00	91410-04	0.80	91410-03	0.60	91410-02	
250	2.00	91410-09	1.80	91410-08	1.12	91410-05	1.00	91410-04	0.75	91410-34	
350	2.80	91410-13	2.50	91410-12	1.60	91410-07	1.25	91410-06	1.00	91410-04	
375	3.00	91410-14	2.80	91410-13	1.80	91410-08	1.60	91410-07	1.25	91410-06	
500	4.00	91410-17	3.50	91410-16	2.15	91410-10	2.00	91410-09	1.50	91410-31	
750	6.00	91410-21	5.60	91410-20	3.50	91410-16	2.80	91410-13	2.50	91410-12	
1,000	8.00	91410-24	7.00	91410-23	4.50	91410-18	3.50	91410-16	3.00	91410-14	
1,500	12.00	91410-27	12.00	91410-26	7.00	91410-23	5.60	91410-20	5.00	91410-19	
2,000	15.00	91410-28	15.00	91410-28	9.00	91410-25	7.00	91410-23	6.00	91410-21	
3,000	20.00	91410-29	20.00	91410-29	12.00	91410-27	10.00	91410-26	9.00	91410-25	

TABLE 5									
TRANSFORMER SECONDARY									
VA	115V			24V					
	AMPS	PART#		AMPS	PART#				
200	2.00	92466-01		10.00	90193-05				
250	2.00	92466-01		15.00	90193-06				
350	3.00	92466-06		20.00	90193-07				
375	4.00	92466-03		20.00	90193-07				
500	6.00	92466-04		25.00	90193-08				
750	8.00	92466-05		X	X				
1,000	12.00	90193-35		X	X				
1,500	20.00	90193-07		X	X				
2,000	25.00	90193-08		X	X				
3,000	30.00	90193-09							

TABLE 3		
HEATER PROTECTION		
600V - ONE TIME		
AMPS / BANK	FUSE SIZE	PART NUMBER
0.0-4.8	6	91816-03
4.9 - 6.4	8	91816-21
6.5 - 8.0	10	91816-04
8.1 - 12	15	91816-05
12.1 - 16	20	91816-06
16.1 - 20	25	91816-07
20.1 - 24	30	91816-08
24.2 - 28	35	91816-09
28.1 - 32	40	91816-10
32.1 - 36	45	91816-11
36.1 - 40	50	91816-12
40.1 - 48	60	91816-13

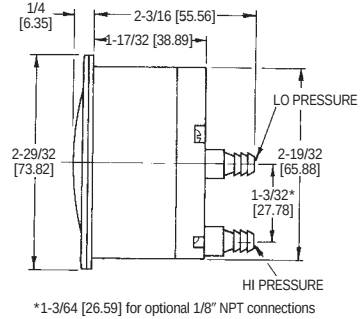
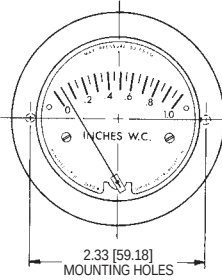
TABLE 2		
MOTOR OVERLOADS		
1, 3 PH MOTORS		
F.L.A. RANGE	PART#	
Replaced W/92902-##	0.1 - 0.16	91682-01
Replaced W/92902-##	0.16 - 0.24	91682-02
Replaced W/92902-##	0.24 - 0.40	91682-03
Replaced W/92902-##	0.40 - 0.60	91682-04
Replaced W/92902-##	0.60 - 1.0	91682-05
Replaced W/92902-##	0.8 - 1.2	91682-06
Replaced W/92902-##	1.0 - 1.6	91682-07
Replaced W/92902-##	1.5 - 2.3	91682-21
Replaced W/92902-##	2.0 - 3.0	91682-08
Replaced W/92902-##	2.8 - 4.2	91682-09
Replaced W/92902-##	4.0 - 6.0	91682-10
Replaced W/92902-##	5.5 - 8.0	91682-11
Replaced W/92902-##	6.0 - 10	91682-12
Replaced W/92902-##	10 - 16	91682-13
Replaced W/92902-##	16 - 24	91682-14
Replaced W/92902-##	22 - 32	91682-15
Replaced W/92902-##	18 - 30	91682-16
Replaced W/92902-##	30 - 45	91682-17
Replaced W/92902-##	45 - 60	91682-18
Replaced W/92902-##	60 - 75	91682-19
	60 - 88	91682-20
	80 - 120	91682-23

Munters Corporation 79 Monroe Street Amesbury, MA 01913	REV 3 - REVISED TABLE 2 2/91 G.G KWC KWC	DFT	DATE	SIZE	DESCRIPTION		
	REV 4 - REVISED TABLE 2 8/91 DMR KWC KWC	JSR	9/11/90		B	FUSE SELECTION AND OVERLOAD PROTECTION	
	REV 5 - REVISED TABLE 2 8/91 JOS KWC KWC	CHK KWC	DATE				
	REV 6 - REVISED TABLE 4-5 6/93 MH KWC KWC		2/91				
	REV 7 - REVISED TABLE 1 10/98 TTM KPW KPW		DATE				
REV 8 - UPDATED MOTOR AMP 12/98 TTM KPW TTM	APP	2/91	SCALE	DRAWING NUMBER		REV	
	REV 9 - REVISED TABLE 1 2/99 TTM KPW KPW	KWC	2/91	NTS	22576		11
	REV 10 - TABLE ADDED 2A 8/00 EAC EAC EAC						
	REV 11- UPDATED TABLE 4 AND 5 RDS CAG CAG						



Series 2-5000 Minihelic® II Differential Pressure Gage

Specifications: Installation & Operating Instructions



Dimensions, Series 2-5000 Minihelic® II Gage.

Series 2-5000 Minihelic® II Differential Pressure Gages have clean design, small size, low cost and sufficient accuracy for all but the most demanding applications. With housing molded from mineral- and glass-filled nylon and a lens molded from polycarbonate, this gage will withstand rough use and exposure, as well as high total pressure up to 30 psig [2.067 bar]. Over-pressure is accommodated by a blow-out membrane molded in conjunction with the diaphragm.

INSTALLATION

1. Select a location free from excessive vibration and where ambient temperature will be between 20°F to 120°F (-6.7°C to 49°C). Sensing lines may be any length necessary without affecting accuracy. However, long runs of tubing will dampen readings slightly and cause a minor increase in response time. If pulsing pressure or vibration cause excessive pointer oscillation, please contact factory for ways to provide additional damping.

2. This gage is calibrated and zeroed in the vertical position at the factory. If the gage is used in any other position, it must be re-zeroed each time the position is changed. Gages with ranges under 5 inches w.c. (1.24 kPa), or the equivalent, should be used only in the vertical position unless special calibration was specified when ordering.

SPECIFICATIONS

Dimensions: 2-29/32" (73.82 mm) x 2- 7/16" (61.93 mm).

Weight: 6 oz. [170 gr].

Rated Total Pressure: 50 psig (3.445 bar) surge; 30 psig (2.067 bar) continuous to either pressure connection.

Ambient Temperature Range: 20°F to 120°F (- 6.7°C to 49°C).

Accuracy: ± 5% of full scale at 70°F (21.1°C).

Connections: standard, barbed for 3/16" I.D. tubing; optional, 1/8" NPT(M).

Housing: glass-filled nylon, polycarbonate lens.

Finish: black

Standard Accessories: (2) 4-40 x 1-5/8" mounting studs, (2) 4-40 hex nuts, (1) .050" hex allen wrench, (1) panel mounting bracket.

CAUTION:

Use only with air or compatible non-corrosive gases.



PANEL MOUNTED INSTALLATION

3. To surface-mount the gage, drill two 5/32" [3.97 mm] holes on a horizontal line, 2-1/3" [59.26 mm] apart for mounting screws. Next, drill two 7/16" [11.11 mm] holes 1-1/32" [26.19 mm] apart on a vertical line for pressure connections. Install mounting studs in back of the gage, insert through holes in the panel, and secure with hex nuts provided. Be careful not to block the slotted hole near the right-hand mounting hole. This provides a path for pressure relief in the event of over-pressurization.

4. To panel-mount gage, cut a 2-5/8" diameter hole. Install the mounting studs in the back of gage, position gage in the panel, and place bracket over the studs. Thread hex nuts over studs and tighten.

5. After installation, the gage may need to be zeroed before placing in operation. If re-zeroing is required, firmly hold the case of gage with one hand and unscrew the front cover with the palm of the other hand in a counterclockwise direction. If difficult to loosen, place a small sheet of rubber between the cover and the palm of the hand. Zero-adjust screw is located behind the scale at the pair marked

"zero." Use the hex allen wrench supplied and adjust until pointer is on zero. This must be done with both pressure connections vented to atmosphere and the gage oriented in the final mounting position. Replace cover.

6. To measure positive pressure, connect tubing to port marked "HI" and vent "LO" port to atmosphere. For negative pressure (vacuum), connect to port marked "LO" and vent "HI" port to atmosphere. For differential pressure, connect higher pressure to port marked "HI" and lower to "LO" port. If gage is supplied with 1/8" NPT connections, be careful not to over-tighten fittings to avoid damage to the gage.

CALIBRATION CHECK

Select a second gage or manometer of known accuracy and in an appropriate range. Use short lengths of rubber or vinyl tubing to connect the high-pressure side of the Minihelic® II gage and the test gage to two legs of a tee. Very slowly, apply pressure through the third leg. Allow enough time for pressure to equalize throughout the system and for fluid to drain. If a manometer is being used. Compare readings. If the gage being tested exceeds rated accuracy, it should be returned to the factory for recalibration.

MAINTENANCE

No lubrication or periodic servicing is required. Keep case exterior and cover clean. Occasionally, disconnect pressure lines to vent both sides of the gage to atmosphere and re-zero per paragraph 5.

SIMATIC S7-1200 Basic Controllers

Introduction

S7-1200

Overview



- Compact controllers for the low to mid-performance ranges
- Large-scale integration, space-saving, powerful
- With exceptional real-time performance and powerful communication options:
 - Controller with integrated PROFINET IO interface for communication between SIMATIC controllers, HMI, programming device or other automation components
- All CPUs can be used in stand-alone mode, in networks and within distributed structures
- Extremely simple installation, programming and operation
- Integrated web server with standard and user-specific web pages
- Data logging functionality for archiving of data at runtime from the user program
- Powerful, integrated technology functions such as counting, measuring, closed-loop control, and motion control
- Integrated digital and analog inputs/outputs
- Flexible expansion facilities
 - Signal boards for direct use in a controller
 - Signal modules for expansion of controllers with input/output channels; including an Energy Meter module for recording and preparing energy data
 - Accessories, e.g. power supply, switch module or SIMATIC Memory Card

Technical specifications

General technical specifications SIMATIC S7-1200

Degree of protection	IP20 acc. to IEC 529
Ambient temperature	
• Operation (95% humidity)	
- Horizontal installation	-20 ... +60 °C
- Vertical installation	-20 ... +50 °C
• Transportation and storage	
- With 95% humidity	-40 ... +70 °C
	25 ... 55 °C
Insulation	
• 5/24 V DC circuits	500 V AC test voltage
• 115/230 V AC circuits to ground	1500 V AC test voltage
• 115/230 V AC circuits to 115/230 V AC circuits	1500 V AC test voltage
• 230 V AC circuits to 5/24 V DC circuits	1500 V AC test voltage
• 115 V AC circuits to 5/24 V DC circuits	1500 V AC test voltage
Electromagnetic compatibility	
• Noise immunity acc. to EN 50082-2	Requirements of the EMC directive Test acc. to: IEC 801-2, IEC 801-3, IEC 801-4, EN 50141, EN 50204, IEC 801-5, VDE 0160
• Emitted interference acc. to EN 50081-1 and EN 50081-2	Test according to EN 55011, Class A, Group 1
Mechanical strength	
• Vibrations, test acc. to / tested with	IEC 68, Part 2-6: 10 ... 57 Hz; constant amplitude 0.3 mm; 58 ... 150 Hz; constant acceleration 1 g (mounted on DIN rail) or 2 g (mounted in switchboard); mode of vibration: frequency sweeps with a sweep rate of 1 octave/minute; duration of vibration: 10 frequency sweeps per axis in each direction of the three mutually perpendicular axes
• Shocks, test acc. to / tested with	IEC 68, Part 2-27/half-sine: magnitude of shock 15 g (peak value), duration 11 ms, 6 shocks in each of the three mutually perpendicular axes

General technical specifications SIPLUS S7-1200

Ambient temperature range	-40/-25/-20 ... +55/+60/+70 °C
Conformal coating	Coating of the printed circuit boards and the electronic components
Technical specifications	The technical specifications of the standard product apply except for the ambient conditions.
Ambient conditions	
Extended range of environmental conditions	
• with reference to ambient temperature, air pressure and altitude	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m) 0° C
• At cold restart, min.	
Relative humidity	
• with condensation, max.	100%; RH incl. bedewing/frost (no commissioning in bedewed state)
Resistance	
• to biologically active substances/ compliance with EN 60721-3-3	Yes; Class 3B2 mold and fungal spores (except fauna); the supplied plug covers must remain in place on the unused interfaces during operation.
• to chemically active substances/ compliance with EN 60721-3-3	Yes; Class 3C4 (RH < 75%) incl. salt spray in accordance with EN 60068-2-52 (severity 3); the supplied plug covers must remain in place on the unused interfaces during operation.
• to mechanically active substances, compliance with EN 60721-3-3	Yes; Class 3S4 incl. sand, dust; the supplied plug covers must remain in place on unused interfaces during operation.

SIMATIC S7-1200 Basic Controllers

Central processing units
Standard CPUs

CPU 1211C

Overview



- Controller for intro to S7
- Expandable by:
 - 1 signal board (SB), battery board (BB) or communication board (CB)
 - Max. 3 communication modules (CM)

Technical specifications

Article number	6ES7211-1BE40-0XB0	6ES7211-1AE40-0XB0	6ES7211-1HE40-0XB0
	CPU 1211C, AC/DC/RELAY, 6DI/4DO/2AI	CPU 1211C, DC/DC/DC, 6DI/4DO/2AI	CPU 1211C, DC/DC/RELAY, 6DI/4DO/2AI
General information			
Product type designation	CPU 1211C AC/DC/Relay	CPU 1211C DC/DC/DC	CPU 1211C DC/DC/Relay
Engineering with			
• Programming package	STEP 7 V14 or higher	STEP 7 V14 or higher	STEP 7 V14 or higher
Supply voltage			
Rated value (DC)			
• 24 V DC		Yes	Yes
Rated value (AC)			
• 120 V AC	Yes		
• 230 V AC	Yes		
Encoder supply			
24 V encoder supply			
• 24 V	20.4 to 28.8V	L+ minus 4 V DC min.	L+ minus 4 V DC min.
Power loss			
Power loss, typ.	10 W	8 W	8 W
Memory			
Work memory			
• integrated	50 kbyte	50 kbyte	50 kbyte
Load memory			
• integrated	1 Mbyte	1 Mbyte	1 Mbyte
• Plug-in (SIMATIC Memory Card), max.	with SIMATIC memory card	with SIMATIC memory card	with SIMATIC memory card
Backup			
• without battery	Yes	Yes	Yes
CPU processing times			
for bit operations, typ.	0.085 µs; / instruction	0.085 µs; / instruction	0.085 µs; / instruction
for word operations, typ.	1.7 µs; / instruction	1.7 µs; / instruction	1.7 µs; / instruction
for floating point arithmetic, typ.	2.3 µs; / instruction	2.3 µs; / instruction	2.3 µs; / instruction
Data areas and their retentivity			
Flag			
• Number, max.	4 kbyte; Size of bit memory address area	4 kbyte; Size of bit memory address area	4 kbyte; Size of bit memory address area
Process image			
• Inputs, adjustable	1 kbyte	1 kbyte	1 kbyte
• Outputs, adjustable	1 kbyte	1 kbyte	1 kbyte
Time of day			
Clock			
• Hardware clock (real-time)	Yes	Yes	Yes

Technical specifications (continued)

Article number	6ES7211-1BE40-0XB0 CPU 1211C, AC/DC/RELAY, 6DI/4DO/2AI	6ES7211-1AE40-0XB0 CPU 1211C, DC/DC/DC, 6DI/4DO/2AI	6ES7211-1HE40-0XB0 CPU 1211C, DC/DC/RELAY, 6DI/4DO/2AI
Digital inputs			
Number of digital inputs	6; Integrated	6; Integrated	6; Integrated
• of which inputs usable for technological functions	3; HSC (High Speed Counting)	3; HSC (High Speed Counting)	3; HSC (High Speed Counting)
Digital outputs			
Number of digital outputs	4; Relays	4	4; Relays
• of which high-speed outputs		4; 100 kHz Pulse Train Output	
Analog inputs			
Number of analog inputs	2	2	2
Input ranges			
• Voltage	Yes	Yes	Yes
Analog outputs			
Number of analog outputs	0	0	0
1. Interface			
Interface type	PROFINET	PROFINET	PROFINET
Physics	Ethernet	Ethernet	Ethernet
Functionality			
• PROFINET IO Controller	Yes	Yes	Yes
• PROFINET IO Device	Yes	Yes	Yes
• SIMATIC communication	Yes	Yes	Yes
• Open IE communication	Yes	Yes	Yes
• Web server	Yes	Yes	Yes
• Media redundancy	No	No	No
Communication functions			
S7 communication			
• supported	Yes	Yes	Yes
Open IE communication			
• TCP/IP	Yes	Yes	Yes
• ISO-on-TCP (RFC1006)	Yes	Yes	Yes
• UDP	Yes	Yes	Yes
Web server			
• supported	Yes	Yes	Yes
Number of connections			
• overall	16; dynamically	16; dynamically	16; dynamically
Integrated Functions			
Number of counters	3	6	3
Counting frequency (counter) max.	100 kHz	100 kHz	100 kHz
Frequency meter	Yes	Yes	Yes
controlled positioning	Yes	Yes	Yes
Number of position-controlled positioning axes, max.	8	8	8
Number of positioning axes via pulse-direction interface	Up to 4 with SB 1222	4; With integrated outputs	Up to 4 with SB 1222
PID controller	Yes	Yes	Yes
Number of alarm inputs	4	4	4
Number of pulse outputs		4	
Limit frequency (pulse)		100 kHz	
Ambient conditions			
Ambient temperature during operation			
• min.	-20 °C	-20 °C	-20 °C
• max.	60 °C	60 °C	60 °C
Pollutant concentrations			
- SO ₂ at RH < 60% without condensation	SO ₂ : < 0.5 ppm; H ₂ S: < 0.1 ppm; RH < 60% condensation-free	SO ₂ : < 0.5 ppm; H ₂ S: < 0.1 ppm; RH < 60% condensation-free	SO ₂ : < 0.5 ppm; H ₂ S: < 0.1 ppm; RH < 60% condensation-free

SIMATIC S7-1200 Basic Controllers

Central processing units
Standard CPUs

CPU 1211C

Technical specifications (continued)

Article number	6ES7211-1BE40-0XB0	6ES7211-1AE40-0XB0	6ES7211-1HE40-0XB0
	CPU 1211C, AC/DC/RELAY, 6DI/4DO/2AI	CPU 1211C, DC/DC/DC, 6DI/4DO/2AI	CPU 1211C, DC/DC/RELAY, 6DI/4DO/2AI
Configuration			
Programming			
Programming language			
- LAD	Yes	Yes	Yes
- FBD	Yes	Yes	Yes
- SCL	Yes	Yes	Yes
Dimensions			
Width	90 mm	90 mm	90 mm
Height	100 mm	100 mm	100 mm
Depth	75 mm	75 mm	75 mm
Weights			
Weight, approx.	420 g	370 g	380 g

Ordering data

Article No.	Article No.
CPU 1211C	
Compact CPU, AC/DC/relay; Integrated program/data memory 50 KB, load memory 1 MB; Wide-range power supply 85 ... 264 V AC; Boolean execution times 0.1 µs per operation; 6 digital inputs, 4 digital outputs (relays), 2 analog inputs; Expandable by up to 3 communication modules and 1 signal board/communication board; Digital inputs can be used as HSC at 100 kHz	6ES7211-1BE40-0XB0
Compact CPU, DC/DC/DC; Integrated program/data memory 50 KB, load memory 1 MB; Power supply 24 V DC; Boolean execution times 0.1 µs per operation; 6 digital inputs, 4 digital outputs, 2 analog inputs; Expandable by up to 3 communication modules and 1 signal board/communication board; Digital inputs can be used as HSC at 100 kHz, 24 V DC digital outputs can be used as pulse outputs (PTO) or pulse-width modulated outputs (PWM) at 100 kHz	6ES7211-1AE40-0XB0
Compact CPU, DC/DC/relay; Integrated program/data memory 50 KB, load memory 1 MB; Power supply 24 V DC; Boolean execution times 0.1 µs per operation; 6 digital inputs, 4 digital outputs (relays), 2 analog inputs; Expandable by up to 3 communication modules and 1 signal board/communication board; Digital inputs can be used as HSC at 100 kHz	6ES7211-1HE40-0XB0
SB 1221 signal board 4 inputs, 5 V DC, 200 kHz 4 inputs, 24 V DC, 200 kHz	6ES7221-3AD30-0XB0 6ES7221-3BD30-0XB0
SB 1222 signal board 4 outputs, 5 V DC, 0.1 A, 200 kHz 4 outputs, 24 V DC, 0.1 A, 200 kHz	6ES7222-1AD30-0XB0 6ES7222-1BD30-0XB0
SB 1223 signal board 2 inputs, 24 V DC, IEC type 1 current sinking; 2 x 24 V DC transistor outputs, 0.5 A, 5 W; can be used as HSC at up to 30 kHz 2 inputs, 5 V DC, 200 kHz 2 outputs 5 V DC, 0.1 A, 200 kHz 2 inputs, 24 V DC, 200 kHz 2 outputs 24 V DC, 0.1 A, 200 kHz	6ES7223-0BD30-0XB0 6ES7223-3AD30-0XB0 6ES7223-3BD30-0XB0
SB 1231 signal board 1 analog input, ±10 V with 12 bits or 0 ... 20 mA with 11 bits	6ES7231-4HA30-0XB0
SB 1231 thermocouple signal board 1 input +/- 80 mV, resolution 15 bits + sign, thermocouples type J, K	6ES7231-5QA30-0XB0
RTD signal board SB 1231 1 input for resistance temperature sensors Pt 100, Pt 200, Pt 500, Pt 1000, resolution 15 bits + sign	6ES7231-5PA30-0XB0
SB 1232 signal board 1 analog output, ±10 V with 12 bits or 0 to 20 mA with 11 bits	6ES7232-4HA30-0XB0
CB 1241 RS 485 communication board For point-to-point connection, with 1 RS 485 interface	6ES7241-1CH30-1XB0

SIMATIC S7-1200 Basic ControllersCentral processing units
Standard CPUs

CPU 1211C

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Ordering data	Article No.	Article No.
Battery board BB1297 For long-term backup of real-time clock, can be plugged into the signal board slot; battery (CR1025) is not included	6ES7297-0AX30-0XA0	STEP 7 Professional / Basic V14 SP1 Target system: SIMATIC S7-1200, S7-1500, S7-300, S7-400, WinAC Requirement: Windows 7 Professional SP1 (64-bit), Windows 7 Enterprise SP1 (64-bit), Windows 7 Ultimate SP1 (64-bit), Windows 8.1 Professional (64-bit), Windows 8.1 Enterprise (64-bit), Windows 10 Professional Version 1607, Windows 10 Enterprise Version 1607, Windows 10 Enterprise 2016 LTSC, Windows 10 Enterprise 2015 LTSC, Windows Server 2008 R2 StdE (full installation), Windows Server 2012 StdE (full installation), Windows Server 2016 Standard (full installation); STEP 7 Basic V14 SP1 in addition: Windows 7 Home Premium SP1 (64-bit), Windows 8.1 (64-bit), Windows 10 Home Version 1607 Type of delivery: German, English, Chinese, Italian, French, Spanish STEP 7 Professional V14 SP1, floating license STEP 7 Professional V14 SP1, floating license, software download incl. license key ¹⁾ Email address required for delivery STEP 7 Basic V14 SP1, floating license STEP 7 Basic V14 SP1, floating license, software download incl. license key ¹⁾ Email address required for delivery
Digital input simulator SIM 1274 simulator module (optional) 8 input switches, for CPU 1211C / CPU 1212C	6ES7274-1XF30-0XA0	
Analog input simulator SIM 1274 simulator module (optional) 2 potentiometers	6ES7274-1XA30-0XA0	
SIMATIC Memory Card (optional) 4 MB 12 MB 24 MB 256 MB 2 GB 32 GB	6ES7954-8LC02-0AA0 6ES7954-8LE02-0AA0 6ES7954-8LF02-0AA0 6ES7954-8LL02-0AA0 6ES7954-8LP02-0AA0 6ES7954-8LT03-0AA0	
Terminal block (spare part) For CPU 1211C AC/DC/relay • For DI, with 14 screws, tin-coated, coded; 4 units • For DO, with 8 screws, tin-coated, coded; 4 units • For AI, with 3 screws, gold-plated; 4 units For CPU 1211C DC/DC/DC • For DI, with 14 screws, tin-coated; 4 units • For DO, with 8 screws, tin-coated; 4 units • For AI, with 3 screws, gold-plated; 4 units For CPU 1211C DC/DC/relay • For DI, with 14 screws, tin-coated; 4 units • For DO, with 8 screws, tin-coated, coded; 4 units • For AI, with 3 screws, gold-plated; 4 units	6ES7292-1AP40-0XA0 6ES7292-1AH40-0XA0 6ES7292-1BC30-0XA0 6ES7292-1AP30-0XA0 6ES7292-1AH30-0XA0 6ES7292-1BC30-0XA0 6ES7292-1AP30-0XA0 6ES7292-1AH40-0XA0 6ES7292-1BC30-0XA0	
RJ45 cable grip 4 units per pack Single port	6ES7290-3AA30-0XA0	
Front flap set (spare part) For CPU 1211C/1212C	6ES7291-1AA30-0XA0	

¹⁾ For up-to-date information and download availability, see:
<http://www.siemens.com/tia-online-software-delivery>

SIMATIC S7-1200 Basic Controllers

Central processing units

Standard CPUs

CPU 1212C

Overview



- Controller for intro to S7 with basic expansion options
- Expandable by:
 - 1 signal board (SB), battery board (BB) or communication board (CB)
 - 2 signal modules (SM)
 - Max. 3 communication modules (CM)

Technical specifications

Article number	6ES7212-1BE40-0XB0 CPU 1212C, AC/DC/RELAY, 8DI/6DO/2AI	6ES7212-1AE40-0XB0 CPU 1212C, DC/DC/DC, 8DI/6DO/2AI	6ES7212-1HE40-0XB0 CPU 1212C, DC/DC/RELAY, 8DI/6DO/2AI
General information			
Product type designation	CPU 1212C AC/DC/Relay	CPU 1212C DC/DC/DC	CPU 1212C DC/DC/Relay
Engineering with			
• Programming package	STEP 7 V14 or higher	STEP 7 V14 or higher	STEP 7 V14 or higher
Supply voltage			
Rated value (DC)			
• 24 V DC		Yes	Yes
Rated value (AC)			
• 120 V AC	Yes		
• 230 V AC	Yes		
Encoder supply			
24 V encoder supply			
• 24 V	20.4 to 28.8V	L+ minus 4 V DC min.	L+ minus 4 V DC min.
Power loss			
Power loss, typ.	11 W	9 W	9 W
Memory			
Work memory			
• integrated	75 kbyte	75 kbyte	75 kbyte
Load memory			
• integrated	2 Mbyte	2 Mbyte	2 Mbyte
• Plug-in (SIMATIC Memory Card), max.	with SIMATIC memory card	with SIMATIC memory card	with SIMATIC memory card
Backup			
• without battery	Yes	Yes	Yes
CPU processing times			
for bit operations, typ.	0.085 µs; / instruction	0.085 µs; / instruction	0.085 µs; / instruction
for word operations, typ.	1.7 µs; / instruction	1.7 µs; / instruction	1.7 µs; / instruction
for floating point arithmetic, typ.	2.3 µs; / instruction	2.3 µs; / instruction	2.3 µs; / instruction
Data areas and their retentivity			
Flag			
• Number, max.	4 kbyte; Size of bit memory address area	4 kbyte; Size of bit memory address area	4 kbyte; Size of bit memory address area
Process image			
• Inputs, adjustable	1 kbyte	1 kbyte	1 kbyte
• Outputs, adjustable	1 kbyte	1 kbyte	1 kbyte

Technical specifications (continued)

Article number	6ES7212-1BE40-0XB0 CPU 1212C, AC/DC/RELAY, 8DI/6DO/2AI	6ES7212-1AE40-0XB0 CPU 1212C, DC/DC/DC, 8DI/6DO/2AI	6ES7212-1HE40-0XB0 CPU 1212C, DC/DC/RELAY, 8DI/6DO/2AI
Time of day			
Clock			
• Hardware clock (real-time)	Yes	Yes	Yes
Digital inputs			
Number of digital inputs	8; Integrated	8; Integrated	8; Integrated
• of which inputs usable for technological functions	4; HSC (High Speed Counting)	4; HSC (High Speed Counting)	4; HSC (High Speed Counting)
Digital outputs			
Number of digital outputs	6; Relays	6	6; Relays
• of which high-speed outputs		4; 100 kHz Pulse Train Output	
Analog inputs			
Number of analog inputs	2	2	2
Input ranges			
• Voltage	Yes	Yes	Yes
Analog outputs			
Number of analog outputs	0	0	0
1. Interface			
Interface type	PROFINET	PROFINET	PROFINET
Physics	Ethernet	Ethernet	Ethernet
Functionality			
• PROFINET IO Controller	Yes	Yes	Yes
• PROFINET IO Device	Yes	Yes	Yes
• SIMATIC communication	Yes	Yes	Yes
• Open IE communication	Yes	Yes	Yes
• Web server	Yes	Yes	Yes
• Media redundancy	No	No	No
Communication functions			
S7 communication			
• supported	Yes	Yes	Yes
Open IE communication			
• TCP/IP	Yes	Yes	Yes
• ISO-on-TCP (RFC1006)	Yes	Yes	Yes
• UDP	Yes	Yes	Yes
Web server			
• supported	Yes	Yes	Yes
Number of connections			
• overall	16; dynamically	16; dynamically	16; dynamically
Integrated Functions			
Number of counters	4	4	4
Counting frequency (counter) max.	100 kHz	100 kHz	100 kHz
Frequency meter	Yes	Yes	Yes
controlled positioning	Yes	Yes	Yes
Number of position-controlled positioning axes, max.	8	8	8
Number of positioning axes via pulse-direction interface	Up to 4 with SB 1222	4; With integrated outputs	Up to 4 with SB 1222
PID controller	Yes	Yes	Yes
Number of alarm inputs	4	4	4
Number of pulse outputs		4	
Limit frequency (pulse)		100 kHz	

SIMATIC S7-1200 Basic Controllers

Central processing units
Standard CPUs

CPU 1212C

Technical specifications (continued)

Article number	6ES7212-1BE40-0XB0 CPU 1212C, AC/DC/RELAY, 8DI/6DO/2AI	6ES7212-1AE40-0XB0 CPU 1212C, DC/DC/DC, 8DI/6DO/2AI	6ES7212-1HE40-0XB0 CPU 1212C, DC/DC/RELAY, 8DI/6DO/2AI
Ambient conditions			
Ambient temperature during operation			
• min.	-20 °C	-20 °C	-20 °C
• max.	60 °C; Number of simultaneously activated inputs or outputs 4 or 3 (no adjacent points) at 60 °C horizontal or 50 °C vertical, 8 or 6 at 55 °C horizontal or 45 °C vertical	60 °C; Number of simultaneously activated inputs or outputs 4 or 3 (no adjacent points) at 60 °C horizontal or 50 °C vertical, 8 or 6 at 55 °C horizontal or 45 °C vertical	60 °C; Number of simultaneously activated inputs or outputs 4 or 3 (no adjacent points) at 60 °C horizontal or 50 °C vertical, 8 or 6 at 55 °C horizontal or 45 °C vertical
Pollutant concentrations			
- SO2 at RH < 60% without condensation	SO2: < 0.5 ppm; H2S: < 0.1 ppm; RH < 60% condensation-free	SO2: < 0.5 ppm; H2S: < 0.1 ppm; RH < 60% condensation-free	SO2: < 0.5 ppm; H2S: < 0.1 ppm; RH < 60% condensation-free
Configuration			
Programming			
Programming language			
- LAD	Yes	Yes	Yes
- FBD	Yes	Yes	Yes
- SCL	Yes	Yes	Yes
Dimensions			
Width	90 mm	90 mm	90 mm
Height	100 mm	100 mm	100 mm
Depth	75 mm	75 mm	75 mm
Weights			
Weight, approx.	425 g	370 g	385 g

Ordering data

Article No.

Article No.

CPU 1212C

Compact CPU, AC/DC/relay;
Integrated program/data memory
75 KB, load memory 2 MB;
Wide-range power supply
85 ... 264 V AC;
Boolean execution times 0.1 µs
per operation;
8 digital inputs,
6 digital outputs (relays),
2 analog inputs;
Expandable by up to
3 communication modules,
2 signal modules and 1 signal
board/communication board;
Digital inputs can be used as HSC
at 100 kHz

6ES7212-1BE40-0XB0

Compact CPU, DC/DC/DC;
Integrated program/data memory
75 KB, load memory 2 MB;
Power supply 24 V DC;
Boolean execution times 0.1 µs
per operation;
8 digital inputs,
6 digital outputs,
2 analog inputs;
Expandable by up to
3 communication modules,
2 signal modules, and 1 signal
board/communication board;
Digital inputs can be used as HSC
at 100 kHz,
24 V DC digital outputs can be
used as pulse outputs (PTO) or
pulse-width modulated outputs
(PWM) at 100 kHz

6ES7212-1AE40-0XB0

Compact CPU, DC/DC/relay;
Integrated program/data memory
75 KB, load memory 2 MB;
power supply 24 V DC;
Boolean execution times 0.1 µs
per operation;
8 digital inputs,
6 digital outputs (relays),
2 analog inputs;
expandable by up to
3 communication modules, 2
signal modules, and 1 signal
board/communication board;
digital inputs can be used as HSC
at 100 kHz

6ES7212-1HE40-0XB0

SB 1221 signal board

4 inputs, 5 V DC, 200 kHz

6ES7221-3AD30-0XB0

4 inputs, 24 V DC, 200 kHz

6ES7221-3BD30-0XB0

SB 1222 signal board

4 outputs, 5 V DC, 0.1 A, 200 kHz

6ES7222-1AD30-0XB0

4 outputs, 24 V DC, 0.1 A, 200 kHz

6ES7222-1BD30-0XB0

SB 1223 signal board

2 inputs, 24 V DC,
IEC type 1 current sinking;
2 x 24 V DC transistor outputs,
0.5 A, 5 W; can be used as HSC at
up to 30 kHz

6ES7223-0BD30-0XB0

2 inputs, 5 V DC, 200 kHz
2 outputs 5 V DC, 0.1 A, 200 kHz

6ES7223-3AD30-0XB0

2 inputs, 24 V DC, 200 kHz
2 outputs 24 V DC, 0.1 A, 200 kHz

6ES7223-3BD30-0XB0

SB 1231 signal board

1 analog input, ±10 V with 12 bits or
0 ... 20 mA with 11 bits

6ES7231-4HA30-0XB0

SIMATIC S7-1200 Basic ControllersCentral processing units
Standard CPUs**CPU 1212C**

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Ordering data	Article No.		Article No.
SB 1231 thermocouple signal board 1 input +/- 80 mV, resolution 15 bits + sign, thermocouples type J, K	6ES7231-5QA30-0XB0	Terminal block (spare part) (cont.) For CPU 1212C DC/DC/DC • For DI, with 14 screws, tin-coated; 4 units • For DO, with 8 screws, tin-coated; 4 units • For AI, with 3 screws, gold-plated; 4 units For CPU 1212C DC/DC/relay • For DI, with 14 screws, tin-coated; 4 units • For DO, with 8 screws, tin-coated, coded; 4 units • For AI, with 3 screws, gold-plated; 4 units	6ES7292-1AP30-0XA0 6ES7292-1AH30-0XA0 6ES7292-1BC30-0XA0
RTD signal board SB 1231 1 input for resistance temperature sensors Pt 100, Pt 200, Pt 500, Pt 1000, resolution 15 bits + sign	6ES7231-5PA30-0XB0		
SB 1232 signal board 1 analog output, ±10 V with 12 bits or 0 to 20 mA with 11 bits	6ES7232-4HA30-0XB0		6ES7292-1AP30-0XA0 6ES7292-1AH40-0XA0 6ES7292-1BC30-0XA0
CB 1241 RS 485 communication board For point-to-point connection, with 1 RS 485 interface	6ES7241-1CH30-1XB0	RJ45 cable grip 4 units per pack Single port	6ES7290-3AA30-0XA0
BB1297 battery board For long-term backup of real-time clock, can be plugged into the signal board slot; battery (CR1025) is not included	6ES7297-0AX30-0XA0	Front flap set (spare part) For CPU 1211C/1212C	6ES7291-1AA30-0XA0
Digital input simulator SIM 1274 simulator module (optional) 8 input switches, for CPU 1211C / CPU 1212C	6ES7274-1XF30-0XA0	STEP 7 Professional / Basic V14 SP1 Target system: SIMATIC S7-1200, S7-1500, S7-300, S7-400, WinAC Requirement: Windows 7 Professional SP1 (64-bit), Windows 7 Enterprise SP1 (64-bit), Windows 7 Ultimate SP1 (64-bit), Windows 8.1 Professional (64-bit), Windows 8.1 Enterprise (64-bit), Windows 10 Professional Version 1607, Windows 10 Enterprise Version 1607, Windows 10 Enterprise 2016 LTSB, Windows 10 Enterprise 2015 LTSB, Windows Server 2008 R2 StdE (full installation), Windows Server 2012 StdE (full installation), Windows Server 2016 Standard (full installation); STEP 7 Basic V14 SP1 in addition: Windows 7 Home Premium SP1 (64-bit), Windows 8.1 (64-bit), Windows 10 Home Version 1607 Type of delivery: German, English, Chinese, Italian, French, Spanish	
Analog input simulator SIM 1274 simulator module (optional) 2 potentiometers	6ES7274-1XA30-0XA0	STEP 7 Professional V14 SP1, floating license	6ES7822-1AA04-0YA5
SIMATIC Memory Card (optional) 4 MB 12 MB 24 MB 256 MB 2 GB 32 GB	6ES7954-8LC02-0AA0 6ES7954-8LE02-0AA0 6ES7954-8LF02-0AA0 6ES7954-8LL02-0AA0 6ES7954-8LP02-0AA0 6ES7954-8LT03-0AA0	STEP 7 Professional V14 SP1, floating license, software download incl. license key ¹⁾ Email address required for delivery	6ES7822-1AE04-0YA5
Extension cable for two-tier configuration For connecting digital/analog signal modules; length 2 m	6ES7290-6AA30-0XA0	STEP 7 Basic V14 SP1, floating license	6ES7822-0AA04-0YA5
Starter box CPU 1212C AC/DC/relay Complete offer SIMATIC S7-1200, starter box, comprising: CPU 1212C AC/DC/relay, simulator, STEP 7 BASIC CD, manual CD, info material, in Systainer	6ES7212-1BD34-4YB0	STEP 7 Basic V14 SP1, floating license, software download incl. license key ¹⁾ Email address required for delivery	6ES7822-0AE04-0YA5
Terminal block (spare part) For CPU 1212C AC/DC/relay • For DI, with 14 screws, tin-coated, coded; 4 units • For DO, with 8 screws, tin-coated, coded; 4 units • For AI, with 3 screws, gold-plated; 4 units	6ES7292-1AP40-0XA0 6ES7292-1AH40-0XA0 6ES7292-1BC30-0XA0		

¹⁾ For up-to-date information and download availability, see:
<http://www.siemens.com/tia-online-software-delivery>

SIMATIC S7-1200 Basic Controllers

Central processing units

Standard CPUs

CPU 1214C

Overview



- Controller for intro to S7 with flexible expansion options
- Expandable by:
 - 1 signal board (SB), battery board (BB) or communication board (CB)
 - 8 signal modules (SM)
 - Max. 3 communication modules (CM)

Technical specifications

Article number	6ES7214-1BG40-0XB0 CPU 1214C, AC/DC/RELAY, 14DI/10DO/2AI	6ES7214-1AG40-0XB0 CPU 1214C, DC/DC/DC, 14DI/10DO/2AI	6ES7214-1HG40-0XB0 CPU 1214C, DC/DC/RELAY, 14DI/10DO/2AI
General information			
Product type designation	CPU 1214C AC/DC/Relay	CPU 1214C DC/DC/DC	CPU 1214C DC/DC/Relay
Engineering with			
• Programming package	STEP 7 V14 or higher	STEP 7 V14 or higher	STEP 7 V14 or higher
Supply voltage			
Rated value (DC)			
• 24 V DC		Yes	Yes
Rated value (AC)			
• 120 V AC	Yes		
• 230 V AC	Yes		
Encoder supply			
24 V encoder supply			
• 24 V	20.4 to 28.8V	L+ minus 4 V DC min.	L+ minus 4 V DC min.
Power loss			
Power loss, typ.	14 W	12 W	12 W
Memory			
Work memory			
• integrated	100 kbyte	100 kbyte	100 kbyte
Load memory			
• integrated	4 Mbyte	4 Mbyte	4 Mbyte
• Plug-in (SIMATIC Memory Card), max.	with SIMATIC memory card	with SIMATIC memory card	with SIMATIC memory card
Backup			
• without battery	Yes	Yes	Yes
CPU processing times			
for bit operations, typ.	0.085 µs; / instruction	0.085 µs; / instruction	0.085 µs; / instruction
for word operations, typ.	1.7 µs; / instruction	1.7 µs; / instruction	1.7 µs; / instruction
for floating point arithmetic, typ.	2.3 µs; / instruction	2.3 µs; / instruction	2.3 µs; / instruction
Data areas and their retentivity			
Flag			
• Number, max.	8 kbyte; Size of bit memory address area	8 kbyte; Size of bit memory address area	8 kbyte; Size of bit memory address area
Process image			
• Inputs, adjustable	1 kbyte	1 kbyte	1 kbyte
• Outputs, adjustable	1 kbyte	1 kbyte	1 kbyte
Time of day			
Clock			
• Hardware clock (real-time)	Yes	Yes	Yes

Technical specifications (continued)

Article number	6ES7214-1BG40-0XB0 CPU 1214C, AC/DC/RELAY, 14DI/10DO/2AI	6ES7214-1AG40-0XB0 CPU 1214C, DC/DC/DC, 14DI/10DO/2AI	6ES7214-1HG40-0XB0 CPU 1214C, DC/DC/RELAY, 14DI/10DO/2AI
Digital inputs			
Number of digital inputs	14; Integrated	14; Integrated	14; Integrated
• of which inputs usable for technological functions	6; HSC (High Speed Counting)	6; HSC (High Speed Counting)	6; HSC (High Speed Counting)
Digital outputs			
Number of digital outputs	10; Relays	10	10; Relays
• of which high-speed outputs		4; 100 kHz Pulse Train Output	
Analog inputs			
Number of analog inputs	2	2	2
Input ranges			
• Voltage	Yes	Yes	Yes
Analog outputs			
Number of analog outputs	0	0	0
1. Interface			
Interface type	PROFINET	PROFINET	PROFINET
Physics	Ethernet	Ethernet	Ethernet
Functionality			
• PROFINET IO Controller	Yes	Yes	Yes
• PROFINET IO Device	Yes	Yes	Yes
• SIMATIC communication	Yes	Yes	Yes
• Open IE communication	Yes	Yes	Yes
• Web server	Yes	Yes	Yes
• Media redundancy	No	No	No
Communication functions			
S7 communication			
• supported	Yes	Yes	Yes
Open IE communication			
• TCP/IP	Yes	Yes	Yes
• ISO-on-TCP (RFC1006)	Yes	Yes	Yes
• UDP	Yes	Yes	Yes
Web server			
• supported	Yes	Yes	Yes
Number of connections			
• overall	16; dynamically	16; dynamically	16; dynamically
Integrated Functions			
Number of counters	6	6	6
Counting frequency (counter) max.	100 kHz	100 kHz	100 kHz
Frequency meter	Yes	Yes	Yes
controlled positioning	Yes	Yes	Yes
Number of position-controlled positioning axes, max.	8	8	8
Number of positioning axes via pulse-direction interface	Up to 4 with SB 1222	4; With integrated outputs	Up to 4 with SB 1222
PID controller	Yes	Yes	Yes
Number of alarm inputs	4	4	4
Number of pulse outputs		4	
Limit frequency (pulse)		100 kHz	
Ambient conditions			
Ambient temperature during operation			
• min.	-20 °C	-20 °C	-20 °C
• max.	60 °C; Number of simultaneously activated inputs or outputs 7 or 5 (no adjacent points) at 60 °C horizontal or 50 °C vertical, 14 or 10 at 55 °C horizontal or 45 °C vertical	60 °C; Number of simultaneously activated inputs or outputs 7 or 5 (no adjacent points) at 60 °C horizontal or 50 °C vertical, 14 or 10 at 55 °C horizontal or 45 °C vertical	60 °C; Number of simultaneously activated inputs or outputs 7 or 5 (no adjacent points) at 60 °C horizontal or 50 °C vertical, 14 or 10 at 55 °C horizontal or 45 °C vertical
Pollutant concentrations			
- SO2 at RH < 60% without condensation	SO2: < 0.5 ppm; H2S: < 0.1 ppm; RH < 60% condensation-free	SO2: < 0.5 ppm; H2S: < 0.1 ppm; RH < 60% condensation-free	SO2: < 0.5 ppm; H2S: < 0.1 ppm; RH < 60% condensation-free

SIMATIC S7-1200 Basic Controllers

Central processing units

Standard CPUs

CPU 1214C

Technical specifications (continued)

Article number	6ES7214-1BG40-0XB0 CPU 1214C, AC/DC/RELAY, 14DI/10DO/2AI	6ES7214-1AG40-0XB0 CPU 1214C, DC/DC/DC, 14DI/10DO/2AI	6ES7214-1HG40-0XB0 CPU 1214C, DC/DC/RELAY, 14DI/10DO/2AI
Configuration			
Programming			
Programming language			
- LAD	Yes	Yes	Yes
- FBD	Yes	Yes	Yes
- SCL	Yes	Yes	Yes
Dimensions			
Width	110 mm	110 mm	110 mm
Height	100 mm	100 mm	100 mm
Depth	75 mm	75 mm	75 mm
Weights			
Weight, approx.	455 g	415 g	435 g

Ordering data

Article No.	Article No.
CPU 1214C	SB 1221 signal board
Compact CPU, AC/DC/relay; Integrated program/data memory 100 KB, load memory 2 MB; Wide-range power supply 85 ... 264 V AC; Boolean execution times 0.1 µs per operation; 14 digital inputs, 10 digital outputs (relays), 2 analog inputs; Expandable by up to 3 communication modules, 8 signal modules and 1 signal board/communication board; Digital inputs can be used as HSC at 100 kHz	4 inputs, 5 V DC, 200 kHz 4 inputs, 24 V DC, 200 kHz 6ES7221-3AD30-0XB0 6ES7221-3BD30-0XB0
Compact CPU, DC/DC/DC; Integrated program/data memory 100 KB, load memory 2 MB; Power supply 24 V DC; Boolean execution times 0.1 µs per operation; 14 digital inputs, 10 digital outputs, 2 analog inputs; expandable by up to 3 communication modules, 8 signal modules, and 1 signal board/communication board; Digital inputs can be used as HSC at 100 kHz, 24 V DC digital outputs can be used as pulse outputs (PTO) or pulse-width modulated outputs (PWM) at 100 kHz	SB 1222 signal board 4 outputs, 5 V DC, 0.1 A, 200 kHz 4 outputs, 24 V DC, 0.1 A, 200 kHz 6ES7222-1AD30-0XB0 6ES7222-1BD30-0XB0
Compact CPU, DC/DC/relay; Integrated program/data memory 100 KB, load memory 2 MB; Power supply 24 V DC; Boolean execution times 0.1 µs per operation; 14 digital inputs, 10 digital outputs (relays), 2 analog inputs; Expandable by up to 3 communication modules, 8 signal modules, and 1 signal board/communication board; Digital inputs can be used as HSC at 100 kHz	SB 1223 signal board 2 inputs, 24 V DC, IEC type 1 current sinking; 2 x 24 V DC transistor outputs, 0.5 A, 5 W; can be used as HSC at up to 30 kHz 6ES7223-0BD30-0XB0 2 inputs, 5 V DC, 200 kHz 2 outputs 5 V DC, 0.1 A, 200 kHz 6ES7223-3AD30-0XB0 2 inputs, 24 V DC, 200 kHz 2 outputs 24 V DC, 0.1 A, 200 kHz 6ES7223-3BD30-0XB0
	SB 1231 signal board 1 analog input, ±10 V with 12 bits or 0 ... 20 mA with 11 bits 6ES7231-4HA30-0XB0
	SB 1231 thermocouple signal board 1 input +/- 80 mV, resolution 15 bits + sign, thermocouples type J, K 6ES7231-5QA30-0XB0
	RTD signal board SB 1231 1 input for resistance temperature sensors Pt 100, Pt 200, Pt 500, Pt 1000, resolution 15 bits + sign 6ES7231-5PA30-0XB0
	SB 1232 signal board 1 analog output, ±10 V with 12 bits or 0 to 20 mA with 11 bits 6ES7232-4HA30-0XB0
	CB 1241 RS 485 communication board For point-to-point connection, with 1 RS 485 interface 6ES7241-1CH30-1XB0

SIMATIC S7-1200 Basic ControllersCentral processing units
Standard CPUs**CPU 1214C****3**

Ordering data	Article No.		Article No.
Battery board BB1297 For long-term backup of real-time clock, can be plugged into the signal board slot; battery (CR1025) is not included	6ES7297-0AX30-0XA0	RJ45 cable grip 4 items per pack Single port	6ES7290-3AA30-0XA0
Digital input simulator SIM 1274 simulator module (optional) 14 input switches, for CPU 1214C/1215C	6ES7274-1XH30-0XA0	Front flap set (spare part) For CPU 1214C	6ES7291-1AB30-0XA0
Analog input simulator SIM 1274 simulator module (optional) 2 potentiometers	6ES7274-1XA30-0XA0	STEP 7 Professional / Basic V14 SP1 Target system: SIMATIC S7-1200, S7-1500, S7-300, S7-400, WinAC Requirement: Windows 7 Professional SP1 (64-bit), Windows 7 Enterprise SP1 (64-bit), Windows 7 Ultimate SP1 (64-bit), Windows 8.1 Professional (64-bit), Windows 8.1 Enterprise (64-bit), Windows 10 Professional Version 1607, Windows 10 Enterprise Version 1607, Windows 10 Enterprise 2016 LTSB, Windows 10 Enterprise 2015 LTSB, Windows Server 2008 R2 StdE (full installation), Windows Server 2012 StdE (full installation), Windows Server 2016 Standard (full installation); STEP 7 Basic V14 SP1 in addition: Windows 7 Home Premium SP1 (64-bit), Windows 8.1 (64-bit), Windows 10 Home Version 1607 Type of delivery: German, English, Chinese, Italian, French, Spanish	
SIMATIC Memory Card (optional) 4 MB 12 MB 24 MB 256 MB 2 GB 32 GB	6ES7954-8LC02-0AA0 6ES7954-8LE02-0AA0 6ES7954-8LF02-0AA0 6ES7954-8LL02-0AA0 6ES7954-8LP02-0AA0 6ES7954-8LT03-0AA0	STEP 7 Professional V14 SP1, floating license STEP 7 Professional V14 SP1, floating license, software download incl. license key ¹⁾ Email address required for delivery STEP 7 Basic V14 SP1, floating license STEP 7 Basic V14 SP1, floating license, software download incl. license key ¹⁾ Email address required for delivery	6ES7822-1AA04-0YA5 6ES7822-1AE04-0YA5 6ES7822-0AA04-0YA5 6ES7822-0AE04-0YA5
Extension cable for two-tier configuration For connecting digital/analog signal modules; length 2 m	6ES7290-6AA30-0XA0		
Terminal block (spare part) For CPU 1214C AC/DC/relay • For DI, with 20 screws, tin-coated, coded; 4 units • For DO, with 12 screws, tin-coated, coded; 4 units • For AI, with 3 screws, gold-plated; 4 units For CPU 1214C DC/DC/DC • For DI, with 20 screws, tin-coated; 4 units • For DO, with 12 screws, tin-coated; 4 units • For AI, with 3 screws, gold-plated; 4 units For CPU 1214C DC/DC/relay • For DI, with 20 screws, tin-coated; 4 units • For DO, with 12 screws, tin-coated, coded; 4 units • For AI, with 3 screws, gold-plated; 4 units	6ES7292-1AV40-0XA0 6ES7292-1AM40-0XA0 6ES7292-1BC30-0XA0 6ES7292-1AV30-0XA0 6ES7292-1AM30-0XA0 6ES7292-1BC30-0XA0 6ES7292-1AV30-0XA0 6ES7292-1AM40-0XA0 6ES7292-1BC30-0XA0		

¹⁾ For up-to-date information and download availability, see:
<http://www.siemens.com/tia-online-software-delivery>

SIMATIC S7-1200 Basic Controllers

Central processing units
Standard CPUs

CPU 1215C

Overview



- Powerful controller with enhanced networking option
- Expandable by:
 - 1 signal board (SB), battery board (BB) or communication board (CB)
 - 8 signal modules (SM)
 - Max. 3 communication modules (CM)

Technical specifications

Article number	6ES7215-1BG40-0XB0	6ES7215-1AG40-0XB0	6ES7215-1HG40-0XB0
	CPU 1215C, AC/DC/RLY, 14DI/10DO/2AI/2AO	CPU 1215C, DC/DC/DC, 14DI/10DO/2AI/2AO	CPU 1215C, DC/DC/RLY, 14DI/10DO/2AI/2AO
General information			
Product type designation	CPU 1215C AC/DC/Relay	CPU 1215C DC/DC/DC	CPU 1215C DC/DC/Relay
Engineering with			
• Programming package	STEP 7 V14 or higher	STEP 7 V14 or higher	STEP 7 V14 or higher
Supply voltage			
Rated value (DC)			
• 24 V DC		Yes	Yes
Rated value (AC)			
• 120 V AC	Yes		
• 230 V AC	Yes		
Encoder supply			
24 V encoder supply			
• 24 V	20.4 to 28.8V	L+ minus 4 V DC min.	L+ minus 4 V DC min.
Power loss			
Power loss, typ.	14 W	12 W	12 W
Memory			
Work memory			
• integrated	125 kbyte	125 kbyte	125 kbyte
Load memory			
• integrated	4 Mbyte	4 Mbyte	4 Mbyte
• Plug-in (SIMATIC Memory Card), max.	with SIMATIC memory card	with SIMATIC memory card	with SIMATIC memory card
Backup			
• without battery	Yes	Yes	Yes
CPU processing times			
for bit operations, typ.	0.085 µs; / instruction	0.085 µs; / instruction	0.085 µs; / instruction
for word operations, typ.	1.7 µs; / instruction	1.7 µs; / instruction	1.7 µs; / instruction
for floating point arithmetic, typ.	2.3 µs; / instruction	2.3 µs; / instruction	2.3 µs; / instruction
Data areas and their retentivity			
Flag			
• Number, max.	8 kbyte; Size of bit memory address area	8 kbyte; Size of bit memory address area	8 kbyte; Size of bit memory address area
Process image			
• Inputs, adjustable	1 kbyte	1 kbyte	1 kbyte
• Outputs, adjustable	1 kbyte	1 kbyte	1 kbyte
Time of day			
Clock			
• Hardware clock (real-time)	Yes	Yes	Yes

Technical specifications (continued)

Article number	6ES7215-1BG40-0XB0 CPU 1215C, AC/DC/RLY, 14DI/10DO/2AI/2AO	6ES7215-1AG40-0XB0 CPU 1215C, DC/DC/DC, 14DI/10DO/2AI/2AO	6ES7215-1HG40-0XB0 CPU 1215C, DC/DC/RLY, 14DI/10DO/2AI/2AO
Digital inputs			
Number of digital inputs	14; Integrated	14; Integrated	14; Integrated
• of which inputs usable for technological functions	6; HSC (High Speed Counting)	6; HSC (High Speed Counting)	6; HSC (High Speed Counting)
Digital outputs			
Number of digital outputs	10; Relays	10	10; Relays
• of which high-speed outputs		4; 100 kHz Pulse Train Output	
Analog inputs			
Number of analog inputs	2	2	2
Input ranges			
• Voltage	Yes	Yes	Yes
Analog outputs			
Number of analog outputs	2	2	2
Output ranges, current			
• 0 to 20 mA	Yes	Yes	Yes
1. Interface			
Interface type	PROFINET	PROFINET	PROFINET
Physics	Ethernet	Ethernet	Ethernet
Functionality			
• PROFINET IO Controller	Yes	Yes	Yes
• PROFINET IO Device	Yes	Yes	Yes
• SIMATIC communication	Yes	Yes	Yes
• Open IE communication	Yes	Yes	Yes
• Web server	Yes	Yes	Yes
• Media redundancy	Yes; as MRP client	Yes; as MRP client	Yes; as MRP client
Communication functions			
S7 communication			
• supported	Yes	Yes	Yes
Open IE communication			
• TCP/IP	Yes	Yes	Yes
• ISO-on-TCP (RFC1006)	Yes	Yes	Yes
• UDP	Yes	Yes	Yes
Web server			
• supported	Yes	Yes	Yes
Number of connections			
• overall	16; dynamically	16; dynamically	16; dynamically
Integrated Functions			
Number of counters	6	6	6
Counting frequency (counter) max.	100 kHz	100 kHz	100 kHz
Frequency meter	Yes	Yes	Yes
controlled positioning	Yes	Yes	Yes
Number of position-controlled positioning axes, max.	8	8	8
Number of positioning axes via pulse-direction interface	Up to 4 with SB 1222	4; With integrated outputs	Up to 4 with SB 1222
PID controller	Yes	Yes	Yes
Number of alarm inputs	4	4	4
Number of pulse outputs		4	
Limit frequency (pulse)		100 kHz	
Ambient conditions			
Ambient temperature during operation			
• min.	-20 °C	-20 °C	-20 °C
• max.	60 °C; Number of simultaneously activated inputs or outputs 7 or 5 (no adjacent points) at 60 °C horizontal or 50 °C vertical, 14 or 10 at 55 °C horizontal or 45 °C vertical	60 °C; Number of simultaneously activated inputs or outputs 7 or 5 (no adjacent points) at 60 °C horizontal or 50 °C vertical, 14 or 10 at 55 °C horizontal or 45 °C vertical	60 °C; Number of simultaneously activated inputs or outputs 7 or 5 (no adjacent points) at 60 °C horizontal or 50 °C vertical, 14 or 10 at 55 °C horizontal or 45 °C vertical
Pollutant concentrations			
- SO ₂ at RH < 60% without condensation	SO ₂ : < 0.5 ppm; H ₂ S: < 0.1 ppm; RH < 60% condensation-free	SO ₂ : < 0.5 ppm; H ₂ S: < 0.1 ppm; RH < 60% condensation-free	SO ₂ : < 0.5 ppm; H ₂ S: < 0.1 ppm; RH < 60% condensation-free

SIMATIC S7-1200 Basic Controllers

Central processing units
Standard CPUs

CPU 1215C

Technical specifications (continued)

Article number	6ES7215-1BG40-0XB0 CPU 1215C, AC/DC/RLY, 14DI/10DO/2AI/2AO	6ES7215-1AG40-0XB0 CPU 1215C, DC/DC/DC, 14DI/10DO/2AI/2AO	6ES7215-1HG40-0XB0 CPU 1215C, DC/DC/RLY, 14DI/10DO/2AI/2AO
Configuration			
Programming			
Programming language			
- LAD	Yes	Yes	Yes
- FBD	Yes	Yes	Yes
- SCL	Yes	Yes	Yes
Dimensions			
Width	130 mm	130 mm	130 mm
Height	100 mm	100 mm	100 mm
Depth	75 mm	75 mm	75 mm
Weights			
Weight, approx.	550 g	500 g	585 g

Ordering data

CPU 1215C

Compact CPU, AC/DC/relay;
Integrated program/data memory
125 KB, load memory 4 MB;
Wide-range power supply
85 ... 264 V AC;
Boolean execution times 0.085 µs
per operation;
14 digital inputs,
10 digital outputs (relays),
2 analog inputs,
2 analog outputs;
Expandable by up to
3 communication modules,
8 signal modules and 1 signal
board/communication board;
Digital inputs can be used as HSC
at 100 kHz

6ES7215-1BG40-0XB0

Compact CPU, DC/DC/DC;
Integrated program/data memory
125 KB, load memory 4 MB;
Power supply 24 V DC;
Boolean execution times 0.085 µs
per operation;
14 digital inputs,
10 digital outputs,
2 analog inputs,
2 analog outputs;
Expandable by up to
3 communication modules,
8 signal modules, and 1 signal
board/communication board;
Digital inputs can be used as HSC
at 100 kHz,
24 V DC digital outputs can be
used as pulse outputs (PTO) or
pulse-width modulated outputs
(PWM) at 100 kHz

6ES7215-1AG40-0XB0

Compact CPU, DC/DC/relay;
Integrated program/data memory
125 KB, load memory 4 MB;
Power supply 24 V DC;
Boolean execution times 0.085 µs
per operation;
14 digital inputs,
10 digital outputs (relays),
2 analog inputs,
2 analog outputs;
Expandable by up to
3 communication modules,
8 signal modules, and 1 signal
board/communication board;
Digital inputs can be used as HSC
at 100 kHz

6ES7215-1HG40-0XB0

SB 1221 signal board

4 inputs, 5 V DC, 200 kHz
4 inputs, 24 V DC, 200 kHz

6ES7221-3AD30-0XB0

6ES7221-3BD30-0XB0

SB 1222 signal board

4 outputs, 5 V DC, 0.1 A, 200 kHz
4 outputs, 24 V DC, 0.1 A, 200 kHz

6ES7222-1AD30-0XB0

6ES7222-1BD30-0XB0

SB 1223 signal board

2 inputs, 24 V DC,
IEC type 1 current sinking;
2 x 24 V DC transistor outputs,
0.5 A, 5 W; can be used as HSC at
up to 30 kHz

6ES7223-0BD30-0XB0

2 inputs, 5 V DC, 200 kHz
2 outputs 5 V DC, 0.1 A, 200 kHz

6ES7223-3AD30-0XB0

6ES7223-3BD30-0XB0

2 inputs, 24 V DC, 200 kHz
2 outputs 24 V DC, 0.1 A, 200 kHz

SB 1231 signal board

1 analog input, ±10 V with 12 bits or
0 ... 20 mA with 11 bits

6ES7231-4HA30-0XB0

SB 1231 thermocouple signal board

1 input +/- 80 mV, resolution 15 bits
+ sign, thermocouples type J, K

6ES7231-5QA30-0XB0

RTD signal board SB 1231

1 input for resistance temperature
sensors Pt 100, Pt 200, Pt 500,
Pt 1000, resolution 15 bits + sign

6ES7231-5PA30-0XB0

SB 1232 signal board

1 analog output, ±10 V with 12 bits
or 0 to 20 mA with 11 bits

6ES7232-4HA30-0XB0

CB 1241 RS 485 communication board

For point-to-point connection,
with 1 RS 485 interface

6ES7241-1CH30-1XB0

BB 1297 battery board

For long-term backup of real-time
clock; can be plugged into the sig-
nal board slot;
battery (CR 1025) is not included

6ES7297-0AX30-0XA0

SIMATIC S7-1200 Basic ControllersCentral processing units
Standard CPUs

CPU 1215C

3

Ordering data	Article No.		Article No.
Digital input simulator SIM 1274 simulator module (optional) 14 input switches, for CPU 1214C/1215C	6ES7274-1XH30-0XA0	Front flap set (spare part) For CPU 1215C	6ES7291-1AC30-0XA0
Analog input simulator SIM 1274 simulator module (optional) 2 potentiometers	6ES7274-1XA30-0XA0	RJ45 cable grip 4 items per pack Dual port	6ES7290-3AB30-0XA0
SIMATIC Memory Card (optional) 4 MB 12 MB 24 MB 256 MB 2 GB 32 GB	6ES7954-8LC02-0AA0 6ES7954-8LE02-0AA0 6ES7954-8LF02-0AA0 6ES7954-8LL02-0AA0 6ES7954-8LP02-0AA0 6ES7954-8LT03-0AA0	STEP 7 Professional / Basic V14 SP1 Target system: SIMATIC S7-1200, S7-1500, S7-300, S7-400, WinAC Requirement: Windows 7 Professional SP1 (64-bit), Windows 7 Enterprise SP1 (64-bit), Windows 7 Ultimate SP1 (64-bit), Windows 8.1 Professional (64-bit), Windows 8.1 Enterprise (64-bit), Windows 10 Professional Version 1607, Windows 10 Enterprise Version 1607, Windows 10 Enterprise 2016 LTSB, Windows 10 Enterprise 2015 LTSB, Windows Server 2008 R2 StdE (full installation), Windows Server 2012 StdE (full installation), Windows Server 2016 Standard (full installation); STEP 7 Basic V14 SP1 in addition: Windows 7 Home Premium SP1 (64-bit), Windows 8.1 (64-bit), Windows 10 Home Version 1607 Type of delivery: German, English, Chinese, Italian, French, Spanish STEP 7 Professional V14 SP1, floating license STEP 7 Professional V14 SP1, floating license, software download incl. license key ¹⁾ Email address required for delivery STEP 7 Basic V14 SP1, floating license STEP 7 Basic V14 SP1, floating license, software download incl. license key ¹⁾ Email address required for delivery	6ES7822-1AA04-0YA5 6ES7822-1AE04-0YA5 6ES7822-0AA04-0YA5 6ES7822-0AE04-0YA5
Extension cable for two-tier configuration For connecting digital/analog signal modules; length 2 m	6ES7290-6AA30-0XA0		
Terminal block (spare part) For CPU 1215C AC/DC/relay • For DI, with 20 screws, tin-coated, coded; 4 units • For DO, with 12 screws, tin-coated, coded; 4 units • For analog units, with 6 screws, gold-plated; 4 units For CPU 1215C DC/DC/DC • For DI, with 20 screws, tin-coated; 4 units • For DO, with 12 screws, tin-coated; 4 units • For analog units, with 6 screws, gold-plated; 4 units For CPU 1215C DC/DC/relay • For DI, with 20 screws, tin-coated; 4 units • For DO, with 12 screws, tin-coated, coded; 4 units • For analog units, with 6 screws, gold-plated; 4 units	6ES7292-1AV40-0XA0 6ES7292-1AM40-0XA0 6ES7292-1BF30-0XB0 6ES7292-1AV30-0XA0 6ES7292-1AM30-0XA0 6ES7292-1BF30-0XB0 6ES7292-1AV30-0XA0 6ES7292-1AM40-0XA0 6ES7292-1BF30-0XB0		

¹⁾ For up-to-date information and download availability, see:
<http://www.siemens.com/tia-online-software-delivery>

SIMATIC S7-1200 Basic Controllers

Central processing units
Standard CPUs

CPU 1217C

Overview



- Powerful controller for extremely fast signal processing
- Expandable by:
 - 1 signal board (SB), battery board (BB) or communication board (CB)
 - 8 signal modules (SM)
 - Max. 3 communication modules (CM)

Technical specifications

Article number	6ES7217-1AG40-0XB0 CPU 1217C, DC/DC/DC, 14DI/10DQ/2AI/2AQ
General information	
Product type designation	CPU 1217C DC/DC/DC
Engineering with	
• Programming package	STEP 7 V14 or higher
Supply voltage	
Rated value (DC)	
• 24 V DC	Yes
Encoder supply	
24 V encoder supply	
• 24 V	L+ minus 4 V DC min.
Power loss	
Power loss, typ.	12 W
Memory	
Work memory	
• integrated	150 kbyte
Load memory	
• integrated	4 Mbyte
• Plug-in (SIMATIC Memory Card), max.	with SIMATIC memory card
Backup	
• without battery	Yes
CPU processing times	
for bit operations, typ.	0.085 µs; / instruction
for word operations, typ.	1.7 µs; / instruction
for floating point arithmetic, typ.	2.3 µs; / Operation
Data areas and their retentivity	
Flag	
• Number, max.	8 kbyte; Size of bit memory address area
Process image	
• Inputs, adjustable	1 kbyte
• Outputs, adjustable	1 kbyte

Article number	6ES7217-1AG40-0XB0 CPU 1217C, DC/DC/DC, 14DI/10DQ/2AI/2AQ
Time of day	
Clock	
• Hardware clock (real-time)	Yes
Digital inputs	
Number of digital inputs	14; Integrated
• of which inputs usable for technological functions	6; HSC (High Speed Counting)
Digital outputs	
Number of digital outputs	10
• of which high-speed outputs	4; 100 kHz Pulse Train Output
Analog inputs	
Number of analog inputs	2
Input ranges	
• Voltage	Yes
Analog outputs	
Number of analog outputs	2
Output ranges, current	
• 0 to 20 mA	Yes
1. Interface	
Interface type	PROFINET
Physics	Ethernet
Functionality	
• PROFINET IO Controller	Yes
• PROFINET IO Device	Yes
• SIMATIC communication	Yes
• Open IE communication	Yes
• Web server	Yes
• Media redundancy	Yes; as MRP client

SIMATIC S7-1200 Basic ControllersCentral processing units
Standard CPUs

CPU 1217C

3

Technical specifications (continued)

Article number	6ES7217-1AG40-0XB0 CPU 1217C, DC/DC/DC, 14DI/10DQ/2AI/2AQ
Communication functions	
S7 communication	
• supported	Yes
Open IE communication	
• TCP/IP	Yes
• ISO-on-TCP (RFC1006)	Yes
• UDP	Yes
Web server	
• supported	Yes
Number of connections	
• overall	16; dynamically
Integrated Functions	
Number of counters	6
Counting frequency (counter) max.	1 MHz
Frequency meter	Yes
controlled positioning	Yes
Number of position-controlled positioning axes, max.	8
Number of positioning axes via pulse-direction interface	4; With integrated outputs
PID controller	Yes
Number of alarm inputs	4
Number of pulse outputs	4
Limit frequency (pulse)	1 MHz
Ambient conditions	
Ambient temperature during operation	
• min.	-20 °C
• max.	60 °C; Number of simultaneously activated inputs or outputs 7 or 5 (no adjacent points) at 60 °C horizontal or 50 °C vertical, 14 or 10 at 55 °C horizontal or 45 °C vertical
Pollutant concentrations	
- SO ₂ at RH < 60% without condensation	SO ₂ : < 0.5 ppm; H ₂ S: < 0.1 ppm; RH < 60% condensation-free
Configuration	
Programming	
Programming language	
- LAD	Yes
- FBD	Yes
- SCL	Yes
Dimensions	
Width	150 mm
Height	100 mm
Depth	75 mm
Weights	
Weight, approx.	530 g

Ordering data**Article No.****CPU 1217C****Compact CPU, DC/DC/DC;**

Integrated program/data memory 150 KB, load memory 4 MB;
Power supply 24 V DC;
Boolean execution times 0.085 µs per operation;
14 digital inputs (10 digital 24 V DC inputs, 4 digital 1.5 V DC differential inputs),
10 digital outputs (6 digital 24 V DC outputs, 4 digital 1.5 V DC differential outputs),
2 analog inputs, 2 analog outputs;
Expandable by up to 3 communication modules, 8 signal modules, and 1 signal board/communication board;
Digital inputs can be used as HSC at 1 MHz,
24 V DC digital outputs can be used as pulse outputs (PTO) or pulse-width modulated outputs (PWM) at 100 kHz

6ES7217-1AG40-0XB0**SB 1221 signal board**

4 inputs, 5 V DC, 200 kHz

6ES7221-3AD30-0XB0

4 inputs, 24 V DC, 200 kHz

6ES7221-3BD30-0XB0**SB 1222 signal board**

4 outputs, 5 V DC, 0.1 A, 200 kHz

6ES7222-1AD30-0XB0

4 outputs, 24 V DC, 0.1 A, 200 kHz

6ES7222-1BD30-0XB0**SB 1223 signal board**

2 inputs, 24 V DC,
IEC type 1 current sinking;
2 x 24 V DC transistor outputs,
0.5 A, 5 W; can be used as HSC at up to 30 kHz

6ES7223-0BD30-0XB0

2 inputs, 5 V DC, 200 kHz
2 outputs 5 V DC, 0.1 A, 200 kHz

6ES7223-3AD30-0XB0

2 inputs, 24 V DC, 200 kHz
2 outputs 24 V DC, 0.1 A, 200 kHz

6ES7223-3BD30-0XB0**SB 1231 signal board**

1 analog input, ±10 V with 12 bits or
0 ... 20 mA with 11 bits

6ES7231-4HA30-0XB0**SB 1231 thermocouple signal board**

1 input +/- 80 mV,
resolution 15 bits + sign,
thermocouples type J, K

6ES7231-5QA30-0XB0**RTD signal board SB 1231**

1 input for resistance temperature sensors Pt 100, Pt 200, Pt 500, Pt 1000, resolution 15 bits + sign

6ES7231-5PA30-0XB0**SB 1232 signal board**

1 analog output, ±10 V with 12 bits
or 0 to 20 mA with 11 bits

6ES7232-4HA30-0XB0

SIMATIC S7-1200 Basic Controllers

Central processing units
Standard CPUs

CPU 1217C

3

Ordering data	Article No.		Article No.
CB 1241 RS 485 communication board For point-to-point connection, with 1 RS 485 interface	6ES7241-1CH30-1XB0	RJ45 cable grip 4 items per pack Dual port	6ES7290-3AB30-0XA0
BB 1297 battery board For long-term backup of real-time clock; can be plugged into the signal board slot; battery (CR 1025) is not included	6ES7297-0AX30-0XA0	STEP 7 Professional / Basic V14 SP1 Target system: SIMATIC S7-1200, S7-1500, S7-300, S7-400, WinAC Requirement: Windows 7 Professional SP1 (64-bit), Windows 7 Enterprise SP1 (64-bit), Windows 7 Ultimate SP1 (64-bit), Windows 8.1 Professional (64-bit), Windows 8.1 Enterprise (64-bit), Windows 10 Professional Version 1607, Windows 10 Enterprise Version 1607, Windows 10 Enterprise 2016 LTSB, Windows 10 Enterprise 2015 LTSB, Windows Server 2008 R2 StdE (full installation), Windows Server 2012 StdE (full installation), Windows Server 2016 Standard (full installation); STEP 7 Basic V14 SP1 in addition: Windows 7 Home Premium SP1 (64-bit), Windows 8.1 (64-bit), Windows 10 Home Version 1607 Type of delivery: German, English, Chinese, Italian, French, Spanish	
Digital input simulator SIM 1274 simulator module (optional) 14 input switches, for CPU 1217C	6ES7274-1XK30-0XA0		
Analog input simulator SIM 1274 simulator module (optional) 2 potentiometers	6ES7274-1XA30-0XA0		
SIMATIC Memory Card (optional) 4 MB 12 MB 24 MB 256 MB 2 GB 32 GB	6ES7954-8LC02-0AA0 6ES7954-8LE02-0AA0 6ES7954-8LF02-0AA0 6ES7954-8LL02-0AA0 6ES7954-8LP02-0AA0 6ES7954-8LT03-0AA0		
Extension cable for two-tier configuration For connecting digital/analog signal modules; length 2 m	6ES7290-6AA30-0XA0		
Terminal block (spare part) For CPU 1217C • For DI, with 10 screws, tin-coated; 4 units • For DI, with 10 screws, tin-coated; 4 units • For DO, with 18 screws, tin-coated; 4 units • For analog units, with 6 screws, gold-plated; 4 units	6ES7292-1AK30-0XA0 6ES7292-1AR30-0XA0 6ES7292-1AT30-0XA0 6ES7292-1BF30-0XB0	STEP 7 Professional V14 SP1, floating license STEP 7 Professional V14 SP1, floating license, software download incl. license key ¹⁾ Email address required for delivery STEP 7 Basic V14 SP1, floating license STEP 7 Basic V14 SP1, floating license, software download incl. license key ¹⁾ Email address required for delivery	6ES7822-1AA04-0YA5 6ES7822-1AE04-0YA5 6ES7822-0AA04-0YA5 6ES7822-0AE04-0YA5
Front flap set (spare part) For CPU 1217C	6ES7291-1AD30-0XA0		

¹⁾ For up-to-date information and download availability, see:
<http://www.siemens.com/tia-online-software-delivery>

SIMATIC S7-1200 Basic Controllers

Central processing units

SIPLUS standard CPUs

SIPLUS CPU 1211C**Overview**

- The clever compact solution
- With 10 integrated I/Os
- Expandable with:
 - 1 signal board (SB) or communication board (CB); not possible with: 6AG1211-1AE31-2XB0, 6AG1211-1BE31-2XB0, 6AG1211-1HE31-2XB0
 - Max. 3 communication modules (CM)

Note:

SIPLUS extreme products are based on SIMATIC standard products. The contents listed here were taken from the respective standard products. SIPLUS extreme-specific information was added.

3**Technical specifications**

Article number	6AG1211-1AE31-4XB0	6AG1211-1AE31-2XB0
Based on	6ES7211-1AE31-0XB0	6ES7211-1AE31-0XB0
	SIPLUS S7-1200 CPU1211 DC/DC/DC	SIPLUS S7-1200 CPU1211 DC/DC/DC
Ambient conditions		
Ambient temperature during operation		
• min. • max.	-20 °C; = Tmin; Startup @ 0 °C 60 °C; = Tmax	-40 °C; = Tmin; Startup @ -25 °C 70 °C; = Tmax; > +60 °C Number of simultaneously controllable inputs and outputs max. 50%; no signal board can be used
Extended ambient conditions		
• relative to ambient temperature-atmospheric pressure-installation altitude • At cold restart, min.	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m) 0 °C	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m) -25 °C
Relative humidity		
- With condensation, tested in accordance with IEC 60068-2-38, max.	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)
Resistance		
- against biologically active substances / conformity with EN 60721-3-3 - against chemically active substances / conformity with EN 60721-3-3 - against mechanically active substances / conformity with EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation! Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation! Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation! Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation! Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!

SIMATIC S7-1200 Basic Controllers

Central processing units
SIPLUS standard CPUs

SIPLUS CPU 1211C

Technical specifications (continued)

Article number	6AG1211-1BE31-4XB0	6AG1211-1BE31-2XB0
Based on	6ES7211-1BE31-0XB0 SIPLUS S7-1200 CPU1211 AC/DC/RLY	6ES7211-1BE31-0XB0 SIPLUS S7-1200 CPU1211 AC/DC/RLY
Ambient conditions		
Ambient temperature during operation		
• min.	-20 °C; = Tmin; Startup @ 0 °C	-40 °C; = Tmin; Startup @ -25 °C
• max.	60 °C; = Tmax	70 °C; = Tmax; > +60 °C Number of simultaneously controllable inputs and outputs max. 50%; no signal board can be used
Extended ambient conditions		
• relative to ambient temperature-atmospheric pressure-installation altitude	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)
• At cold restart, min.	0 °C	-25 °C
Relative humidity		
- With condensation, tested in accordance with IEC 60068-2-38, max.	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)
Resistance		
- against biologically active substances / conformity with EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!
- against chemically active substances / conformity with EN 60721-3-3	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!
Article number	6AG1211-1HE31-4XB0	6AG1211-1HE31-2XB0
Based on	6ES7211-1HE31-0XB0 SIPLUS S7-1200 CPU1211 DC/DC/RLY	6ES7211-1HE31-0XB0 SIPLUS S7-1200 CPU1211 DC/DC/RLY
Ambient conditions		
Ambient temperature during operation		
• min.	-20 °C; = Tmin; Startup @ 0 °C	-40 °C; = Tmin; Startup @ -25 °C
• max.	60 °C; = Tmax	70 °C; = Tmax; > +60 °C Number of simultaneously controllable inputs and outputs max. 50%; no signal board can be used
Extended ambient conditions		
• relative to ambient temperature-atmospheric pressure-installation altitude	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)
• At cold restart, min.	0 °C	-25 °C
Relative humidity		
- With condensation, tested in accordance with IEC 60068-2-38, max.	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)
Resistance		
- against biologically active substances / conformity with EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!
- against chemically active substances / conformity with EN 60721-3-3	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!
- against mechanically active substances / conformity with EN 60721-3-3	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!

Ordering data	Article No.		Article No.
SIPLUS CPU 1211C compact CPU, AC/DC/relay (Extended temperature range and exposure to media) Integrated program and data memory of 25 KB; load memory of 1 MB; wide-range alternating voltage supply 85 ... 264 V AC; Boolean execution times of 0.1 ms per operation; 6 digital inputs, 4 digital outputs (relay), 2 analog inputs; expandable with up to 3 communication modules and 1 signal board/communication board; digital inputs usable as HSC with 100 kHz - For areas with extreme medial exposure (conformal coating); ambient temperature -20 ... +60 °C - For areas with extreme medial exposure (conformal coating); ambient temperature -40 ... +70 °C	6AG1211-1BE31-4XB0 6AG1211-1BE31-2XB0	SIPLUS CPU 1211C compact CPU, DC/DC/relay (Extended temperature range and exposure to media) Integrated program and data memory of 25 KB; load memory of 1 MB; power supply 24 V DC; Boolean execution times of 0.1 ms per operation; 6 digital inputs, 4 digital outputs (relay), 2 analog inputs; expandable with up to 3 communication modules and 1 signal board/communication board; digital inputs usable as HSC with 100 kHz - For areas with extreme medial exposure (conformal coating); ambient temperature -20 ... +60 °C - For areas with extreme medial exposure (conformal coating); ambient temperature -40 ... +70 °C	6AG1211-1HE31-4XB0 6AG1211-1HE31-2XB0
SIPLUS CPU 1211C compact CPU, DC/DC/DC (Extended temperature range and exposure to media) Integrated program and data memory of 25 KB; load memory of 1 MB; power supply 24 V DC; Boolean execution times of 0.1 ms per operation; 6 digital inputs, 4 digital outputs, 2 analog inputs; expandable with up to 3 communication modules and 1 signal board/communication board; digital inputs usable as HSC with 100 kHz, 24 V DC digital outputs usable as pulse outputs (PTO) or pulse-width-modulated outputs (PWM) with 100 kHz - For areas with extreme medial exposure (conformal coating); ambient temperature -20 ... +60 °C - For areas with extreme medial exposure (conformal coating); ambient temperature -40 ... +70 °C	6AG1211-1AE31-4XB0 6AG1211-1AE31-2XB0	Accessories SIPLUS SB 1221 digital input signal board (Extended temperature range and exposure to media; cannot be used with 6AG1211-1.....-2XB0) 4 inputs, 5 V DC, 200 kHz, sourcing 4 inputs, 24 V DC, 200 kHz, sourcing SIPLUS SB 1222 digital output signal board (Extended temperature range and exposure to media; cannot be used with 6AG1211-1.....-2XB0) 4 outputs, 5 V DC, 0.1 A, 200 kHz 4 outputs, 24 V DC, 0.1 A, 200 kHz	6AG1221-3AD30-5XB0 6AG1221-3BD30-5XB0 6AG1222-1AD30-5XB0 6AG1222-1BD30-5XB0

SIMATIC S7-1200 Basic Controllers

Central processing units
SIPLUS standard CPUs

SIPLUS CPU 1211C

3

Ordering data

Article No.

SIPLUS SB 1223 digital input/ output signal board

(Extended temperature range and exposure to media; cannot be used with 6AG1211-1.....-2XB0)

2 inputs, 24 V DC,
IEC type 1 current sinking;
2 transistor outputs 24 V DC,
0.5 A, 5 W;
can be used as HSC at up to
30 kHz

- Suitable for areas with extreme medial exposure (conformal coating)
- Ambient temperature
-25 ... +55 °C

2 inputs, 5 V DC, 200 kHz
2 outputs 5 V DC, 0.1 A, 200 kHz

2 inputs, 24 V DC, 200 kHz
2 outputs 24 V DC, 0.1 A, 200 kHz

6AG1223-0BD30-4XB0

6AG1223-0BD30-5XB0

6AG1223-3AD30-5XB0

6AG1223-3BD30-5XB0

Article No.

SIPLUS SB 1232 analog output signal board

(Extended temperature range and exposure to media; cannot be used with 6AG1211-1.....-2XB0)

Ambient temperature range
-25 ... +55 °C

1 analog output, ± 10 V with 12 bits
or 0 ... 20 mA with 11 bits

Ambient temperature range
0 ... +55 °C

1 analog output, ± 10 V with 12 bits
or 0 ... 20 mA with 11 bits

6AG1232-4HA30-5XB0

6AG1232-4HA30-4XB0

SIPLUS CB 1241 RS 485 communication board

(Extended temperature range and exposure to media; cannot be used with 6AG1211-1.....-2XB0)

For point-to-point connection,
with 1 RS 485 interface

6AG1241-1CH30-5XB1

Additional accessories

See SIMATIC S7-1200
CPU 1211C, page 3/6

SIMATIC S7-1200 Basic Controllers

Central processing units

SIPLUS standard CPUs

SIPLUS CPU 1212C**Overview**

- The superior compact solution
- With 14 integral input/outputs
- Expandable by:
 - 1 signal board (SB) or communication board (CB); not possible with: 6AG1212-1AE31-2XB0, 6AG1212-1BE31-2XB0, 6AG1212-1HE31-2XB0
 - 2 signal modules (SM)
 - Max. 3 communication modules (CM)

Note:

SIPLUS extreme products are based on SIMATIC standard products. The contents listed here were taken from the respective standard products. SIPLUS extreme-specific information was added.

3**Technical specifications**

Article number	6AG1212-1AE40-4XB0	6AG1212-1AE40-2XB0
Based on	6ES7212-1AE40-0XB0 SIPLUS S7-1200 CPU 1212C DC/DC/DC	6ES7212-1AE40-0XB0 SIPLUS S7-1200 CPU 1212C DC/DC/DC
Ambient conditions		
Ambient temperature during operation		
• min. • max.	-20 °C; = Tmin; Startup @ 0 °C 60 °C; Number of simultaneously activated inputs or outputs 4 or 3 (no adjacent points) at 60 °C horizontal or 50 °C vertical, 8 or 6 at 55 °C horizontal or 45 °C vertical	-40 °C; = Tmin; Startup @ -25 °C 70 °C; = Tmax; Tmax > +55 °C number of simultaneously switched-on digital inputs 4, digital outputs 3, analog inputs 2 (no adjacent points) with horizontal mounting position; Tmax > +60 °C number of simultaneously switched-on digital inputs 3, digital outputs 2, analog inputs 0 (no adjacent points) with horizontal mounting position
Extended ambient conditions		
• relative to ambient temperature-atmospheric pressure-installation altitude	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)
Relative humidity		
- With condensation, tested in accordance with IEC 60068-2-38, max.	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)
Resistance		
- against biologically active substances / conformity with EN 60721-3-3 - against chemically active substances / conformity with EN 60721-3-3 - against mechanically active substances / conformity with EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation! Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation! Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation! Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation! Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!

SIMATIC S7-1200 Basic Controllers

Central processing units
SIPLUS standard CPUs

SIPLUS CPU 1212C

Technical specifications (continued)

Article number	6AG1212-1BE40-4XB0	6AG1212-1BE40-2XB0
Based on	6ES7212-1BE40-0XB0 SIPLUS S7-1200 CPU 1212C AC/DC/RLY	6ES7212-1BE40-0XB0 SIPLUS S7-1200 CPU 1212C AC/DC/RLY
Ambient conditions		
Ambient temperature during operation		
• min.	-20 °C; = Tmin; Startup @ 0 °C	-40 °C; = Tmin; Startup @ -25 °C
• max.	60 °C; Number of simultaneously activated inputs or outputs 4 or 3 (no adjacent points) at 60 °C horizontal or 50 °C vertical, 8 or 6 at 55 °C horizontal or 45 °C vertical	70 °C; = Tmax; Tmax > +55 °C number of simultaneously switched-on digital inputs 4, digital outputs 3, analog inputs 2 (no adjacent points) with horizontal mounting position; Tmax > +60 °C number of simultaneously switched-on digital inputs 3, digital outputs 2, analog inputs 0 (no adjacent points) with horizontal mounting position
Extended ambient conditions		
• relative to ambient temperature-atmospheric pressure-installation altitude	Tmin ... Tmax at 1 080 hPa ... 795 hPa (-1 000 m ... +2 000 m) // Tmin ... (Tmax - 10 K) at 795 hPa ... 658 hPa (+2 000 m ... +3 500 m) // Tmin ... (Tmax - 20 K) at 658 hPa ... 540 hPa (+3 500 m ... +5 000 m); from 2 000 m max. 132 V AC	Tmin ... Tmax at 1 080 hPa ... 795 hPa (-1 000 m ... +2 000 m) // Tmin ... (Tmax - 10 K) at 795 hPa ... 658 hPa (+2 000 m ... +3 500 m) // Tmin ... (Tmax - 20 K) at 658 hPa ... 540 hPa (+3 500 m ... +5 000 m); from 2 000 m max. 132 V AC
Relative humidity		
- With condensation, tested in accordance with IEC 60068-2-38, max.	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)
Resistance		
- against biologically active substances / conformity with EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!
- against chemically active substances / conformity with EN 60721-3-3	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!
- against mechanically active substances / conformity with EN 60721-3-3	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!
Article number	6AG1212-1HE40-4XB0	6AG1212-1HE40-2XB0
Based on	6ES7212-1HE40-0XB0 SIPLUS S7-1200 CPU 1212C DC/DC/RLY	6ES7212-1HE40-0XB0 SIPLUS S7-1200 CPU 1212C DC/DC/RLY
Ambient conditions		
Ambient temperature during operation		
• min.	-20 °C; = Tmin; Startup @ 0 °C	-40 °C; = Tmin; Startup @ -25 °C
• max.	60 °C; Number of simultaneously activated inputs or outputs 4 or 3 (no adjacent points) at 60 °C horizontal or 50 °C vertical, 8 or 6 at 55 °C horizontal or 45 °C vertical	70 °C; = Tmax; Tmax > +55 °C number of simultaneously switched-on digital inputs 4, digital outputs 3, analog inputs 2 (no adjacent points) with horizontal mounting position; Tmax > +60 °C number of simultaneously switched-on digital inputs 3, digital outputs 2, analog inputs 0 (no adjacent points) with horizontal mounting position
Extended ambient conditions		
• relative to ambient temperature-atmospheric pressure-installation altitude	Tmin ... Tmax at 1 080 hPa ... 795 hPa (-1 000 m ... +2 000 m) // Tmin ... (Tmax - 10 K) at 795 hPa ... 658 hPa (+2 000 m ... +3 500 m) // Tmin ... (Tmax - 20 K) at 658 hPa ... 540 hPa (+3 500 m ... +5 000 m); from 2 000 m max. 132 V AC	Tmin ... Tmax at 1 080 hPa ... 795 hPa (-1 000 m ... +2 000 m) // Tmin ... (Tmax - 10 K) at 795 hPa ... 658 hPa (+2 000 m ... +3 500 m) // Tmin ... (Tmax - 20 K) at 658 hPa ... 540 hPa (+3 500 m ... +5 000 m); from 2 000 m max. 132 V AC
Relative humidity		
- With condensation, tested in accordance with IEC 60068-2-38, max.	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)
Resistance		
- against biologically active substances / conformity with EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!
- against chemically active substances / conformity with EN 60721-3-3	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!
- against mechanically active substances / conformity with EN 60721-3-3	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!

Ordering data	Article No.		Article No.	
SIPLUS CPU 1212C compact CPU, AC/DC/relay (Extended temperature range and exposure to media) Integrated program/data memory 75 KB, load memory 1 MB; Wide-range power supply 85 ... 264 V AC; Boolean execution times 0.1 µs per operation; 8 digital inputs, 6 digital outputs (relays), 2 analog inputs; Expandable by up to 3 communication modules, 2 signal modules and 1 signal board/communication board; Digital inputs can be used as HSC at 100 kHz • For areas with extreme medial exposure (conformal coating); ambient temperature -20 ... +60 °C • For areas with extreme medial exposure (conformal coating); ambient temperature -40 ... +70 °C	6AG1212-1BE40-4XB0 6AG1212-1BE40-2XB0		SIPLUS CPU 1212C compact CPU, DC/DC/relay (Extended temperature range and exposure to media) Integrated program/data memory 75 KB, load memory 1 MB; Power supply 24 V DC; Boolean execution times 0.1 µs per operation; 8 digital inputs, 6 digital outputs (relays), 2 analog inputs; Expandable by up to 3 communication modules, 2 signal modules, and 1 signal board/communication board; Digital inputs can be used as HSC at 100 kHz • For areas with extreme medial exposure (conformal coating); ambient temperature -20 ... +60 °C • For areas with extreme medial exposure (conformal coating); ambient temperature -40 ... +70 °C	6AG1212-1HE40-4XB0 6AG1212-1HE40-2XB0
SIPLUS CPU 1212C compact CPU, DC/DC/DC (Extended temperature range and exposure to media) Integrated program/data memory 75 KB, load memory 1 MB; Power supply 24 V DC; Boolean execution times 0.1 µs per operation; 8 digital inputs, 6 digital outputs, 2 analog inputs; Expandable by up to 3 communication modules, 2 signal modules, and 1 signal board/communication board; Digital inputs can be used as HSC at 100 kHz, 24 V DC digital outputs can be used as pulse outputs (PTO) or pulse-width modulated outputs (PWM) at 100 kHz • For areas with extreme medial exposure (conformal coating); ambient temperature -20 ... +60 °C • For areas with extreme medial exposure (conformal coating); ambient temperature -40 ... +70 °C	6AG1212-1AE40-4XB0 6AG1212-1AE40-2XB0	Accessories SIPLUS SB 1221 digital input signal board (Extended temperature range and exposure to media; cannot be used with 6AG1212-1.....-2XB0) 4 inputs, 5 V DC, 200 kHz, sourcing 4 inputs, 24 V DC, 200 kHz, sourcing	6AG1221-3AD30-5XB0 6AG1221-3BD30-5XB0	
		SIPLUS SB 1222 digital output signal board (Extended temperature range and exposure to media; cannot be used with 6AG1212-1.....-2XB0) 4 outputs, 5 V DC, 0.1 A, 200 kHz 4 outputs, 24 V DC, 0.1 A, 200 kHz	6AG1222-1AD30-5XB0 6AG1222-1BD30-5XB0	

SIMATIC S7-1200 Basic Controllers

Central processing units
SIPLUS standard CPUs

SIPLUS CPU 1212C

Ordering data

Article No.

Digital input/output SIPLUS signal board SB 1223

(Extended temperature range and exposure to media; cannot be used with 6AG1212-1.....-2XB0)

2 inputs, 24 V DC,
IEC type 1 current sinking;
2 transistor outputs 24 V DC,
0.5 A, 5 W; can be used as HSC at
up to 30 kHz

- Suitable for areas with extreme medial exposure (conformal coating)
- Ambient temperature -25 ... +55 °C

2 inputs, 5 V DC, 200 kHz
2 outputs 5 V DC, 0.1 A, 200 kHz

2 inputs, 24 V DC, 200 kHz
2 outputs 24 V DC, 0.1 A, 200 kHz

6AG1223-0BD30-4XB0

6AG1223-0BD30-5XB0

6AG1223-3AD30-5XB0

6AG1223-3BD30-5XB0

Article No.

SIPLUS SB 1232 analog output signal board

(Extended temperature range and exposure to media; cannot be used with 6AG1212-1.....-2XB0)

Ambient temperature range
-25 ... +55 °C

1 analog output, ± 10 V with 12 bits
or 0 ... 20 mA with 11 bits

Ambient temperature range
0 ... +55 °C

1 analog output, ± 10 V with 12 bits
or 0 ... 20 mA with 11 bits

6AG1232-4HA30-5XB0

6AG1232-4HA30-4XB0

SIPLUS CB 1241 RS 485 communication board

(Extended temperature range and exposure to media; cannot be used with 6AG1212-1.....-2XB0)

For point-to-point connection,
with 1 RS 485 interface

6AG1241-1CH30-5XB1

Additional accessories

See SIMATIC S7-1200
CPU 1212C, page 3/10

SIMATIC S7-1200 Basic Controllers

Central processing units

SIPLUS standard CPUs

SIPLUS CPU 1214C

Overview

- The compact high-performance CPU
- With 24 integrated I/Os
- Expandable with:
 - 1 signal board (SB) or communication board (CB); not possible with: 6AG1214-1AG40-2XB0, 6AG1214-1BG40-2XB0, 6AG1214-1HG40-2XB0
 - 8 signal modules (SM)
 - Max. 3 communication modules (CM)

Note:

SIPLUS extreme products are based on SIMATIC standard products. The contents listed here were taken from the respective standard products. SIPLUS extreme-specific information was added.

3

Technical specifications

Article number	6AG1214-1AG40-4XB0	6AG1214-1AG40-5XB0	6AG1214-1AG40-2XB0
Based on	6ES7214-1AG40-0XB0 SIPLUS S7-1200 CPU 1214C DC/DC/DC	6ES7214-1AG40-0XB0 SIPLUS S7-1200 CPU 1214C DC/DC/DC	6ES7214-1AG40-0XB0 SIPLUS S7-1200 CPU 1214C DC/DC/DC
Ambient conditions			
Ambient temperature during operation			
• min.	-20 °C; = Tmin; Startup @ 0 °C	-40 °C; = Tmin; Startup @ -25 °C	-40 °C; = Tmin; Startup @ -25 °C
• max.	60 °C; Number of simultaneously activated inputs or outputs 7 or 5 (no adjacent points) at 60 °C horizontal or 50 °C vertical, 14 or 10 at 55 °C horizontal or 45 °C vertical	60 °C; = Tmax; Tmax > +55 °C number of simultaneously switched-on digital inputs 7, digital outputs 5, analog inputs 2 (no adjacent points) with horizontal mounting position	70 °C; = Tmax; Tmax > +55 °C number of simultaneously switched-on digital inputs 7, digital outputs 5, analog inputs 2 (no adjacent points) with horizontal mounting position; Tmax > +60 °C number of simultaneously switched-on digital inputs 7, digital outputs 5, analog inputs 1 (no adjacent points) with horizontal mounting position
Extended ambient conditions			
• relative to ambient temperature-atmospheric pressure-installation altitude	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)
Relative humidity			
- With condensation, tested in accordance with IEC 60068-2-38, max.	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)
Resistance			
- against biologically active substances / conformity with EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!
- against chemically active substances / conformity with EN 60721-3-3	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!
- against mechanically active substances / conformity with EN 60721-3-3	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!

SIMATIC S7-1200 Basic Controllers

Central processing units
SIPLUS standard CPUs

SIPLUS CPU 1214C

Technical specifications (continued)

Article number	6AG1214-1BG40-4XB0	6AG1214-1BG40-5XB0	6AG1214-1BG40-2XB0
Based on	6ES7214-1BG40-0XB0 SIPLUS S7-1200 CPU 1214C AC/DC/RLY	6ES7214-1BG40-0XB0 SIPLUS S7-1200 CPU 1214C AC/DC/RLY	6ES7214-1BG40-0XB0 SIPLUS S7-1200 CPU 1214C AC/DC/RLY
Ambient conditions			
Ambient temperature during operation			
• min.	-20 °C; = Tmin; Startup @ 0 °C	-40 °C; = Tmin; Startup @ -25 °C	-40 °C; = Tmin; Startup @ -25 °C
• max.	60 °C; = Tmax	60 °C; = Tmax; Tmax > +55 °C number of simultaneously switched-on digital inputs 7, digital outputs 5, analog inputs 2 (no adjacent points) with horizontal mounting position	70 °C; = Tmax; Tmax > +55 °C number of simultaneously switched-on digital inputs 7, digital outputs 5, analog inputs 2 (no adjacent points) with horizontal mounting position; Tmax > +60 °C number of simultaneously switched-on digital inputs 7, digital outputs 5, analog inputs 1 (no adjacent points) with horizontal mounting position
Extended ambient conditions			
• relative to ambient temperature-atmospheric pressure-installation altitude	Tmin ... Tmax at 1 080 hPa ... 795 hPa (-1 000 m ... +2 000 m) // Tmin ... (Tmax - 10 K) at 795 hPa ... 658 hPa (+2 000 m ... +3 500 m) // Tmin ... (Tmax - 20 K) at 658 hPa ... 540 hPa (+3 500 m ... +5 000 m); from 2 000 m max. 132 V AC	Tmin ... Tmax at 1 080 hPa ... 795 hPa (-1 000 m ... +2 000 m) // Tmin ... (Tmax - 10 K) at 795 hPa ... 658 hPa (+2 000 m ... +3 500 m) // Tmin ... (Tmax - 20 K) at 658 hPa ... 540 hPa (+3 500 m ... +5 000 m); from 2 000 m max. 132 V AC	Tmin ... Tmax at 1 080 hPa ... 795 hPa (-1 000 m ... +2 000 m) // Tmin ... (Tmax - 10 K) at 795 hPa ... 658 hPa (+2 000 m ... +3 500 m) // Tmin ... (Tmax - 20 K) at 658 hPa ... 540 hPa (+3 500 m ... +5 000 m); from 2 000 m max. 132 V AC
Relative humidity			
- With condensation, tested in accordance with IEC 60068-2-38, max.	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)
Resistance			
- against biologically active substances / conformity with EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!
- against chemically active substances / conformity with EN 60721-3-3	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!
- against mechanically active substances / conformity with EN 60721-3-3	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!

SIMATIC S7-1200 Basic Controllers

Central processing units

SIPLUS standard CPUs

SIPLUS CPU 1214C**Technical specifications (continued)**

Article number	6AG1214-1HG40-4XB0	6AG1214-1HG40-5XB0	6AG1214-1HG40-2XB0
Based on	6ES7214-1HG40-0XB0 SIPLUS S7-1200 CPU 1214C DC/DC/RLY	6ES7214-1HG40-0XB0 SIPLUS S7-1200 CPU 1214C DC/DC/RLY	6ES7214-1HG40-0XB0 SIPLUS S7-1200 CPU 1214C DC/DC/RLY
Ambient conditions			
Ambient temperature during operation			
• min. • max.	-20 °C; = Tmin; Startup @ 0 °C 60 °C; Number of simultaneously activated inputs or outputs 7 or 5 (no adjacent points) at 60 °C horizontal or 50 °C vertical, 14 or 10 at 55 °C horizontal or 45 °C vertical	-40 °C; = Tmin; Startup @ -25 °C 60 °C; = Tmax; Tmax > +55 °C number of simultaneously switched-on digital inputs 7, digital outputs 5, analog inputs 2 (no adjacent points) with horizontal mounting position	-40 °C; = Tmin; Startup @ -25 °C 70 °C; = Tmax; Tmax > +55 °C number of simultaneously switched-on digital inputs 7, digital outputs 5, analog inputs 2 (no adjacent points) with horizontal mounting position; Tmax > +60 °C number of simultaneously switched-on digital inputs 7, digital outputs 5, analog inputs 1 (no adjacent points) with horizontal mounting position
Extended ambient conditions			
• relative to ambient temperature-atmospheric pressure-installation altitude	Tmin ... Tmax at 1 080 hPa ... 795 hPa (-1 000 m ... +2 000 m) // Tmin ... (Tmax - 10 K) at 795 hPa ... 658 hPa (+2 000 m ... +3 500 m) // Tmin ... (Tmax - 20 K) at 658 hPa ... 540 hPa (+3 500 m ... +5 000 m); from 2 000 m max. 132 V AC	Tmin ... Tmax at 1 080 hPa ... 795 hPa (-1 000 m ... +2 000 m) // Tmin ... (Tmax - 10 K) at 795 hPa ... 658 hPa (+2 000 m ... +3 500 m) // Tmin ... (Tmax - 20 K) at 658 hPa ... 540 hPa (+3 500 m ... +5 000 m); from 2 000 m max. 132 V AC	Tmin ... Tmax at 1 080 hPa ... 795 hPa (-1 000 m ... +2 000 m) // Tmin ... (Tmax - 10 K) at 795 hPa ... 658 hPa (+2 000 m ... +3 500 m) // Tmin ... (Tmax - 20 K) at 658 hPa ... 540 hPa (+3 500 m ... +5 000 m); from 2 000 m max. 132 V AC
Relative humidity			
- With condensation, tested in accordance with IEC 60068-2-38, max.	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)
Resistance			
- against biologically active substances / conformity with EN 60721-3-3 - against chemically active substances / conformity with EN 60721-3-3 - against mechanically active substances / conformity with EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation! Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation! Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation! Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation! Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation! Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation! Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!

SIMATIC S7-1200 Basic Controllers

Central processing units
SIPLUS standard CPUs

SIPLUS CPU 1214C

Ordering data

Article No.

Article No.

SIPLUS CPU 1214C compact CPU, AC/DC/relay

(Extended temperature range and exposure to media)

Integrated program/data memory
100 KB, load memory 2 MB;
Wide-range power supply
85 ... 264 V AC;
Boolean execution times 0.1 µs
per operation;
14 digital inputs, 10 digital outputs
(relays), 2 analog inputs;
Expandable by up to
3 communication modules,
8 signal modules and 1 signal
board/communication board;
Digital inputs can be used as HSC
at 100 kHz

- For areas with extreme medial exposure (conformal coating); ambient temperature -20 ... +60 °C
- For areas with extreme medial exposure (conformal coating); ambient temperature -40 ... +60 °C
- For areas with extreme medial exposure (conformal coating); ambient temperature -40 ... +70 °C

6AG1214-1BG40-4XB0

6AG1214-1BG40-5XB0

6AG1214-1BG40-2XB0

SIPLUS CPU 1214C compact CPU, DC/DC/DC

(Extended temperature range and exposure to media)

Integrated program/data memory
100 KB, load memory 2 MB;
Power supply 24 V DC;
Boolean execution times 0.1 µs
per operation;
14 digital inputs,
10 digital outputs,
2 analog inputs;
expandable by up to
3 communication modules,
8 signal modules, and 1 signal
board/communication board;
Digital inputs can be used as HSC
at 100 kHz,
24 V DC digital outputs can be
used as pulse outputs (PTO) or
pulse-width modulated outputs
(PWM) at 100 kHz

- For areas with extreme medial exposure (conformal coating); ambient temperature -20 ... +60 °C
- For areas with extreme medial exposure (conformal coating); ambient temperature -40 ... +60 °C
- For areas with extreme medial exposure (conformal coating); ambient temperature -40 ... +70 °C

6AG1214-1AG40-4XB0

6AG1214-1AG40-5XB0

6AG1214-1AG40-2XB0

SIPLUS CPU 1214C compact CPU, DC/DC/relay

(Extended temperature range and exposure to media)

Integrated program/data memory
100 KB, load memory 2 MB;
Power supply 24 V DC;
Boolean execution times 0.1 µs
per operation;
14 digital inputs, 10 digital outputs
(relays), 2 analog inputs;
Expandable by up to
3 communication modules, 8
signal modules, and 1 signal
board/communication board;
Digital inputs can be used as HSC
at 100 kHz

- For areas with extreme medial exposure (conformal coating); ambient temperature -20 ... +60 °C
- For areas with extreme medial exposure (conformal coating); ambient temperature -40 ... +60 °C
- For areas with extreme medial exposure (conformal coating); ambient temperature -40 ... +70 °C

6AG1214-1HG40-4XB0

6AG1214-1HG40-5XB0

6AG1214-1HG40-2XB0

Accessories

SIPLUS SB 1221 digital input signal board

(Extended temperature range and exposure to media; cannot be used with 6AG1214-1.....-2XB0)

4 inputs, 5 V DC, 200 kHz, sourcing

6AG1221-3AD30-5XB0

4 inputs, 24 V DC, 200 kHz,
sourcing

6AG1221-3BD30-5XB0

SIPLUS SB 1222 digital output signal board

(Extended temperature range and exposure to media; cannot be used with 6AG1214-1.....-2XB0)

4 outputs, 5 V DC, 0.1 A, 200 kHz

6AG1222-1AD30-5XB0

4 outputs, 24 V DC, 0.1 A, 200 kHz

6AG1222-1BD30-5XB0

SIMATIC S7-1200 Basic Controllers

Central processing units

SIPLUS standard CPUs

SIPLUS CPU 1214C**3**

Ordering data	Article No.		Article No.	
SIPLUS SB 1223 digital input/output signal board (Extended temperature range and exposure to media; cannot be used with 6AG1214-1.....-2XB0) 2 inputs, 24 V DC, IEC type 1 current sinking; 2 transistor outputs 24 V DC, 0.5 A, 5 W; can be used as HSC at up to 30 kHz • Suitable for areas with extreme medial exposure (conformal coating) • Ambient temperature -25 ... +55 °C 2 inputs, 5 V DC, 200 kHz 2 outputs 5 V DC, 0.1 A, 200 kHz 2 inputs, 24 V DC, 200 kHz 2 outputs 24 V DC, 0.1 A, 200 kHz	6AG1223-0BD30-4XB0 6AG1223-0BD30-5XB0 6AG1223-3AD30-5XB0 6AG1223-3BD30-5XB0		SIPLUS SB 1232 analog output signal board (Extended temperature range and exposure to media; cannot be used with 6AG1214-1.....-2XB0) <u>Ambient temperature range</u> <u>-25 ... +55 °C</u> 1 analog output, ±10 V with 12 bits or 0 ... 20 mA with 11 bits <u>Ambient temperature range</u> <u>0 ... +55 °C</u> 1 analog output, ±10 V with 12 bits or 0 ... 20 mA with 11 bits SIPLUS CB 1241 RS 485 communication board (Extended temperature range and exposure to media; cannot be used with 6AG1214-1.....-2XB0) For point-to-point connection, with 1 RS 485 interface Additional accessories	6AG1232-4HA30-5XB0 6AG1232-4HA30-4XB0 6AG1241-1CH30-5XB1 See SIMATIC S7-1200 CPU 1214C, page 3/14

SIMATIC S7-1200 Basic Controllers

Central processing units
SIPLUS standard CPUs

SIPLUS CPU 1215C

Overview



- The compact high-performance CPU
- With 24 integrated I/Os
- Expandable with:
 - 1 signal board (SB) or communication board (CB); not possible with: 6AG1215-1AG40-2XB0, 6AG1215-1BG40-2XB0, 6AG1215-1HG40-2XB0
 - 8 signal modules (SM)
 - Max. 3 communication modules (CM)

Note:

SIPLUS extreme products are based on SIMATIC standard products. The contents listed here were taken from the respective standard products. SIPLUS extreme-specific information was added.

Technical specifications

Article number	6AG1215-1AG40-4XB0	6AG1215-1AG40-5XB0	6AG1215-1AG40-2XB0
Based on	6ES7215-1AG40-0XB0	6ES7215-1AG40-0XB0	6ES7215-1AG40-0XB0
	SIPLUS S7-1200 CPU 1215C DC/DC/DC	SIPLUS S7-1200 CPU 1215C DC/DC/DC	SIPLUS S7-1200 CPU 1215C DC/DC/DC
Ambient conditions			
Ambient temperature during operation			
• min.	-20 °C; = Tmin; Startup @ 0 °C	-40 °C; = Tmin; Startup @ -25 °C	-40 °C; = Tmin; Startup @ -25 °C
• max.	60 °C; Number of simultaneously activated inputs or outputs 7 or 5 (no adjacent points) at 60 °C horizontal or 50 °C vertical, 14 or 10 at 55 °C horizontal or 45 °C vertical	60 °C; = Tmax; Tmax > +55 °C number of simultaneously switched-on digital inputs 7, digital outputs 5, analog inputs 2, analog outputs 2 (no adjacent points) with horizontal mounting position	70 °C; = Tmax; Tmax > +55 °C number of simultaneously switched-on digital inputs 7, digital outputs 5, analog inputs 2, analog outputs 2 (no adjacent points) with horizontal mounting position; Tmax > +60 °C number of simultaneously switched-on digital inputs 7, digital outputs 5, analog inputs 1, analog outputs 1 (no adjacent points) with horizontal mounting position
Extended ambient conditions			
• relative to ambient temperature-atmospheric pressure-installation altitude	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)
Relative humidity			
- With condensation, tested in accordance with IEC 60068-2-38, max.	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)
Resistance			
- against biologically active substances / conformity with EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!
- against chemically active substances / conformity with EN 60721-3-3	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!
- against mechanically active substances / conformity with EN 60721-3-3	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!

SIMATIC S7-1200 Basic Controllers

Central processing units

SIPLUS standard CPUs

SIPLUS CPU 1215C**Technical specifications (continued)**

Article number	6AG1215-1BG40-4XB0	6AG1215-1BG40-5XB0	6AG1215-1BG40-2XB0
Based on	6ES7215-1BG40-0XB0 SIPLUS S7-1200 CPU 1215C AC/DC/RLY	6ES7215-1BG40-0XB0 SIPLUS S7-1200 CPU 1215C AC/DC/RLY	6ES7215-1BG40-0XB0 SIPLUS S7-1200 CPU 1215C AC/DC/RLY
Ambient conditions			
Ambient temperature during operation			
• min. • max.	-20 °C; = Tmin; Startup @ 0 °C 60 °C; Number of simultaneously activated inputs or outputs 7 or 5 (no adjacent points) at 60 °C horizontal or 50 °C vertical, 14 or 10 at 55 °C horizontal or 45 °C vertical	-40 °C; = Tmin; Startup @ -25 °C 60 °C; = Tmax; Tmax > +55 °C number of simultaneously switched-on digital inputs 7, digital outputs 5, analog inputs 2, analog outputs 2 (no adjacent points) with horizontal mounting position	-40 °C; = Tmin; Startup @ -25 °C 70 °C; = Tmax; Tmax > +55 °C number of simultaneously switched-on digital inputs 7, digital outputs 5, analog inputs 2, analog outputs 2 (no adjacent points) with horizontal mounting position; Tmax > +60 °C number of simultaneously switched-on digital inputs 7, digital outputs 5, analog inputs 1, analog outputs 1 (no adjacent points) with horizontal mounting position
Extended ambient conditions			
• relative to ambient temperature-atmospheric pressure-installation altitude	Tmin ... Tmax at 1 080 hPa ... 795 hPa (-1 000 m ... +2 000 m) // Tmin ... (Tmax - 10 K) at 795 hPa ... 658 hPa (+2 000 m ... +3 500 m) // Tmin ... (Tmax - 20 K) at 658 hPa ... 540 hPa (+3 500 m ... +5 000 m); from 2 000 m max. 132 V AC	Tmin ... Tmax at 1 080 hPa ... 795 hPa (-1 000 m ... +2 000 m) // Tmin ... (Tmax - 10 K) at 795 hPa ... 658 hPa (+2 000 m ... +3 500 m) // Tmin ... (Tmax - 20 K) at 658 hPa ... 540 hPa (+3 500 m ... +5 000 m); from 2 000 m max. 132 V AC	Tmin ... Tmax at 1 080 hPa ... 795 hPa (-1 000 m ... +2 000 m) // Tmin ... (Tmax - 10 K) at 795 hPa ... 658 hPa (+2 000 m ... +3 500 m) // Tmin ... (Tmax - 20 K) at 658 hPa ... 540 hPa (+3 500 m ... +5 000 m); from 2 000 m max. 132 V AC
Relative humidity			
- With condensation, tested in accordance with IEC 60068-2-38, max.	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)
Resistance			
- against biologically active substances / conformity with EN 60721-3-3 - against chemically active substances / conformity with EN 60721-3-3 - against mechanically active substances / conformity with EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation! Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation! Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation! Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation! Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation! Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation! Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!

SIMATIC S7-1200 Basic Controllers

Central processing units
SIPLUS standard CPUs

SIPLUS CPU 1215C

Technical specifications (continued)

Article number	6AG1215-1HG40-4XB0	6AG1215-1HG40-5XB0	6AG1215-1HG40-2XB0
Based on	6ES7215-1HG40-0XB0 SIPLUS S7-1200 CPU 1215C DC/DC/RLY	6ES7215-1HG40-0XB0 SIPLUS S7-1200 CPU 1215C DC/DC/RLY	6ES7215-1HG40-0XB0 SIPLUS S7-1200 CPU 1215C DC/DC/RLY
Ambient conditions			
Ambient temperature during operation			
- min. - max.	-20 °C; = Tmin; Startup @ 0 °C 60 °C; Number of simultaneously activated inputs or outputs 7 or 5 (no adjacent points) at 60 °C horizontal or 50 °C vertical, 14 or 10 at 55 °C horizontal or 45 °C vertical	-40 °C; = Tmin; Startup @ -25 °C 60 °C; = Tmax; Tmax > +55 °C number of simultaneously switched-on digital inputs 7, digital outputs 5, analog inputs 2, analog outputs 2 (no adjacent points) with horizontal mounting position	-40 °C; = Tmin; Startup @ -25 °C 70 °C; = Tmax; Tmax > +55 °C number of simultaneously switched-on digital inputs 7, digital outputs 5, analog inputs 2, analog outputs 2 (no adjacent points) with horizontal mounting position; Tmax > +60 °C number of simultaneously switched-on digital inputs 7, digital outputs 5, analog inputs 1, analog outputs 1 (no adjacent points) with horizontal mounting position
Extended ambient conditions			
relative to ambient temperature-atmospheric pressure-installation altitude	Tmin ... Tmax at 1 080 hPa ... 795 hPa (-1 000 m ... +2 000 m) // Tmin ... (Tmax - 10 K) at 795 hPa ... 658 hPa (+2 000 m ... +3 500 m) // Tmin ... (Tmax - 20 K) at 658 hPa ... 540 hPa (+3 500 m ... +5 000 m); from 2 000 m max. 132 V AC	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1 080 hPa ... 795 hPa (-1 000 m ... +2 000 m) // Tmin ... (Tmax - 10 K) at 795 hPa ... 658 hPa (+2 000 m ... +3 500 m) // Tmin ... (Tmax - 20 K) at 658 hPa ... 540 hPa (+3 500 m ... +5 000 m); from 2 000 m max. 132 V AC
Relative humidity			
With condensation, tested in accordance with IEC 60068-2-38, max.	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)
Resistance			
- against biologically active substances / conformity with EN 60721-3-3 - against chemically active substances / conformity with EN 60721-3-3 - against mechanically active substances / conformity with EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation! Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation! Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation! Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation! Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation! Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation! Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!

Ordering data	Article No.		Article No.	
SIPLUS CPU 1215C compact CPU, AC/DC/relay (Extended temperature range and exposure to media) Integrated program and data memory 125 KB; load memory 4 MB; wide-range power supply 85 ... 264 V AC; Boolean execution times 0.085 µs per operation; 14 digital inputs, 10 digital outputs (relay), 2 analog inputs, 2 analog outputs; expandable by up to 3 communication modules, 8 signal modules and 1 signal board/communication board; digital inputs usable as HSC with 100 kHz • For areas with extreme medial exposure (conformal coating); ambient temperature -20 ... +60 °C • For areas with extreme medial exposure (conformal coating); ambient temperature -40 ... +60 °C • For areas with extreme medial exposure (conformal coating); ambient temperature -40 ... +70 °C	6AG1215-1BG40-4XB0 6AG1215-1BG40-5XB0 6AG1215-1BG40-2XB0		SIPLUS CPU 1215C compact CPU, DC/DC/relay (Extended temperature range and exposure to media) Integrated program and data memory 125 KB; load memory 4 MB; power supply 24 V DC; Boolean execution times 0.085 µs per operation; 14 digital inputs, 10 digital outputs (relay), 2 analog inputs, 2 analog outputs; expandable by up to 3 communication modules, 8 signal modules and 1 signal board/communication board; digital inputs usable as HSC with 100 kHz • For areas with extreme medial exposure (conformal coating); ambient temperature -20 ... +60 °C • For areas with extreme medial exposure (conformal coating); ambient temperature -40 ... +60 °C • For areas with extreme medial exposure (conformal coating); ambient temperature -40 ... +70 °C	6AG1215-1HG40-4XB0 6AG1215-1HG40-5XB0 6AG1215-1HG40-2XB0
SIPLUS CPU 1215C compact CPU, DC/DC/DC (Extended temperature range and exposure to media) Integrated program and data memory 125 KB; load memory 4 MB; power supply 24 V DC; Boolean execution times 0.085 µs per operation; 14 digital inputs, 10 digital outputs, 2 analog inputs, 2 analog outputs; expandable by up to 3 communication modules, 8 signal modules and 1 signal board/communication board; digital inputs usable as HSC with 100 kHz; 24 V DC digital outputs usable as pulse outputs (PTO) or pulse-width-modulated outputs (PWM) with 100 kHz • For areas with extreme medial exposure (conformal coating); ambient temperature -20 ... +60 °C • For areas with extreme medial exposure (conformal coating); ambient temperature -40 ... +60 °C • For areas with extreme medial exposure (conformal coating); ambient temperature -40 ... +70 °C	6AG1215-1AG40-4XB0 6AG1215-1AG40-5XB0 6AG1215-1AG40-2XB0	Accessories SIPLUS SB 1221 digital input signal board (Extended temperature range and exposure to media; cannot be used with 6AG1215-1.....2XB0) 4 inputs, 5 V DC, 200 kHz, sourcing 4 inputs, 24 V DC, 200 kHz, sourcing SIPLUS SB 1222 digital output signal board (Extended temperature range and exposure to media; cannot be used with 6AG1215-1.....2XB0) 4 outputs, 5 V DC, 0.1 A, 200 kHz 4 outputs, 24 V DC, 0.1 A, 200 kHz	6AG1221-3AD30-5XB0 6AG1221-3BD30-5XB0 6AG1222-1AD30-5XB0 6AG1222-1BD30-5XB0	

SIMATIC S7-1200 Basic Controllers

Central processing units
SIPLUS standard CPUs

SIPLUS CPU 1215C

Ordering data

Article No.

Digital input/output SIPLUS signal board SB 1223

(Extended temperature range and exposure to media; cannot be used with 6AG1215-1.....-2XB0)

2 inputs, 24 V DC,
IEC type 1 current sinking;
2 transistor outputs 24 V DC,
0.5 A, 5 W;
can be used as HSC at up to
30 kHz

- Suitable for areas with extreme medial exposure (conformal coating)
- Ambient temperature -25 ... +55 °C

2 inputs, 5 V DC, 200 kHz
2 outputs 5 V DC, 0.1 A, 200 kHz

2 inputs, 24 V DC, 200 kHz
2 outputs 24 V DC, 0.1 A, 200 kHz

6AG1223-0BD30-4XB0

6AG1223-0BD30-5XB0

6AG1223-3AD30-5XB0

6AG1223-3BD30-5XB0

Article No.

SIPLUS SB 1232 analog output signal board

(Extended temperature range and exposure to media; cannot be used with 6AG1215-1.....-2XB0)

Ambient temperature range
-25 ... +55 °C

1 analog output, ± 10 V with 12 bits
or 0 ... 20 mA with 11 bits

Ambient temperature range
0 ... +55 °C

1 analog output, ± 10 V with 12 bits
or 0 ... 20 mA with 11 bits

6AG1232-4HA30-5XB0

6AG1232-4HA30-4XB0

SIPLUS CB 1241 RS 485 communication board

(Extended temperature range and exposure to media; cannot be used with 6AG1215-1.....-2XB0)

For point-to-point connection,
with 1 RS 485 interface

6AG1241-1CH30-5XB1

Additional accessories

See SIMATIC S7-1200
CPU 1215C, page 3/18

SIMATIC S7-1200 Basic Controllers

Central processing units

Fail-safe CPUs

CPU 1212 FC, CPU 1214 FC, CPU 1215 FC

Overview

The fail-safe SIMATIC S7-1200 Controllers are based on the S7-1200 standard CPUs and offer additional safety-related functions.

They can be used for safety-oriented tasks according to IEC 61508 up to SIL 3 and ISO 13849-1 up to PL e.

Safety-related programs are created in the TIA Portal. The STEP 7 Safety engineering tool offers commands, operations and blocks for safety-related programs in the LAD and FBD languages. To this end, there is a library with pre-configured, TÜV-approved blocks for safety-related functions.

- Standard controller with integrated safety functions:
 - Standardized and convenient diagnostic functions for standard and safety
 - Uniform symbols, data consistency, ...
- Modular system with scalable range of CPUs and expandable I/O quantity structure:
 - One engineering for standard and fail-safe automation
 - Use of the standard I/O modules together with the fail-safe I/O modules in the central system
 - Integrated standard PROFINET functionalities for PROFINET controllers and PROFINET iDevice services
 - Connection of distributed standard I/O via field bus such as PROFINET or PROFIBUS
 - F-library certified by the German Technical Inspectorate (TÜV) for all common safety functions
 - Free programming of the safety logic using FBD and LAD
 - Standard-compliant printout of the F-program
- One integrated engineering for both standard and safety from S7-1200 to S7-300/400/1500 and WinAC RTX F:
 - STEP 7 Safety Basic for easy engineering of the CPU 1200 FC
 - STEP 7 Safety Advanced for the entire fail-safe SIMATIC S7 portfolio
- Integrated system diagnosis of the CPUs, for standard and safety:
 - Consistent plain text display of system diagnostic information in the TIA Portal, HMI and web server
 - Messages are updated even if the CPU is in STOP state
 - System diagnostics integrated in the CPU firmware. Configuration by user not required
 - The diagnostics is automatically updated on configuration changes.
- 2 fail-safe compact controllers with graded performances in the versions DC/DC/DC and DC/DC/relay

Characteristics	CPU 1212 FC	CPU 1214 FC	CPU 1215 FC
Variants	DC/DC/DC, DC/DC/relay	DC/DC/DC, DC/DC/relay	DC/DC/DC, DC/DC/relay
Main memory, integrated	100 KB	125 KB	150 KB
Load memory, integrated	2 MB	4 MB	4 MB
Memory card	SIMATIC Memory Card (optional)	SIMATIC Memory Card (optional)	SIMATIC Memory Card (optional)
Standard digital inputs/outputs, integrated	8/6	14/10	14/10
Standard analog inputs, integrated	2	2	2
Standard analog outputs, integrated	-	-	2
Process image	1024 bytes for inputs, 1024 bytes for outputs	1024 bytes for inputs, 1024 bytes for outputs	1024 bytes for inputs, 1024 bytes for outputs
Expansion by signal board	Max. 1	Max. 1	Max. 1
Expansion by signal modules	Max. 2	Max. 8	Max. 8
Expansion by communication modules	Max. 3	Max. 3	Max. 3

SIMATIC S7-1200 Basic Controllers

Central processing units

Fail-safe CPUs

CPU 1212 FC, CPU 1214 FC, CPU 1215 FC

Technical specifications

Article number	6ES7212-1AF40-0XB0	6ES7212-1HF40-0XB0	6ES7214-1AF40-0XB0	6ES7214-1HF40-0XB0	6ES7215-1AF40-0XB0	6ES7215-1HF40-0XB0
	CPU 1212FC , DC/DC/DC, 8DI/6DO/2AI	CPU 1212FC, DC/DC/RELAY, 8DI/6DO/2AI	CPU 1214 FC, DC/DC/DC, 14DI/10DO/2AI	CPU 1214 FC, DC/DC/RELAY, 14DI/10DO/2AI	CPU 1215 FC, DC/DC/DC, 14DI/ 10DO/2AI/2AO	CPU 1215 FC, DC/DC/RLY, 14DI/ 10DO/2AI/2AO
General information						
Product type designation	CPU 1212FC DC/DC/DC	CPU 1212FC DC/DC/relay	CPU 1214FC DC/DC	CPU 1214FC DC/DC/Relay	CPU 1215FC DC/DC/DC	CPU 1215C DC/DC/Relay
Engineering with						
• Programming package	STEP 7 V14 or higher	STEP 7 V14 or higher	STEP 7 V14 or higher	STEP 7 V14 or higher	STEP 7 V14 or higher	STEP 7 V14 or higher
Supply voltage						
Rated value (DC)						
• 24 V DC	Yes	Yes	Yes	Yes	Yes	Yes
Encoder supply						
24 V encoder supply						
• 24 V	Permissible range: 20.4V to 28.8V	Permissible range: 20.4V to 28.8V	L+ minus 4 V DC min.	L+ minus 4 V DC min.	L+ minus 4 V DC min.	L+ minus 4 V DC min.
Power loss						
Power loss, typ.	9 W	9 W	12 W	12 W	12 W	12 W
Memory						
Work memory						
• integrated	100 kbyte	100 kbyte	125 kbyte	125 kbyte	150 kbyte	150 kbyte
Load memory						
• integrated	2 Mbyte	2 Mbyte	4 Mbyte	4 Mbyte	4 Mbyte	4 Mbyte
• Plug-in (SIMATIC Memory Card), max.	with SIMATIC memory card	with SIMATIC memory card	with SIMATIC memory card	with SIMATIC memory card	with SIMATIC memory card	with SIMATIC memory card
Backup						
• without battery	Yes	Yes	Yes	Yes	Yes	Yes
CPU processing times						
for bit operations, typ.	0.085 µs; / instruction	0.085 µs; / instruction	0.085 µs; / instruction	0.085 µs; / instruction	0.085 µs; / instruction	0.085 µs; / instruction
for word operations, typ.	1.7 µs; / instruction	1.7 µs; / instruction	1.7 µs; / instruction	1.7 µs; / instruction	1.7 µs; / instruction	1.7 µs; / instruction
for floating point arithmetic, typ.	2.5 µs; / instruction	2.5 µs; / instruction	2.3 µs; / instruction	2.3 µs; / instruction	2.3 µs; / instruction	2.3 µs; / instruction
Data areas and their retentivity						
Flag						
• Number, max.	4 kbyte; Size of bit memory address area	4 kbyte; Size of bit memory address area	8 kbyte; Size of bit memory address area	8 kbyte; Size of bit memory address area	8 kbyte; Size of bit memory address area	8 kbyte; Size of bit memory address area
Address area						
I/O address area						
• Inputs	1 024 byte	1 024 byte	1 024 byte			
• Outputs	1 024 byte	1 024 byte	1 024 byte			
Process image						
• Inputs, adjustable	1 kbyte	1 kbyte	1 kbyte	1 kbyte	1 kbyte	1 kbyte
• Outputs, adjustable	1 kbyte	1 kbyte	1 kbyte	1 kbyte	1 kbyte	1 kbyte
Time of day						
Clock						
• Hardware clock (real-time)	Yes	Yes	Yes		Yes	Yes
Digital inputs						
Number of digital inputs	8; Integrated	8; Integrated	14		14; Integrated	14; Integrated
• of which inputs usable for technological functions	4; HSC (High Speed Counting)	4; HSC (High Speed Counting)	6; HSC (High Speed Counting)		6; HSC (High Speed Counting)	6; HSC (High Speed Counting)
Digital outputs						
Number of digital outputs	6	6	10		10; Relays	10; Relays
• of which high-speed outputs	4; 100 kHz Pulse Train Output		4; 100 kHz Pulse Train Output			
Analog inputs						
Number of analog inputs	2	2	2	2	2	2
Input ranges						
• Voltage	Yes	Yes	Yes	Yes	Yes	Yes
Analog outputs						
Number of analog outputs		0	0	0	2	2
Output ranges, current						
• 0 to 20 mA				Yes	Yes	Yes

SIMATIC S7-1200 Basic Controllers

Central processing units

Fail-safe CPUs

CPU 1212 FC, CPU 1214 FC, CPU 1215 FC

Technical specifications (continued)

Article number	6ES7212-1AF40-0XB0 CPU 1212FC , DC/DC/DC, 8DI/6DO/2AI	6ES7212-1HF40-0XB0 CPU 1212FC, DC/DC/RELAY, 8DI/6DO/2AI	6ES7214-1AF40-0XB0 CPU 1214 FC, DC/DC/DC, 14DI/10DO/2AI	6ES7214-1HF40-0XB0 CPU 1214 FC, DC/DC/RELAY, 14DI/10DO/2AI	6ES7215-1AF40-0XB0 CPU 1215 FC, DC/DC/DC, 14DI/ 10DO/2AI/2AO	6ES7215-1HF40-0XB0 CPU 1215 FC, DC/DC/RLY, 14DI/ 10DO/2AI/2AO
1. Interface						
Interface type	PROFINET	PROFINET	PROFINET	PROFINET	PROFINET	PROFINET
Physics	Ethernet	Ethernet	Ethernet	Ethernet	Ethernet	Ethernet
Functionality						
• PROFINET IO Controller	Yes	Yes	Yes	Yes	Yes	Yes
• PROFINET IO Device	Yes	Yes	Yes	Yes	Yes	Yes
• SIMATIC communication	Yes	Yes	Yes	Yes	Yes	Yes
• Open IE communication	Yes	Yes	Yes	Yes	Yes	Yes
• Web server	Yes	Yes	Yes	Yes	Yes	Yes
• Media redundancy			Yes; as MRP client	Yes; as MRP client	Yes; as MRP client	Yes; as MRP client
Communication functions						
S7 communication						
• supported	Yes	Yes	Yes	Yes	Yes	Yes
Open IE communication						
• TCP/IP	Yes	Yes	Yes	Yes	Yes	Yes
• ISO-on-TCP (RFC1006)	Yes	Yes	Yes	Yes	Yes	Yes
• UDP	Yes	Yes	Yes	Yes	Yes	Yes
Web server						
• supported	Yes	Yes	Yes	Yes	Yes	Yes
Number of connections						
• overall			16; dynamically	16; dynamically	16; dynamically	16; dynamically
Integrated Functions						
Number of counters	4	4	6	6	6	6
Counting frequency (counter) max.	100 kHz	100 kHz	100 kHz	100 kHz	100 kHz	100 kHz
Frequency meter	Yes	Yes	Yes	Yes	Yes	Yes
controlled positioning	Yes	Yes	Yes	Yes	Yes	Yes
Number of position-controlled positioning axes, max.	8	8	8	8	8	8
Number of positioning axes via pulse-direction interface	Up to 4 with SB 1222	Up to 4 with SB 1222	Up to 4 with SB 1222	Up to 4 with SB 1222	Up to 4 with SB 1222	Up to 4 with SB 1222
PID controller	Yes	Yes	Yes	Yes	Yes	Yes
Number of alarm inputs	4		4	4	4	4
Number of pulse outputs	4	4				
Limit frequency (pulse)	100 kHz					
Ambient conditions						
Ambient temperature during operation						
• min.	-20 °C	-20 °C	-20 °C	-20 °C	-20 °C	-20 °C
• max.	60 °C; Number of simultaneously activated inputs or outputs 7 or 5 (no adjacent points) at 60 °C horizontal or 50 °C vertical, 14 or 10 at 55 °C horizontal or 45 °C vertical	60 °C; Number of simultaneously activated inputs or outputs 7 or 5 (no adjacent points) at 60 °C horizontal or 50 °C vertical, 14 or 10 at 55 °C horizontal or 45 °C vertical	60 °C; Number of simultaneously activated inputs or outputs 7 or 5 (no adjacent points) at 60 °C horizontal or 50 °C vertical, 14 or 10 at 55 °C horizontal or 45 °C vertical	60 °C; Number of simultaneously activated inputs or outputs 7 or 5 (no adjacent points) at 60 °C horizontal or 50 °C vertical, 14 or 10 at 55 °C horizontal or 45 °C vertical	60 °C; Number of simultaneously activated inputs or outputs 7 or 5 (no adjacent points) at 60 °C horizontal or 50 °C vertical, 14 or 10 at 55 °C horizontal or 45 °C vertical	60 °C; Number of simultaneously activated inputs or outputs 7 or 5 (no adjacent points) at 60 °C horizontal or 50 °C vertical, 14 or 10 at 55 °C horizontal or 45 °C vertical
Pollutant concentrations						
- SO ₂ at RH < 60% without condensation	SO ₂ : < 0.5 ppm; H ₂ S: < 0.1 ppm; RH < 60% condensation-free	SO ₂ : < 0.5 ppm; H ₂ S: < 0.1 ppm; RH < 60% condensation-free	SO ₂ : < 0.5 ppm; H ₂ S: < 0.1 ppm; RH < 60% condensation-free	SO ₂ : < 0.5 ppm; H ₂ S: < 0.1 ppm; RH < 60% condensation-free	SO ₂ : < 0.5 ppm; H ₂ S: < 0.1 ppm; RH < 60% condensation-free	SO ₂ : < 0.5 ppm; H ₂ S: < 0.1 ppm; RH < 60% condensation-free

SIMATIC S7-1200 Basic Controllers

Central processing units

Fail-safe CPUs

CPU 1212 FC, CPU 1214 FC, CPU 1215 FC

Technical specifications (continued)

Article number	6ES7212-1AF40-0XB0	6ES7212-1HF40-0XB0	6ES7214-1AF40-0XB0	6ES7214-1HF40-0XB0	6ES7215-1AF40-0XB0	6ES7215-1HF40-0XB0
	CPU 1212FC , DC/DC/DC, 8DI/6DO/2AI	CPU 1212FC, DC/DC/RELAY, 8DI/6DO/2AI	CPU 1214 FC, DC/DC/DC, 14DI/10DO/2AI	CPU 1214 FC, DC/DC/RELAY, 14DI/10DO/2AI	CPU 1215 FC, DC/DC/DC, 14DI/ 10DO/2AI/2AO	CPU 1215 FC, DC/DC/RLY, 14DI/ 10DO/2AI/2AO
Configuration						
Programming						
Programming language						
- LAD	Yes; incl. failsafe	Yes; incl. failsafe	Yes; incl. failsafe	Yes; incl. failsafe	Yes; incl. failsafe	Yes; incl. failsafe
- FBD	Yes; incl. failsafe	Yes; incl. failsafe	Yes; incl. failsafe	Yes; incl. failsafe	Yes; incl. failsafe	Yes; incl. failsafe
- SCL	Yes	Yes	Yes	Yes	Yes	Yes
Dimensions						
Width	90 mm	90 mm	110 mm	110 mm	130 mm	130 mm
Height	100 mm	100 mm	100 mm	100 mm	100 mm	100 mm
Depth	75 mm	75 mm	75 mm	75 mm	75 mm	75 mm
Weights						
Weight, approx.	370 g	385 g	435 g	435 g	585 g	585 g

Ordering data

Article No.

Article No.

CPU 1212 FC

Fail-safe compact CPU, DC/DC/DC;
integrated program/data memory 100 KB, load memory 2 MB;
power supply 24 V DC;
Boolean execution times 0.085 µs per operation;
8 digital inputs,
6 digital outputs,
2 analog inputs;
expandable by up to
3 communication modules,
2 signal modules, and 1 signal board/communication board;
digital inputs can be used as HSC at 100 kHz,
24 V DC digital outputs can be used as pulse outputs (PTO) or pulse-width modulated outputs (PWM) at 100 kHz

6ES7212-1AF40-0XB0

Fail-safe compact CPU, DC/DC/relay;
integrated program/data memory 125 KB, load memory 2 MB;
power supply 24 V DC;
Boolean execution times 0.085 µs per operation;
8 digital inputs, 6 digital outputs (relays),
2 analog inputs;
expandable by up to
3 communication modules, 2 signal modules, and 1 signal board/communication board;
digital inputs can be used as HSC at 100 kHz

6ES7212-1HF40-0XB0

CPU 1214 FC

Fail-safe compact CPU, DC/DC/DC;
integrated program/data memory 125 KB, load memory 4 MB;
power supply 24 V DC;
Boolean execution times 0.085 µs per operation;
14 digital inputs,
10 digital outputs,
2 analog inputs;
expandable by up to
3 communication modules,
8 signal modules, and 1 signal board/communication board;
digital inputs can be used as HSC at 100 kHz,
24 V DC digital outputs can be used as pulse outputs (PTO) or pulse-width modulated outputs (PWM) at 100 kHz

6ES7214-1AF40-0XB0

Fail-safe compact CPU, DC/DC/relay;
integrated program/data memory 125 KB, load memory 4 MB;
power supply 24 V DC;
Boolean execution times 0.085 µs per operation;
14 digital inputs,
10 digital outputs (relays),
2 analog inputs;
expandable by up to
3 communication modules,
8 signal modules, and 1 signal board/communication board;
digital inputs can be used as HSC at 100 kHz

6ES7214-1HF40-0XB0

SIMATIC S7-1200 Basic Controllers

Central processing units

Fail-safe CPUs

CPU 1212 FC, CPU 1214 FC, CPU 1215 FC

3

Ordering data	Article No.		Article No.
CPU 1215 FC		Terminal block (spare part) (cont.)	
Fail-safe compact CPU, DC/DC/DC; integrated program/data memory 150 KB, load memory 4 MB; power supply 24 V DC; Boolean execution times 0.085 µs per operation; 14 digital inputs, 10 digital outputs, 2 analog inputs; 2 analog outputs; expandable by up to 3 communication modules, 8 signal modules, and 1 signal board/communication board; digital inputs can be used as HSC at 100 kHz, 24 V DC digital outputs can be used as pulse outputs (PTO) or pulse-width modulated outputs (PWM) at 100 kHz	6ES7215-1AF40-0XB0	For CPU 1214FC, DC/DC/relay • For DI, with 20 screws, tin-coated; 4 units • For DO, with 12 screws, tin-coated, coded; 4 units • For AI, with 3 screws, gold-plated; 4 units	6ES7292-1AV30-0XA0 6ES7292-1AM40-0XA0 6ES7292-1BC30-0XA0
Fail-safe compact CPU, DC/DC/relay; integrated program/data memory 150 KB, load memory 4 MB; power supply 24 V DC; Boolean execution times 0.085 µs per operation; 14 digital inputs, 10 digital outputs (relays), 2 analog inputs; 2 analog outputs; expandable by up to 3 communication modules, 8 signal modules, and 1 signal board/communication board; digital inputs can be used as HSC at 100 kHz	6ES7215-1HF40-0XB0	For CPU 1215FC, DC/DC/DC • For DI, with 20 screws, tin-coated; 4 units • For DO, with 12 screws, tin-coated; 4 units • For AI, with 6 screws, gold-plated; 4 units	6ES7292-1AV30-0XA0 6ES7292-1AM30-0XA0 6ES7292-1BF30-0XA0
Accessories		For CPU 1215FC, DC/DC/relay • For DI, with 20 screws, tin-coated; 4 units • For DO, with 12 screws, tin-coated, coded; 4 units • For AI, with 6 screws, gold-plated; 4 units	6ES7292-1AV30-0XA0 6ES7292-1AM40-0XA0 6ES7292-1BF30-0XA0
SIMATIC S7-1200 Fail-Safe Starter Kit Consisting of: CPU 1212FC DC/DC relay, F digital input SM 1226 16 x 24 V DC, F digital output SM 1226 4 x 24 V DC, input simulator, STEP 7 Basic and STEP 7 Safety Basic on CD, manual on CD, info material; in Systainer	6ES7212-1HF41-4YB0	Front flap set (spare part) For CPU 1214 FC For CPU 1215 FC	6ES7291-1AB30-0XA0 6ES7291-1AC30-0XA0
Simulator (optional) 14 incoming circuit breakers	6ES7274-1XH30-0XA0	RJ45 cable grip 4 items per pack Single port Dual port	6ES7290-3AA30-0XA0 6ES7290-3AB30-0XA0
SIMATIC Memory Card (optional) 4 MB 12 MB 24 MB 256 MB 2 GB 32 GB	6ES7954-8LC02-0AA0 6ES7954-8LE02-0AA0 6ES7954-8LF02-0AA0 6ES7954-8LL02-0AA0 6ES7954-8LP02-0AA0 6ES7954-8LT03-0AA0	STEP 7 Safety Advanced V14 SP1 Task: Engineering tool for configuring and programming fail-safe user programs for SIMATIC S7-1200 FC, S7-1500F, S7-1500F Software Controller, S7-300F, S7-400F, WinAC RTX F, ET 200SP F Controller and the fail-safe ET 200SP, ET 200S, ET 200M, ET 200iSP, ET 200pro and ET 200eco I/O Requirement: STEP 7 Professional V14 SP1 Floating license for 1 user, software and documentation on DVD, license key on USB flash drive Floating license for 1 user, software, documentation and license key for download ¹⁾ ; email address required for delivery	6ES7833-1FA14-0YA5 6ES7833-1FA14-0YH5
Extension cable for two-tier configuration For connecting digital/analog signal modules; length 2 m	6ES7290-6AA30-0XA0	STEP 7 Safety Basic V14 SP1 Task: Engineering tool for configuring fail-safe user programs for SIMATIC S7-1200 FC Requirement: STEP 7 Basic V14 SP1 and higher Floating license for 1 user; software and documentation on DVD, license key on USB flash drive Floating license for 1 user, software, documentation and license key for download ¹⁾ ; email address required for delivery	6ES7833-1FB14-0YA5 6ES7833-1FB14-0YH5
Terminal block (spare part) For CPU 1214FC, DC/DC/DC • For DI, with 20 screws, tin-coated; 4 units • For DO, with 12 screws, tin-coated; 4 units • For AI, with 3 screws, gold-plated; 4 units	6ES7292-1AV30-0XA0 6ES7292-1AM30-0XA0 6ES7292-1BC30-0XA0		

¹⁾ For up-to-date information and download availability, see:
<http://www.siemens.com/tia-online-software-delivery>

SIMATIC S7-1200 Basic Controllers

Central processing units

SIPLUS fail-safe CPUs

Overview



The fail-safe SIPLUS S7-1200 Controllers are based on the S7-1200 standard CPUs and offer additional safety-related functions.

They can be used for safety-oriented tasks according to IEC 61508 up to SIL 3 and ISO 13849-1 up to PL e.

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- Standard controller with integrated safety functions:
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 - Uniform symbols, data consistency, ...
- Modular system with scalable range of CPUs and expandable I/O quantity structure:
 - One engineering for standard and fail-safe automation
 - Use of the standard I/O modules together with the fail-safe I/O modules in the central system
 - Integrated standard PROFINET functionalities for PROFINET controllers and PROFINET iDevice services
 - Connection of distributed standard I/O via field bus such as PROFINET or PROFIBUS
 - F-library certified by the German Technical Inspectorate (TÜV) for all common safety functions
 - Free programming of the safety logic using FBD and LAD
 - Standard-compliant printout of the F-program
- One integrated engineering for both standard and safety from S7-1200 to S7-300/400/1500 and WinAC RTX F:
 - STEP 7 Safety Basic for easy engineering of the CPU 1200 FC
 - STEP 7 Safety Advanced for the entire fail-safe SIMATIC S7 portfolio
- Integrated system diagnosis of the CPUs, for standard and safety:
 - Consistent plain text display of system diagnostic information in the TIA Portal, HMI and web server
 - Messages are updated even if the CPU is in STOP state
 - System diagnostics integrated in the CPU firmware. Configuration by user not required
 - The diagnostics is automatically updated on configuration changes.
- 2 fail-safe compact controllers with graded performances in the versions DC/DC/DC and DC/DC/relay

Characteristics	SIPLUS CPU 1214 FC
Variants	DC/DC/DC, DC/DC/relay
Main memory, integrated	125 KB
Load memory, integrated	4 MB
Memory card	SIMATIC memory card (optional)
Standard digital inputs/outputs, integrated	14/10
Standard analog inputs, integrated	2
Standard analog outputs, integrated	-
Process image	1024 bytes for inputs, 1024 bytes for outputs
Expansion by signal board	Max. 1
Expansion by signal modules	Max. 8
Expansion by communication modules	Max. 3

Note:

SIPLUS extreme products are based on SIMATIC standard products. The contents listed here were taken from the corresponding standard products. SIPLUS extreme-specific information was added.

SIMATIC S7-1200 Basic Controllers

Central processing units

SIPLUS fail-safe CPUs

Technical specifications

Article number	6AG1214-1AF40-5XB0	6AG1214-1HF40-5XB0
Based on	6ES7214-1AF40-0XB0 SIPLUS S7-1200 CPU 1214FC DC/DC/DC	6ES7214-1HF40-0XB0 SIPLUS S7-1200 CPU 1214FC DC/DC/RLY
Ambient conditions		
Ambient temperature during operation		
• min.	-25 °C; = Tmin	-25 °C; = Tmin
• max.	55 °C; = Tmax	55 °C; = Tmax
Extended ambient conditions		
• relative to ambient temperature-atmospheric pressure-installation altitude	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m)
Relative humidity		
- With condensation, tested in accordance with IEC 60068-2-38, max.	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)
Pollutant concentrations		
- SO ₂ at RH < 60% without condensation	SO ₂ : < 0.5 ppm; H ₂ S: < 0.1 ppm; RH < 60% condensation-free	SO ₂ : < 0.5 ppm; H ₂ S: < 0.1 ppm; RH < 60% condensation-free
Resistance		
- against biologically active substances / conformity with EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!
- against chemically active substances / conformity with EN 60721-3-3	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!
- against mechanically active substances / conformity with EN 60721-3-3	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!

Ordering data

	Article No.	Article No.
CPU 1214 FC (Extended temperature range and environmental stress)	6AG1214-1AF40-5XB0	6AG1214-1HF40-5XB0
Fail-safe compact CPU, DC/DC/DC; integrated program/data memory 125 KB, load memory 4 MB; power supply 24 V DC; Boolean execution times 0.085 µs per operation; 14 digital inputs, 10 digital outputs, 2 analog inputs; expandable by up to 3 communication modules, 8 signal modules, and 1 signal board/communication board; digital inputs can be used as HSC at 100 kHz, 24 V DC digital outputs can be used as pulse outputs (PTO) or pulse-width modulated outputs (PWM) at 100 kHz		
Fail-safe compact CPU, DC/DC/relay; integrated program/data memory 125 KB, load memory 4 MB; power supply 24 V DC; Boolean execution times 0.085 µs per operation; 14 digital inputs, 10 digital outputs (relays), 2 analog inputs; expandable by up to 3 communication modules, 8 signal modules, and 1 signal board/communication board; digital inputs can be used as HSC at 100 kHz		
Accessories		See SIMATIC CPU 1214 FC, page 3/45

SIMATIC S7-1200 Basic Controllers

I/O modules

Digital modules

SM 1221 digital input modules

Overview



- Digital inputs as supplement to the integral I/O of the CPUs
- For flexible adaptation of the controller to the relevant task
- For subsequent expansion of the system with additional inputs

Technical specifications

Article number	6ES7221-1BF32-0XB0 DIGITAL INPUT SM 1221, 8DI, 24V DC	6ES7221-1BH32-0XB0 DIGITAL INPUT SM 1221, 16DI, 24V DC
General information		
Product type designation	SM 1221 DI 8x24 VDC	SM 1221 DI 16x24 VDC
Supply voltage		
Rated value (DC)		
• 24 V DC	Yes	Yes
Input current		
from backplane bus 5 V DC, max.	105 mA	130 mA
Digital inputs		
• from load voltage L+ (without load), max.	4 mA; per channel	4 mA; per channel
Output voltage		
Power supply to the transmitters		
• present	Yes	Yes
Power loss		
Power loss, typ.	1.5 W	2.5 W
Digital inputs		
Number of digital inputs	8	16
• in groups of	2	4
Input characteristic curve in accordance with IEC 61131, type 1	Yes	Yes
Number of simultaneously controllable inputs		
all mounting positions		
- up to 40 °C, max.	8	16
horizontal installation		
- up to 40 °C, max.	8	16
- up to 50 °C, max.	8	16
vertical installation		
- up to 40 °C, max.	8	16

Technical specifications (continued)

Article number	6ES7221-1BF32-0XB0 DIGITAL INPUT SM 1221, 8DI, 24V DC	6ES7221-1BH32-0XB0 DIGITAL INPUT SM 1221, 16DI, 24V DC
Input voltage		
• Type of input voltage	DC	DC
• Rated value (DC)	24 V	24 V
• for signal "0"	5 V DC at 1 mA	5 V DC at 1 mA
• for signal "1"	15 V DC at 2.5 mA	15 V DC at 2.5 mA
Input current		
• for signal "0", max. (permissible quiescent current)	1 mA	1 mA
• for signal "1", min.	2.5 mA	2.5 mA
• for signal "1", typ.	4 mA	4 mA
Input delay (for rated value of input voltage)		
for standard inputs		
- parameterizable	Yes; 0.2 ms, 0.4 ms, 0.8 ms, 1.6 ms, 3.2 ms, 6.4 ms and 12.8 ms, selectable in groups of four	Yes; 0.2 ms, 0.4 ms, 0.8 ms, 1.6 ms, 3.2 ms, 6.4 ms and 12.8 ms, selectable in groups of four
for interrupt inputs		
- parameterizable	Yes	Yes
Cable length		
• shielded, max.	500 m	500 m
• unshielded, max.	300 m	300 m
Interrupts/diagnostics/ status information		
Alarms		
• Diagnostic alarm	Yes	Yes
Diagnostics indication LED		
• for status of the inputs	Yes	Yes
Potential separation		
Potential separation digital inputs		
• between the channels, in groups of	2	4
Degree and class of protection		
Degree of protection acc. to EN 60529		
• IP20	Yes	Yes
Standards, approvals, certificates		
CE mark	Yes	Yes
CSA approval	Yes	Yes
cULus	Yes	Yes
FM approval	Yes	Yes
RCM (formerly C-TICK)	Yes	Yes
Marine approval		
• Marine approval		Yes
Ambient conditions		
Free fall		
• Fall height, max.	0.3 m; five times, in product package	0.3 m; five times, in product package
Ambient temperature during operation		
• permissible temperature range	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting, 95% humidity, non-condensing	
• min.	-20 °C	-20 °C
• max.	60 °C	60 °C
Connection method		
required front connector	Yes	Yes
Mechanics/material		
Enclosure material (front)		
• Plastic	Yes	Yes
Dimensions		
Width	45 mm	45 mm
Height	100 mm	100 mm
Depth	75 mm	75 mm
Weights		
Weight, approx.	170 g	210 g

SIMATIC S7-1200 Basic Controllers

I/O modules

Digital modules

SM 1221 digital input modules**Ordering data****Article No.****SM 1221 digital input
signal module**8 inputs, 24 V DC, isolated,
current sourcing/sinking**6ES7221-1BF32-0XB0**16 inputs, 24 V DC, isolated,
current sourcing/sinking**6ES7221-1BH32-0XB0****Extension cable for
two-tier configuration**For connecting digital/analog
signal modules;
length 2 m**6ES7290-6AA30-0XA0****Article No.****Terminal block (spare part)**For 6ES7221-1BF32-0XB0,
6ES7221-1BH32-0XB0

With 7 screws, zinc-plated; 4 pcs.

6ES7292-1AG30-0XA0**Front flap set (spare part)**

For modules with a width of 45 mm

6ES7291-1BA30-0XA0

Overview

- Digital inputs as a supplement to the integral I/O of SIMATIC S7-1200 CPUs
- Can be plugged directly into the CPU

3**Technical specifications**

Article number	6ES7221-3AD30-0XB0	6ES7221-3BD30-0XB0
	SIGNAL BOARD SB 1221, 4 DI 5VDC 200KHZ	SIGNAL BOARD SB 1221, 4 DI 24VDC 200KHZ
General information		
Product type designation	SB 1221 4xDI 5VDC 200kHz	SB 1221 4xDI 24VDC 200kHz
Input current		
from backplane bus 5 V DC, typ.	40 mA	40 mA
Power loss		
Power loss, typ.	1 W	1 W
Digital inputs		
Number of digital inputs	4; Current-sourcing	4; Current-sourcing
• in groups of	4	4
Input voltage		
• Type of input voltage	DC	DC
• Rated value (DC)	5 V	24 V
• for signal "0"	(L+ minus 1.0 V DC) ... L+ (2.2 ... 0 mA)	(L+ minus 5.0 V DC) ... L+ (1.4 ... 0 mA)
• for signal "1"	0 V ... (L+ minus 2.0 V DC (20 ... 5.1 mA))	0 V ... (L+ minus 10 V DC (10 ... 2.9 mA))
Input current		
• for signal "0", max. (permissible quiescent current)	2.2 mA	1.4 mA
• for signal "1", min.	5.1 mA	2.9 mA
• for signal "1", typ.		7 mA
Input delay (for rated value of input voltage)		
for standard inputs		
- parameterizable	Yes; 0.1 / 0.2 / 0.4 / 0.8 / 1.6 / 3.2 / 6.4 / 10.0 / 12.8 / 20.0 μs; 0.05 / 0.1 / 0.2 / 0.4 / 0.8 / 1.6 / 3.2 / 6.4 / 10.0 / 12.8 / 20.0 ms	Yes; 0.1 / 0.2 / 0.4 / 0.8 / 1.6 / 3.2 / 6.4 / 10.0 / 12.8 / 20.0 μs; 0.05 / 0.1 / 0.2 / 0.4 / 0.8 / 1.6 / 3.2 / 6.4 / 10.0 / 12.8 / 20.0 ms
for interrupt inputs		
- parameterizable	Yes	Yes
for counter/technological functions		
- parameterizable	Yes	Yes
Cable length		
• shielded, max.	50 m; shielded, twisted pair	50 m; shielded, twisted pair
Diagnostics indication LED		
• for status of the inputs	Yes	Yes
Degree and class of protection		
Degree of protection acc. to EN 60529		
• IP20	Yes	Yes

SIMATIC S7-1200 Basic Controllers

I/O modules

Digital modules

SB 1221 digital input modules**Technical specifications** (continued)

Article number	6ES7221-3AD30-0XB0 SIGNAL BOARD SB 1221, 4 DI 5VDC 200KHZ	6ES7221-3BD30-0XB0 SIGNAL BOARD SB 1221, 4 DI 24VDC 200KHZ
Ambient conditions		
Free fall		
• Fall height, max.	0.3 m; five times, in product package	0.3 m; five times, in product package
Ambient temperature during operation		
• permissible temperature range	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting with 2 simultaneously activated, non-adjacent inputs; otherwise up to 55 °C with horizontal mounting or 45 °C with vertical mounting
• min.	-20 °C	-20 °C
• max.	60 °C	60 °C
Mechanics/material		
Enclosure material (front)		
• Plastic	Yes	Yes
Dimensions		
Width	38 mm	38 mm
Height	62 mm	62 mm
Depth	21 mm	21 mm
Weights		
Weight, approx.	35 g	35 g

Ordering data**Article No.****Article No.****SB 1221 signal board digital input modules**

4 inputs, 5 V DC, 200 kHz, sourcing

4 inputs, 24 V DC, 200 kHz, sourcing

6ES7221-3AD30-0XB0
6ES7221-3BD30-0XB0**Terminal block (spare part)**

For signal board

With 6 screws, gold-plated; 4 pcs.

6ES7292-1BF30-0XA0

Overview

- Digital outputs as supplement to the integral I/O of the CPUs
- For flexible adaptation of the controller to the relevant task
- For subsequent expansion of the system with additional outputs

3**Technical specifications**

Article number	6ES7222-1BF32-0XB0 DIGITAL OUTPUT SM1222, 8 DO, 24V DC	6ES7222-1BH32-0XB0 DIGITAL OUTPUT SM1222, 16 DO, 24V DC	6ES7222-1HF32-0XB0 DIGITAL OUTPUT SM 1222, 8 DO, RELAY	6ES7222-1HH32-0XB0 DIGITAL OUTPUT SM1222, 16 DO, RELAY	6ES7222-1XF32-0XB0 DIGITAL OUTPUT SM 1222, 8 DO, CHANGEOVER
General information					
Product type designation	SM 1222 DQ 8x24VDC	SM 1222 DQ 16x24VDC	SM 1222 DQ 8xRelay	SM 1222 DQ 16xRelay	SM 1222 DQ 8x relay changeover contact
Input current					
from backplane bus 5 V DC, max.	120 mA	140 mA	120 mA	135 mA	140 mA
Digital outputs					
• from load voltage L+, max.			11 mA/relay coil	11 mA/relay coil	16.7 mA/relay coil
Power loss					
Power loss, typ.	1.5 W	2.5 W	4.5 W	8.5 W	5 W
Digital outputs					
Number of digital outputs	8	16	8	16	8
• in groups of	1	1	2	1	1
Short-circuit protection	No; to be provided externally	No; to be provided externally	No; to be provided externally	No; to be provided externally	No; to be provided externally
Limitation of inductive shutdown voltage to	typ. (L+) -48 V	typ. (L+) -48 V			
Switching capacity of the outputs					
• with resistive load, max.	0.5 A	0.5 A	2 A	2 A	2 A
• on lamp load, max.	5 W	5 W	30 W with DC, 200 W with AC	30 W with DC, 200 W with AC	30 W with DC, 200 W with AC
Output voltage					
• Rated value (DC)	24 V	24 V	5 V DC to 30 V DC	5 V DC to 30 V DC	5 V DC to 30 V DC
• Rated value (AC)			5 V AC to 250 V AC	5 V AC to 250 V AC	5 V AC to 250 V AC
• for signal "0", max.	0.1 V; with 10 kOhm load	0.1 V; with 10 kOhm load			
• for signal "1", min.	20 V DC	20 V DC			
Output current					
• for signal "1" rated value	0.5 A	0.5 A			
• for signal "1" permissible range, max.			2 A	2 A	2 A
• for signal "0" residual current, max.	10 µA	10 µA			
Output delay with resistive load					
• "0" to "1", max.	50 µs	50 µs	10 ms	10 ms	10 ms
• "1" to "0", max.	200 µs	200 µs	10 ms	10 ms	10 ms
Total current of the outputs (per group)					
horizontal installation					
- up to 50 °C, max.	4 A; Current per mass	8 A; Current per mass	10 A; Current per mass	10 A; Current per mass	2 A; Current per mass

SIMATIC S7-1200 Basic Controllers

I/O modules

Digital modules

SM 1222 digital output modules

Technical specifications (continued)

Article number	6ES7222-1BF32-0XB0 DIGITAL OUTPUT SM1222, 8 DO, 24V DC	6ES7222-1BH32-0XB0 DIGITAL OUTPUT SM1222, 16 DO, 24V DC	6ES7222-1HF32-0XB0 DIGITAL OUTPUT SM 1222, 8 DO, RELAY	6ES7222-1HH32-0XB0 DIGITAL OUTPUT SM1222, 16 DO, RELAY	6ES7222-1XF32-0XB0 DIGITAL OUTPUT SM 1222, 8 DO, CHANGEOVER
Relay outputs					
• Number of relay outputs			8	16	8
• Rated supply voltage of relay coil L+ (DC)			24 V	24 V	24 V
• Number of operating cycles, max.			mechanically 10 million, at rated load voltage 100 000	mechanically 10 million, at rated load voltage 100 000	mechanically 10 million, at rated load voltage 100 000
Switching capacity of contacts					
- with inductive load, max.	0.5 A	0.5 A	2 A	2 A	2 A
- on lamp load, max.	5 W	5 W	30 W with DC, 200 W with AC	30 W with DC, 200 W with AC	30 W with DC, 200 W with AC
- with resistive load, max.	0.5 A	0.5 A	2 A	2 A	2 A
Cable length					
• shielded, max.	500 m	500 m	500 m	500 m	500 m
• unshielded, max.	150 m	150 m	150 m	150 m	150 m
Interrupts/diagnostics/ status information					
Alarms					
• Diagnostic alarm	Yes	Yes	Yes	Yes	Yes
Diagnostics indication LED					
• for status of the outputs	Yes	Yes	Yes	Yes	Yes
Potential separation					
Potential separation digital outputs					
• between the channels			Relays	Relays	Relays
• between the channels, in groups of	1	1	2	4	1
• between the channels and backplane bus	500 V AC	500 V AC	1500 V AC for 1 minute	1500 V AC for 1 minute	1500 V AC for 1 minute
Permissible potential difference					
between different circuits			750 V AC for 1 minute	750 V AC for 1 minute	750 V AC for 1 minute
Degree and class of protection					
Degree of protection acc. to EN 60529					
• IP20	Yes	Yes	Yes	Yes	Yes
Standards, approvals, certificates					
CE mark	Yes	Yes	Yes	Yes	Yes
CSA approval	Yes	Yes	Yes	Yes	Yes
cULus	Yes	Yes	Yes	Yes	Yes
FM approval	Yes	Yes	Yes	Yes	Yes
RCM (formerly C-TICK)	Yes	Yes	Yes	Yes	Yes
Marine approval					
• Marine approval	Yes	Yes	Yes	Yes	Yes
Ambient conditions					
Free fall					
• Fall height, max.	0.3 m; five times, in product package	0.3 m; five times, in product package	0.3 m; five times, in product package	0.3 m; five times, in product package	0.3 m; five times, in product package
Ambient temperature during operation					
• permissible temperature range	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting, 95% humidity, non-condensing	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting, 95% humidity, non-condensing	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting, 95% humidity, non-condensing	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting, 95% humidity, non-condensing	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting, 95% humidity, non-condensing
• min.	-20 °C	-20 °C	-20 °C	-20 °C	-20 °C
• max.	60 °C	60 °C	60 °C	60 °C; Number of simultaneously activated outputs: 8 (no adjacent points) at 60 °C horizontal or 50 °C vertical, 16 at 55 °C horizontal or 45 °C vertical	60 °C; Number of simultaneously activated outputs: 4 (no adjacent points) at 60 °C horizontal or 50 °C vertical, 8 at 55 °C horizontal or 45 °C vertical

Technical specifications (continued)

Article number	6ES7222-1BF32-0XB0 DIGITAL OUTPUT SM1222, 8 DO, 24V DC	6ES7222-1BH32-0XB0 DIGITAL OUTPUT SM1222, 16 DO, 24V DC	6ES7222-1HF32-0XB0 DIGITAL OUTPUT SM 1222, 8 DO, RELAY	6ES7222-1HH32-0XB0 DIGITAL OUTPUT SM1222, 16 DO, RELAY	6ES7222-1XF32-0XB0 DIGITAL OUTPUT SM 1222, 8 DO, CHANGEOVER
Connection method					
required front connector	Yes	Yes	Yes	Yes	Yes
Mechanics/material					
Enclosure material (front)					
• Plastic	Yes	Yes	Yes	Yes	Yes
Dimensions					
Width	45 mm	45 mm	45 mm	45 mm	70 mm
Height	100 mm	100 mm	100 mm	100 mm	100 mm
Depth	75 mm	75 mm	75 mm	75 mm	75 mm
Weights					
Weight, approx.	180 g	220 g	190 g	260 g	310 g

Ordering data

Article No.

Article No.

SM 1222 digital output signal module8 outputs, 24 V DC;
0.5 A, 5 W, isolated16 outputs, 24 V DC;
0.5 A, 5 W, isolated8 relay outputs,
5 ... 30 V DC / 5 ... 250 V AC, 2 A,
30 W DC / 200 W AC8 relay outputs,
change-over contact,
5 ... 30 V DC / 5 ... 250 V AC, 2 A,
30 W DC / 200 W AC16 relay outputs,
5 ... 30 V DC / 5 ... 250 V AC, 2 A,
30 W DC / 200 W AC**Extension cable for two-tier configuration**For connecting digital/analog
signal modules;
length 2 m**6ES7222-1BF32-0XB0****6ES7222-1BH32-0XB0****6ES7222-1HF32-0XB0****6ES7222-1XF32-0XB0****6ES7222-1HH32-0XB0****6ES7290-6AA30-0XA0****Terminal block (spare part)**For 6ES7222-1BF32-0XB0,
6ES7222-1BH32-0XB0

- With 7 screws, zinc-plated; 4 pcs.

For 6ES7222-1HF32-0XB0

- With 7 screws, tin-coated,
left coded; 4 units

For 6ES7222-1HH32-0XB0

- With 7 screws, tin-coated,
right coded; 4 units

For 6ES7222-1XF32-0XB0

- With 11 screws, tin-coated; 4 units

Front flap set (spare part)

For modules with a width of 45 mm

For modules with a width of 70 mm

6ES7292-1AG30-0XA0**6ES7292-1AG40-0XA1****6ES7292-1AG40-0XA0****6ES7292-1AL30-0XA0****6ES7291-1BA30-0XA0****6ES7291-1BB30-0XA0**

SIMATIC S7-1200 Basic Controllers

I/O modules

Digital modules

SB 1222 digital output modules

Overview



- Digital outputs as a supplement to the integral I/O of SIMATIC S7-1200 CPUs
- Can be plugged directly into the CPU

Technical specifications

Article number	6ES7222-1AD30-0XB0	6ES7222-1BD30-0XB0
	SIGNAL BOARD SB1222, 4 DQ 5VDC 200KHZ	SIGNAL BOARD SB1222, 4 DQ 24VDC 200KHZ
General information		
Product type designation	SB 1222 4xDQ 5VDC 200kHz	SB 1222 4xDQ 24VDC 200kHz
Input current		
from backplane bus 5 V DC, typ.	35 mA	35 mA
Power loss		
Power loss, typ.	0.5 W	0.5 W
Digital outputs		
Number of digital outputs	4; MOSFET, solid-state (current-sinking/current-sourcing)	4; MOSFET, solid-state (current-sinking/current-sourcing)
• in groups of	4	4
Short-circuit protection	No	No
Switching capacity of the outputs		
• with resistive load, max.	0.1 A	0.1 A
Load resistance range		
• upper limit	7 Ω	11 Ω
Output voltage		
• Rated value (DC)	5 V	24 V
• for signal "0", max.	0.2 V	1 V; with 10 kOhm load
• for signal "1", min.	L+ minus 0.7 V DC	L+ (-1.5 V)
• for signal "1", max.	6 V	
Output current		
• for signal "1" rated value	0.1 A	0.1 A
• for signal "1" permissible range, max.	0.1 A	
Cable length		
• shielded, max.	50 m	50 m
Diagnostics indication LED		
• for status of the outputs	Yes	Yes
Degree and class of protection		
Degree of protection acc. to EN 60529		
• IP20	Yes	Yes

Technical specifications (continued)

Article number	6ES7222-1AD30-0XB0 SIGNAL BOARD SB1222, 4 DQ 5VDC 200KHZ	6ES7222-1BD30-0XB0 SIGNAL BOARD SB1222, 4 DQ 24VDC 200KHZ
Ambient conditions		
Free fall		
• Fall height, max.	0.3 m; five times, in product package	0.3 m; five times, in product package
Ambient temperature during operation		
• permissible temperature range	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting with 2 simultaneously activated, non-adjacent inputs; otherwise up to 55 °C with horizontal mounting or 45 °C with vertical mounting	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting
• min.	-20 °C	-20 °C
• max.	60 °C	60 °C
Mechanics/material		
Enclosure material (front)		
• Plastic	Yes	Yes
Dimensions		
Width	38 mm	38 mm
Height	62 mm	62 mm
Depth	21 mm	21 mm
Weights		
Weight, approx.	35 g	35 g

Ordering data**Article No.****Article No.****SB 1222 signal board
digital output modules**

4 outputs, 5 V DC, 0.1 A, 200 kHz
4 outputs, 24 V DC, 0.1 A, 200 kHz

6ES7222-1AD30-0XB0
6ES7222-1BD30-0XB0

Terminal block (spare part)

For signal board
With 6 screws, gold-plated; 4 pcs.

6ES7292-1BF30-0XA0

SIMATIC S7-1200 Basic Controllers

I/O modules

Digital modules

SM 1223 digital input/output modules

Overview



- Digital inputs and outputs as supplement to the integral I/O of the CPUs
- For flexible adaptation of the controller to the relevant task
- For subsequent expansion of the system with additional inputs and outputs

Technical specifications

Article number	6ES7223-1BH32-0XB0	6ES7223-1BL32-0XB0	6ES7223-1PH32-0XB0	6ES7223-1PL32-0XB0	6ES7223-1QH32-0XB0
	DIGITAL I/O SM 1223, 8 DI / 8 DO	DIGITAL I/O SM 1223, 16DI/16DO	DIGITAL I/O SM 1223, 8DI/8DO	DIGITAL I/O SM 1223, 16DI/16DO	DIGITAL I/O SM 1223, 8DI AC/ 8DO RLY
General information					
Product type designation	SM 1223 DI 8x24 VDC, DQ 8x24 VDC	SM 1223 DI 16x24 VDC, DQ 16x24 VDC	SM 1223 DI 8x24 VDC, DQ 8xRelay	SM 1223 DI 16x24 VDC, DQ 16xRelay	120/230 V AC SM223 DIx8/DQx8 RLY
Supply voltage					
Rated value (DC)					
• 24 V DC	Yes	Yes	Yes	Yes	Yes
Input current					
from backplane bus 5 V DC, max.	145 mA	185 mA	145 mA	180 mA	120 mA
Digital inputs					
• from load voltage L+ (without load), max.	4 mA; per channel	4 mA; per channel	4 mA/input 11 mA/relay	4 mA/input 11 mA/relay	
Output voltage					
Power supply to the transmitters					
• present	Yes	Yes	Yes	Yes	Yes
Power loss					
Power loss, typ.	2.5 W	4.5 W	5.5 W	10 W	7.5 W
Digital inputs					
Number of digital inputs	8	16	8	16	8
• in groups of	2	2	2	2	4
Input characteristic curve in accordance with IEC 61131, type 1	Yes	Yes	Yes	Yes	Yes
Number of simultaneously controllable inputs					
all mounting positions					
- up to 40 °C, max.	8	16	8	16	8
horizontal installation					
- up to 40 °C, max.	8	16	8	16	8
- up to 50 °C, max.	8	16	8	16	8
vertical installation					
- up to 40 °C, max.	8	16	8	16	8
Input voltage					
• Type of input voltage	DC	DC	DC	DC	AC
• Rated value (DC)	24 V	24 V	24 V	24 V	
• Rated value (AC)					120/230V AC
• for signal "0"	5 V DC at 1 mA	5 V DC at 1 mA	5 V DC at 1 mA	5 V DC at 1 mA	20 V AC at 1 mA
• for signal "1"	15 V DC at 2.5 mA	15 V DC at 2.5 mA	15 V DC at 2.5 mA	15 V DC at 2.5 mA	79 V AC at 2.5 mA
Input current					
• for signal "0", max. (permissible quiescent current)	1 mA	1 mA	1 mA	1 mA	1 mA
• for signal "1", min.	2.5 mA	2.5 mA	2.5 mA	2.5 mA	2.5 mA
• for signal "1", typ.	4 mA	4 mA	4 mA	4 mA	9 mA

Technical specifications (continued)

Article number	6ES7223-1BH32-0XB0 DIGITAL I/O SM 1223, 8 DI / 8 DO	6ES7223-1BL32-0XB0 DIGITAL I/O SM 1223, 16DI/16DO	6ES7223-1PH32-0XB0 DIGITAL I/O SM 1223, 8DI/8DO	6ES7223-1PL32-0XB0 DIGITAL I/O SM 1223, 16DI/16DO	6ES7223-1QH32-0XB0 DIGITAL I/O SM 1223, 8DI AC/ 8DO RLY
Input delay (for rated value of input voltage)					
for standard inputs					
- parameterizable	Yes; 0.2 ms, 0.4 ms, 0.8 ms, 1.6 ms, 3.2 ms, 6.4 ms and 12.8 ms, selectable in groups of four	Yes; 0.2 ms, 0.4 ms, 0.8 ms, 1.6 ms, 3.2 ms, 6.4 ms and 12.8 ms, selectable in groups of four	Yes; 0.2 ms, 0.4 ms, 0.8 ms, 1.6 ms, 3.2 ms, 6.4 ms and 12.8 ms, selectable in groups of four	Yes; 0.2 ms, 0.4 ms, 0.8 ms, 1.6 ms, 3.2 ms, 6.4 ms and 12.8 ms, selectable in groups of four	Yes; 0.2 ms, 0.4 ms, 0.8 ms, 1.6 ms, 3.2 ms, 6.4 ms and 12.8 ms, selectable in groups of four
for interrupt inputs					
- parameterizable	Yes	Yes	Yes	Yes	Yes
Cable length					
• shielded, max.	500 m	500 m	500 m	500 m	500 m
• unshielded, max.	300 m	300 m	300 m	300 m	300 m
Digital outputs					
Number of digital outputs	8	16	8	16	8
• in groups of	1	1	2	4	4
Short-circuit protection	No; to be provided externally	No; to be provided externally	No; to be provided externally	No; to be provided externally	No; to be provided externally
Limitation of inductive shutdown voltage to	L+ (-48 V)	L+ (-48 V)			
Switching capacity of the outputs					
• with resistive load, max.	0.5 A	0.5 A	2 A	2 A	2 A
• on lamp load, max.	5 W	5 W	30 W with DC, 200 W with AC	30 W with DC, 200 W with AC	30 W with DC, 200 W with AC
Output voltage					
• Rated value (DC)	24 V	24 V	5 V DC to 30 V DC	5 V DC to 30 V DC	5 V DC to 30 V DC
• Rated value (AC)			5 V AC to 250 V AC	5 V AC to 250 V AC	5 V AC to 250 V AC
• for signal "0", max.	0.1 V; with 10 kOhm load	0.1 V; with 10 kOhm load			
• for signal "1", min.	20 V DC	20 V DC			
Output current					
• for signal "1" rated value	0.5 A	0.5 A			
• for signal "1" permissible range, max.	0.5 A	0.5 A	2 A	2 A	2 A
• for signal "0" residual current, max.	10 µA	10 µA			
Output delay with resistive load					
• "0" to "1", max.	50 µs	50 µs	10 ms	10 ms	10 ms
• "1" to "0", max.	200 µs	200 µs	10 ms	10 ms	10 ms
Total current of the outputs (per group)					
horizontal installation					
- up to 50 °C, max.	4 A; Current per mass	8 A; Current per mass	10 A; Current per mass	8 A; Current per mass	8 A; Current per mass
Relay outputs					
• Number of relay outputs			8	16	8
• Rated supply voltage of relay coil L+ (DC)			24 V	24 V	24 V
• Number of operating cycles, max.			mechanically 10 million, at rated load voltage 100 000	mechanically 10 million, at rated load voltage 100 000	mechanically 10 million, at rated load voltage 100 000
Switching capacity of contacts					
- with inductive load, max.		0.5 A	2 A	2 A	2 A
- on lamp load, max.		5 W	30 W with DC, 200 W with AC	30 W with DC, 200 W with AC	30 W with DC, 200 W with AC
- with resistive load, max.		0.5 A	2 A	2 A	2 A
Cable length					
• shielded, max.	500 m	500 m	500 m	500 m	500 m
• unshielded, max.	150 m	150 m	150 m	150 m	150 m

SIMATIC S7-1200 Basic Controllers

I/O modules

Digital modules

SM 1223 digital input/output modules

Technical specifications (continued)

Article number	6ES7223-1BH32-0XB0 DIGITAL I/O SM 1223, 8 DI / 8 DO	6ES7223-1BL32-0XB0 DIGITAL I/O SM 1223, 16DI/16DO	6ES7223-1PH32-0XB0 DIGITAL I/O SM 1223, 8DI/8DO	6ES7223-1PL32-0XB0 DIGITAL I/O SM 1223, 16DI/16DO	6ES7223-1QH32-0XB0 DIGITAL I/O SM 1223, 8DI AC/ 8DO RLY
Interrupts/diagnostics/ status information					
Alarms					
• Diagnostic alarm	Yes	Yes	Yes	Yes	Yes
Diagnostics indication LED					
• for status of the inputs	Yes	Yes	Yes	Yes	Yes
• for status of the outputs	Yes	Yes	Yes	Yes	Yes
Potential separation					
Potential separation digital inputs					
• between the channels, in groups of	2	2	2	2	2
Potential separation digital outputs					
• between the channels			Relays	Relays	Relays
• between the channels, in groups of	1	1	2	4	2
• between the channels and backplane bus	500 V AC	500 V AC	1500 V AC for 1 minute	1500 V AC for 1 minute	1500 V AC for 1 minute
Permissible potential difference					
between different circuits			750 V AC for 1 minute	750 V AC for 1 minute	750 V AC for 1 minute
Degree and class of protection					
Degree of protection acc. to EN 60529					
• IP20	Yes	Yes	Yes	Yes	Yes
Standards, approvals, certificates					
CE mark	Yes	Yes	Yes	Yes	Yes
CSA approval	Yes	Yes	Yes	Yes	Yes
cULus	Yes	Yes	Yes	Yes	Yes
FM approval	Yes	Yes	Yes	Yes	Yes
RCM (formerly C-TICK)	Yes	Yes	Yes	Yes	Yes
Marine approval					
• Marine approval	Yes		Yes	Yes	Yes
Ambient conditions					
Free fall					
• Fall height, max.	0.3 m; five times, in product package	0.3 m; five times, in product package	0.3 m; five times, in product package	0.3 m; five times, in product package	0.3 m; five times, in product package
Ambient temperature during operation					
• permissible temperature range	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting, 95% humidity, non-condensing	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting, 95% humidity, non-condensing	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting, 95% humidity, non-condensing	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting, 95% humidity, non-condensing	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting, 95% humidity, non-condensing
• min.	-20 °C	-20 °C	-20 °C	-20 °C	-20 °C
• max.	60 °C	60 °C	60 °C	60 °C; Number of simultaneously activated outputs: 8 (no adjacent points) at 60 °C horizontal or 50 °C vertical, 16 at 55 °C horizontal or 45 °C vertical	60 °C; Number of simultaneously activated outputs: 4 (no adjacent points) at 60 °C horizontal or 50 °C vertical, 8 at 55 °C horizontal or 45 °C vertical
Connection method					
required front connector	Yes	Yes	Yes	Yes	Yes
Mechanics/material					
Enclosure material (front)					
• Plastic	Yes	Yes	Yes	Yes	Yes
Dimensions					
Width	45 mm	70 mm	45 mm	70 mm	45 mm
Height	100 mm	100 mm	100 mm	100 mm	100 mm
Depth	75 mm	75 mm	75 mm	75 mm	75 mm
Weights					
Weight, approx.	210 g	310 g	230 g	350 g	230 g

Ordering data	Article No.		Article No.
SM 1223 digital input/output signal module		Terminal block (spare part)	
8 inputs, 24 V DC, IEC type 1 current sinking; 8 24 V DC transistor outputs, 0.5 A, 5 W	6ES7223-1BH32-0XB0	For 6ES7223-1BH32-0XB0 • With 7 screws, tin-coated; 4 units	6ES7292-1AG30-0XA0
16 inputs, 24 V DC, IEC type 1 current sinking; 16 24 V DC transistor outputs, 0.5 A, 5 W	6ES7223-1BL32-0XB0	For 6ES7223-1BL32-0XB0 • With 11 screws, tin-coated; 4 units	6ES7292-1AL30-0XA0
8 inputs, 24 V DC, IEC type 1 current sinking; 8 relay outputs, 5 ... 30 V DC/5 ... 250 V AC, 2 A, 30 W DC/200 W AC	6ES7223-1PH32-0XB0	For 6ES7223-1PH32-0XB0 • With 7 screws, zinc-plated; 4 pcs. • With 7 screws, tin-coated, right coded; 4 units	6ES7292-1AG30-0XA0 6ES7292-1AG40-0XA0
16 inputs, 24 V DC, IEC type 1 current sinking; 16 relay outputs, 5 ... 30 V DC/5 ... 250 V AC, 2 A, 30 W DC/200 W AC	6ES7223-1PL32-0XB0	For 6ES7223-1PL32-0XB0 • With 11 screws, tin-coated; 4 units • With 11 screws, tin-coated, coded; 4 units	6ES7292-1AL30-0XA0 6ES7292-1AL40-0XA0
8 inputs, 120/230 V AC; 8 relay outputs, 5 ... 30 V DC/5 ... 250 V AC, 2 A, 30 W DC/200 W AC	6ES7223-1QH32-0XB0	For 6ES7223-1PL32-0XB0 • With 7 screws, tin-coated, right coded; 4 units	6ES7292-1AG40-0XA0
Extension cable for two-tier configuration	6ES7290-6AA30-0XA0	Front flap set (spare part)	
For connecting digital/analog signal modules; length 2 m		For modules with a width of 45 mm	6ES7291-1BA30-0XA0
		For modules with a width of 70 mm	6ES7291-1BB30-0XA0

SIMATIC S7-1200 Basic Controllers

I/O modules

Digital modules

SB 1223 digital input/output modules

Overview



- Digital inputs and outputs as supplement to the integral I/O of the SIMATIC S7-1200 CPUs
- Can be plugged direct into the CPU

Technical specifications

Article number	6ES7223-0BD30-0XB0	6ES7223-3AD30-0XB0	6ES7223-3BD30-0XB0
	SIGNAL BOARD SB 1223, 2 DI/2 DO	SIGNAL BOARD SB 1223, 2DI/2DQ 5V 200KHZ	SIGNAL BOARD SB 1223, 2DI/2DQ 24V 200KHZ
General information			
Product type designation	SB 1223 DI2x24 VDC, DO 2x24 VDC	SB 1223 2xDI / 2xDQ 5VDC 200kHz	SB 1223 2xDI / 2xDQ 24VDC 200kHz
Input current			
from backplane bus 5 V DC, typ.	50 mA	35 mA	35 mA
Output voltage			
Power supply to the transmitters			
• Supply current, max.	4 mA; per channel		
Power loss			
Power loss, typ.	1 W	0.5 W	0.5 W
Digital inputs			
Number of digital inputs	2; Current-sinking	2; Current-sourcing	2; Current-sourcing
• in groups of	1	2	2
Input characteristic curve in accordance with IEC 61131, type 1	Yes		
Number of simultaneously controllable inputs all mounting positions			
- up to 40 °C, max.	2		2
Input voltage			
• Type of input voltage	DC	DC	DC
• Rated value (DC)	24 V	5 V	24 V
• for signal "0"	0 to 5 V	(L+ minus 1.0 V DC) ... L+	(L+ minus 5.0 V DC) ... L+
• for signal "1"	+15 to +30V	0 V ... (L+ minus 2.0 V DC)	0 V ... (L+ minus 10 V DC)
Input current			
• for signal "0", max. (permissible quiescent current)	1 mA	2.2 mA	1.4 mA
• for signal "1", min.		5.1 mA	2.9 mA
• for signal "1", typ.	0.5 A		7 mA

Technical specifications (continued)

Article number	6ES7223-0BD30-0XB0 SIGNAL BOARD SB1223, 2 DI/2 DO	6ES7223-3AD30-0XB0 SIGNAL BOARD SB 1223, 2DI/2DQ 5V 200KHZ	6ES7223-3BD30-0XB0 SIGNAL BOARD SB 1223, 2DI/2DQ 24V 200KHZ
Input delay (for rated value of input voltage)			
for standard inputs			
- parameterizable	Yes; 0.2 ms, 0.4 ms, 0.8 ms, 1.6 ms, 3.2 ms, 6.4 ms and 12.8 ms, selectable in groups of four	Yes; 0.1 / 0.2 / 0.4 / 0.8 / 1.6 / 3.2 / 6.4 / 10.0 / 12.8 / 20.0 μ s; 0.05 / 0.1 / 0.2 / 0.4 / 0.8 / 1.6 / 3.2 / 6.4 / 10.0 / 12.8 / 20.0 ms	Yes; 0.1 / 0.2 / 0.4 / 0.8 / 1.6 / 3.2 / 6.4 / 10.0 / 12.8 / 20.0 μ s; 0.05 / 0.1 / 0.2 / 0.4 / 0.8 / 1.6 / 3.2 / 6.4 / 10.0 / 12.8 / 20.0 ms
- at "0" to "1", max.	2 μ s		
- at "1" to "0", max.	10 μ s		
for interrupt inputs			
- parameterizable	Yes	Yes	Yes
for counter/technological functions			
- parameterizable	Yes	Yes	Yes
Cable length			
• shielded, max.	500 m	50 m; shielded, twisted pair	50 m; shielded, twisted pair
• unshielded, max.	300 m		
Digital outputs			
Number of digital outputs	2; MOSFET, solid-state (current-sinking/current-sourcing)	2; MOSFET, solid-state (current-sinking/current-sourcing)	2; MOSFET, solid-state (current-sinking/current-sourcing)
• in groups of	1	2	2
Short-circuit protection	No	No	No
Switching capacity of the outputs			
• with resistive load, max.	0.5 A	0.1 A	0.1 A
• on lamp load, max.	5 W		
Load resistance range			
• upper limit	0.6 Ω	7 Ω	
Output voltage			
• Rated value (DC)	24 V	5 V	24 V
• for signal "0", max.	0.1 V; with 10 kOhm load	0.2 V	1 V
• for signal "1", min.	20 V	L+ minus 0.7 V DC	L+ (-1.5 V)
• for signal "1", max.		6 V	
Output current			
• for signal "1" rated value	0.5 A	0.1 A	0.1 A
• for signal "1" permissible range, max.		0.1 A	
• for signal "0" residual current, max.	10 μ A		
Cable length			
• shielded, max.	500 m	50 m	50 m
• unshielded, max.	150 m		
Interrupts/diagnostics/ status information			
Alarms	Yes		
Diagnostic functions	Yes		
Diagnostics indication LED			
• for status of the inputs	Yes	Yes	Yes
• for status of the outputs	Yes	Yes	Yes
Degree and class of protection			
Degree of protection acc. to EN 60529			
• IP20	Yes	Yes	Yes

SIMATIC S7-1200 Basic Controllers

I/O modules

Digital modules

SB 1223 digital input/output modules**Technical specifications** (continued)

Article number	6ES7223-0BD30-0XB0 SIGNAL BOARD SB1223, 2 DI/2 DO	6ES7223-3AD30-0XB0 SIGNAL BOARD SB 1223, 2DI/2DQ 5V 200KHZ	6ES7223-3BD30-0XB0 SIGNAL BOARD SB 1223, 2DI/2DQ 24V 200KHZ
Ambient conditions			
Free fall			
• Fall height, max.	0.3 m; five times, in product package	0.3 m; five times, in product package	0.3 m; five times, in product package
Ambient temperature during operation			
• permissible temperature range	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting
• min.	0 °C	-20 °C	-20 °C
• max.	55 °C	60 °C	60 °C
Mechanics/material			
Enclosure material (front)			
• Plastic	Yes	Yes	Yes
Dimensions			
Width	38 mm	38 mm	38 mm
Height	62 mm	62 mm	62 mm
Depth	21 mm	21 mm	21 mm
Weights			
Weight, approx.	40 g	35 g	35 g

Ordering data**Article No.****Article No.****SB 1223 digital input/output signal board**

2 inputs, 24 V DC,
IEC type 1 current sinking;
2 24 V DC transistor outputs,
0.5 A, 5 W; can be used as HSC at
up to 30 kHz

6ES7223-0BD30-0XB0

2 inputs, 5 V DC, 200 kHz
2 outputs 5 V DC, 0.1 A, 200 kHz

6ES7223-3AD30-0XB0

2 inputs, 24 V DC, 200 kHz
2 outputs 24 V DC, 0.1 A, 200 kHz

6ES7223-3BD30-0XB0**Terminal block (spare part)**

For signal board
With 6 screws, gold-plated; 4 pcs.

6ES7292-1BF30-0XA0

Overview

- Digital inputs as supplement to the integral I/O of the CPUs
- For flexible adaptation of the controller to the corresponding task
- For subsequent expansion of the system with additional inputs
- From +60 °C to +70 °C, max. 50% of the inputs can be controlled simultaneously

Note:

SIPLUS extreme products are based on SIMATIC standard products. The contents listed here were taken from the respective standard products. SIPLUS extreme-specific information was added.

3**Technical specifications**

Article number	6AG1221-1BF32-2XB0	6AG1221-1BF32-4XB0	6AG1221-1BH32-2XB0	6AG1221-1BH32-4XB0
Based on	6ES7221-1BF32-0XB0 SIPLUS S7-1200 SM 1221 8DI	6ES7221-1BF32-0XB0 SIPLUS S7-1200 SM 1221 8DI	6ES7221-1BH32-0XB0 SIPLUS S7-1200 SM 1221 16DI	6ES7221-1BH32-0XB0 SIPLUS S7-1200 SM 1221 16DI
Ambient conditions				
Free fall				
• Fall height, max.	0.3 m; five times, in product package	0.3 m; five times, in product package	0.3 m; five times, in product package	0.3 m; five times, in product package
Ambient temperature during operation				
• min.	-40 °C; = Tmin; Startup @ -25 °C	-20 °C; = Tmin; Startup @ 0 °C	-40 °C; = Tmin; Startup @ -25 °C	-20 °C; = Tmin; Startup @ 0 °C
• max.	70 °C; = Tmax; Tmax > +60 °C number of simultaneously activated inputs 4 (no adjacent points) for horizontal mounting position	60 °C; = Tmax	70 °C; = Tmax; Tmax > +60 °C number of simultaneously activated inputs 8 (no adjacent points) for horizontal mounting position	60 °C; = Tmax
Extended ambient conditions				
• relative to ambient temperature-atmospheric pressure-installation altitude	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)
Relative humidity				
- With condensation, tested in accordance with IEC 60068-2-38, max.	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)
Resistance				
- against biologically active substances / conformity with EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!
- against chemically active substances / conformity with EN 60721-3-3	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!
- against mechanically active substances / conformity with EN 60721-3-3	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!

SIMATIC S7-1200 Basic Controllers

I/O modules

SIPLUS digital modules

SIPLUS SM 1221 digital input modules**Ordering data****Article No.****Article No.****Digital input SIPLUS
signal module SM 1221**(Extended temperature range and
exposure to media)8 inputs, 24 V DC, isolated,
current sourcing/sinking

- Suitable for areas with extraordinary medial exposure (conformal coating)
- -25 ... +70 °C, from +60 ... +70 °C number of simultaneously controllable inputs and outputs max. 50%

16 inputs, 24 V DC, isolated,
current sourcing/sinking

- Suitable for areas with extraordinary medial exposure (conformal coating)
- -25 ... +70 °C, from +60 ... +70 °C number of simultaneously controllable inputs and outputs max. 50%

6AG1221-1BF32-4XB0**6AG1221-1BF32-2XB0****6AG1221-1BH32-4XB0****6AG1221-1BH32-2XB0****Accessories**See SIMATIC S7-1200 digital input
SM 1221, page 3/50

Overview

- Digital inputs as a supplement to the integral I/O of SIMATIC S7-1200 CPUs
- Can be plugged directly into the CPU

Note:

SIPLUS extreme products are based on SIMATIC standard products. The contents listed here were adopted from the respective standard products. SIPLUS extreme specific information was added.

3**Technical specifications**

Article number	6AG1221-3AD30-5XB0	6AG1221-3BD30-5XB0
Based on	6ES7221-3AD30-0XB0 SIPLUS S7-1200 SB 1221 4DI 5VDC	6ES7221-3BD30-0XB0 SIPLUS S7-1200 SB 1221 4DI 24VDC
Ambient conditions		
Ambient temperature during operation		
• min. • max.	-40 °C; = Tmin; Startup @ -25 °C 60 °C; = Tmax; Tmax > 55 °C number of simultaneously activated inputs 2 (no adjacent points) for horizontal mounting position	-40 °C; = Tmin; Startup @ -25 °C 60 °C; = Tmax; Tmax > 55 °C number of simultaneously activated inputs 2 (no adjacent points) for horizontal mounting position
Extended ambient conditions		
• relative to ambient temperature-atmospheric pressure-installation altitude	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)
Relative humidity		
- With condensation, tested in accordance with IEC 60068-2-38, max.	100%; RH incl. condensation / frost (no commissioning in bedewed state), horizontal installation	100%; RH incl. condensation / frost (no commissioning in bedewed state), horizontal installation
Resistance		
- against biologically active substances / conformity with EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!
- against chemically active substances / conformity with EN 60721-3-3	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!
- against mechanically active substances / conformity with EN 60721-3-3	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!

Ordering data**Article No.****Article No.****SIPLUS SB 1221 digital input signal board**

(Extended temperature range and exposure to media)

4 inputs, 5 V DC, 200 kHz, sourcing

4 inputs, 24 V DC, 200 kHz, sourcing

6AG1221-3AD30-5XB0

6AG1221-3BD30-5XB0

Accessories

See SIMATIC S7-1200 digital input SB 1221, page 3/52

SIMATIC S7-1200 Basic Controllers

I/O modules

SIPLUS digital modules

SIPLUS SM 1222 digital output modules

Overview



- Digital outputs as a supplement to the integral I/O of the CPUs
- For flexible adaptation of the controller to the corresponding task
- For subsequent expansion of the system with additional outputs
- From +60 °C to +70 °C, max. 50% of the inputs can be controlled simultaneously

Note:

SIPLUS extreme products are based on SIMATIC standard products. The contents listed here were taken from the respective standard products. SIPLUS extreme-specific information was added.

Technical specifications

Article number	6AG1222-1BF32-2XB0	6AG1222-1BF32-4XB0	6AG1222-1BH32-2XB0	6AG1222-1BH32-4XB0
Based on	6ES7222-1BF32-0XB0 SIPLUS S7-1200 SM 1222 8DQ	6ES7222-1BF32-0XB0 SIPLUS S7-1200 SM 1222 8DQ	6ES7222-1BH32-0XB0 SIPLUS S7-1200 SM 1222 16DQ	6ES7222-1BH32-0XB0 SIPLUS S7-1200 SM 1222 16DQ
Ambient conditions				
Free fall				
• Fall height, max.	0.3 m; five times, in product package	0.3 m; five times, in product package	0.3 m; five times, in product package	0.3 m; five times, in product package
Ambient temperature during operation				
• min.	-40 °C; = Tmin; Startup @ -25 °C	-20 °C; = Tmin; Startup @ 0 °C	-40 °C; = Tmin; Startup @ -25 °C	-20 °C; = Tmin; Startup @ 0 °C
• max.	70 °C; = Tmax; Tmax > +60 °C number of simultaneously activated outputs 4 (no adjacent points) for horizontal mounting position	60 °C; = Tmax	70 °C; = Tmax; Tmax > +60 °C number of simultaneously activated outputs 8 (no adjacent points) for horizontal mounting position	60 °C; = Tmax
Extended ambient conditions				
• relative to ambient temperature-atmospheric pressure-installation altitude	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)
Relative humidity				
- With condensation, tested in accordance with IEC 60068-2-38, max.	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)
Resistance				
- against biologically active substances / conformity with EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!
- against chemically active substances / conformity with EN 60721-3-3	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!
- against mechanically active substances / conformity with EN 60721-3-3	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!

Technical specifications (continued)

Article number	6AG1222-1HF32-2XB0	6AG1222-1HF32-4XB0	6AG1222-1HH32-2XB0	6AG1222-1HH32-4XB0
Based on	6ES7222-1HF32-0XB0	6ES7222-1HF32-0XB0	6ES7222-1HH32-0XB0	6ES7222-1HH32-0XB0
	SIPLUS S7-1200 SM 1222 8DQ RLY	SIPLUS S7-1200 SM 1222 8DQ RLY	SIPLUS S7-1200 SM 1222 16DQ RLY	SIPLUS S7-1200 SM 1222 16DQ RLY
Ambient conditions				
Free fall				
• Fall height, max.	0.3 m; five times, in product package	0.3 m; five times, in product package	0.3 m; five times, in product package	0.3 m; five times, in product package
Ambient temperature during operation				
• min.	-40 °C; = Tmin; Startup @ -25 °C	-20 °C; = Tmin; Startup @ 0 °C	-40 °C; = Tmin; Startup @ -25 °C	-20 °C; = Tmin; Startup @ 0 °C
• max.	70 °C; = Tmax; Tmax > +60 °C number of simultaneously activated outputs 4 (no adjacent points) for horizontal mounting position	60 °C; = Tmax	70 °C; = Tmax; Tmax > +60 °C number of simultaneously activated outputs 8 (no adjacent points) for horizontal mounting position	60 °C; = Tmax
Extended ambient conditions				
• relative to ambient temperature-atmospheric pressure-installation altitude	Tmin ... Tmax at 1 080 hPa ... 795 hPa (-1 000 m ... +2 000 m) // Tmin ... (Tmax - 10 K) at 795 hPa ... 658 hPa (+2 000 m ... +3 500 m) // Tmin ... (Tmax - 20 K) at 658 hPa ... 540 hPa (+3 500 m ... +5 000 m); from 2 000 m max. 132 V AC	Tmin ... Tmax at 1 080 hPa ... 795 hPa (-1 000 m ... +2 000 m) // Tmin ... (Tmax - 10 K) at 795 hPa ... 658 hPa (+2 000 m ... +3 500 m) // Tmin ... (Tmax - 20 K) at 658 hPa ... 540 hPa (+3 500 m ... +5 000 m); from 2 000 m max. 132 V AC	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)
• At cold restart, min.	-25 °C			
Relative humidity				
- With condensation, tested in accordance with IEC 60068-2-38, max.	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)
Resistance				
- against biologically active substances / conformity with EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!
- against chemically active substances / conformity with EN 60721-3-3	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!
- against mechanically active substances / conformity with EN 60721-3-3	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!

SIMATIC S7-1200 Basic Controllers

I/O modules

SIPLUS digital modules

SIPLUS SM 1222 digital output modules

3

Ordering data**Article No.****Article No.****Digital output SIPLUS
signal module SM 1222**

(Extended temperature range and exposure to media)

8 outputs, 24 V DC;
0.5 A, 5 W, isolated

- Suitable for areas with extraordinary medial exposure (conformal coating)
- -25 ... +70 °C, from +60 ... +70 °C number of simultaneously controllable inputs and outputs max. 50%

16 outputs, 24 V DC;
0.5 A, 5 W, isolated

- Suitable for areas with extraordinary medial exposure (conformal coating)
- -25 ... +70 °C, from +60 ... +70 °C number of simultaneously controllable inputs and outputs max. 50%

8 outputs,
5 ... 30 V DC/5 ... 250 V AC,
relay 2 A, 30 W DC/200 W AC

- Suitable for areas with extraordinary medial exposure (conformal coating)
- -25 ... +70 °C, from +60 ... +70 °C number of simultaneously controllable inputs and outputs max. 50%

16 outputs,
5 ... 30 V DC/5 ... 250 V AC,
relay 2 A, 30 W DC/200 W AC

- Suitable for areas with extraordinary medial exposure (conformal coating)
- -25 ... +70 °C, from +60 ... +70 °C number of simultaneously controllable inputs and outputs max. 50%

6AG1222-1BF32-4XB0**6AG1222-1BF32-2XB0****6AG1222-1BH32-4XB0****6AG1222-1BH32-2XB0****6AG1222-1HF32-4XB0****6AG1222-1HF32-2XB0****6AG1222-1HH32-4XB0****6AG1222-1HH32-2XB0****Accessories**See SIMATIC S7-1200 digital output
SM 1222, page 3/55

Overview

- Digital outputs as a supplement to the integral I/O of the CPUs
- For flexible adaptation of the controller to the respective task
- For subsequent expansion of the system with additional outputs
- Can be plugged directly into the CPU
- From +60 °C to +70 °C, max. 50% of the inputs can be controlled simultaneously

Note:

SIPLUS extreme products are based on SIMATIC standard products. The contents listed here were taken from the respective standard products. SIPLUS extreme specific information was added.

3**Technical specifications**

Article number	6AG1222-1AD30-5XB0	6AG1222-1BD30-5XB0
Based on	6ES7222-1AD30-0XB0 SIPLUS S7-1200 SB 1222 4DQ 5VDC	6ES7222-1BD30-0XB0 SIPLUS S7-1200 SB 1222 4DQ 24VDC
Ambient temperature during operation • min. • max.	-40 °C; = Tmin; Startup @ -25 °C 60 °C; = Tmax; Tmax > 55 °C number of simultaneously activated outputs 2 (no adjacent points) for horizontal mounting position	-40 °C; = Tmin; Startup @ -25 °C 60 °C; = Tmax; Tmax > 55 °C number of simultaneously activated outputs 2 (no adjacent points) for horizontal mounting position
Extended ambient conditions • relative to ambient temperature-atmospheric pressure-installation altitude	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)
Relative humidity - With condensation, tested in accordance with IEC 60068-2-38, max.	100%; RH incl. condensation / frost (no commissioning in bedewed state), horizontal installation	100%; RH incl. condensation / frost (no commissioning in bedewed state), horizontal installation
Resistance - against biologically active substances / conformity with EN 60721-3-3 - against chemically active substances / conformity with EN 60721-3-3 - against mechanically active substances / conformity with EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation! Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation! Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation! Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation! Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!

Ordering data**Article No.****Article No.****SIPLUS SB 1222 digital output signal board**

(Extended temperature range and exposure to media)

4 outputs, 5 V DC, 0.1 A, 200 kHz

4 outputs, 24 V DC, 0.1 A, 200 kHz

6AG1222-1AD30-5XB0

6AG1222-1BD30-5XB0

Accessories

See SIMATIC S7-1200 digital output module SB 1222, page 3/57

SIMATIC S7-1200 Basic Controllers

I/O modules

SIPLUS digital modules

SIPLUS SM 1223 digital input/output modules

Overview

3



- Digital inputs and outputs as supplement to the integral I/O of the CPUs
- For flexible adaptation of the controller to the corresponding task
- For subsequent expansion of the system with additional inputs and outputs
- From +60 °C to +70 °C, max. 50% of the inputs can be controlled simultaneously

Note:

SIPLUS extreme products are based on SIMATIC standard products. The contents listed here were taken from the respective standard products. SIPLUS extreme-specific information was added.

Technical specifications

Article number	6AG1223-1BH32-2XB0	6AG1223-1BH32-4XB0	6AG1223-1PH32-2XB0	6AG1223-1PH32-4XB0
Based on	6ES7223-1BH32-0XB0 SIPLUS S7-1200 SM 1223 8DI/8DQ	6ES7223-1BH32-0XB0 SIPLUS S7-1200 SM 1223 8DI/8DQ	6ES7223-1PH32-0XB0 SIPLUS S7-1200 SM 1223 8DI/8DQ RLY	6ES7223-1PH32-0XB0 SIPLUS S7-1200 SM 1223 8DI/8DQ RLY
Ambient conditions				
Free fall				
• Fall height, max.	0.3 m; five times, in product package	0.3 m; five times, in product package	0.3 m; five times, in product package	0.3 m; five times, in product package
Ambient temperature during operation				
• min.	-40 °C; = Tmin; Startup @ -25 °C	-20 °C; = Tmin; Startup @ 0 °C	-40 °C; = Tmin; Startup @ -25 °C	-20 °C; = Tmin; Startup @ 0 °C
• max.	70 °C; = Tmax; Tmax > +60 °C number of simultaneously activated outputs 4, inputs 4 (no adjacent points) for horizontal mounting position	60 °C; = Tmax	70 °C; = Tmax; Tmax > +60 °C number of simultaneously activated outputs 4, inputs 4 (no adjacent points) for horizontal mounting position	60 °C; = Tmax
Extended ambient conditions				
• relative to ambient temperature-atmospheric pressure-installation altitude	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)
• At cold restart, min.	-25 °C			
Relative humidity				
- With condensation, tested in accordance with IEC 60068-2-38, max.	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)
Resistance				
- against biologically active substances / conformity with EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!
- against chemically active substances / conformity with EN 60721-3-3	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!
- against mechanically active substances / conformity with EN 60721-3-3	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!

Technical specifications (continued)

Article number	6AG1223-1PL32-2XB0	6AG1223-1PL32-4XB0	6AG1223-1BL32-2XB0	6AG1223-1BL32-4XB0
Based on	6ES7223-1PL32-0XB0 SIPLUS S7-1200 SM 1223 16DI/16DQ RLY	6ES7223-1PL32-0XB0 SIPLUS S7-1200 SM 1223 16DI/16DQ RLY	6ES7223-1BL32-0XB0 SIPLUS S7-1200 SM 1223 16DI/16DQ	6ES7223-1BL32-0XB0 SIPLUS S7-1200 SM 1223 16DI/16DQ
Ambient conditions				
Free fall				
• Fall height, max.	0.3 m; five times, in product package	0.3 m; five times, in product package	0.3 m; five times, in product package	0.3 m; five times, in product package
Ambient temperature during operation				
• min.	-40 °C; = Tmin; Startup @ -25 °C	-20 °C; = Tmin; Startup @ 0 °C	-40 °C; = Tmin; Startup @ -25 °C	-20 °C; = Tmin; Startup @ 0 °C
• max.	70 °C; = Tmax; Tmax > +60 °C number of simultaneously activated outputs 8, inputs 8 (no adjacent points) for horizontal mounting position	60 °C; = Tmax	70 °C; = Tmax; Tmax > +60 °C number of simultaneously activated outputs 8, inputs 8 (no adjacent points) for horizontal mounting position	60 °C; = Tmax
Extended ambient conditions				
• relative to ambient temperature-atmospheric pressure-installation altitude	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)
Relative humidity				
- With condensation, tested in accordance with IEC 60068-2-38, max.	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)
Resistance				
- against biologically active substances / conformity with EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!
- against chemically active substances / conformity with EN 60721-3-3	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!
- against mechanically active substances / conformity with EN 60721-3-3	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!

SIMATIC S7-1200 Basic Controllers

I/O modules

SIPLUS digital modules

SIPLUS SM 1223 digital input/output modules**Ordering data****Article No.****Article No.****Digital input/output
SIPLUS signal module SM 1223**

(Extended temperature range and exposure to media)

8 inputs, 24 V DC,
IEC type 1 current sinking
8 transistor outputs,
24 V DC, 0.5 A, 5 W

- Suitable for areas with extraordinary medial exposure (conformal coating)
- -25 ... +70 °C, from +60 ... +70 °C number of simultaneously controllable inputs and outputs max. 50%

6AG1223-1BH32-4XB0**6AG1223-1BH32-2XB0**16 inputs, 24 V DC,
IEC type 1 current sinking
16 transistor outputs,
24 V DC, 0.5 A, 5 W

- Suitable for areas with extraordinary medial exposure (conformal coating)
- -25 ... +70 °C, from +60 ... +70 °C number of simultaneously controllable inputs and outputs max. 50%

6AG1223-1BL32-4XB0**6AG1223-1BL32-2XB0**8 inputs, 24 V DC,
IEC type 1 current sinking
8 relay outputs,
5 ... 30 V DC / 5 ... 250 V AC, 2 A,
30 W DC / 200 W AC

- Suitable for areas with extraordinary medial exposure (conformal coating)
- -25 ... +70 °C, from +60 ... +70 °C number of simultaneously controllable inputs and outputs max. 50%

6AG1223-1PH32-4XB0**6AG1223-1PH32-2XB0**16 inputs, 24 V DC,
IEC type 1 current sinking
16 relay outputs,
5 ... 30 V DC / 5 ... 250 V AC, 2 A,
30 W DC / 200 W AC

- Suitable for areas with extraordinary medial exposure (conformal coating)
- -25 ... +70 °C, from +60 ... +70 °C number of simultaneously controllable inputs and outputs max. 50%

6AG1223-1PL32-4XB0**6AG1223-1PL32-2XB0****Accessories**

See SIMATIC S7-1200 digital input/output SM 1223, page 3/61

Overview



- Digital inputs and outputs as supplement to the integral I/O of the SIPLUS S7-1200-CPU's
- Can be plugged directly into the CPU (cannot be used for +70 °C version)

Note:

SIPLUS extreme products are based on SIMATIC standard products. The contents listed here were taken from the respective standard products. SIPLUS extreme-specific information was added.

3

Technical specifications

Article number	6AG1223-0BD30-4XB0	6AG1223-0BD30-5XB0	6AG1223-3AD30-5XB0	6AG1223-3BD30-5XB0
Based on	6ES7223-0BD30-0XB0 SIPLUS S7-1200 SB 1223 2DI/2DO 24VDC	6ES7223-0BD30-0XB0 SIPLUS S7-1200 SB 1223 2DI/2DO 24VDC	6ES7223-3AD30-0XB0 SIPLUS S7-1200 SB 1223 2DI/2DO 5VDC	6ES7223-3BD30-0XB0 SIPLUS S7-1200 SB 1223 2DI/2DO, 24VDC
Ambient conditions				
Free fall				
• Fall height, max.	0.3 m; five times, in product package			
Ambient temperature during operation				
• min.	0 °C; = Tmin	-40 °C; = Tmin; Startup @ -25 °C	-40 °C; = Tmin; Startup @ -25 °C	-40 °C; = Tmin; Startup @ -25 °C
• max.	55 °C; = Tmax	60 °C; = Tmax	60 °C; = Tmax	60 °C; = Tmax
Extended ambient conditions				
• relative to ambient temperature-atmospheric pressure-installation altitude	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)
Relative humidity				
- With condensation, tested in accordance with IEC 60068-2-38, max.	100%; Relative humidity, incl. condensation / frost permitted (no commissioning under condensation conditions)	100%; RH incl. condensation / frost (no commissioning in bedewed state), horizontal installation	100%; RH incl. condensation / frost (no commissioning in bedewed state), horizontal installation	100%; RH incl. condensation / frost (no commissioning in bedewed state), horizontal installation
Resistance				
- against biologically active substances / conformity with EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!
- against chemically active substances / conformity with EN 60721-3-3	Yes; Class 3C4 incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!
- against mechanically active substances / conformity with EN 60721-3-3	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!

SIMATIC S7-1200 Basic Controllers

I/O modules

SIPLUS digital modules

SIPLUS SB 1223 digital input/output modules**Ordering data****Article No.****Article No.****Digital input/output
SIPLUS signal board SB 1223**

(Extended temperature range and exposure to media)

2 inputs, 24 V DC,
IEC type 1 current sinking;
2 transistor outputs
24 V DC, 0.5 A, 5 W;
can be used as HSC at up to
30 kHz

- Suitable for areas with extreme medial exposure (conformal coating)
- Ambient temperature -25 ... +55 °C

2 inputs, 5 V DC, 200 kHz
2 outputs 5 V DC, 0.1 A, 200 kHz

2 inputs, 24 V DC, 200 kHz
2 outputs 24 V DC, 0.1 A, 200 kHz

6AG1223-0BD30-4XB0**6AG1223-0BD30-5XB0****6AG1223-3AD30-5XB0****6AG1223-3BD30-5XB0****Accessories**

See SIMATIC S7-1200
digital input/output SB 1223,
page 3/64

Overview

- Analog inputs for SIMATIC S7-1200
- With extremely short conversion times
- For connecting analog sensors without additional amplifiers
- For solving even more complex automation tasks

3**Technical specifications**

Article number	6ES7231-4HD32-0XB0 ANALOG INPUT SM 1231, 4AI	6ES7231-4HF32-0XB0 ANALOG INPUT SM 1231, 8AI	6ES7231-5ND32-0XB0 ANALOG INPUT SM 1231, 4AI 16BIT
General information			
Product type designation	SM 1231 AI 4x13 bit	SM 1231 AI 8 x 13 bit	SM 1231 AI 4 x 16 bit
Supply voltage			
Rated value (DC)			
• 24 V DC	Yes	Yes	Yes
Input current			
Current consumption, typ.	45 mA	45 mA	65 mA
from backplane bus 5 V DC, typ.	80 mA	90 mA	80 mA
Power loss			
Power loss, typ.	1.5 W	1.5 W	1.8 W
Analog inputs			
Number of analog inputs	4; Current or voltage differential inputs	8; Current or voltage differential inputs	4; Current or voltage differential inputs
permissible input voltage for current input (destruction limit), max.	±35 V	±35 V	±15 V
permissible input voltage for voltage input (destruction limit), max.	35 V	35 V	±35 V
permissible input current for voltage input (destruction limit), max.	40 mA	40 mA	40 mA; limited by input resistance 1 MOhm
permissible input current for current input (destruction limit), max.	40 mA	40 mA	40 mA
Cycle time (all channels) max.	625 µs	625 µs	100 µs
Input ranges			
• Voltage	Yes; ±10V, ±5V, ±2.5V	Yes; ±10V, ±5V, ±2.5V	Yes; ±10V, ±5V, ±2.5V or ±1.25V
• Current	Yes; 4 to 20 mA, 0 to 20 mA	Yes; 4 to 20 mA, 0 to 20 mA	Yes; 4 to 20 mA, 0 to 20 mA
• Thermocouple	No	No	No
• Resistance thermometer	No	No	No
• Resistance	No	Yes	No
Input ranges (rated values), voltages			
• -1.25 V to +1.25 V			Yes
• -10 V to +10 V	Yes	Yes	Yes
• -2.5 V to +2.5 V	Yes	Yes	Yes
• -5 V to +5 V	Yes	Yes	Yes
Input ranges (rated values), currents			
• 0 to 20 mA	Yes	Yes	Yes
• 4 mA to 20 mA	Yes	Yes	Yes
Thermocouple (TC)			
Temperature compensation			
- parameterizable		No	

SIMATIC S7-1200 Basic Controllers

I/O modules

Analog modules

SM 1231 analog input modules

Technical specifications (continued)

Article number	6ES7231-4HD32-0XB0 ANALOG INPUT SM 1231, 4AI	6ES7231-4HF32-0XB0 ANALOG INPUT SM 1231, 8AI	6ES7231-5ND32-0XB0 ANALOG INPUT SM 1231, 4AI 16BIT
Analog value generation for the inputs			
Integration and conversion time/ resolution per channel			
• Resolution with overrange (bit including sign), max.	12 bit; + sign	12 bit; + sign	15 bit; + sign
• Integration time, parameterizable	Yes	Yes	Yes
• Interference voltage suppression for interference frequency f1 in Hz	40 dB, DC to 60 V for interference frequency 50 / 60 Hz	40 dB, DC to 60 V for interference frequency 50 / 60 Hz	40 dB, DC to 60 V for interference frequency 50 / 60 Hz
Smoothing of measured values			
• parameterizable	Yes	Yes	Yes
• Step: None	Yes	Yes	Yes
• Step: low	Yes	Yes	Yes
• Step: Medium	Yes	Yes	Yes
• Step: High	Yes	Yes	Yes
Errors/accuracies			
Temperature error (relative to input range), (+/-)	25 °C ±0.1%, to 55 °C ±0.2% total measurement range	25 °C ±0.1%, to 55 °C ±0.2% total measurement range	25 °C ±0.1% / ±0.3% total measurement range
Basic error limit (operational limit at 25 °C)			
• Voltage, relative to input range, (+/-)	0.1%	0.1%	0.1%
• Current, relative to input range, (+/-)	0.1%	0.1%	0.1%
Interference voltage suppression for $f = n \times (f1 \pm 1\%)$, f1 = interference frequency			
• Common mode voltage, max.	12 V	12 V	12 V
Interrupts/diagnostics/ status information			
Alarms	Yes	Yes	Yes
Diagnostic functions	Yes	Yes	Yes
Alarms			
• Diagnostic alarm	Yes	Yes	Yes
Diagnostic messages			
• Monitoring the supply voltage	Yes	Yes	Yes
• Wire-break	Yes	Yes	Yes
Diagnostics indication LED			
• for status of the inputs	Yes	Yes	Yes
• for maintenance	Yes	Yes	Yes
Degree and class of protection			
Degree of protection acc. to EN 60529			
• IP20	Yes	Yes	Yes
Standards, approvals, certificates			
CE mark	Yes	Yes	Yes
CSA approval	Yes	Yes	Yes
FM approval	Yes	Yes	Yes
RCM (formerly C-TICK)	Yes	Yes	Yes
Marine approval			
• Marine approval	Yes	Yes	Yes

Technical specifications (continued)

Article number	6ES7231-4HD32-0XB0 ANALOG INPUT SM 1231, 4AI	6ES7231-4HF32-0XB0 ANALOG INPUT SM 1231, 8AI	6ES7231-5ND32-0XB0 ANALOG INPUT SM 1231, 4AI 16BIT
Ambient conditions			
Free fall			
• Fall height, max.	0.3 m; five times, in product package	0.3 m; five times, in product package	0.3 m; five times, in product package
Ambient temperature during operation			
• permissible temperature range	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting, 95% humidity, non-condensing	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting, 95% humidity, non-condensing	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting, 95% humidity, non-condensing
• min.	-20 °C	-20 °C	-20 °C
• max.	60 °C	60 °C	60 °C
Pollutant concentrations			
- SO ₂ at RH < 60% without condensation	SO ₂ : < 0.5 ppm; H ₂ S: < 0.1 ppm; RH < 60% condensation-free	SO ₂ : < 0.5 ppm; H ₂ S: < 0.1 ppm; RH < 60% condensation-free	SO ₂ : < 0.5 ppm; H ₂ S: < 0.1 ppm; RH < 60% condensation-free
Connection method			
required front connector	Yes	Yes	Yes
Mechanics/material			
Enclosure material (front)			
• Plastic	Yes	Yes	Yes
Dimensions			
Width	45 mm	45 mm	45 mm
Height	100 mm	100 mm	100 mm
Depth	75 mm	75 mm	75 mm
Weights			
Weight, approx.	180 g	180 g	180 g

Ordering data

Article No.

Article No.

SM 1231 analog input signal module

4 analog inputs,
±10V, ±5V, ±2.5V, or
0 ... 20 mA, 16 bits

4 analog inputs,
±10V, ±5V, ±2.5V, or
0 ... 20 mA, 12 bits + sign

8 analog inputs,
±10V, ±5V, ±2.5V, or
0 ... 20 mA, 12 bits + sign

6ES7231-5ND32-0XB0

6ES7231-4HD32-0XB0

6ES7231-4HF32-0XB0

Extension cable for two-tier configuration

For connecting digital/analog
signal modules;
length 2 m

6ES7290-6AA30-0XA0

Terminal block (spare part)

For 6ES7231-5ND32-0XB0,
6ES7231-4HD32-0XB0,
6ES7231-4HF32-0XB0

- With 7 screws, gold-plated; 4 pcs.

6ES7292-1BG30-0XA0

Front flap set (spare part)

For modules with a width of 45 mm

6ES7291-1BA30-0XA0

SIMATIC S7-1200 Basic Controllers

I/O modules

Analog modules

SB 1231 analog input modules

Overview

- Analog input module for the SIMATIC S7-1200
- With extremely short conversion times
- For the connection of analog sensors without additional amplifiers
- For the solution of more complex automation tasks as well
- Can be plugged directly into the CPU

Technical specifications

Article number	6ES7231-4HA30-0XB0 SIGNAL BOARD SB 1231, 1 AI
General information	
Product type designation	SB1231 AI 1x12 BIT
Supply voltage	
Rated value (DC)	
• 24 V DC	Yes
Input current	
from backplane bus 5 V DC, typ.	55 mA
Power loss	
Power loss, typ.	0.4 W
Analog inputs	
Number of analog inputs	1; Current or voltage differential inputs
permissible input voltage for current input (destruction limit), max.	±35 V
permissible input voltage for voltage input (destruction limit), max.	35 V
permissible input current for voltage input (destruction limit), max.	40 mA
permissible input current for current input (destruction limit), max.	40 mA
Cycle time (all channels) max.	156.25 µs; 400 Hz suppression
Input ranges	
• Voltage	Yes; ±10V, ±5V, ±2.5V
• Current	Yes; 0 to 20 mA
• Thermocouple	No
• Resistance thermometer	No
• Resistance	No
Input ranges (rated values), voltages	
• -10 V to +10 V	Yes
• -2.5 V to +2.5 V	Yes
• -5 V to +5 V	Yes
Input ranges (rated values), currents	
• 0 to 20 mA	Yes
Analog outputs	
Number of analog outputs	0

Article number	6ES7231-4HA30-0XB0 SIGNAL BOARD SB 1231, 1 AI
Analog value generation for the inputs	
Measurement principle	integrating
Integration and conversion time/ resolution per channel	
• Resolution with overrange (bit including sign), max.	11 bit; + sign
• Integration time, parameterizable	Yes
• Interference voltage suppression for interference frequency f1 in Hz	40 dB, DC to 60 Hz
Smoothing of measured values	
• parameterizable	Yes
• Step: None	Yes
• Step: low	Yes
• Step: Medium	Yes
• Step: High	Yes
Errors/accuracies	
Temperature error (relative to input range), (+/-)	25 °C ±0.3%, to 55 °C ±0.6% total measurement range
Interrupts/diagnostics/ status information	
Alarms	Yes
Diagnostic functions	Yes
Alarms	
• Diagnostic alarm	Yes
Diagnostic messages	
• Wire-break	No
Diagnostics indication LED	
• for status of the inputs	Yes
• for maintenance	Yes
Degree and class of protection	
Degree of protection acc. to EN 60529	
• IP20	Yes
Standards, approvals, certificates	
CE mark	Yes
CSA approval	Yes
FM approval	Yes
RCM (formerly C-TICK)	Yes

Technical specifications (continued)

Article number	6ES7231-4HA30-0XB0 SIGNAL BOARD SB 1231, 1 AI
Ambient conditions	
Free fall	
• Fall height, max.	0.3 m; five times, in product package
Ambient temperature during operation	
• permissible temperature range	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting, 95% humidity, non-condensing
• min.	-20 °C
• max.	60 °C
Pollutant concentrations	
- SO ₂ at RH < 60% without condensation	SO ₂ : < 0.5 ppm; H ₂ S: < 0.1 ppm; RH < 60% condensation-free
Connection method	
required front connector	Yes
Mechanics/material	
Enclosure material (front)	
• Plastic	Yes
Dimensions	
Width	38 mm
Height	62 mm
Depth	21 mm
Weights	
Weight, approx.	35 g

Ordering data**Article No.****SB 1231 signal board analog input module**

1 analog input,
±10 V with 12 bits or
0 ... 20 mA with 11 bits

6ES7231-4HA30-0XB0**Terminal block (spare part)**

For signal board
With 6 screws, gold-plated; 4 pcs.

6ES7292-1BF30-0XA0

SIMATIC S7-1200 Basic Controllers

I/O modules

Analog modules

SM 1232 analog output modules

Overview



- Analog outputs for SIMATIC S7-1200
- With extremely short conversion times
- For connecting analog actuators without additional amplifiers
- For solving even more complex automation tasks

Technical specifications

Article number	6ES7232-4HB32-0XB0 ANALOG OUTPUT SM 1232, 2AO	6ES7232-4HD32-0XB0 ANALOG OUTPUT SM 1232, 4AO
General information		
Product type designation	SM 1232 AQ 2x14 bit	SM 1232 AQ 4 x 14bit
Supply voltage		
Rated value (DC)		
• 24 V DC	Yes	Yes
Input current		
Current consumption, typ.	45 mA	45 mA
from backplane bus 5 V DC, typ.	80 mA	80 mA
Power loss		
Power loss, typ.	1.5 W	1.5 W
Analog outputs		
Number of analog outputs	2; Current or voltage	4; Current or voltage
Output ranges, voltage		
• -10 V to +10 V	Yes	Yes
Output ranges, current		
• 0 to 20 mA	Yes	Yes
Load impedance (in rated range of output)		
• with voltage outputs, min.	1 000 Ω	1 000 Ω
• with current outputs, max.	600 Ω	600 Ω
Analog value generation for the outputs		
Integration and conversion time/ resolution per channel		
• Resolution (incl. overrange)	Voltage: 14 bits; Current : 13 bits	Voltage: 14 bits; Current : 13 bits
Errors/accuracies		
Temperature error (relative to output range), (+/-)	25 °C ±0.3%, to 55 °C ±0.6% total measurement range	25 °C ±0.3%, to 55 °C ±0.6% total measurement range
Basic error limit (operational limit at 25 °C)		
• Voltage, relative to output range, (+/-)	0.3%	0.3%
• Current, relative to output range, (+/-)	0.3%	0.3%
Interference voltage suppression for f = n x (f1 +/- 1%), f1 = interference frequency		
• Common mode voltage, max.	12 V	12 V

Technical specifications (continued)

Article number	6ES7232-4HB32-0XB0 ANALOG OUTPUT SM 1232, 2AO	6ES7232-4HD32-0XB0 ANALOG OUTPUT SM 1232, 4AO
Interrupts/diagnostics/ status information		
Alarms	Yes	Yes
Diagnostic functions	Yes	Yes
Alarms		
• Diagnostic alarm	Yes	Yes
Diagnostic messages		
• Monitoring the supply voltage	Yes	Yes
• Wire-break	Yes	Yes
• Short-circuit	Yes	Yes
Diagnostics indication LED		
• for status of the outputs	Yes	Yes
• for maintenance	Yes	Yes
Degree and class of protection		
Degree of protection acc. to EN 60529		
• IP20	Yes	Yes
Standards, approvals, certificates		
CE mark	Yes	Yes
CSA approval	Yes	Yes
FM approval	Yes	Yes
RCM (formerly C-TICK)	Yes	Yes
Ambient conditions		
Free fall		
• Fall height, max.	0.3 m; five times, in product package	0.3 m; five times, in product package
Ambient temperature during operation		
• permissible temperature range	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting, 95% humidity, non-condensing	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting, 95% humidity, non-condensing
• min.	-20 °C	-20 °C
• max.	60 °C	60 °C
Pollutant concentrations		
- SO ₂ at RH < 60% without condensation	SO ₂ : < 0.5 ppm; H ₂ S: < 0.1 ppm; RH < 60% condensation-free	SO ₂ : < 0.5 ppm; H ₂ S: < 0.1 ppm; RH < 60% condensation-free
Connection method		
required front connector	Yes	Yes
Mechanics/material		
Enclosure material (front)		
• Plastic	Yes	Yes
Dimensions		
Width	45 mm	45 mm
Height	100 mm	100 mm
Depth	75 mm	75 mm
Weights		
Weight, approx.	180 g	180 g

SIMATIC S7-1200 Basic Controllers

I/O modules

Analog modules

SM 1232 analog output modules**Ordering data****Article No.****SM 1232 analog output signal module**

2 analog outputs,
±10 V with 14 bits or
0 ... 20 mA with 13 bits

6ES7232-4HB32-0XB0

4 analog outputs,
±10 V with 14 bits or
0 ... 20 mA with 13 bits

6ES7232-4HD32-0XB0**Terminal block (spare part)**

For 6ES7232-4HB32-0XB0,
6ES7232-4HD32-0XB0

With 7 screws, gold-plated; 4 units

6ES7292-1BG30-0XA0**Article No.****Extension cable for two-tier configuration**

For connecting digital/analog
signal modules;
length 2 m

6ES7290-6AA30-0XA0**Front flap set (spare part)**

For modules with a width of 45 mm

6ES7291-1BA30-0XA0

Overview



- Analog output for the SIMATIC S7-1200
- Can be plugged direct into the CPU

3

Technical specifications

Article number	6ES7232-4HA30-0XB0 SIGNAL BOARD SB 1232, 1 AO
General information	
Product type designation	SB 1232 1x AO
Input current	
from backplane bus 5 V DC, typ.	15 mA
Output voltage	
Power supply to the transmitters	
• Supply current, max.	25 mA
Power loss	
Power loss, typ.	1.5 W
Analog inputs	
Number of analog inputs	0
Analog outputs	
Number of analog outputs	1
Cycle time (all channels) max.	Voltage: 300 μ S (R), 750 μ S (1 μ F) Current: 600 ms (1 mH); 2 ms (10 mH)
Output ranges, voltage	
• -10 V to +10 V	Yes
Output ranges, current	
• 0 to 20 mA	Yes
Load impedance (in rated range of output)	
• with voltage outputs, min.	1 000 Ω
• with current outputs, max.	600 Ω
Cable length	
• shielded, max.	10 m; shielded, twisted pair

Article number	6ES7232-4HA30-0XB0 SIGNAL BOARD SB 1232, 1 AO
Errors/accuracies	
Temperature error (relative to output range), (+/-)	25 °C $\pm$ 0.5%, to 55 °C $\pm$ 1%
Interrupts/diagnostics/status information	
Alarms	Yes
Diagnostic functions	Yes
Diagnostics indication LED	
• for status of the outputs	Yes
Degree and class of protection	
Degree of protection acc. to EN 60529	
• IP20	Yes
Standards, approvals, certificates	
CE mark	Yes
CSA approval	Yes
FM approval	Yes
RCM (formerly C-TICK)	Yes

SIMATIC S7-1200 Basic Controllers

I/O modules

Analog modules

SB 1232 analog output modules**Technical specifications** (continued)

Article number	6ES7232-4HA30-0XB0 SIGNAL BOARD SB 1232, 1 AO
Ambient conditions	
Free fall	
• Fall height, max.	0.3 m; five times, in product package
Ambient temperature during operation	
• permissible temperature range	0 °C to 55 °C horizontal installation, 0 °C to 45 °C vertical installation
• min.	0 °C
• max.	55 °C
Pollutant concentrations	
- SO ₂ at RH < 60% without condensation	SO ₂ : < 0.5 ppm; H ₂ S: < 0.1 ppm; RH < 60% condensation-free
Mechanics/material	
Enclosure material (front)	
• Plastic	Yes
Dimensions	
Width	38 mm
Height	62 mm
Depth	21 mm
Weights	
Weight, approx.	40 g

Ordering data**Article No.****SB 1232 analog output signal board**

1 analog output,
±10 V with 12 bits or
0 ... 20 mA with 11 bits

6ES7232-4HA30-0XB0**Terminal block (spare part)**

For signal board
With 6 screws, gold-plated; 4 pcs.

6ES7292-1BF30-0XA0

Overview



- Analog inputs and outputs for the SIMATIC S7-1200
- With extremely short conversion times
- For connecting analog actuators and sensors without additional amplifiers
- For solving even more complex automation tasks

3

Technical specifications

Article number	6ES7234-4HE32-0XB0 ANALOG I/O SM 1234, 4AI/2AO
General information	
Product type designation	SM 1234 A I4x13 bit AQ 2x14 bit
Supply voltage	
Rated value (DC)	
• 24 V DC	Yes
Input current	
Current consumption, typ.	60 mA
from backplane bus 5 V DC, typ.	80 mA
Power loss	
Power loss, typ.	2 W
Analog inputs	
Number of analog inputs	4; Current or voltage differential inputs
permissible input voltage for current input (destruction limit), max.	±35 V
permissible input voltage for voltage input (destruction limit), max.	35 V
permissible input current for voltage input (destruction limit), max.	40 mA
permissible input current for current input (destruction limit), max.	40 mA
Cycle time (all channels) max.	625 µs
Input ranges	
• Voltage	Yes; ±10V, ±5V, ±2.5V
• Current	Yes; 4 to 20 mA, 0 to 20 mA
Input ranges (rated values), voltages	
• -10 V to +10 V	Yes
• -2.5 V to +2.5 V	Yes
• -5 V to +5 V	Yes
Input ranges (rated values), currents	
• 0 to 20 mA	Yes
• 4 mA to 20 mA	Yes

Article number	6ES7234-4HE32-0XB0 ANALOG I/O SM 1234, 4AI/2AO
Analog outputs	
Number of analog outputs	2; Current or voltage
Output ranges, voltage	
• -10 V to +10 V	Yes
Output ranges, current	
• 0 to 20 mA	Yes
• 4 mA to 20 mA	Yes
Load impedance (in rated range of output)	
• with voltage outputs, min.	1 000 Ω
• with current outputs, max.	600 Ω
Analog value generation for the inputs	
Measurement principle	Differential
Integration and conversion time/ resolution per channel	
• Resolution with overrange (bit including sign), max.	12 bit; + sign
• Integration time, parameterizable	Yes
• Interference voltage suppression for interference frequency f1 in Hz	40 dB, DC to 60 V for interference frequency 50 / 60 Hz
Smoothing of measured values	
• parameterizable	Yes
• Step: None	Yes
• Step: low	Yes
• Step: Medium	Yes
• Step: High	Yes
Analog value generation for the outputs	
Integration and conversion time/ resolution per channel	
• Resolution (incl. overrange)	Voltage: 14 bits; Current : 13 bits

SIMATIC S7-1200 Basic Controllers

I/O modules

Analog modules

SM 1234 analog input/output modules

Technical specifications (continued)

Article number	6ES7234-4HE32-0XB0 ANALOG I/O SM 1234, 4AI/2AO
Errors/accuracies	
Temperature error (relative to input range), (+/-)	25 °C ±0.1%, to 55 °C ±0.2% total measurement range
Temperature error (relative to output range), (+/-)	25 °C ±0.3%, to 55 °C ±0.6% total measurement range
Basic error limit (operational limit at 25 °C)	
• Voltage, relative to input range, (+/-)	0.1%
• Current, relative to input range, (+/-)	0.1%
• Voltage, relative to output range, (+/-)	0.3%
• Current, relative to output range, (+/-)	0.3%
Interference voltage suppression for $f = n \times (f1 \pm 1\%)$, $f1 =$ interference frequency	
• Common mode voltage, max.	12 V
Interrupts/diagnostics/status information	
Alarms	Yes
Diagnostic functions	Yes
Alarms	
• Diagnostic alarm	Yes
Diagnostic messages	
• Monitoring the supply voltage	Yes
• Wire-break	Yes
• Short-circuit	Yes
Diagnostics indication LED	
• for status of the inputs	Yes
• for status of the outputs	Yes
• for maintenance	Yes
Potential separation analog outputs	
• between the channels and the power supply of the electronics	No
Degree and class of protection	
Degree of protection acc. to EN 60529	
• IP20	Yes

Article number	6ES7234-4HE32-0XB0 ANALOG I/O SM 1234, 4AI/2AO
Standards, approvals, certificates	
CE mark	Yes
CSA approval	Yes
FM approval	Yes
RCM (formerly C-TICK)	Yes
Marine approval	
• Marine approval	Yes
Ambient conditions	
Free fall	
• Fall height, max.	0.3 m; five times, in product package
Ambient temperature during operation	
• permissible temperature range	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting, 95% humidity, non-condensing
• min.	-20 °C
• max.	60 °C
Pollutant concentrations	
- SO2 at RH < 60% without condensation	SO2: < 0.5 ppm; H2S: < 0.1 ppm; RH < 60% condensation-free
Connection method	
required front connector	Yes
Mechanics/material	
Enclosure material (front)	
• Plastic	Yes
Dimensions	
Width	45 mm
Height	100 mm
Depth	75 mm
Weights	
Weight, approx.	220 g

Ordering data

SM 1234 analog input/output signal module

4 analog inputs,
±10 V, ±5 V, ±2.5 V, or
0 ... 20 mA, 12 bits + sign;
2 analog outputs,
±10 V with 14 bits or
0 ... 20 mA with 13 bits

6ES7234-4HE32-0XB0

Terminal block (spare part)

For 6ES7234-4HE32-0XB0
With 7 screws, gold-plated; 4 pcs.

6ES7292-1BG30-0XA0

Extension cable for two-tier configuration

For connecting digital/analog
signal modules;
length 2 m

6ES7290-6AA30-0XA0

Front flap set (spare part)

For modules with a width of 45 mm

6ES7291-1BA30-0XA0

Overview

- For the convenient recording of temperatures with great accuracy
- 7 common thermocouple types can be used
- Also for the measurement of analog signals with a low level (± 80 mV)
- Can easily be retrofitted to existing plant

Technical specifications

Article number	6ES7231-5QD32-0XB0 S7-1200, ANALOG INPUT SM 1231 TC, 4 AI	6ES7231-5QF32-0XB0 S7-1200, ANALOG INPUT SM 1231 TC, 8 AI
General information		
Product type designation	SM 1231 TC 4x16 bit	SM 1231 TC 8x16bit
Supply voltage		
Rated value (DC)		
• 24 V DC	Yes	Yes
Input current		
Current consumption, typ.	40 mA	40 mA
from backplane bus 5 V DC, typ.	80 mA	80 mA
Power loss		
Power loss, typ.	1.5 W	1.5 W
Analog inputs		
Number of analog inputs	4; Thermocouples	8; Thermocouples
permissible input voltage for current input (destruction limit), max.	± 35 V	± 35 V
permissible input voltage for voltage input (destruction limit), max.	± 35 V	± 35 V
Technical unit for temperature measurement adjustable	Degrees Celsius/degrees Fahrenheit	Degrees Celsius/degrees Fahrenheit
Input ranges		
• Voltage	Yes	Yes
• Current	No	No
• Thermocouple	Yes; J, K, T, E, R, S, N, C, TXK/XK(L); voltage range: ± 80 mV	Yes; J, K, T, E, R & S, B, N, C, TXK/XK(L); voltage range: ± 80 mV
• Resistance thermometer	No	No
• Resistance	No	No
Input ranges (rated values), voltages		
• -80 mV to +80 mV	Yes	Yes
Input ranges (rated values), thermocouples		
• Type B	Yes	Yes
• Type C	Yes	Yes
• Type E	Yes	Yes
• Type J	Yes	Yes
• Type K	Yes	Yes
• Type N	Yes	Yes
• Type R	Yes	Yes
• Type S	Yes	Yes
• Type T	Yes	Yes
• Type TXK/TXK(L) to GOST	Yes	Yes
Thermocouple (TC)		
Temperature compensation		
- parameterizable	No	No

SIMATIC S7-1200 Basic Controllers

I/O modules

Analog modules

SM 1231 thermocouple modules

Technical specifications (continued)

Article number	6ES7231-5QD32-0XB0 S7-1200, ANALOG INPUT SM 1231 TC, 4 AI	6ES7231-5QF32-0XB0 S7-1200, ANALOG INPUT SM 1231 TC, 8 AI
Analog value generation for the inputs		
Measurement principle	integrating	integrating
Integration and conversion time/resolution per channel		
• Resolution with overrange (bit including sign), max.	15 bit; + sign	15 bit; + sign
• Integration time, parameterizable	No	No
• Interference voltage suppression for interference frequency f1 in Hz	85 dB at 50 / 60 / 400 Hz	85 dB at 50 / 60 / 400 Hz
Smoothering of measured values		
• parameterizable	Yes	Yes
Errors/accuracies		
Temperature error (relative to input range), (+/-)	25 °C ±0.1%, to 55 °C ±0.2% total measurement range	25 °C ±0.1%, to 55 °C ±0.2% total measurement range
Repeat accuracy in steady state at 25 °C (relative to output range), (+/-)	0.5%	0.5%
Interference voltage suppression for f = n x (f1 +/- 1%), f1 = interference frequency		
• Common mode interference, min.	120 dB	120 dB
Interrupts/diagnostics/status information		
Alarms	Yes	Yes
Diagnostic functions	Yes; Can be read out	Yes; Can be read out
Alarms		
• Diagnostic alarm	Yes	Yes
Diagnostic messages		
• Monitoring the supply voltage	Yes	Yes
• Wire-break	Yes	Yes
Diagnostics indication LED		
• for status of the inputs	Yes	Yes
• for maintenance	Yes	Yes
Degree and class of protection		
Degree of protection acc. to EN 60529		
• IP20	Yes	Yes
Standards, approvals, certificates		
CE mark	Yes	Yes
CSA approval	Yes	Yes
FM approval	Yes	Yes
RCM (formerly C-TICK)	Yes	Yes
Ambient conditions		
Free fall		
• Fall height, max.	0.3 m; five times, in product package	0.3 m; five times, in product package
Ambient temperature during operation		
• permissible temperature range	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting, 95% humidity, non-condensing	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting, 95% humidity, non-condensing
• min.	-20 °C	-20 °C
• max.	60 °C	60 °C
Pollutant concentrations		
- SO2 at RH < 60% without condensation	SO2: < 0.5 ppm; H2S: < 0.1 ppm; RH < 60% condensation-free	SO2: < 0.5 ppm; H2S: < 0.1 ppm; RH < 60% condensation-free

SIMATIC S7-1200 Basic Controllers

I/O modules

Analog modules

SM 1231 thermocouple modules**Technical specifications** (continued)

Article number	6ES7231-5QD32-0XB0 S7-1200, ANALOG INPUT SM 1231 TC, 4 AI	6ES7231-5QF32-0XB0 S7-1200, ANALOG INPUT SM 1231 TC, 8 AI
Connection method		
required front connector	Yes	Yes
Mechanics/material		
Enclosure material (front)		
• Plastic	Yes	Yes
Dimensions		
Width	45 mm	45 mm
Height	100 mm	100 mm
Depth	75 mm	75 mm
Weights		
Weight, approx.	180 g	220 g

Ordering data**Article No.****Article No.****SM 1231 thermocouple module**

4 inputs +/- 80 mV,
resolution 15 bits + sign,
thermocouple types
J, K, S, T, R, E, N

8 inputs +/- 80 mV,
resolution 15 bits + sign,
thermocouple types
J, K, T, E, R, S, N, C, TXK/XK(L)

6ES7231-5QD32-0XB0**6ES7231-5QF32-0XB0****Accessories****Terminal block (spare part)**

For 6ES7231-5QD32-0XB0,
6ES7231-5QF32-0XB0

With 7 screws, gold-plated; 4 units

6ES7292-1BG30-0XA0**Extension cable for two-tier configuration**

For connecting digital/analog
signal modules;
length 2 m

6ES7290-6AA30-0XA0**Front flap set (spare part)**

For modules with a width of 45 mm

6ES7291-1BA30-0XA0

SIMATIC S7-1200 Basic Controllers

I/O modules

Analog modules

SB 1231 thermocouple signal boards

Overview

- For the convenient recording of temperatures with great accuracy
- 1 input with 16-bit resolution
- Common thermocouple types can be used
- Also for the measurement of analog signals with a low level (± 80 mV)
- Can easily be retrofitted to existing plant
- Can be plugged directly into the CPU

Technical specifications

Article number	6ES7231-5QA30-0XB0 SIGNAL BOARD SB 1231 TC, 1 AI
General information	
Product type designation	SB1231 AI 1xTC
Supply voltage	
Rated value (DC)	
• 24 V DC	Yes
Input current	
Current consumption, typ.	5 mA
from backplane bus 5 V DC, typ.	20 mA
Power loss	
Power loss, typ.	0.5 W
Analog inputs	
Number of analog inputs	1; Thermocouples
permissible input voltage for current input (destruction limit), max.	± 35 V
permissible input voltage for voltage input (destruction limit), max.	± 35 V
Technical unit for temperature measurement adjustable	Degrees Celsius/degrees Fahrenheit
Input ranges	
• Voltage	Yes
• Current	No
• Thermocouple	Yes; J, K; voltage range ± 80 mV
• Resistance thermometer	No
• Resistance	No
Input ranges (rated values), voltages	
• -80 mV to +80 mV	Yes
Input ranges (rated values), thermocouples	
• Type J	Yes
• Type K	Yes
Thermocouple (TC)	
Temperature compensation	
- parameterizable	No
Analog outputs	
Number of analog outputs	0

Article number	6ES7231-5QA30-0XB0 SIGNAL BOARD SB 1231 TC, 1 AI
Analog value generation for the inputs	
Measurement principle	integrating
Integration and conversion time/resolution per channel	
• Resolution with overrange (bit including sign), max.	15 bit; + sign
• Integration time, parameterizable	No
• Interference voltage suppression for interference frequency f1 in Hz	85 dB at 10 / 50 / 60 / 400 Hz
Smoothing of measured values	
• parameterizable	Yes
Errors/accuracies	
Temperature error (relative to input range), (+/-)	25 °C $\pm 0.1\%$, to 55 °C $\pm 0.2\%$ total measurement range
Repeat accuracy in steady state at 25 °C (relative to output range), (+/-)	0.5%
Interference voltage suppression for $f = n \times (f1 \pm 1\%)$, f1 = interference frequency	
• Common mode interference, min.	120 dB
Interrupts/diagnostics/status information	
Alarms	Yes
Diagnostic functions	Yes; Can be read out
Alarms	
• Diagnostic alarm	Yes
Diagnostic messages	
• Wire-break	Yes
Diagnostics indication LED	
• for status of the inputs	Yes
• for maintenance	Yes
Degree and class of protection	
Degree of protection acc. to EN 60529	
• IP20	Yes
Standards, approvals, certificates	
CE mark	Yes
CSA approval	Yes
FM approval	Yes
RCM (formerly C-TICK)	Yes

Technical specifications (continued)

Article number	6ES7231-5QA30-0XB0 SIGNAL BOARD SB 1231 TC, 1 AI
Ambient conditions	
Free fall	
• Fall height, max.	0.3 m; five times, in product package
Ambient temperature during operation	
• permissible temperature range	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting, 95% humidity, non-condensing
• min.	-20 °C
• max.	60 °C
Pollutant concentrations	
- SO ₂ at RH < 60% without condensation	SO ₂ : < 0.5 ppm; H ₂ S: < 0.1 ppm; RH < 60% condensation-free
Connection method	
required front connector	Yes
Mechanics/material	
Enclosure material (front)	
• Plastic	Yes
Dimensions	
Width	38 mm
Height	62 mm
Depth	21 mm
Weights	
Weight, approx.	35 g

Ordering data**Article No.****SB 1231 thermocouple signal board**

1 input +/- 80 mV, resolution 15 bits
+ sign, thermocouples type J, K

Accessories**Terminal block (spare part)**

For signal board
With 6 screws, gold-plated; 4 pcs.

6ES7231-5QA30-0XB0**6ES7292-1BF30-0XA0**

SIMATIC S7-1200 Basic Controllers

I/O modules

Analog modules

SM 1231 RTD signal modules

Overview

- For the convenient recording of temperatures with great accuracy
- 4 inputs
- Most popular resistance temperature detectors can be used
- Can easily be retrofitted to existing installation

Technical specifications

Article number	6ES7231-5PD32-0XB0 S7-1200, ANALOG INPUT SM 1231 RTD, 4 AI	6ES7231-5PF32-0XB0 S7-1200, ANALOG INPUT SM 1231 RTD, 8 AI
General information		
Product type designation	SM 1231 RTD 4x16bit	SM 1231 RTD 8x16bit
Supply voltage		
Rated value (DC)		
• 24 V DC	Yes	Yes
Input current		
Current consumption, typ.	40 mA	40 mA
from backplane bus 5 V DC, typ.	80 mA	80 mA
Power loss		
Power loss, typ.	1.5 W	1.5 W
Analog inputs		
Number of analog inputs	4; Resistance thermometer	8; Resistance thermometer
permissible input voltage for current input (destruction limit), max.	±35 V	±35 V
Technical unit for temperature measurement adjustable	Degrees Celsius/degreess Fahrenheit	Degrees Celsius/degreess Fahrenheit
Input ranges		
• Voltage	No	No
• Current	No	No
• Thermocouple	No	No
• Resistance thermometer	Yes; Resistance-type transmitter: Pt10, Pt50, Pt100, Pt200, Pt500, Pt1000, Ni100, Ni120, Ni200, Ni500, Ni1000, Cu10, Cu50, Cu100, LG-Ni1000	Yes; Resistance-type transmitter: Pt10, Pt50, Pt100, Pt200, Pt500, Pt1000, Ni100, Ni120, Ni200, Ni500, Ni1000, Cu10, Cu50, Cu100, LG-Ni1000
• Resistance	Yes; 150 Ω, 300 Ω, 600 Ω	Yes; 150 Ω, 300 Ω, 600 Ω
Input ranges (rated values), resistance thermometer		
• Cu 10	Yes	Yes
• Ni 100	Yes	Yes
• Ni 1000	Yes	Yes
• LG-Ni 1000	Yes	Yes
• Ni 120	Yes	Yes
• Ni 200	Yes	Yes
• Ni 500	Yes	Yes
• Pt 100	Yes	Yes
• Pt 1000	Yes	Yes
• Pt 200	Yes	Yes
• Pt 500	Yes	Yes
Input ranges (rated values), resistors		
• 0 to 150 ohms	Yes	Yes
• 0 to 300 ohms	Yes	Yes
• 0 to 600 ohms	Yes	Yes
Thermocouple (TC)		
Temperature compensation		
- parameterizable	No	No

Technical specifications (continued)

Article number	6ES7231-5PD32-0XB0 S7-1200, ANALOG INPUT SM 1231 RTD, 4 AI	6ES7231-5PF32-0XB0 S7-1200, ANALOG INPUT SM 1231 RTD, 8 AI
Analog value generation for the inputs		
Measurement principle	integrating	integrating
Integration and conversion time/resolution per channel		
• Resolution with overrange (bit including sign), max.	15 bit; + sign	15 bit; + sign
• Integration time, parameterizable	No	No
• Interference voltage suppression for interference frequency f1 in Hz	85 dB at 50 / 60 / 400 Hz	85 dB at 50 / 60 / 400 Hz
Errors/accuracies		
Temperature error (relative to input range), (+/-)	25 °C ±0.1%, to 55 °C ±0.2% total measurement range	25 °C ±0.1%, to 55 °C ±0.2% total measurement range
Repeat accuracy in steady state at 25 °C (relative to output range), (+/-)	0.05%	0.05%
Interference voltage suppression for f = n x (f1 +/- 1%), f1 = interference frequency		
• Common mode interference, min.	120 dB	120 dB
Interrupts/diagnostics/status information		
Alarms	Yes	Yes
Diagnostic functions	Yes; Can be read out	Yes; Can be read out
Alarms		
• Diagnostic alarm	Yes	Yes
Diagnostic messages		
• Monitoring the supply voltage	Yes	Yes
• Wire-break	Yes	Yes
Diagnostics indication LED		
• for status of the inputs	Yes	Yes
• for maintenance	Yes	Yes
Degree and class of protection		
Degree of protection acc. to EN 60529		
• IP20	Yes	Yes
Standards, approvals, certificates		
CE mark	Yes	Yes
CSA approval	Yes	Yes
FM approval	Yes	Yes
RCM (formerly C-TICK)	Yes	Yes
Ambient conditions		
Free fall		
• Fall height, max.	0.3 m; five times, in product package	0.3 m; five times, in product package
Ambient temperature during operation		
• permissible temperature range	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting, 95% humidity, non-condensing	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting, 95% humidity, non-condensing
• min.	-20 °C	-20 °C
• max.	60 °C	60 °C
Pollutant concentrations		
- SO2 at RH < 60% without condensation	SO2: < 0.5 ppm; H2S: < 0.1 ppm; RH < 60% condensation-free	SO2: < 0.5 ppm; H2S: < 0.1 ppm; RH < 60% condensation-free
Connection method		
required front connector	Yes	Yes
Mechanics/material		
Enclosure material (front)		
• Plastic	Yes	Yes
Dimensions		
Width	45 mm	70 mm
Height	100 mm	100 mm
Depth	75 mm	75 mm
Weights		
Weight, approx.	220 g	220 g

SIMATIC S7-1200 Basic Controllers

I/O modules

Analog modules

SM 1231 RTD signal modules**Ordering data****Article No.****Article No.****SM 1231 RTD signal module**

4 inputs for resistance temperature detectors Pt10/50/100/200/500/1000, Ni100/120/200/500/1000, Cu10/50/100, LG-Ni1000; resistance 150/300/600 Ohm, resolution 15 bits + sign

6ES7231-5PD32-0XB0

8 inputs for resistance temperature detectors Pt10/50/100/200/500/1000, Ni100/120/200/500/1000, Cu10/50/100, LG-Ni1000; resistance 150/300/600 Ohm, resolution 15 bits + sign

6ES7231-5PF32-0XB0**Accessories****Terminal block (spare part)**

For 6ES7231-5PD32-0XB0

- With 7 screws, gold-plated; 4 units

6ES7292-1BG30-0XA0

For 6ES7231-5PF32-0XB0

- With 11 screws, gold-plated; 4 units

6ES7292-1BL30-0XA0**Extension cable for two-tier configuration**

For connecting digital/analog signal modules; length 2 m

6ES7290-6AA30-0XA0**Front flap set (spare part)**

For modules with a width of 45 mm

6ES7291-1BA30-0XA0

For modules with a width of 70 mm

6ES7291-1BB30-0XA0

Overview

- For the convenient recording of temperatures with great accuracy
- 1 input with 16-bit resolution
- Common resistance-type temperature detectors can be used
- Can easily be retrofitted to existing plant
- Can be plugged directly into the CPU

Technical specifications

Article number	6ES7231-5PA30-0XB0 SIGNAL BOARD SB 1231 RTD
General information	
Product type designation	SB1231 AI 1xRTD
Supply voltage	
Rated value (DC)	
• 24 V DC	Yes
Input current	
Current consumption, typ.	5 mA
from backplane bus 5 V DC, typ.	20 mA
Power loss	
Power loss, typ.	0.5 W
Analog inputs	
Number of analog inputs	1; Resistance thermometer
permissible input voltage for current input (destruction limit), max.	±35 V
Technical unit for temperature measurement adjustable	Degrees Celsius/degrees Fahrenheit
Input ranges	
• Voltage	Yes
• Current	No
• Thermocouple	No
• Resistance thermometer	Yes; Platinum (Pt)
• Resistance	Yes; 150 Ω, 300 Ω, 600 Ω
Input ranges (rated values), resistance thermometer	
• Pt 100	Yes
• Pt 1000	Yes
• Pt 200	Yes
• Pt 500	Yes
Input ranges (rated values), resistors	
• 0 to 150 ohms	Yes
• 0 to 300 ohms	Yes
• 0 to 600 ohms	Yes
Thermocouple (TC)	
Temperature compensation	
- parameterizable	No
Analog outputs	
Number of analog outputs	0

Article number	6ES7231-5PA30-0XB0 SIGNAL BOARD SB 1231 RTD
Analog value generation for the inputs	
Measurement principle	integrating
Integration and conversion time/resolution per channel	
• Resolution with overrange (bit including sign), max.	15 bit; + sign
• Integration time, parameterizable	No
• Interference voltage suppression for interference frequency f1 in Hz	85 dB at 10 / 50 / 60 / 400 Hz
Errors/accuracies	
Temperature error (relative to input range), (+/-)	25 °C ±0.1%, to 55 °C ±0.2% total measurement range
Repeat accuracy in steady state at 25 °C (relative to output range), (+/-)	0.05%
Interference voltage suppression for f = n x (f1 +/- 1%), f1 = interference frequency	
• Common mode interference, min.	120 dB
Interrupts/diagnostics/status information	
Alarms	Yes
Diagnostic functions	Yes; Can be read out
Alarms	
• Diagnostic alarm	Yes
Diagnostic messages	
• Wire-break	Yes
Diagnostics indication LED	
• for status of the inputs	Yes
• for maintenance	Yes
Degree and class of protection	
Degree of protection acc. to EN 60529	
• IP20	Yes
Standards, approvals, certificates	
CE mark	Yes
CSA approval	Yes
FM approval	Yes
RCM (formerly C-TICK)	Yes

SIMATIC S7-1200 Basic Controllers

I/O modules

Analog modules

SB 1231 RTD signal boards**Technical specifications** (continued)

Article number	6ES7231-5PA30-0XB0 SIGNAL BOARD SB 1231 RTD
Ambient conditions	
Free fall	
• Fall height, max.	0.3 m; five times, in product package
Ambient temperature during operation	
• permissible temperature range	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting, 95% humidity, non-condensing
• min.	-20 °C
• max.	60 °C
Pollutant concentrations	
- SO ₂ at RH < 60% without condensation	SO ₂ : < 0.5 ppm; H ₂ S: < 0.1 ppm; RH < 60% condensation-free
Connection method	
required front connector	Yes
Mechanics/material	
Enclosure material (front)	
• Plastic	Yes
Dimensions	
Width	38 mm
Height	62 mm
Depth	21 mm
Weights	
Weight, approx.	35 g

Ordering data**Article No.****RTD signal board SB 1231**

1 input for resistance temperature sensors Pt 100, Pt 200, Pt 500, Pt 1000, resolution 15 bits + sign

Accessories**Terminal block (spare part)**

For signal board
With 6 screws, gold-plated; 4 pcs.

6ES7231-5PA30-0XB0**6ES7292-1BF30-0XA0**

Overview

- Energy management based on SIMATIC S7-1200
- Data acquisition of electrical characteristics in 1 and 3-phase networks up to 480 V AC
- Direct connection of voltage inputs
- Current measurement performed by 1 A and 5 A current transformers
- Can be used in TN and TT networks
- Data backup of measurement data in the event of a power failure

Technical specifications

Article number	6ES7238-5XA32-0XB0 SM 1238 Energy Meter 480V AC
General information	
Product type designation	SM 1238 AI Energy Meter 480 VAC
Product function	
• Voltage measurement	Yes
• Voltage measurement with voltage transformers	Yes
• Current measurement	Yes
• Phase current measurement without current transformers	No
• Phase current measurement with current transformers	Yes
• Energy measurement	Yes
• Frequency measurement	Yes
• Power measurement	Yes
• Active power measurement	Yes
• Reactive power measurement	Yes
• I&M data	Yes; I&M 0
• Isochronous mode	No
Engineering with	
• STEP 7 TIA Portal configurable/ integrated as of version	V13 SP1
Operating mode	
• cyclic measurement	Yes
• acyclic measurement	Yes
• Acyclic measured value access	Yes
• Fixed measured value sets	Yes
• Freely definable measured value sets	Yes
CiR – Configuration in RUN	
Reparameterization possible in RUN	Yes
Calibration possible in RUN	Yes
Installation type/mounting	
Mounting position	Horizontal, vertical
Supply voltage	
Design of the power supply	Supply via voltage measurement channel L1
Type of supply voltage	AC 100 - 277 V
permissible range, lower limit (AC)	90 V
permissible range, upper limit (AC)	293 V
Line frequency	
• permissible range, lower limit	47 Hz
• permissible range, upper limit	63 Hz

Article number	6ES7238-5XA32-0XB0 SM 1238 Energy Meter 480V AC
Power loss	
Power loss, typ.	0.6 W
Address area	
Address space per module	
• Address space per module, max.	124 byte; 112 byte input / 12 byte output
Analog inputs	
Cycle time (all channels), typ.	50 ms; Time for consistent update of all measured and calculated values (cyclic und acyclic data)
Interrupts/diagnostics/status information	
Alarms	
• Diagnostic alarm	Yes
• Limit value alarm	Yes
• Hardware interrupt	No
Diagnostics indication LED	
• Monitoring of the supply voltage (PWR-LED)	Yes
• Channel status display	Yes; Green LED
• for channel diagnostics	Yes; red Fn LED
• for module diagnostics	Yes; green/red DIAG LED
Integrated Functions	
Measuring functions	
• Measuring procedure for voltage measurement	TRMS
• Measuring procedure for current measurement	TRMS
• Type of measured value acquisition	seamless
• Curve shape of voltage	Sinusoidal or distorted
• Buffering of measured variables	Yes
• Parameter length	74 byte
• Bandwidth of measured value acquisition	2 kHz; Harmonics: 39 / 50 Hz, 32 / 60 Hz
Operating mode for measured value acquisition	
- automatic detection of line frequency	No; Parameterizable
Measuring range	
- Frequency measurement, min.	45 Hz
- Frequency measurement, max.	65 Hz

SIMATIC S7-1200 Basic Controllers

I/O modules

Analog modules

SM 1238 Energy Meter 480 V AC analog input modules

Technical specifications (continued)

Article number	6ES7238-5XA32-0XB0 SM 1238 Energy Meter 480V AC
Measuring inputs for voltage	
- Measurable line voltage between phase and neutral conductor	277 V
- Measurable line voltage between the line conductors	480 V
- Measurable line voltage between phase and neutral conductor, min.	90 V
- Measurable line voltage between phase and neutral conductor, max.	293 V
- Measurable line voltage between the line conductors, min.	155 V
- Measurable line voltage between the line conductors, max.	508 V
- Measurement category for voltage measurement in accordance with IEC 61010-2-030	CAT II; CAT III in case of guaranteed protection level of 1.5 kV
- Internal resistance line conductor and neutral conductor	3.4 MΩ
- Power consumption per phase	20 mW
- Impulse voltage resistance 1,2/50μs	1 kV
Measuring inputs for current	
- measurable relative current (AC), min.	1%; Relative to the secondary rated current 5 A
- measurable relative current (AC), max.	100%; Relative to the secondary rated current 5 A
- Continuous current with AC, maximum permissible	5 A
- Apparent power consumption per phase for measuring range 5 A	0.6 V·A
- Rated value short-time withstand current restricted to 1 s	100 A
- Input resistance measuring range 0 to 5 A	25 mΩ; At the terminal
- Zero point suppression	Parameterizable: 2 - 250 mA, default 50 mA
- Surge strength	10 A; for 1 minute

Article number	6ES7238-5XA32-0XB0 SM 1238 Energy Meter 480V AC
Accuracy class according to IEC 61557-12	
- Measured variable voltage	0,2
- Measured variable current	0,2
- Measured variable apparent power	0.5
- Measured variable active power	0.5
- Measured variable reactive power	1
- Measured variable power factor	0.5
- Measured variable active energy	0.5
- Measured variable reactive energy	1
- Measured variable neutral current	0.5; calculated
- Measured variable phase angle	± 1 °; not covered by IEC 61557-12
- Measured variable frequency	0.05
Potential separation	
Potential separation channels	
• between the channels and backplane bus	Yes; 3 700V AC (type test) CAT III
Isolation	
Isolation tested with	2 300V AC for 1 min. (type test)
Ambient conditions	
Ambient temperature during operation	
• horizontal installation, min.	-20 °C
• horizontal installation, max.	60 °C
• vertical installation, min.	-20 °C
• vertical installation, max.	50 °C
Dimensions	
Width	45 mm
Height	100 mm
Depth	75 mm
Weights	
Weight (without packaging)	165 g
Data for selecting a current transformer	
• Burden power current transformer x/1A, min.	As a function of cable length and cross section, see device manual
• Burden power current transformer x/5A, min.	As a function of cable length and cross section, see device manual

Ordering data

SM 1238 Energy Meter 480 V AC analog input

Energy measurement module for data acquisition in 1 and 3-phase networks (TN, TT) up to 480 V AC; current range: 1 A, 5 A; recording of voltage, current, phase angles, power ratings, energy values, frequencies; with channel diagnostics

Article No.

6ES7238-5XA32-0XB0

Extension cable for two-tier configuration

For connecting digital/analog signal modules; length 2 m

6ES7290-6AA30-0XA0

Terminal block (spare part)

For voltage input (top), 7-pole, tinned, coded in middle

Article No.

6ES7292-1AG40-0XA2

For current input (bottom), 7-pole, tinned

6ES7292-1AG30-0XA0

Front flap set (spare part)

For modules with a width of 45 mm

6ES7291-1BA30-0XA0

Overview

- Analog inputs for SIPLUS S7-1200
- With extremely short conversion times
- For connecting analog actuators and sensors without additional amplifiers
- Even solves more complex automation tasks
- From +60 °C to +70 °C, max. 50% of the inputs can be controlled simultaneously

Note:

SIPLUS extreme products are based on SIMATIC standard products. The contents listed here were taken from the respective standard products. SIPLUS extreme-specific information was added.

3**Technical specifications**

Article number	6AG1231-4HD32-4XB0
Based on	6ES7231-4HD32-0XB0 SIPLUS S7-1200 SM 1231 4AI 13Bit
Ambient conditions	
Free fall	
• Fall height, max.	0.3 m; five times, in product package
Ambient temperature during operation	
• min.	-20 °C; = Tmin; Startup @ 0 °C
• max.	60 °C; = Tmax
Extended ambient conditions	
• relative to ambient temperature-atmospheric pressure-installation altitude	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)

Article number	6AG1231-4HD32-4XB0
Based on	6ES7231-4HD32-0XB0 SIPLUS S7-1200 SM 1231 4AI 13Bit
Relative humidity	
- With condensation, tested in accordance with IEC 60068-2-38, max.	100%; RH incl. condensation/frost (no commissioning under condensation conditions)
Resistance	
- against biologically active substances / conformity with EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!
- against chemically active substances / conformity with EN 60721-3-3	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!
- against mechanically active substances / conformity with EN 60721-3-3	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!

Ordering data**Article No.****Analog input
SIPLUS signal module SM 1231**

(Extended temperature range and exposure to media)

Ambient temperature range
0 ... +55 °C

4 analog inputs
±10 V, ±5 V, ±2.5 V, or
0 ... 20 mA; 12 bits + sign

6AG1231-4HD32-4XB0**Accessories****Article No.**

See SIMATIC S7-1200
analog input SM 1231,
page 3/79

SIMATIC S7-1200 Basic Controllers

I/O modules

SIPLUS analog modules

SIPLUS SM 1232 analog output modules**Overview**

- Analog outputs for SIPLUS S7-1200
- With extremely short conversion times
- For connecting analog actuators without additional amplifiers
- Even solves more complex automation tasks
- From +60 °C to +70 °C, max. 50% of the outputs can be controlled simultaneously

Note:

SIPLUS extreme products are based on SIMATIC standard products. The contents listed here were taken from the respective standard products. SIPLUS extreme-specific information was added.

Technical specifications

Article number	6AG1232-4HB32-4XB0
Based on	6ES7232-4HB32-0XB0 SIPLUS S7-1200 SM 1232 2AQ 13Bit
Ambient conditions	
Free fall	
• Fall height, max.	0.3 m; five times, in product package
Ambient temperature during operation	
• min.	-20 °C; = Tmin; Startup @ 0 °C
• max.	60 °C; = Tmax
Extended ambient conditions	
• relative to ambient temperature-atmospheric pressure-installation altitude	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)

Article number	6AG1232-4HB32-4XB0
Based on	6ES7232-4HB32-0XB0 SIPLUS S7-1200 SM 1232 2AQ 13Bit
Relative humidity	
- With condensation, tested in accordance with IEC 60068-2-38, max.	100%; RH incl. condensation/frost (no commissioning under condensation conditions)
Resistance	
- against biologically active substances / conformity with EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!
- against chemically active substances / conformity with EN 60721-3-3	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!
- against mechanically active substances / conformity with EN 60721-3-3	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!

Ordering data**Article No.****Analog output SIPLUS signal module SM 1232**

(Extended temperature range and exposure to media)

Ambient temperature range-20 ... +60 °C

2 analog outputs,
±10 V with 14 bits or
0 ... 20 mA with 13 bits

6AG1232-4HB32-4XB0**Article No.****Accessories**

See SIMATIC S7-1200
analog output SM 1232,
page 3/84

Overview

- Analog output for SIPLUS S7-1200
- Can be plugged directly into the CPU
(cannot be used for +70 °C version)

Note:

SIPLUS extreme products are based on SIMATIC standard products. The contents listed here were taken from the respective standard products. SIPLUS extreme-specific information was added.

3**Technical specifications**

Article number	6AG1232-4HA30-4XB0	6AG1232-4HA30-5XB0
Based on	6ES7232-4HA30-0XB0 SIPLUS S7-1200 SB 1232 1AQ	6ES7232-4HA30-0XB0 SIPLUS S7-1200 SB 1232 1AQ
Ambient conditions		
Free fall		
• Fall height, max.	0.3 m; five times, in product package	0.3 m; five times, in product package
Ambient temperature during operation		
• min.	0 °C; = Tmin	-40 °C; = Tmin; Startup @ -25 °C
• max.	55 °C; = Tmax	55 °C; = Tmax
Extended ambient conditions		
• relative to ambient temperature-atmospheric pressure-installation altitude	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)
Relative humidity		
- With condensation, tested in accordance with IEC 60068-2-38, max.	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)
Resistance		
- against biologically active substances / conformity with EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!
- against chemically active substances / conformity with EN 60721-3-3	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!
- against mechanically active substances / conformity with EN 60721-3-3	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!

SIMATIC S7-1200 Basic Controllers

I/O modules

SIPLUS analog modules

SIPLUS SB 1232 analog output modules

3

Ordering data	Article No.	Accessories	Article No.
SIPLUS SB 1232 analog output signal board (Extended temperature range and exposure to media) <u>Ambient temperature range</u> -25 ... +55 °C 1 analog output, ±10 V with 12 bits or 0 ... 20 mA with 11 bits <u>Ambient temperature range</u> 0 ... +55 °C 1 analog output, ±10 V with 12 bits or 0 ... 20 mA with 11 bits	6AG1232-4HA30-5XB0 6AG1232-4HA30-4XB0		See SIMATIC S7-1200 analog output SB 1232, page 3/86

Overview

- Analog inputs and outputs for SIPLUS S7-1200
- With extremely short conversion times
- For connecting analog actuators and sensors without additional amplifiers
- Even solves more complex automation tasks
- From +60 °C to +70 °C, max. 50% of the inputs and outputs can be controlled simultaneously

Note:

SIPLUS extreme products are based on SIMATIC standard products. The contents listed here were taken from the respective standard products. SIPLUS extreme-specific information was added.

3**Technical specifications**

Article number	6AG1234-4HE32-2XB0	6AG1234-4HE32-4XB0
Based on	6ES7234-4HE32-0XB0 SIPLUS S7-1200 SM 1234 4AI/2AQ 13Bit	6ES7234-4HE32-0XB0 SIPLUS S7-1200 SM 1234 4AI/2AQ 13Bit
Ambient conditions		
Free fall		
• Fall height, max.	0.3 m; five times, in product package	0.3 m; five times, in product package
Ambient temperature during operation		
• min.	-40 °C; = Tmin; Startup @ -25 °C	-20 °C; = Tmin; Startup @ 0 °C
• max.	70 °C; = Tmax; Tmax > +60 °C number of simultaneously used outputs 1, inputs 2 (no adjacent points) for horizontal mounting position	60 °C; = Tmax
Extended ambient conditions		
• relative to ambient temperature-atmospheric pressure-installation altitude	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)
Relative humidity		
- With condensation, tested in accordance with IEC 60068-2-38, max.	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)
Resistance		
- against biologically active substances / conformity with EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!
- against chemically active substances / conformity with EN 60721-3-3	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!
- against mechanically active substances / conformity with EN 60721-3-3	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!

SIMATIC S7-1200 Basic Controllers

I/O modules

SIPLUS analog modules

SIPLUS SM 1234 analog input/output modules**Ordering data****Article No.****Article No.****Analog input/output SIPLUS signal module SM 1234**

(Extended temperature range and exposure to media)

Ambient temperature range-25 ... +70 °C,

from +60 ... +70 °C number of simultaneously controllable inputs and outputs max. 50%

4 analog inputs,
±10 V, ±5 V, ±2.5 V, or 0 ... 20 mA,
12 bits + sign;
2 analog outputs,
±10 V with 14 bits or 0 ... 20 mA
with 13 bits**6AG1234-4HE32-2XB0**Ambient temperature range0 ... +55 °C4 analog inputs,
±10 V, ±5 V, ±2.5 V, or
0 ... 20 mA, 12 bits + sign;
2 analog outputs,
±10 V with 14 bits or
0 ... 20 mA with 13 bits**6AG1234-4HE32-4XB0****Accessories**See SIMATIC S7-1200
analog input/output SM 1234,
page 3/88

Overview

- For the convenient recording of temperatures with great accuracy
- 7 common thermocouple types can be used
- Also for the measurement of analog signals with a low level (± 80 mV)
- Can easily be retrofitted to existing plant

Note:

SIPLUS extreme products are based on SIMATIC standard products. The contents listed here were taken from the respective standard products. SIPLUS extreme specific information was added.

Technical specifications

Article number	6AG1231-5QF32-4XB0	6AG1231-5QD32-4XB0
Based on	6ES7231-5QF32-0XB0 SIPLUS S7-1200 SM 1231 8AI TC 16Bit	6ES7231-5QD32-0XB0 SIPLUS S7-1200 SM 1231 4AI TC 16Bit
Ambient conditions		
Free fall		
• Fall height, max.	0.3 m; five times, in product package	0.3 m; five times, in product package
Ambient temperature during operation		
• min.	-20 °C; = Tmin; Startup @ 0 °C	-20 °C; = Tmin; Startup @ 0 °C
• max.	60 °C; = Tmax	60 °C; = Tmax
Extended ambient conditions		
• relative to ambient temperature-atmospheric pressure-installation altitude	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)
Relative humidity		
- With condensation, tested in accordance with IEC 60068-2-38, max.	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)
Resistance		
- against biologically active substances / conformity with EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!
- against chemically active substances / conformity with EN 60721-3-3	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!
- against mechanically active substances / conformity with EN 60721-3-3	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!

Ordering data**Article No.****Article No.****SM 1231 thermocouple module**

(Extended temperature range and exposure to media)

Ambient temperature range
-40 ... +70 °C

8 inputs +/- 80 mV,
resolution 15 bits + sign,
thermocouple types
J, K, T, E, R, S, N, C, TXK/XK(L)

4 inputs +/- 80 mV,
resolution 15 bits + sign,
thermocouple types
J, K, T, E, R, S, N, C, TXK/XK(L)

6AG1231-5QF32-4XB0**6AG1231-5QD32-4XB0****Accessories**

See SIMATIC S7-1200
thermocouple module SM 1231,
page 3/91

SIMATIC S7-1200 Basic Controllers

I/O modules

SIPLUS analog modules

SIPLUS RTD SM 1231 signal modules

Overview

- For the convenient recording of temperatures with great accuracy
- 4 inputs
- Most popular resistance temperature detectors can be used
- Can easily be retrofitted to existing plant

Note:

SIPLUS extreme products are based on SIMATIC standard products. The contents listed here were taken from the respective standard products. SIPLUS extreme specific information was added.

Technical specifications

Article number	6AG1231-5PD32-4XB0	6AG1231-5PD32-2XB0	6AG1231-5PF32-4XB0	6AG1231-5PF32-2XB0
Based on	6ES7231-5PD32-0XB0	6ES7231-5PD32-0XB0	6ES7231-5PF32-0XB0	6ES7231-5PF32-0XB0
	SIPLUS S7-1200 SM 1231 4AI RTD 16Bit	SIPLUS S7-1200 SM 1231 4AI RTD 16Bit	SIPLUS S7-1200 SM 1231 8AI RTD 16Bit	SIPLUS S7-1200 SM 1231 8AI RTD 16Bit
Ambient conditions				
Free fall				
• Fall height, max.	0.3 m; five times, in product package	0.3 m; five times, in product package	0.3 m; five times, in product package	0.3 m; five times, in product package
Ambient temperature during operation				
• min.	-20 °C; = Tmin; Startup @ 0 °C	-40 °C; = Tmin; Startup @ -25 °C	-20 °C; = Tmin; Startup @ 0 °C	-40 °C; = Tmin; Startup @ -25 °C
• max.	60 °C; = Tmax	70 °C; = Tmax	60 °C; = Tmax	70 °C; = Tmax
Extended ambient conditions				
• relative to ambient temperature-atmospheric pressure-installation altitude	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)
Relative humidity				
- With condensation, tested in accordance with IEC 60068-2-38, max.	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)
Resistance				
- against biologically active substances / conformity with EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!
- against chemically active substances / conformity with EN 60721-3-3	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!
- against mechanically active substances / conformity with EN 60721-3-3	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!

Ordering data	Article No.		Article No.
SIPLUS RTD signal module SM 1231 (Extended temperature range and exposure to media) 4 inputs for resistance temperature detectors Pt10/50/100/200/500/1000, Ni100/120/200/500/1000, Cu10/50/100, LG-Ni1000; resistance 150/300/600 Ohm, resolution 15 bits + sign • For areas with extreme medial exposure (conformal coating); ambient temperature -20 ... +60 °C • For areas with extreme medial exposure (conformal coating); ambient temperature -40 ... +70 °C 8 inputs for resistance temperature detectors Pt10/50/100/200/500/1000, Ni100/120/200/500/1000, Cu10/50/100, LG-Ni1000; resistance 150/300/600 Ohm, resolution 15 bits + sign • For areas with extreme medial exposure (conformal coating); ambient temperature -20 ... +60 °C • For areas with extreme medial exposure (conformal coating); ambient temperature -40 ... +70 °C			Accessories See SIMATIC S7-1200 RTD signal module SM 1231, page 3/93
	6AG1231-5PD32-4XB0		
	6AG1231-5PD32-2XB0		
	6AG1231-5PF32-4XB0		
	6AG1231-5PF32-2XB0		

SIMATIC S7-1200 Basic Controllers

I/O modules

Special modules

SM 1278 4xIO-Link Master

Overview



- Module for connecting up to 4 IO-Link devices according to IO Link Specification V1.1. The IO-Link parameters are configured using the Port Configuration Tool (PCT), version V3.2 and higher.

Technical specifications

Article number	6ES7278-4BD32-0XB0 S7-1200, SM1278, 4 X IO-LINK MASTER
General information	
Product type designation	SM 1278, IO-Link Master
Supply voltage	
Rated value (DC)	
• 24 V DC	Yes
Power loss	
Power loss, typ.	1 W
Interrupts/diagnostics/ status information	
Diagnostic functions	Yes
Degree and class of protection	
Degree of protection acc. to EN 60529	
• IP20	Yes
Standards, approvals, certificates	
FM approval	Yes
RCM (formerly C-TICK)	Yes
Ambient conditions	
Free fall	
• Fall height, max.	0.3 m; five times, in product package
Ambient temperature during operation	
• permissible temperature range	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting, 95% humidity, non-condensing
Connection method	
required front connector	Yes
Mechanics/material	
Enclosure material (front)	
• Plastic	Yes
Dimensions	
Width	45 mm
Height	100 mm
Depth	75 mm
Weights	
Weight, approx.	150 g

Ordering data	Article No.
SM 1278 signal module 4xIO-Link master	6ES7278-4BD32-0XB0
For the connection of up to 4 IO-Link devices according to IO Link Specification V1.1	
Terminal block (spare part)	
With 7 screws, tin-coated; 4 units	6ES7292-1AG30-0XA0

Overview

SIPLUS CMS1200 SM 1281 Condition Monitoring forms part of SIMATIC S7-1200 and is used for the:

- Monitoring of motors, generators, pumps, fans, or other mechanical components
- Recording and analysis of vibrations
- Expansion capability of up to 7 modules

3**Technical specifications**

Article number	6AT8007-1AA10-0AA0 SM1281_Condition_Monitoring
General information	
Product brand name	SIPLUS
Product category	Condition Monitoring IEPE
Product designation	CMS1200 SM 1281 Condition Monitoring
Product description	S7-1200 module for the monitoring of vibrations on mechanical components based on parameters and frequency-selective analysis functions
Physical measuring principle	Vibration acceleration
Measurement range vibration frequency, min.	0.1 Hz
Measurement range vibration frequency, max.	10 000 Hz
Installation type/mounting	
Mounting type	Rail or wall mounting
Mounting position	Horizontal, vertical
Recommended mounting position	Horizontal
Supply voltage	
Type of supply voltage	DC
Rated value (DC)	
• 24 V DC	Yes
permissible range, lower limit (DC)	20.4 V
permissible range, upper limit (DC)	28.8 V
Input current	
Current consumption, typ.	200 mA
Current consumption, max.	250 mA
from backplane bus 5 V DC, typ.	80 mA
from backplane bus 5 V DC, max.	85 mA
Memory	
Total memory capacity	1 Gbyte
Hardware configuration	
Design of hardware configuration	Modular, up to 7 modules per CPU

Article number	6AT8007-1AA10-0AA0 SM1281_Condition_Monitoring
Speed input	
Number of speed inputs	1
Protocols	
• 24 V DC digital	Yes
Sensor input	
Number of IEPE sensor inputs	4
Sampling frequency, max.	46 875 Hz
Interfaces	
Type of data transmission	Exporting of raw data as WAV file for further analyses (e.g. using CMS X-Tools) can be downloaded via browser
Ethernet interface	Yes
Protocols	
Bus communication	Yes
Web server	
• HTTP	Yes
Interrupts/diagnostics/status information	
Alarms	
• Diagnostic alarm	Yes
Diagnostics indication LED	
• Status indicator digital input (green)	No
• for status of the inputs	Yes
• for maintenance	Yes
Integrated Functions	
Monitoring functions	
• Monitoring of the sensor inputs	Yes; Cable break and short-circuit
• Vibration characteristic monitoring via RMS value of the vibration speed	Yes
• Vibration characteristic monitoring via RMS value of the vibration acceleration	Yes
• Frequency-selective monitoring via vibration speed spectrum	Yes
• Frequency-selective monitoring via vibration acceleration spectrum	Yes
• Frequency-selective monitoring via envelope curve analysis	Yes

SIMATIC S7-1200 Basic Controllers

I/O modules

Special modules

SIPLUS CMS1200 SM 1281 Condition Monitoring

Technical specifications (continued)

Article number	6AT8007-1AA10-0AA0 SM1281_Condition_Monitoring
Degree and class of protection	
Degree of protection acc. to EN 60529	
• IP20	Yes
Standards, approvals, certificates	
Certificate of suitability	CE
CE mark	Yes
UL approval	Yes
EAC (formerly Gost-R)	Yes
China RoHS compliance	Yes
Device tag according to DIN EN 81346-2	P
Ambient conditions	
Free fall	
• Fall height, max.	0.3 m; five times, in product package
Ambient temperature during operation	
• horizontal installation, min.	-20 °C
• horizontal installation, max.	60 °C
• vertical installation, min.	-20 °C
• vertical installation, max.	45 °C
Ambient temperature during storage/transportation	
• min.	-40 °C
• max.	70 °C
Air pressure acc. to IEC 60068-2-13	
• Operation, min.	795 hPa
• Operation, max.	1 080 hPa
• Storage/transport, min.	660 hPa
• Storage/transport, max.	1 080 hPa
Relative humidity	
• Operation without condensation, min.	5%
• Operation without condensation, max.	95%
Software	
Browser software required	Web browser Mozilla Firefox (ESR31) or Microsoft Internet Explorer (10/11)
Connection method	
required front connector	Yes
Design of electrical connection	Screw connection
Mechanics/material	
Material of housing	Plastic: polycarbonate, abbreviation: PC- GF 10 FR
Enclosure material (front)	
• Plastic	Yes
Dimensions	
Width	70 mm
Height	112 mm
Depth	75 mm
Weights	
Weight	260 g

Ordering data

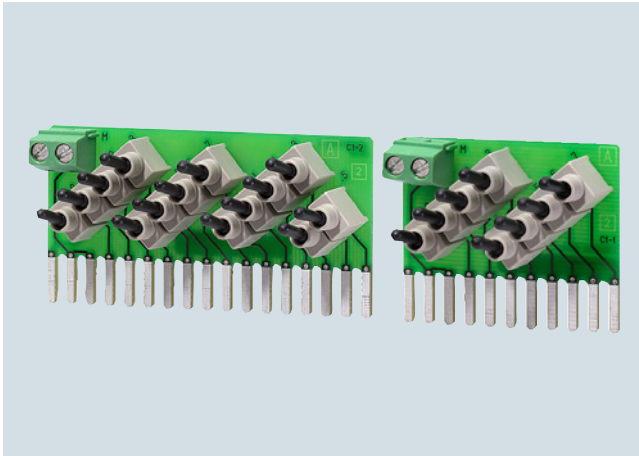
SIPLUS CMS1200 SM 1281 Condition Monitoring

Module for SIMATIC S7-1200 for monitoring vibrations in mechanical components based on characteristic values and frequency-selective analysis functions.

Article No.

6AT8007-1AA10-0AA0

Overview



- Simulator module for program testing during commissioning and ongoing operation
- Simulation of 8 or 14 inputs

Technical specifications

Article number	6ES7274-1XF30-0XA0 S7-1200 SIMULATOR MODULE SIM1274, 8 INP	6ES7274-1XH30-0XA0 S7-1200 SIMULATOR MODULE SIM1274, 14 INP
General information		
Product type designation	SIM 1274 8Ch DI Simulator	SIM 1274 14Ch DI Simulator
Supply voltage		
Rated value (DC)	24 V	24 V
Digital inputs		
Number of digital inputs	8	14
Digital outputs		
Number of digital outputs	0	0
Degree and class of protection		
Degree of protection acc. to EN 60529		
• IP20	Yes	Yes
Dimensions		
Width	43 mm	67 mm
Height	35 mm	35 mm
Depth	23 mm	23 mm

Ordering data

Article No.

**Digital input simulator
SIM 1274 simulator module**With 8 input switches,
for CPU 1211C/1212C**6ES7274-1XF30-0XA0**With 14 input switches,
for CPU 1214C/1215C**6ES7274-1XH30-0XA0**With 14 input switches,
for CPU 1217C**6ES7274-1XK30-0XA0****Analog input simulator
SIM 1274 simulator module**

2 potentiometers

6ES7274-1XA30-0XA0

SIMATIC S7-1200 Basic Controllers

I/O modules

Special modules

Battery Board BB 1297

Overview

- Battery board for extending the power reserve for the S7-1200 real-time clock

Technical specifications

Article number	6ES7297-0AX30-0XA0 BATTERY BOARD BB 1297 F. CPU 12XX
General information	
Product type designation	BB 1297 battery board
Interrupts/diagnostics/status information	
Alarms	Yes
Diagnostic functions	Yes
Diagnostics indication LED	
• for maintenance	Yes; The maintenance LED (MAINT) of the PLC signals that the battery needs to be replaced.
Degree and class of protection	
Degree of protection acc. to EN 60529	
• IP20	Yes
Standards, approvals, certificates	
CE mark	Yes
CSA approval	Yes
FM approval	Yes
RCM (formerly C-TICK)	Yes
Marine approval	
• Marine approval	Yes
Ambient conditions	
Free fall	
• Fall height, max.	0.3 m; five times, in product package
Ambient temperature during operation	
• min.	-20 °C
• max.	60 °C
Ambient temperature during storage/transportation	
• min.	-40 °C
• max.	70 °C
Air pressure acc. to IEC 60068-2-13	
• Operation, min.	795 hPa
• Operation, max.	1 080 hPa
• Storage/transport, min.	660 hPa
• Storage/transport, max.	1 080 hPa
Relative humidity	
• permissible range (without condensation) at 25 °C	95%
Mechanics/material	
Enclosure material (front)	
• Plastic	Yes
Dimensions	
Width	38 mm
Height	62 mm
Depth	21 mm
Weights	
Weight, approx.	40 g

Ordering data

Article No.

BB 1297 battery board

For long-term backup of real-time clock; can be plugged into the signal board slot of an S7-1200 CPU in FW version 3.0 or higher; battery (CR 1025) is not included

6ES7297-0AX30-0XA0

Terminal block (spare part)

For signal board
With 6 screws, gold-plated; 4 units

6ES7292-1BF30-0XA0

Overview



SIWAREX WP231 is a versatile, legal for trade weighing module for all simple weighing and force measuring tasks. The compact module is easy to install in the SIMATIC S7-1200 automation system. It can also be operated without a SIMATIC CPU.

3

Technical specifications

SIWAREX WP231		SIWAREX WP231	
Integration in automation systems		Load cell powering	
S7-1200	SIMATIC S7-1200 system bus	Supply voltage (regulated via feedback)	4.85 V DC
Operator panel and/or automation systems from other vendors	Via Ethernet (Modbus TCP/IP) or RS 485 (Modbus RTU)	Permissible load resistance	
Communication interfaces		• R_{Lmin}	$> 40 \Omega$
	• SIMATIC S7-1200 backplane bus • RS 485 (Modbus RTU, Siebert remote display) • Ethernet (SIWATOOL V7, Modbus TCP/IP) • Analog output 0/4 - 20 mA • 4 x digital outputs, 24 V DC floating, short-circuit proof • 4 x digital outputs, 24 V DC, floating	• R_{Lmax}	$< 4\ 100 \Omega$
Commissioning options		With SIWAREX IS Ex interface	
	• Using SIWATOOL V7 • Using function block in SIMATIC S7-1200 CPU / Touch Panel • Using Modbus TCP/IP • Using Modbus RTU	• R_{Lmin}	$> 50 \Omega$
Measuring accuracy		• R_{Lmax}	$< 4\ 100 \Omega$
EU type approval as non-automatic weighing instrument, trade class III	$3000\ d \geq 0.5\ \mu V/e$	Load cell characteristic	1 ... 4 mV/V
Error limit according to DIN 1319-1 of full-scale value at $20\ ^\circ C \pm 10\ K$ ($68\ ^\circ F \pm 10\ K$)	0.05%	Permissible range of the measurement signal (with 4 mV/V sensors)	-21.3 ... +21.3 mV
Internal resolution	up to ± 4 million parts	Max. distance of load cells	500 m (229.66 ft)
Measuring frequency	100 / 120 Hz	Connection to load cells in Ex zone 1	Optionally via SIWAREX IS Ex interface (compatibility of the load cells must be checked)
Digital filter	Variable adjustable low-pass and average filter	Approvals/certificates	• ATEX Zone 2 • UL • EAC • KCC • RCM • OIML R76 • Design approval 2009/23/EC (NAWI)
Typical applications	• Non-automatic weighing instruments • Force measurements • Fill-level monitoring • Belt tension monitors	Calibration approval	EU type approval OIML R76
Weighing functions		Auxiliary power supply	
Weight values	• Gross • Net • Tare	Rated voltage	24 V DC
Limit values	• 2 x min/max • Empty	Max. power consumption	200 mA
Zeroing	Per command	Max. power consumption SIMATIC Bus	3 mA
Tare	Per command	IP degree of protection according to DIN EN 60529; IEC 60529	IP20
Tare specification	Per command	Climatic requirements $T_{min}\ (IND) \dots T_{max}\ (IND)$ (operating temperature)	
Load cells	Full-bridge strain gauges in 4-wire or 6-wire system	• Vertical installation	-10 ... +40 °C (14 ... 104 °F)
		• Horizontal installation	-10 ... +55 °C (14 ... 131 °F)
		EMC requirements	according to EN 45501
		Dimensions	70 x 75 x 100 mm (2.76 x 2.95 x 3.94 in)

SIMATIC S7-1200 Basic Controllers

I/O modules

Special modules

SIWAREX WP231

3

Ordering data

Article No.

Article No.

SIWAREX WP231 weighing module

Single-channel, legal-for-trade, for NAWI non-automatic weighing instruments (e.g. platform or hopper scales) with analog load cells / full-bridge strain gauges (1 - 4 mV/V), 1 x LC, 4 x DQ, 4 x DI, 1 x AQ, 1 x RS 485, Ethernet port.

7MH4960-2AA01

SIWAREX S7-1200 device manual

Available in a range of languages

Free download from the Internet at: <http://www.siemens.com/weighing-technology>

SIWAREX WP231 "Ready for Use"

Complete software package for non-automatic weighing instrument (for S7-1200 and a directly connected operator panel).

Free download from the Internet at: <http://www.siemens.com/weighing-technology>

SIWAREX WP231

"Ready for Use - legal-for-trade"

Software package for legal for trade non-automatic weighing instruments for S7-1200.

Free download from the Internet at: <http://www.siemens.com/weighing-technology>

Software SecureDisplay

Software for a legal trade display on Windows CE-based Panel. SIMATIC Basic and Key Panels are excluded.

Free download from the Internet at: <http://www.siemens.com/weighing-technology>

Configuration package SIWAREX WP231 for TIA Portal

- "Ready for use" software for operating a scale with SIWAREX WP231 and a touch panel (in a variety of languages)
- SIWATOOL V7.0
- Device manuals (PDF files in a variety of languages)

7MH4960-2AK01

Calibration set for SIWAREX WP2xx

For verification of up to 3 scales comprising:

- 3 x inscription foil for labeling
- 1 x protection foil
- Guidelines for verification, certificates and approvals, adaptable label, SIWAREX WP

7MH4960-0AY10

Ethernet cable patch cord 2 m (7 ft)

For connecting SIWAREX WP231 to a PC (SIWATOOL), SIMATIC CPU, panel, etc.

6XV1850-2GH20

Remote display (optional)

The digital remote displays can be connected directly to the SIWAREX WP231 via the RS 485 interface.

Suitable remote display:

S102

Siebert Industrieelektronik GmbH
Postfach 1180
D-66565 Eppelborn, Germany
Tel.: +49 6806/980-0
Fax: +49 6806/980-999

Internet: <http://www.siebert.de>

Detailed information is available from the manufacturer.

Accessories

SIWAREX JB junction box, aluminum housing

7MH4710-1BA

For connecting up to 4 load cells in parallel, and for connecting several junction boxes

SIWAREX JB junction box, stainless steel housing

7MH4710-1EA

For connecting up to 4 load cells in parallel.

SIWAREX JB junction box, stainless steel housing (ATEX)

7MH4710-1EA01

For connecting up to 4 load cells in parallel.
(For zone allocation, see manual or type examination certificate).

Ex interface SIWAREX IS

For intrinsically-safe connection of load cells. With ATEX approval (not UL/FM). Suitable for SIWAREX electronic weighing system. Compatibility of load cells must be checked.

- Short-circuit current < 199 mA DC
- Short-circuit current < 137 mA DC

7MH4710-5BA
7MH4710-5CA

Ordering data	Article No.		Article No.
Cable (optional) Cable Li2Y 1 x 2 x 0.75 ST + 2 x (2 x 0.34 ST) – CY For connecting SIWAREX electronic weighing systems to junction box (JB), extension box (EB) and Ex interface or between two JBs. For permanent installation. Occasional bending is possible. External diameter: approx. 10.8 mm (0.43 in) Permissible ambient temperature -40 ... +80 °C (-40 ... +176 °F). Sold by the meter. • Sheath color: orange • For potentially explosive atmospheres. Sheath color: blue	7MH4702-8AG 7MH4702-8AF	Ground terminal for connecting the load cell cable shield to the grounded DIN rail	6ES5728-8MA11

SIMATIC S7-1200 Basic Controllers

I/O modules

Special modules

SIWAREX WP241

Overview



SIWAREX WP241

SIWAREX WP241 is a flexible weighing module for belt scales. The compact module is easy to install in the SIMATIC S7-1200 automation system. It can also be operated as a stand-alone module, i.e. without a SIMATIC CPU.

Technical specifications

SIWAREX WP241	
Integration in automation systems	
S7-1200	SIMATIC S7-1200 system bus
Operator panel and/or automation systems from other vendors	Via Ethernet (Modbus TCP/IP) or RS 485 (Modbus RTU)
Communication interfaces	• SIMATIC S7-1200 backplane bus • RS 485 (Modbus RTU) • Ethernet (SIWATOOL V7, Modbus TCP/IP) • Analog output 0/4 - 20 mA • 4 x digital outputs, 24 V DC floating, short-circuit proof • 4 x digital outputs, 24 V DC, floating
Commissioning options	• Using SIWATOOL V7 • Using function block in SIMATIC S7-1200 CPU / Touch Panel • Using Modbus TCP/IP • Using Modbus RTU
Measuring accuracy	
Error limit according to DIN 1319-1 of full-scale value at 20 °C ± 10 K (68 °F ± 10 K)	0.05%
Internal resolution	up to ±4 million parts
Measuring frequency	100 / 120 Hz
Digital filter	Separate, variable adjustable low-pass and average filter for loading and speed
Filter for conveyor load	Low-pass filter (limit frequency 0.05 ... 50 Hz)
Filter for belt speed	Low-pass filter (limit frequency 0.05 ... 50 Hz)
Weighing functions	
Readout data	• Weight • Belt load • Material flow rate • Accumulated total • Main total • Free totals 1 ... 4 • Belt speed
Limits (min/max)	• Belt load • Material flow rate • Belt speed

SIWAREX WP241	
Load cells	Full-bridge strain gauges in 4-wire or 6-wire system
Load cell excitation	
Supply voltage (regulated via feedback)	4.85 V DC
Permissible load resistance	
• R_{Lmin}	> 40 Ω
• R_{Lmax}	< 4100 Ω
With SIWAREX IS Ex interface	
• R_{Lmin}	> 50 Ω
• R_{Lmax}	< 4100 Ω
Load cell characteristic	1 ... 4 mV/V
Permissible measurement signal range	-21.3 ... +21.3 mV
Max. distance of load cells	500 m (229.66 ft)
Connection to load cells in Ex zone 1	Optionally via SIWAREX IS Ex interface (compatibility of the load cells must be checked)
Approvals/certificates	• ATEX Zone 2 • UL • EAC • KCC • RCM
Calibration approvals	• EU-type examination certificate 2014/31/EU (NAWI) acc. to OIML R76 • EU-type examination certificates 2014/32/EU (MID) acc. to OIML R61 and OIML R51
Auxiliary power supply	
Rated voltage	24 V DC
Max. power consumption	200 mA
Max. power consumption SIMATIC Bus	3 mA
IP degree of protection to DIN EN 60529; IEC 60529	IP20
Climatic requirements	
$T_{min} (IND) ... T_{max} (IND)$ (operating temperature)	
• Vertical installation	-10 ... +40 °C (14 ... 104 °F)
• Horizontal installation	-10 ... +55 °C (14 ... 131 °F)
EMC requirements	according to EN 45501
Dimensions	70 x 75 x 100 mm (2.76 x 2.95 x 3.94 in)

Ordering data	Article No.		Article No.
SIWAREX WP241 weighing module Single-channel, for conveyor scales with analog load cells / full-bridge strain gauges (1 - 4 mV/V), 1 x LC, 4 x DQ, 4 x DI, 1 x AQ, 1 x RS 485, Ethernet port.	7MH4960-4AA01		
SIWAREX S7-1200 device manual Available in a range of languages Free-of-charge download from the Internet at: http://www.siemens.com/weighing-technology		Accessories SIWAREX JB junction box, aluminum housing For connecting up to 4 load cells in parallel, and for connecting several junction boxes	7MH4710-1BA
SIWAREX WP241 "Ready for Use" Complete software package for belt scales (for S7-1200 and a directly connected operator panel) Free download on the Internet at: http://www.siemens.com/weighing-technology		SIWAREX JB junction box, stainless steel housing For connecting up to 4 load cells in parallel.	7MH4710-1EA
SIWAREX WP241 configuration package for TIA Portal "Ready for Use" software for operating a scale with SIWAREX WP241 and a touch panel (in a variety of languages) - SIWATOOL V7.0 - Device manuals (PDF files in a variety of languages)	7MH4960-4AK01	SIWAREX JB junction box, stainless steel housing (ATEX) For parallel connection of up to 4 load cells (for zone allocation, see manual or type examination certificate).	7MH4710-1EA01
Ethernet cable patch cord 2 m (7 ft) For connecting SIWAREX WP241 to a PC (SIWATOOL), SIMATIC CPU, panel, etc.	6XV1850-2GH20	Ex interface SIWAREX IS For intrinsically-safe connection of load cells. With ATEX approval (not UL/FM). Suitable for SIWAREX electronic weighing system. Compatibility of load cells must be checked. - Short-circuit current < 199 mA DC - Short-circuit current < 137 mA DC	7MH4710-5BA 7MH4710-5CA
		Cable (optional) Cable Li2Y 1 x 2 x 0.75 ST + 2 x (2 x 0.34 ST) – CY For connecting SIWAREX electronic weighing systems to junction box (JB), extension box (EB) and Ex interface or between two JB's. For permanent installation. Occasional bending is possible. External diameter: approx. 10.8 mm (0.43 inch) Permissible ambient temperature -40 ... +80 °C (-40 ... +176 °F). Sold by the meter. - Sheath color: orange - For potentially explosive atmospheres. Sheath color: blue	7MH4702-8AG 7MH4702-8AF
		Ground terminal for connecting the load cell cable shield to the grounded DIN rail	6ES5728-8MA11

SIMATIC S7-1200 Basic Controllers

I/O modules

Special modules

SIWAREX WP251

Overview



SIWAREX WP251 electronic weighing module

SIWAREX WP251 is a flexible weighing module for dosing and filling processes. The compact module can be installed seamlessly in the SIMATIC S7-1200 automation system. It can also be used without a SIMATIC CPU in stand-alone mode.

Technical specifications

SIWAREX WP251

Weighing modes	• Non-automatic scales (filling + removal) (in accordance with OIML R-76) • Automatic scales for single weighing (filling + removal) (in accordance with OIML R-51) • Automatic scales for batching (in accordance with OIML R-61)
Integration in automation systems	
S7-1200	SIMATIC S7-1200 system bus
Operator panel and/or automation systems from other vendors	Via Ethernet (Modbus TCP/IP) or RS 485 (Modbus RTU)
Ports	• 1 x SIMATIC S7-1200 system bus • 1 x Ethernet (SIWATOOL and Modbus TCP/IP) • 1 x RS 485 (Modbus RTU or remote display) • 1 x analog output (0/4 ... 20 mA) • 4 x digital inputs (24 V DC, floating) • 4 x digital outputs (24 V DC, floating, short-circuit proof)
Functions	• 3 limits • Tare • Tare specification • Set to zero • Zero adjustment • Statistics • Automatic correction of the shut-off points • Internal protocol memory for 550 000 entries • Trace function for signal analysis • Internal restore point • Stand-alone mode or SIMATIC S7-1200 integrated

SIWAREX WP251

Parameter assignment	• Full access using function block in SIMATIC S7-1200 • Full access using Modbus TCP/IP • Full access using Modbus RTU
Remote display	
Connection	via RS 485
Setting the scales	PC software SIWATOOL (Ethernet), S7-1200 function block and touch panel or directly connected operator panel (Modbus)
Measuring accuracy	
Error limit according to DIN 1319-1 of full-scale value at 20 °C ± 10 K (68 °F ± 10 K)	0.05%
Internal resolution	Up to ±4 million parts
Number of measurements/second	100 or 120 (selectable)
Filter	• Low-pass filter 0.1 ... 50 Hz • Average value filter
Load cells	Full-bridge strain gauges in 4-wire or 6-wire system
Load cell powering	
Supply voltage (regulated via feedback)	4.85 V DC
Permissible load resistance	• R_{Lmin} > 40 Ω • R_{Lmax} < 4 100 Ω
With SIWAREX IS Ex interface	• R_{Lmin} > 50 Ω • R_{Lmax} < 4 100 Ω
Load cell characteristic	1 ... 4 mV/V
Permissible range of the measurement signal (with 4 mV/V sensors)	-21.3 ... +21.3 mV
Max. distance of load cells	500 m (229.66 ft)
Connection to load cells in Ex zone 1	Optionally via SIWAREX IS Ex interface
Certificates	• ATEX Zone 2 • UL • KCC • EAC • RCM
Auxiliary power supply	
Rated voltage	24 V DC
Max. power consumption	200 mA
Max. power consumption SIMATIC Bus	3 mA
IP degree of protection according to DIN EN 60529; IEC 60529	IP20
Climatic requirements $T_{min(IND)} ... T_{max(IND)}$ (operating temperature)	• Vertical installation -10 ... +40 °C (14 ... 104 °F) • Horizontal installation -10 ... +55 °C (14 ... 131 °F)
EMC requirements	according to EN 45501
Dimensions	70 x 75 x 100 mm (2.76 x 2.95 x 3.94 in)

Ordering data	Article No.		Article No.
SIWAREX WP251 weighing module Single-channel, legal-for-trade, for automatic dosing and batching scales (GFI, ACI, NAWI) with analog load cells / full-bridge strain gauges (1 - 4 mV/V), 1 x LC, 4 x DQ, 4 x DI, 1 x AQ, 1 x RS 485, Ethernet port.	7MH4960-6AA01		
SIWAREX WP251 device manual Available in a range of languages Free download from the Internet at: http://www.siemens.com/weighing-technology		Accessories SIWAREX JB junction box, aluminum housing For connecting up to 4 load cells in parallel, and for connecting several junction boxes	7MH4710-1BA
SIWAREX WP251 "Ready for Use" Free download from the Internet at: http://www.siemens.com/weighing-technology		SIWAREX JB junction box, stainless steel housing For connecting up to 4 load cells in parallel	7MH4710-1EA
Configuration package SIWAREX WP251 on CD-ROM for TIA Portal V12 • "Ready for use" software for operating a scale with SIWAREX WP251 and a touch panel (in a variety of languages) • SIWATOOL V7.0 • Device manuals (PDF files in a variety of languages)	7MH4960-6AK01	SIWAREX JB junction box, stainless steel housing (ATEX) For parallel connection of up to 4 load cells (for zone allocation, see manual or type examination certificate).	7MH4710-1EA01
Calibration set for SIWAREX WP2xx For verification of up to 3 scales comprising: • 3 x inscription foil for labeling • 1 x protection foil • 10 x EU verification marks (black M on green background) • Guidelines for verification, certificates and approvals, adaptable label, SIWAREX WP	7MH4960-0AY10	Ex interface SIWAREX IS For intrinsically-safe connection of load cells. With ATEX approval (not UL/FM). Suitable for SIWAREX electronic weighing system. Compatibility of load cells must be checked. • Short-circuit current < 199 mA DC • Short-circuit current < 137 mA DC	7MH4710-5BA 7MH4710-5CA
Ethernet cable patch cord 2 m (7 ft) For connecting SIWAREX WP251 to a PC (SIWATOOL), SIMATIC CPU, panel, etc.	6XV1850-2GH20	Cable (optional) Cable Li2Y 1 x 2 x 0.75 ST + 2 x (2 x 0.34 ST) – CY For connecting SIWAREX electronic weighing systems to junction box (JB), extension box (EB) and Ex interface or between two JB's. For permanent installation. Occasional bending is possible. External diameter: approx. 10.8 mm (0.43 in) Permissible ambient temperature -40 ... +80 °C (-40 ... +176 °F). Sold by the meter. • Sheath color: orange • For potentially explosive atmospheres. Sheath color: blue	7MH4702-8AG 7MH4702-8AF
Remote display (optional) The digital remote displays can be connected directly to the SIWAREX WP251 via the RS 485 interface. Suitable remote display: S102 Siebert Industrieelektronik GmbH Postfach 1180 D-66565 Eppelborn, Germany Tel.: +49 6806/980-0 Fax: +49 6806/980-999 Internet: http://www.siebert.de Detailed information is available from the manufacturer.		Ground terminal for connecting the load cell cable shield to the grounded DIN rail	6ES5728-8MA11

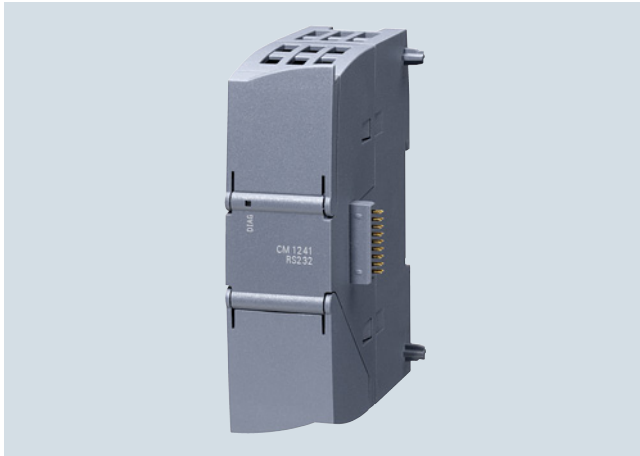
SIMATIC S7-1200 Basic Controllers

I/O modules

Communication

CM 1241 communication modules

Overview



- For quick, high-performance serial data exchange via point-to-point connection
- Implemented protocols: ASCII, USS drive protocol, Modbus RTU, 3964(R)
- Additional protocols can also be loaded
- Simple parameterization with STEP 7 Basic

Technical specifications

Article number	6ES7241-1CH32-0XB0 COMMUNICATION MODULE CM 1241, RS 422/485	6ES7241-1AH32-0XB0 COMMUNICATION MODULE CM 1241, RS 232
General information		
Product type designation	CM 1241 RS 422/485	CM 1241 RS 232
Supply voltage		
Rated value (DC)		
• 24 V DC	Yes	Yes
Input current		
Current consumption, max.	220 mA; From backplane bus 5 V DC	200 mA; From backplane bus 5 V DC
Power loss		
Power loss, typ.	1.1 W	1.1 W
Interfaces		
Number of interfaces	1	1
Interface physics, RS 232C (V.24)		Yes
Interface (physical) RS 422/485 (X.27)	Yes	
Point-to-point		
• Cable length, max.	1 000 m	10 m
Integrated protocol driver		
- Freeport	Yes	Yes
- ASCII	Yes; Available as library function	Yes; Available as library function
- Modbus	Yes	Yes
- Modbus RTU master	Yes	Yes
- MODBUS RTU slave	Yes	Yes
- USS	Yes; Available as library function	
Protocols		
Integrated protocols		
Freeport		
- Telegram length, max.	1 kbyte	1 kbyte
- Bits per character	7 or 8	7 or 8
- Number of stop bits	1 (Standard), 2	1 (Standard), 2
- Parity	No parity (standard); even, uneven, mark (parity bit always 1); space (parity bit always 0)	No parity (standard); even, uneven, mark (parity bit always 1); space (parity bit always 0)
3964 (R)		
- Telegram length, max.	1 kbyte	1 kbyte
- Bits per character	7 or 8	7 or 8
- Number of stop bits	1 (Standard), 2	1 (Standard), 2
- Parity	No parity (standard); even, uneven, mark (parity bit always 1); space (parity bit always 0)	No parity (standard); even, uneven, mark (parity bit always 1); space (parity bit always 0)

Technical specifications (continued)

Article number	6ES7241-1CH32-0XB0 COMMUNICATION MODULE CM 1241, RS 422/485	6ES7241-1AH32-0XB0 COMMUNICATION MODULE CM 1241, RS 232
Modbus RTU master		
- Address area	1 through 49 999 (Standard Modbus addressing)	1 through 49 999 (Standard Modbus addressing)
- Number of slaves, max.	247; slave numbers 1 through 247, per MODBUS network segment maximum 32 devices, additional repeaters needed to expand the network to maximum configuration	247; slave numbers 1 through 247, per MODBUS network segment maximum 32 devices, additional repeaters needed to expand the network to maximum configuration
MODBUS RTU slave		
- Address area	1 through 49 999 (Standard Modbus addressing)	1 through 49 999 (Standard Modbus addressing)
Interrupts/diagnostics/ status information		
Diagnostic functions	Yes	Yes
Diagnostics indication LED		
• for status of the outputs	Yes	Yes
Degree and class of protection		
Degree of protection acc. to EN 60529		
• IP20	Yes	Yes
Standards, approvals, certificates		
CE mark	Yes	Yes
cULus	Yes	Yes
FM approval	Yes	Yes
RCM (formerly C-TICK)	Yes	Yes
KC approval	Yes	Yes
Ambient conditions		
Free fall		
• Fall height, max.	0.3 m; five times, in product package	0.3 m; five times, in product package
Ambient temperature during operation		
• permissible temperature range	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting, 95% humidity, non-condensing	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting, 95% humidity, non-condensing
Dimensions		
Width	30 mm	30 mm
Height	100 mm	100 mm
Depth	75 mm	75 mm
Weights		
Weight, approx.	155 g	150 g

Ordering data

Article No.

Article No.

CM 1241 communication module

Communication module for point-to-point connection, with one RS 422/485 interface

6ES7241-1CH32-0XB0

Communication module for point-to-point connection, with one RS 232 interface

6ES7241-1AH32-0XB0

Accessories

Front flap set (spare part)

For communication modules

6ES7291-1CC30-0XA0

SIMATIC S7-1200 Basic Controllers

I/O modules

Communication

CB 1241 RS 485 communication boards

Overview

- For fast, high-performance serial data exchange via point-to-point connection
- Implemented protocols: ASCII, USS drive protocol, Modbus RTU
- Additional protocols can be loaded later
- Simple parameterization with STEP 7 Basic
- Can be plugged directly into the CPU

Technical specifications

Article number	6ES7241-1CH30-1XB0 COMMUNICATION BOARD CB 1241, RS 485
General information	
Product type designation	CB 1241 RS 485
Input current	
from backplane bus 5 V DC, typ.	50 mA
Power loss	
Power loss, typ.	1.5 W
Interfaces	
Point-to-point	
• Cable length, max.	1 000 m
Integrated protocol driver	
- Freeport	Yes
- ASCII	Yes; Available as library function
- Modbus	Yes
- Modbus RTU master	Yes
- MODBUS RTU slave	Yes
- USS	Yes; Available as library function
Protocols	
Integrated protocols	
Freeport	
- Telegram length, max.	1 kbyte
- Bits per character	7 or 8
- Number of stop bits	1 (Standard), 2
- Parity	No parity (standard); even, uneven, mark (parity bit always 1); space (parity bit always 0)
3964 (R)	
- Telegram length, max.	1 kbyte
- Bits per character	7 or 8
- Number of stop bits	1 (Standard), 2
- Parity	No parity (standard); even, uneven, mark (parity bit always 1); space (parity bit always 0)
Modbus RTU master	
- Address area	1 through 49 999 (Standard Modbus addressing)
- Number of slaves, max.	247; slave numbers 1 through 247, per MODBUS network segment maximum 32 devices, additional repeaters needed to expand the network to maximum configuration
MODBUS RTU slave	
- Address area	1 through 49 999 (Standard Modbus addressing)

Article number	6ES7241-1CH30-1XB0 COMMUNICATION BOARD CB 1241, RS 485
Interrupts/diagnostics/ status information	
Diagnostic functions	Yes
Degree and class of protection	
Degree of protection acc. to EN 60529	
• IP20	Yes
Standards, approvals, certificates	
CE mark	Yes
cULus	Yes
FM approval	Yes
RCM (formerly C-TICK)	Yes
KC approval	Yes
Ambient conditions	
Free fall	
• Fall height, max.	0.3 m; five times, in product package
Ambient temperature during operation	
• permissible temperature range	-20 °C to +60 °C horizontal mounting, -20 °C to 50 °C vertical mounting, 95% humidity, non- condensing
Mechanics/material	
Enclosure material (front)	
• Plastic	Yes
Dimensions	
Width	38 mm
Height	62 mm
Depth	21 mm
Weights	
Weight, approx.	40 g

Ordering data

CB 1241 RS 485 communication board

For point-to-point connection,
with 1 RS 485 interface

Article No.

6ES7241-1CH30-1XB0

Article No.

Accessories

Terminal block (spare part)

For signal board
With 6 screws, gold-plated; 4 pcs.**6ES7292-1BF30-0XA0**

Overview



The CM 1242-5 communication module is used to connect a SIMATIC S7-1200 to PROFIBUS as a DP slave and has the following characteristics:

- PROFIBUS DPV1 slave in accordance with IEC 61158
- Module replacement without PG supported
- Power is supplied via the backplane bus so that no extra cabling is required
- Support of all standard baud rates from 9.6 Kbit/s to 12 Mbit/s
- Compact industry-standard enclosure in S7-1200 design for mounting on a standard mounting rail
- Fast commissioning thanks to easy configuration using STEP 7 without additional programming overhead

The CM 1242-5 is intended for use in factory automation. Low-cost PROFIBUS-based automation solutions can be created on the basis of the S7-1200 for optimal production.

DP-M	DP-S	FMS	PG/OP	S7
	●			

Technical specifications

Article number	6GK7242-5DX30-0XE0
Product type designation	CM 1242-5
Transmission rate	
Transfer rate	
• at the 1st interface acc. to PROFIBUS	9.6 kbit/s ... 12 Mbit/s
Interfaces	
Number of interfaces acc. to Industrial Ethernet	0
Number of electrical connections	
• at the 1st interface acc. to PROFIBUS	1
• for power supply	0
Type of electrical connection	
• at the 1st interface acc. to PROFIBUS	9-pin Sub-D socket (RS 485)
Supply voltage, current consumption, power loss	
Type of voltage of the supply voltage	DC
Supply voltage 1 from backplane bus	5 V
Consumed current	
• from backplane bus at DC at 5 V typical	0.15 A
Power loss [W]	0.75 W

Article number	6GK7242-5DX30-0XE0
Product type designation	CM 1242-5
Permitted ambient conditions	
Ambient temperature	
• for vertical installation during operation	0 ... 45 °C
• for horizontally arranged busbars during operation	0 ... 55 °C
• during storage	-40 ... +70 °C
• during transport	-40 ... +70 °C
Relative humidity at 25 °C without condensation during operation maximum	95 %
Protection class IP	IP20
Design, dimensions and weight	
Module format	Compact module S7-1200 single width
Width	30 mm
Height	100 mm
Depth	75 mm
Net weight	0.115 kg
Mounting type	
• 35 mm DIN rail mounting	Yes
• S7-300 rail mounting	No
• wall mounting	Yes

SIMATIC S7-1200 Basic Controllers

I/O modules

Communication

CM 1242-5

Technical specifications (continued)

Article number	6GK7242-5DX30-0XE0
Product type designation	CM 1242-5
Product properties, functions, components general	
Number of units	
• per CPU maximum	3
Performance data PROFIBUS DP	
Service as DP slave	
• DPV0	Yes
• DPV1	Yes
Amount of data	
• of the address area of the inputs as DP slave total	240 byte
• of the address area of the outputs as DP slave total	240 byte
Performance data telecontrol	
Protocol is supported	
• TCP/IP	No
Configuration software	
• required	STEP 7 Basic/Professional V11 (TIA Portal) or higher

Ordering data

Article No.

CM 1242-5 communication module

Communication module for electrical connection of SIMATIC S7-1200 to PROFIBUS as a DP slave module

6GK7242-5DX30-0XE0

Accessories

PROFIBUS FastConnect connection plug RS 485

With 90° cable outlet; insulation displacement technology, max. transmission rate 12 Mbit/s

- Without PG interface
- With PG interface

6ES7972-0BA52-0XA0
6ES7972-0BB52-0XA0

PROFIBUS FC Standard Cable

2-core bus cable, shielded, special design for fast mounting, sold by the meter; delivery unit: max. 1000 m, minimum order 20 m

6XV1830-0EH10

PROFIBUS FastConnect Stripping Tool

Stripping tool for fast stripping of the PROFIBUS FastConnect bus cable

6GK1905-6AA00

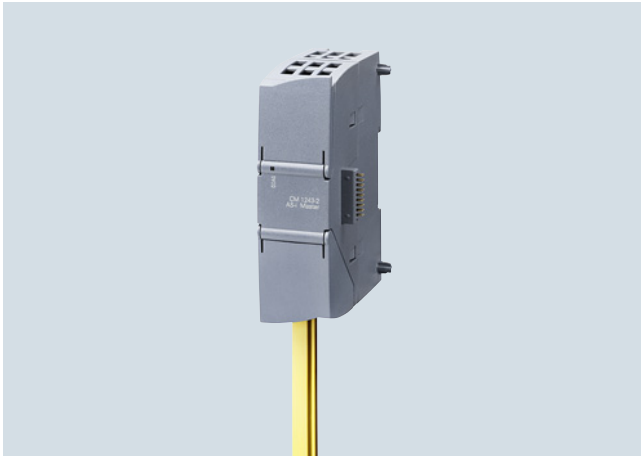
PROFIBUS bus terminal 12M

Bus terminal for connection of PROFIBUS nodes at up to 12 Mbit/s with connecting cable

6GK1500-0AA10

Note:

You can find order information for software in the Catalog IK PI or in the Industry Mall.

Overview

CM 1243-2 communication module for S7-1200

The CM 1243-2 communication module is the AS-Interface master for the SIMATIC S7-1200 and has the following features:

- As many as 62 AS-Interface slaves can be connected
- Integrated analog value transmission
- Supports all AS-Interface master functions in accordance with the AS-Interface specification V3.0
- Indication of the operating state on the front of the device displayed via LED
- Display of operating mode, AS-Interface voltage faults, configuration faults and peripheral faults via LED behind the front panel
- Compact enclosure in the design of the SIMATIC S7-1200
- Suitable for AS-i Power24V: In combination with the optional DCM 1271 data decoupling module, a standard 24 V power supply unit can be used
- Configuration and diagnostics via the TIA portal

Design

The CM 1243-2 communication module is positioned to the left of the S7-1200 CPU and linked to the S7-1200 via lateral contacts.

It has:

- Terminals for two AS-i cables (internally jumpered) via two screw terminals each
- One terminal for connection to the functional ground
- LEDs for indication of the operating state and fault statuses of the connected slaves

The screw terminals (included in scope of supply) can be removed to facilitate installation.

Function

The CM 1243-2 supports all specified functions of the AS-Interface Specification V3.0.

The values of the digital AS-i slaves can be activated via the process image of the S7-1200. During configuration of the slaves in the TIA Portal, the values of the analog AS-i slaves can also be accessed directly in the process image.

It is also possible to exchange all data of the AS-i master and the connected AS-i slaves with the S7-1200 via the data record interface.

Changeover of the operating mode, automatic application of the slave configuration and the re-addressing of a connected AS-i slave can be implemented via the control panel of the CM 1243-2 in the TIA Portal.

The optional DCM 1271 data decoupling module (see "Accessories" and Catalog IC 10) has an integrated detection unit for detecting ground faults on the AS-Interface cable. The integrated overload protection also disconnects the AS-Interface cable if the drive current required exceeds 4 A.

Notes on safety

In order to protect plants, systems, machines and networks against cyber threats, it is necessary to implement – and continuously maintain – a holistic, state-of-the-art industrial security concept. Siemens products and solutions only represent one component of such a concept.

For more information on the topic of Industrial Security, see <http://www.siemens.com/industrialsecurity>.

Configuration

To configure CM 1243-2, you require STEP 7 V11 + SP2 or higher.

For STEP 7 V11 + SP2 or higher, the additional Hardware Support Package for CM 1243-2 is required. This is available from the Industry Online Support Portal. See <https://support.industry.siemens.com/cs/ww/en/view/54164095>.

The software enables user-friendly configuration and diagnostics of the AS-i master and any connected slaves.

Alternatively, you can also apply the AS-Interface ACTUAL configuration at the "touch of a button" via the control panel integrated in the TIA portal/STEP7.

When operated on a S7-1200 CPU with firmware version V4.0 or higher, the firmware version V1.1 (or higher) is required for the CM 1243-2.

SIMATIC S7-1200 Basic Controllers

I/O modules

Communication

CM 1243-2

Ordering data

Article No.

Article No.

CM 1243-2 communication module

- AS-Interface master for SIMATIC S7-1200
- Corresponds to AS-Interface Specification V3.0
- With screw terminals, removable terminals (included in the scope of supply)
- Dimensions (W × H × D / mm): 30 × 100 × 75

3RK7243-2AA30-0XB0

Accessories

DCM 1271 data decoupling module

- With screw terminals, removable terminals (included in the scope of supply)
- Dimensions (W × H × D / mm): 30 × 100 × 75

3RK7271-1AA30-0AA0

Screw terminals (replacement)

- 5-pole for AS-i master CM 1243-2 and AS-i DCM 1271 data decoupling unit
- With screw terminals
- 3-pole for AS-i DCM 1271 data decoupling unit for connecting the power supply unit
- With screw terminals

3RK1901-3MA00

3RK1901-3MB00

AS-interface addressing unit V3.0

- For AS-Interface modules and sensors and actuators with integrated AS-Interface according to AS-i Specification V3.0
- For setting the AS-i address of standard slaves, and slaves with extended addressing mode (A/B slaves)
- With input/output test function and many other commissioning functions
- Battery operation with four type AA batteries (IEC LR6, NEDA 15)
- Degree of protection IP40
- Dimensions (W × H × D / mm): 84 × 195 × 35
- Scope of supply:
 - Addressing unit with 4 batteries
 - Addressing cable, with M12 plug to addressing plug (hollow plug), length 1.5 m

3RK1904-2AB02

More information

For manuals, see <https://support.industry.siemens.com/cs/ww/en/ps/15750/man>.

Overview

DP-M	DP-S	FMS	PG/OP	S7
●			●	●

The CM 1243-5 communication module is used to connect a SIMATIC S7-1200 to PROFIBUS as a DP master and has the following characteristics:

- PROFIBUS DPV1 master in accordance with IEC 61158
- Support of up to 16 PROFIBUS DP slaves
- Communication with other S7 controllers based on S7 communication
- Allows the connection of programming devices and operator panels with a PROFIBUS interface to the S7-1200
- Module replacement without PG supported
- Support of all standard baud rates from 9.6 Kbit/s to 12 Mbit/s
- Compact industry-standard enclosure in S7-1200 design for mounting on a standard mounting rail
- Fast commissioning thanks to easy configuration using STEP 7 without additional programming overhead

The CM 1243-5 is intended for use in factory automation. Low-cost PROFIBUS-based automation solutions can be created on the basis of the S7-1200 for optimal production.

Technical specifications

Article number	6GK7243-5DX30-0XE0
Product type designation	CM 1243-5
Transmission rate	
Transfer rate	
• at the 1st interface acc. to PROFIBUS	9.6 kbit/s ... 12 Mbit/s
Interfaces	
Number of interfaces acc. to Industrial Ethernet	0
Number of electrical connections	
• at the 1st interface acc. to PROFIBUS	1
• for power supply	1
Type of electrical connection	
• at the 1st interface acc. to PROFIBUS	9-pin Sub-D socket (RS 485)
• for power supply	3-pole terminal block
Supply voltage, current consumption, power loss	
Type of voltage of the supply voltage	DC
Supply voltage external	24 V
Supply voltage external at DC Rated value	24 V
Relative positive tolerance at DC at 24 V	20%
Relative negative tolerance at DC at 24 V	20%
Consumed current	
• from external supply voltage at DC at 24 V typical	0.1 A
Power loss [W]	2.4 W

Article number	6GK7243-5DX30-0XE0
Product type designation	CM 1243-5
Permitted ambient conditions	
Ambient temperature	
• for vertical installation during operation	0 ... 45 °C
• for horizontally arranged busbars during operation	0 ... 55 °C
• during storage	-40 ... +70 °C
• during transport	-40 ... +70 °C
Relative humidity at 25 °C without condensation during operation maximum	95%
Protection class IP	IP20
Design, dimensions and weight	
Module format	Compact module S7-1200 single width
Width	30 mm
Height	100 mm
Depth	75 mm
Net weight	0.134 kg
Mounting type	
• 35 mm DIN rail mounting	Yes
• S7-300 rail mounting	No
• wall mounting	Yes
Product properties, functions, components general	
Number of units	
• per CPU maximum	3

SIMATIC S7-1200 Basic Controllers

I/O modules

Communication

CM 1243-5

Technical specifications (continued)

Article number	6GK7243-5DX30-0XE0
Product type designation	CM 1243-5
Performance data PROFIBUS DP	
Service as DP master	
• DPV1	Yes
Number of DP slaves on DP master usable	16
Amount of data	
• of the address area of the inputs as DP master total	512 byte
• of the address area of the outputs as DP master total	512 byte
• of the address area of the inputs per DP slave	244 byte
• of the address area of the outputs per DP slave	244 byte
• of the address area of the diagnostic data per DP slave	240 byte
Service as DP slave	
• DPV0	No
• DPV1	No
Performance data S7 communication	
Number of possible connections for S7 communication	
• maximum	8
• with PG connections maximum	1
• with PG/OP connections maximum	3
• Note	max. 4 connections to other S7 stations
Performance data multi-protocol mode	
Number of active connections with multi-protocol mode	
• without DP maximum	8
• with DP maximum	8
Performance data telecontrol	
Protocol is supported	
• TCP/IP	No
Configuration software	
• required	STEP 7 Basic/Professional V11 (TIA Portal) or higher

Ordering data

Article No.

CM 1243-5 communication module

Communication module for electrical connection of SIMATIC S7-1200 to PROFIBUS as a DPV1 master

6GK7243-5DX30-0XE0

Accessories

PROFIBUS FastConnect connection plug RS 485

With 90° cable outlet; insulation displacement technology, max. transmission rate 12 Mbit/s

- Without PG interface
- With PG interface

6ES7972-0BA52-0XA0
6ES7972-0BB52-0XA0

PROFIBUS FC Standard Cable

2-core bus cable, shielded, special design for fast mounting, delivery unit: max. 1000 m, minimum order 20 m, sold by the meter

6XV1830-0EH10

PROFIBUS FastConnect Stripping Tool

Stripping tool for fast stripping of the PROFIBUS FastConnect bus cable

6GK1905-6AA00

PROFIBUS bus terminal 12M

Bus terminal for connection of PROFIBUS nodes up to 12 Mbps with connecting cable

6GK1500-0AA10

Note:

You can find order information for software in the Catalog IK PI or in the Industry Mall.

Overview

- Unmanaged switch for connecting a SIMATIC S7-1200 to an Industrial Ethernet network with a line, tree or star topology
- Multiplication of Ethernet interfaces on a SIMATIC S7-1200 for additional connection of up to three programming devices, operator controls, and further Ethernet nodes
- Simple, space-saving mounting on the SIMATIC S7-1200 mounting rail
- Low-cost solution for implementing small, local Ethernet networks
- Connection without any problems using RJ45 plugs
- Simple and fast status display via LEDs on the device
- Integral autocrossover function permits use of uncrossed connecting cables

3**Technical specifications**

Article number	6GK7277-1AA10-0AA0
Product type designation	CSM 1277
Transmission rate	
Transfer rate	10 Mbit/s, 100 Mbit/s
Interfaces for communication integrated	
Number of electrical connections	4
• for network components or terminal equipment	
Number of 100 Mbit/s SC ports	
• for multimode	
Number of 1000 Mbit/s LC ports	
• for multimode	0
• for single mode (LD)	0
Interfaces others	
Number of electrical connections	1
• for power supply	
Type of electrical connection	
• for power supply	3-pole terminal block
Supply voltage, current consumption, power loss	
Type of voltage of the supply voltage	DC
Supply voltage	24 V
• external	
• external	19.2 ... 28.8 V
Product component fusing at power supply input	Yes
Fuse protection type at input for supply voltage	0.5 A / 60 V
Consumed current maximum	0.07 A
Power loss [W]	1.6 W
• at DC at 24 V	

Article number	6GK7277-1AA10-0AA0
Product type designation	CSM 1277
Permitted ambient conditions	
Ambient temperature	
• during operation	0 ... 60 °C
• during storage	-40 ... +70 °C
• during transport	-40 ... +70 °C
Relative humidity	
• at 25 °C without condensation during operation maximum	95%
Protection class IP	IP20
Design, dimensions and weight	
Design	SIMATIC S7-1200 device design
Width	45 mm
Height	100 mm
Depth	75 mm
Net weight	0.15 kg
Mounting type	
• 35 mm DIN rail mounting	Yes
• wall mounting	Yes
• S7-300 rail mounting	No
• S7-1500 rail mounting	No
Product functions management, configuration	
Product function	
• multiport mirroring	No
• switch-managed	No
Product functions Redundancy	
Product function	
• Parallel Redundancy Protocol (PRP)/ operation in the PRP-network	Yes
• Parallel Redundancy Protocol (PRP)/ Redundant Network Access (RNA)	No

SIMATIC S7-1200 Basic Controllers

I/O modules

Communication

CSM 1277 unmanaged**Technical specifications** (continued)

Article number	6GK7277-1AA10-0AA0
Product type designation	CSM 1277
Standards, specifications, approvals	
Standard	
• for FM	FM3611: Class 1, Division 2, Group A, B, C, D / T., CL.1, Zone 2, GP. IIC, T. Ta
• for hazardous zone	EN 60079-15:2005, EN 60079-0:2006, II 3 G Ex nA II T4, KEMA 08 ATEX 0003 X
• for safety from CSA and UL	UL 508, CSA C22.2 No. 142
• for emitted interference	EN 61000-6-4 (Class A)
• for interference immunity	EN 61000-6-2
Certificate of suitability CE marking	Yes
Certificate of suitability	EN 61000-6-2, EN 61000-6-4
• C-Tick	Yes
• KC approval	No
Marine classification association	
• American Bureau of Shipping Europe Ltd. (ABS)	Yes
• Bureau Veritas (BV)	Yes
• Det Norske Veritas (DNV)	Yes
• Germanische Lloyd (GL)	No
• Lloyds Register of Shipping (LRS)	Yes
• Nippon Kaiji Kyokai (NK)	Yes
• Polski Rejestr Statkow (PRS)	No
• Royal Institution of Naval Architects (RINA)	No
MTBF at 40 °C	273 y

Ordering data**Article No.****CSM 1277 compact switch module**

Unmanaged switch for connecting a SIMATIC S7-1200 and up to three further nodes to Industrial Ethernet with 10/100 Mbit/s; 4 x RJ45 ports; external 24 V DC power supply, diagnostics on LEDs, S7-1200 module including electronic manual on CD-ROM

6GK7277-1AA10-0AA0**Accessories****IE FC TP Trailing Cable 2 x 2 (Type C)**

4-core, shielded TP installation cable for connection to IE FC outlet RJ45/ IE FC RJ45 Plug 180/90 for use as trailing cable; PROFINET-compatible; with UL approval; sold by the meter; max. delivery unit 1000 m, minimum order quantity 20 m

6XV1840-3AH10**IE FC RJ45 Plug 180 2 x 2**

RJ45 plug connector for Industrial Ethernet with a rugged metal housing and integrated insulation displacement contacts for connecting Industrial Ethernet FC installation cables; 180° cable outlet; for network components and CPUs/CPUs with Industrial Ethernet interface

- 1 pack = 1 unit
- 1 pack = 10 units
- 1 pack = 50 units

6GK1901-1BB10-2AA0
6GK1901-1BB10-2AB0
6GK1901-1BB10-2AE0

IE FC outlet RJ45

For connecting Industrial Ethernet FC cables and TP cords; graduated prices for 10 and 50 units or more

6GK1901-1FC00-0AA0**IE TP Cord RJ45/RJ45**

- TP cord pre-assembled with 2 RJ45 connectors; length: 0.5 m
- TP cable 4 x 2 with 2 RJ45 connectors; length: 0.5 m

6XV1850-2GE50**6XV1870-3QE50**

Overview

The CP 1243-1 communications processor is used for connecting a SIMATIC S7-1200 to the TeleControl Server Basic control center software via Ethernet, and for safe communication via IP-based networks.

The CP has the following features:

- Ethernet-based connection to TeleControl Server Basic, e.g. via Internet
- Data transfer of measured values, control variables, or alarms optimized for telecontrol systems
- Automatic sending of alert emails
- Data buffering of up to 64,000 values ensures a secure database even with temporary connection failures
- Secure communication via VPN connections based on IPSec
- Access protection via Stateful Inspection Firewall
- Clearly laid out LED signaling for fast and easy diagnostics
- Compact industrial enclosure in S7-1200 design for mounting on a standard mounting rail
- Fast commissioning thanks to easy configuration using STEP 7

3**Technical specifications**

Article number	6GK7243-1BX30-0XE0
Product type designation	CP 1243-1
Transmission rate	
Transfer rate	
• at the 1st interface	10 ... 100 Mbit/s
Interfaces	
Number of interfaces acc. to Industrial Ethernet	1
Number of electrical connections	
• at the 1st interface acc. to Industrial Ethernet	1
• for power supply	0
Type of electrical connection	
• at the 1st interface acc. to Industrial Ethernet	RJ45 port
Supply voltage, current consumption, power loss	
Type of voltage of the supply voltage	DC
Supply voltage 1 from backplane bus	5 V
Consumed current	
• from backplane bus at DC at 5 V typical	0.25 A
Power loss [W]	1.25 W

Article number	6GK7243-1BX30-0XE0
Product type designation	CP 1243-1
Permitted ambient conditions	
Ambient temperature	
• for vertical installation during operation	-20 ... +60 °C
• for horizontally arranged busbars during operation	-20 ... +70 °C
• during storage	-40 ... +70 °C
• during transport	-40 ... +70 °C
Relative humidity at 25 °C without condensation during operation maximum	95%
Protection class IP	IP20
Design, dimensions and weight	
Module format	Compact module S7-1200 single width
Width	30 mm
Height	110 mm
Depth	75 mm
Net weight	0.122 kg
Mounting type	
• 35 mm DIN rail mounting	Yes
• wall mounting	Yes

SIMATIC S7-1200 Basic Controllers

I/O modules

Communication

CP 1243-1

Technical specifications (continued)

Article number	6GK7243-1BX30-0XE0
Product type designation	CP 1243-1
Product properties, functions, components general	
Number of units	
• per CPU maximum	3
Performance data S7 communication	
Number of possible connections for S7 communication	
• Note	like CPU
Performance data telecontrol	
Suitability for use	
• Node station	No
• substation	Yes
• TIM control center	No
Control center connection	to be used with Telecontrol Server Basic
• by means of a permanent connection	supported
• Note	Connection to SCADA system via Telecontrol Server Basic
Protocol is supported	
• DNP3	No
• IEC 60870-5	No
Product function data buffering if connection is aborted	Yes; 64,000 values
Number of data points per station maximum	200
Performance data Teleservice	
Diagnostics function online diagnostics with SIMATIC STEP 7	Yes
Product function	
• program download with SIMATIC STEP 7	Yes
• Remote firmware update	Yes
Configuration software	
• required	STEP 7 Basic/Professional V13 Update 2 + HSP (TIA Portal) or higher

Article number	6GK7243-1BX30-0XE0
Product type designation	CP 1243-1
Product functions Security	
Firewall version	stateful inspection
Product function with VPN connection	IPSec
Type of encryption algorithms with VPN connection	AES-256, AES-192, AES-128, 3DES-168
Type of authentication procedure with VPN connection	Preshared key (PSK), X.509v3 certificates
Type of hashing algorithms with VPN connection	MD5, SHA-1
Number of possible connections with VPN connection	8
Product function	
• password protection for Web applications	No
• password protection for teleservice access	No
• encrypted data transmission	Yes
• ACL - IP-based	No
• ACL - IP-based for PLC/routing	No
• switch-off of non-required services	Yes
• Blocking of communication via physical ports	No
• log file for unauthorized access	No
Product functions Time	
Protocol is supported	
• NTP	Yes
time synchronization	
• from control center	Yes

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SIMATIC S7-1200 Basic Controllers

I/O modules

Communication

CP 1242-7 V2 GPRS

Overview



The CP 1242-7 GPRS V2 communications processor is used to connect a SIMATIC S7-1200 to the globally available GSM/GPRS mobile radio network and has the following characteristics:

- Worldwide wireless exchange of data between S7-1200 controllers and/or between S7-1200 controllers and control centers with an Internet connection
- Communication based on the GPRS (**G**eneral **P**acket **R**adio **S**ervice) mobile wireless service with data transmission speeds of up to 86 Kbit/s in the downlink and 43 Kbit/s in the uplink
- GPRS mode with fixed IP addresses and dynamic IP addresses with standard mobile phone contract
- Time synchronization on the basis of NTP (**N**etwork **T**ime **P**rotocol)
- Sending and receiving of text messages
- LED signaling for fast diagnostics
- Compact industrial enclosure in S7-1200 design for mounting on a standard mounting rail
- Fast commissioning thanks to easy configuration using STEP 7

In conjunction with the TeleControl Server Basic software, the CP 1242-7 forms a telecontrol system with additional properties:

- Connection of up to 5000 telecontrol stations to the control center via an OPC interface
- Data buffering in the substations in the event of connection failures
- Central status monitoring of the substations
- No special provider services required for fixed IP addresses
- Teleservice access with STEP 7 to the substations via the Internet

The CP 1242-7 V2 is a new product version of the CP 1242-7. The concept for process data transmission has been expanded with a simple data point configuration, which enables substantially easier commissioning without high programming overhead and minimizes susceptibility to errors during the projects implementation phase. CP 1242-7 has also been equipped with new functions, such as access to the internal Web server of the S7-1200. This opens up numerous new application areas.

Technical specifications

Article number	6GK7242-7KX31-0XE0
Product type designation	CP 1242-7 V2
Transmission rate	
Transfer rate	
• for GPRS transmission	
- with downlink maximum	86 kbit/s
- with uplink maximum	43 kbit/s
Interfaces	
Number of interfaces acc. to Industrial Ethernet	0
Number of electrical connections	
• for external antenna(s)	1
• for power supply	1
Number of slots	
• for SIM cards	1
Type of electrical connection	
• for external antenna(s)	SMA socket (50 ohms)
• for power supply	3-pole terminal block
Slot version	
• for SIM card	Standard

Article number	6GK7242-7KX31-0XE0
Product type designation	CP 1242-7 V2
Wireless technology	
Type of mobile wireless service	
• is supported SMS	Yes
• is supported GPRS	Yes
• Note	GPRS (Multislot Class 10)
Type of mobile network is supported	
• GSM	Yes
• UMTS	No
• LTE	No
Operating frequency	
• 850 MHz	Yes
• 900 MHz	Yes
• 1800 MHz	Yes
• 1900 MHz	Yes
Transmit power	
• at operating frequency 900 MHz	2 W
• at operating frequency 1800 MHz	1 W
• at operating frequency 1900 MHz	1 W

Technical specifications (continued)

Article number	6GK7242-7KX31-0XE0
Product type designation	CP 1242-7 V2
Supply voltage, current consumption, power loss	
Type of voltage of the supply voltage	DC
Supply voltage external	24 V
Supply voltage external at DC Rated value	24 V
Relative positive tolerance at DC at 24 V	20%
Relative negative tolerance at DC at 24 V	20%
Consumed current	
• from external supply voltage at DC at 24 V typical	0.1 A
• from external supply voltage at DC at 24 V maximum	0.22 A
Power loss [W]	2.4 W
Permitted ambient conditions	
Ambient temperature	
• for vertical installation during operation	-20 ... +60 °C
• for horizontally arranged busbars during operation	-20 ... +70 °C
• during storage	-40 ... +70 °C
• during transport	-40 ... +70 °C
Relative humidity at 25 °C without condensation during operation maximum	95%
Protection class IP	IP20
Design, dimensions and weight	
Module format	Compact module S7-1200 single width
Width	30 mm
Height	100 mm
Depth	75 mm
Net weight	0.133 kg
Mounting type	
• 35 mm DIN rail mounting	Yes
• S7-300 rail mounting	No
• wall mounting	Yes
Product properties, functions, components general	
Number of units	
• per CPU maximum	3
Performance data	
Number of users/telephone numbers definable maximum	10

Article number	6GK7242-7KX31-0XE0
Product type designation	CP 1242-7 V2
Performance data open communication	
Number of possible connections for open communication	
• by means of T blocks maximum	like CPU
Performance data IT functions	
Number of possible connections	
• as e-mail client maximum	1
Performance data telecontrol	
Control center connection	Telecontrol Server Basic supported
• by means of a permanent connection	
• by means of demand-oriented connection	supported
• Note	Connection to SCADA system using OPC interface
Protocol is supported	
• DNP3	No
• IEC 60870-5	No
Product function data buffering if connection is aborted	Yes; 64,000 values
Performance data Teleservice	
Diagnostics function online diagnostics with SIMATIC STEP 7	Yes
Product function	
• program download with SIMATIC STEP 7	Yes
• Remote firmware update	Yes
Configuration software	
• required	STEP 7 Basic/Professional V13 SP1 or higher
Product functions Diagnosis	
Product function Web-based diagnostics	Yes
Product functions Security	
Product function	
• password protection for teleservice access	Yes
• encrypted data transmission	Yes
Product functions Time	
Protocol is supported	
• NTP	Yes
time synchronization	
• from control center	Yes

SIMATIC S7-1200 Basic Controllers

I/O modules

Communication

CP 1242-7 V2 GPRS

Ordering data

Article No.

Article No.

Communications processor CP 1242-7 V2¹⁾

Communications processor
CP 1242-7 GPRS V2 for connecting
SIMATIC S7-1200 to TeleControl
Server Basic via GSM/GPRS mobile
radio network

6GK7242-7KX31-0XE0

Accessories

TeleControl Server Basic V3.0

Software for 8 to 5000 stations;
single license for one installation;
OPC (UA) server for GPRS and
Ethernet/Internet communication
with SIMATIC S7-1200 and
SIMATIC S7-200 (GPRS only);
connection management to remote
stations; routing for connections
between S7 stations; German and
English operator interface;
for Windows 7 Professional
32/64-bit + Service Pack 1
Windows 7 Enterprise 32/64-bit +
Service Pack 1
Windows 7 Ultimate 32/64-bit +
Service Pack 1
Windows Server 2008 32-bit +
Service Pack 2
Windows Server 2008 R2 Standard
64-bit Service Pack 1

• TeleControl Server Basic 8 V3

Connection management for
8 SIMATIC S7-1200 or
S7-200 stations

6NH9910-0AA21-0AA0

• TeleControl Server Basic 32 V3

Connection management for
32 SIMATIC S7-1200 or
S7-200 stations

6NH9910-0AA21-0AF0

• TeleControl Server Basic 64 V3

Connection management for
64 SIMATIC S7-1200 or
S7-200 stations

6NH9910-0AA21-0AB0

• TeleControl Server Basic 256 V3

Connection management for
256 SIMATIC S7-1200 or
S7-200 stations

6NH9910-0AA21-0AC0

• TeleControl Server Basic 1000 V3

Connection management for
1000 SIMATIC S7-1200 or
S7-200 stations

6NH9910-0AA21-0AD0

• TeleControl Server Basic 5000 V3

Connection management for
5000 SIMATIC S7-1200 or
S7-200 stations

6NH9910-0AA21-0AE0

• TeleControl Server Basic UPGR V3

Upgrade package from Version
V2.x to V3 for all license sizes

6NH9910-0AA21-0GA0

ANT794-4MR antenna

Omnidirectional antenna for
GSM (2G), UMTS (3G) and
LTE (4G) networks;
weather-resistant for indoor and
outdoor use; 5 m connecting cable
with fixed connection to antenna;
SMA connector; including mounting
bracket, screws, wall plugs

6NH9860-1AA00

ANT794-3M antenna

Flat panel antenna for GSM (2G)
networks, for triband with
900/1800/1900 MHz;
weather-resistant for indoor/
outdoor use, 1.2 m connecting
cable with fixed connection to
antenna;
SMA connector, incl. assembly
adhesive tape

6NH9870-1AA00

STEP 7 Basic Engineering Software V13 SP1 (TIA Portal)

See catalog section 11

¹⁾ Note national approvals under
<http://www.siemens.com/mobilenetwork-approvals>

Overview

CP 1243-7 LTE is used to connect the S7-1200 to a mobile wireless network of the 4th Generation LTE (Long Term Evolution). The increased data rates compared to GPRS and widespread introduction of LTE open up new areas of application. The CP1243-7 is characterized by the following properties:

- 1 connection to LTE (4G) mobile wireless network (various versions for EU and North America)
- Data transfer of measured values, control variables, or alarms optimized for telecontrol systems
- Operation with fixed IP addresses and dynamic IP addresses with standard cellular phone contract
- Time synchronization based on NTP (Network Time Protocol)
- On-demand connection buildup via voice call or text message
- Sending and receiving of text messages
- TeleService access with STEP 7 to substations via mobile wireless networks
- Compact industrial enclosure in S7-1200 design for mounting on a standard mounting rail
- Temperature range in operation: -20 °C to +70 °C
- Installation on standard mounting rails
- Diagnostics LEDs (overall status and details)
- Integrated security functions (VPN and firewall)
- Access to the CPU Web server
- Fast commissioning due to simplified configuration with STEP 7
- Data buffering of up to 64,000 values ensures a secure database even with temporary connection failures

3**Technical specifications**

Article number	6GK7243-7KX30-0XE0	6GK7243-7SX30-0XE0
Product type designation	CP 1243-7 LTE EU	CP 1243-7 LTE US
Transmission rate		
Transfer rate		
• for LTE transmission		
- with downlink maximum	42 Mbit/s	42 Mbit/s
- with uplink maximum	5.76 Mbit/s	5.76 Mbit/s
Interfaces		
Number of interfaces acc. to Industrial Ethernet	0	0
Number of electrical connections		
• for external antenna(s)	1	1
• for power supply	1	1
Number of slots		
• for SIM cards	1	1
Type of electrical connection		
• for external antenna(s)	SMA socket (50 ohms)	SMA socket (50 ohms)
• for power supply	3-pole terminal block	3-pole terminal block
Slot version		
• for SIM card	Standard	Standard

Article number	6GK7243-7KX30-0XE0	6GK7243-7SX30-0XE0
Product type designation	CP 1243-7 LTE EU	CP 1243-7 LTE US
Wireless technology		
Type of mobile wireless service		
• is supported SMS	Yes	Yes
• is supported GPRS	Yes	Yes
• Note	GPRS (Multislot Class 10)	GPRS (Multislot Class 10)
Type of mobile network is supported		
• GSM	Yes	Yes
• UMTS	Yes	Yes
• LTE	Yes	Yes
Operating frequency		
• 850 MHz		Yes
• 1900 MHz		Yes
Operating frequency		
• for GSM transmission 900 MHz	Yes	
• for GSM transmission 1800 MHz	Yes	
• with UMTS transmission 900 MHz	Yes	
• with UMTS transmission 2100 MHz	Yes	
• for LTE transmission 700 MHz		Yes
• for LTE transmission 800 MHz	Yes	
• for LTE transmission 1700 MHz		Yes
• for LTE transmission 1800 MHz	Yes	
• for LTE transmission 2600 MHz	Yes	

SIMATIC S7-1200 Basic Controllers

I/O modules

Communication

CP 1243-7 LTE

Technical specifications (continued)

Article number	6GK7243-7KX30-0XE0	6GK7243-7SX30-0XE0
Product type designation	CP 1243-7 LTE EU	CP 1243-7 LTE US
Supply voltage, current consumption, power loss		
Type of voltage of the supply voltage	DC	DC
Supply voltage external	24 V	24 V
Supply voltage external at DC Rated value	24 V	24 V
Relative positive tolerance at DC at 24 V	20%	20%
Relative negative tolerance at DC at 24 V	20%	20%
Consumed current		
• from external supply voltage at DC at 24 V typical	0.1 A	0.1 A
• from external supply voltage at DC at 24 V maximum	0.22 A	0.22 A
Permitted ambient conditions		
Ambient temperature		
• for vertical installation during operation	-20 ... +60 °C	-20 ... +60 °C
• for horizontally arranged busbars during operation	-20 ... +70 °C	-20 ... +70 °C
• during storage	-40 ... +70 °C	-40 ... +70 °C
• during transport	-40 ... +70 °C	-40 ... +70 °C
Relative humidity at 25 °C without condensation during operation maximum	95%	95%
Protection class IP	IP20	IP20
Design, dimensions and weight		
Module format	Compact module S7-1200 single width	Compact module S7-1200 single width
Width	30 mm	30 mm
Height	100 mm	100 mm
Depth	75 mm	75 mm
Net weight	0.133 kg	0.133 kg
Mounting type		
• 35 mm DIN rail mounting	Yes	Yes
• S7-300 rail mounting	No	No
• wall mounting	Yes	Yes
Product properties, functions, components general		
Number of units		
• per CPU maximum	3	3
Performance data		
Number of users/telephone numbers definable maximum	10	10
Performance data open communication		
Number of possible connections for open communication		
• by means of T blocks maximum	like CPU	like CPU
Performance data IT functions		
Number of possible connections		
• as e-mail client maximum	1	1

Article number	6GK7243-7KX30-0XE0	6GK7243-7SX30-0XE0
Product type designation	CP 1243-7 LTE EU	CP 1243-7 LTE US
Performance data telecontrol		
Suitability for use		
• substation	Yes	Yes
Control center connection	Telecontrol Server Basic	Telecontrol Server Basic
• by means of a permanent connection	supported	supported
• by means of demand-oriented connection	supported	supported
• Note	Connection to SCADA system using OPC interface	Connection to SCADA system using OPC interface
Protocol is supported		
• DNP3	No	No
• IEC 60870-5	No	No
Product function data buffering if connection is aborted	Yes; 64,000 values	Yes; 64,000 values
Performance data Teleservice		
Diagnostics function online diagnostics with SIMATIC STEP 7	Yes	Yes
Product function		
• program download with SIMATIC STEP 7	Yes	Yes
• Remote firmware update	Yes	Yes
Configuration software		
• required	STEP 7 Basic/ Professional V13 SP1 or higher	STEP 7 Basic/ Professional V13 SP1 + HSP or higher
Product functions Diagnosis		
Product function Web-based diagnostics	Yes	Yes
Product functions Security		
Firewall version	stateful inspection IPSec	stateful inspection IPSec
Product function with VPN connection		
Type of encryption algorithms with VPN connection	AES-256, AES-192, AES-128, 3DES-168, DES-56	AES-256, AES-192, AES-128, 3DES-168, DES-56
Type of authentication procedure with VPN connection	Preshared key (PSK), X.509v3 certificates	Preshared key (PSK), X.509v3 certificates
Type of hashing algorithms with VPN connection	MD5, SHA-1	MD5, SHA-1
Number of possible connections with VPN connection	1	1
Product function		
• password protection for teleservice access	Yes	Yes
• encrypted data transmission	Yes	Yes
Product functions Time		
Protocol is supported		
• NTP	Yes	Yes
time synchronization		
• from control center	Yes	Yes

Ordering data	Article No.	Article No.	
Communications processor CP 1243-7 LTE Communications processor for connecting SIMATIC S7-1200 to the TeleControl Server Basic via the LTE mobile wireless network • CP 1243-7 LTE EU Frequencies in European band: 700, 1700 MHz • CP 1243-7 LTE US Frequencies in North American band: 800, 1800, 2600 MHz	6GK7243-7KX30-0XE0 6GK7243-7SX30-0XE0	ANT794-4MR antenna Omnidirectional antenna for GSM (2G), UMTS (3G) and LTE (4G) networks; weather-resistant for indoor and outdoor use; 5 m cable with fixed connection to antenna; SMA connector; including mounting bracket, screws, wall plugs	6NH9860-1AA00
STEP 7 Basic Engineering Software (TIA Portal)		See catalog section 11	
Accessories			
TeleControl Server Basic V3.0 Software for 8 to 5000 stations; single license for one installation; OPC (UA) server for GPRS and Ethernet/Internet communication with SIMATIC S7-1200 and SIMATIC S7-200 (GPRS only); connection management to remote stations; routing for connections between S7 stations; German and English operator interface; for Windows 7 Professional 32/64-bit + Service Pack 1 Windows 7 Enterprise 32/64-bit + Service Pack 1 Windows 7 Ultimate 32/64-bit + Service Pack 1 Windows Server 2008 32-bit + Service Pack 2 Windows Server 2008 R2 Standard 64-bit Service Pack 1 • TeleControl Server Basic 8 V3 Connection management for 8 SIMATIC S7-1200 or S7-200 stations • TeleControl Server Basic 32 V3 Connection management for 32 SIMATIC S7-1200 or S7-200 stations • TeleControl Server Basic 64 V3 Connection management for 64 SIMATIC S7-1200 or S7-200 stations • TeleControl Server Basic 256 V3 Connection management for 256 SIMATIC S7-1200 or S7-200 stations • TeleControl Server Basic 1000 V3 Connection management for 1000 SIMATIC S7-1200 or S7-200 stations • TeleControl Server Basic 5000 V3 Connection management for 5000 SIMATIC S7-1200 or S7-200 stations • TeleControl Server Basic UPGR V3 Upgrade package from Version V2.x to V3 for all license sizes	6NH9910-0AA21-0AA0 6NH9910-0AA21-0AF0 6NH9910-0AA21-0AB0 6NH9910-0AA21-0AC0 6NH9910-0AA21-0AD0 6NH9910-0AA21-0AE0 6NH9910-0AA21-0GA0		

SIMATIC S7-1200 Basic Controllers

I/O modules

Communication

CP 1243-8 IRC

Overview



The CP 1243-8 IRC communications processor (Industrial Remote Communication) is used for connecting a SIMATIC S7-1200 in the TeleControl Professional system via the SINAUT ST7 telecontrol protocol to higher-level ST7 stations or to an ST7 control center.

The CP has the following features:

- SINAUT ST7 telecontrol protocol supported
- Two WAN connections for selecting the communication paths:
 - Ethernet-based connection: RJ45 port on the module for connecting external routers, e.g. SCALANCE M
 - Additional connection configurable via plug-in TS modules

- Both WAN interfaces can also be operated simultaneously: Route redundancy
- Data transfer optimized for telecontrol systems on the basis of SINAUT ST7
- Automatic transmission of alarms per email or SMS
- Time synchronization based on NTP (Network Time Protocol) or via the SINAUT system
- Data buffering of up to 16,000 data messages prevents data loss in the event of temporary connection failures
- Secure communication via VPN connections based on IPSec
- Access protection via Stateful Inspection Firewall
- Fast and simple diagnostics via clear LED indicators, STEP 7 and web browser
- Compact industrial enclosure in S7-1200 design for mounting on a standard mounting rail

The integrated Ethernet interface and the option of using the TS modules provide flexible connection options for the CP. The following TS modules are available:

- TS module RS 232
- TS module MODEM
- TS module ISDN and
- TS module GSM.

Technical specifications

Article number	6GK7243-8RX30-0XE0
Product type designation	CP 1243-8 IRC
Transmission rate	
Transfer rate	
• at the 1st interface	10 ... 100 Mbit/s
• at the 2nd interface	0.3 ... 115.2 kbit/s
Interfaces	
Number of interfaces acc. to Industrial Ethernet	1
Number of electrical connections	
• at the 1st interface acc. to Industrial Ethernet	1
• for power supply	1
Type of electrical connection	
• at the 1st interface acc. to Industrial Ethernet	RJ45 port
• at interface 2 for external data transmission	Interface to the TS Module
• for power supply	3-pole terminal block
Supply voltage, current consumption, power loss	
Type of voltage of the supply voltage	DC
Supply voltage 1 from backplane bus	5 V
Supply voltage external	24 V
Supply voltage external	19.2 ... 28.8 V
Supply voltage external at DC Rated value	24 V
Supply voltage external at DC rated value	19.2 ... 28.8 V

Article number	6GK7243-8RX30-0XE0
Product type designation	CP 1243-8 IRC
Consumed current	
• from backplane bus at DC at 5 V typical	0.25 A
• from external supply voltage at DC at 24 V typical	0.1 A
Power loss [W] Note	1.25 W from S7-1200 backplane without TS module. 2.4 W from 24 V DC external with TS module
Power loss [W]	2.4 W
Permitted ambient conditions	
Ambient temperature	
• for vertical installation during operation	-20 ... +60 °C
• for horizontally arranged busbars during operation	-20 ... +70 °C
• during storage	-40 ... -70 °C
• during transport	-40 ... +70 °C
Relative humidity at 25 °C without condensation during operation maximum	95%
Protection class IP	IP20

Technical specifications (continued)

Article number	6GK7243-8RX30-0XE0
Product type designation	CP 1243-8 IRC
Design, dimensions and weight	
Module format	Compact module S7-1200 single width
Width	30 mm
Height	110 mm
Depth	75 mm
Net weight	0.122 kg
Mounting type	
• 35 mm DIN rail mounting	Yes
• S7-300 rail mounting	No
• wall mounting	Yes
Product properties, functions, components general	
Number of units	1
• per CPU maximum	
• Note	One CP pluggable on left side of CPU, one TS Module pluggable left side of CP.
Performance data open communication	
Number of possible connections for open communication	
• by means of T blocks maximum	like CPU
Performance data S7 communication	
Number of possible connections for S7 communication	
• with PG connections maximum	2
• with OP connections maximum	1
• Note	Configured S7-Connection for S7-Communication
Service	
• SINAUT ST7 via S7 communication	Yes
Performance data IT functions	
Number of possible connections	
• as e-mail client maximum	1
Performance data telecontrol	
Suitability for use	
• Node station	No
• substation	Yes
• TIM control center	No
• Note	Ethernet and TS Module can be operated in parallel
Control center connection	control center with ST7 function supported
• by means of a permanent connection	
Protocol is supported	
• SINAUT ST7 protocol	Yes
Product function data buffering if connection is aborted	Yes; 16,000 data messages
Number of data points per station maximum	200
Transmission format	
• for SINAUT ST7 protocol with multi-master polling 10-bit	Yes
• for SINAUT ST7 protocol with polling or spontaneous 10-bit or 11-bit	Yes
Operating mode for scanning of data transmission	
• with dedicated line/radio link with SINAUT ST7 protocol	Polling
• with dial-up network with SINAUT ST7 protocol	spontaneous
Hamming distance	
• for SINAUT ST7 protocol	4

Article number	6GK7243-8RX30-0XE0
Product type designation	CP 1243-8 IRC
Performance data Teleservice	
Diagnostics function online diagnostics with SIMATIC STEP 7	Yes
Product function	
• program download with SIMATIC STEP 7	Yes
• Remote firmware update	Yes
Protocol is supported	
• SNMP v3	Yes
• DCP	Yes
Configuration software	
• required	SINAUT ES V5.5 and STEP7 V13 SP1 or higher
• for PG configuring required SINAUT ST7 configuration software for PG	Yes
Product functions Diagnosis	
Product function Web-based diagnostics	Yes
Product functions Security	
Firewall version	stateful inspection
Suitability for operation Virtual Private Network	Yes
Product function with VPN connection	IPSec
Type of encryption algorithms with VPN connection	AES-256, AES-192, AES-128, 3DES-168, DES-56
Type of authentication procedure with VPN connection	Preshared key (PSK), X.509v3 certificates
Type of hashing algorithms with VPN connection	MD5, SHA-1
Number of possible connections with VPN connection	8
Product function	
• password protection for teleservice access	No
• encrypted data transmission	Yes
• MSC client via GPRS modem with MSC capability	Yes
Protocol	
• is supported MSC protocol	Yes
• with Virtual Private Network MSC is supported	TCP/IP
Key length for MSC with Virtual Private Network	128 bit
Number of possible connections	
• as MSC client with VPN connection	1
• as MSC server with VPN connection	0
Product functions Time	
Protocol is supported	
• NTP	Yes
time synchronization	
• from NTP-server	Yes
• from control center	Yes
Accessories	
accessories	TS Module RS 232 or TS Module MODEM or TS Module ISDN or TS Module GSM pluggable

SIMATIC S7-1200 Basic Controllers

I/O modules

Communication

CP 1243-8 IRC

3

Ordering data	Article No.		Article No.
CP 1243-8 IRC communications processor Communications processor for connecting a SIMATIC S7-1200 in the TeleControl Professional system via the SINAUT ST7 telecontrol protocol to higher-level ST7 stations or to an ST7 control center.	6GK7243-8RX30-0XE0	SINAUT Engineering Software V5.5 Upgrade from V5.0, V5.1, V5.2, V5.3 or V5.4 For upgrading functional expansions; on CD ROM / DVD	6NH7997-0CA55-0GA0
Accessories		TeleService module Connection to TS Adapter IE Basic/Advanced or CP 1243-8 IRC. Power supply via TS Adapter IE Basic/Advanced or CP 1243-8 IRC.	
STEP 7 Professional 2010 SR4/V14, Floating Combo License; on DVD	6ES7810-5CC11-0YA5	TS module RS 232	6ES7972-0MS00-0XA0
SINAUT Engineering Software V5.5 On CD-ROM, comprising: • SINAUT ST7/DNP3 configuration and diagnostic software for STEP 7 V5.5 incl. SP4; • SINAUT TD7 block library • Electronic manual in German and English	6NH7997-0CA55-0AA0	TS module modem	6ES7972-0MM00-0XA0
		TS module ISDN	6ES7972-0MD00-0XA0
		TS module GSM GSM/GPRS modem for SIMATIC Teleservice ¹⁾	6GK7972-0MG00-0XA0

¹⁾ Note national approvals under
<http://www.siemens.com/mobilenetwork-approvals>

Overview

The CP 1243-1 DNP3 communications processor is used to connect a SIMATIC S7-1200 to a control center system via the DNP3 protocol and has the following characteristics:

- Support for the established DNP3 telecontrol protocol for standardized linking of the SIMATIC S7-1200 to WinCC, PCS 7, or other commercially available control center systems
- Data transfer of measured values, control variables, or alarms optimized for telecontrol systems
- Automatic sending of alert emails
- Clearly laid out LED signaling for fast and easy diagnostics
- Compact industrial enclosure in S7-1200 design for mounting on a standard mounting rail
- Fast commissioning thanks to easy configuration using STEP 7
- Data buffering of up to 64,000 values ensures a secure database even with temporary connection failures

3**Technical specifications**

Article number	6GK7243-1JX30-0XE0
Product type designation	CP 1243-1 DNP3
Transmission rate	
Transfer rate	
• at the 1st interface	10 ... 100 Mbit/s
Interfaces	
Number of interfaces acc. to Industrial Ethernet	1
Number of electrical connections	
• at the 1st interface acc. to Industrial Ethernet	1
• for power supply	0
Type of electrical connection	
• at the 1st interface acc. to Industrial Ethernet	RJ45 port
Supply voltage, current consumption, power loss	
Type of voltage of the supply voltage	DC
Supply voltage 1 from backplane bus	5 V
Consumed current	
• from backplane bus at DC at 5 V typical	0.25 A
Power loss [W]	1.25 W
Permitted ambient conditions	
Ambient temperature	
• for vertical installation during operation	-20 ... +60 °C
• for horizontally arranged busbars during operation	-20 ... +70 °C
• during storage	-40 ... +70 °C
• during transport	-40 ... +70 °C
Relative humidity at 25 °C without condensation during operation maximum	95%
Protection class IP	IP20

Article number	6GK7243-1JX30-0XE0
Product type designation	CP 1243-1 DNP3
Design, dimensions and weight	
Module format	Compact module S7-1200 single width
Width	30 mm
Height	110 mm
Depth	75 mm
Net weight	0.122 kg
Mounting type	
• 35 mm DIN rail mounting	Yes
• wall mounting	Yes
Product properties, functions, components general	
Number of units	
• per CPU maximum	3
Performance data S7 communication	
Number of possible connections for S7 communication	
• Note	like CPU
Performance data IT functions	
Number of possible connections	
• as e-mail client maximum	1

SIMATIC S7-1200 Basic Controllers

I/O modules

Communication

CP 1243-1 DNP3

Technical specifications (continued)

Article number	6GK7243-1JX30-0XE0
Product type designation	CP 1243-1 DNP3
Performance data telecontrol	
Suitability for use	
• Node station	No
• substation	Yes
• TIM control center	No
Control center connection	control center with DNP3 function supported
• by means of a permanent connection	
• Note	Connection to SCADA system using DNP3 services
Protocol is supported	
• DNP3	Yes
• IEC 60870-5	No
Product function data buffering if connection is aborted	Yes; 64,000 values
Number of data points per station maximum	200

Article number	6GK7243-1JX30-0XE0
Product type designation	CP 1243-1 DNP3
Performance data Teleservice	
Diagnostics function online diagnostics with SIMATIC STEP 7	Yes
Product function	
• program download with SIMATIC STEP 7	Yes
• Remote firmware update	Yes
Configuration software	
• required	STEP 7 Basic/Professional V12 SP1 (TIA Portal) or higher
Product functions Time	
Protocol is supported	
• NTP	No
time synchronization	
• from control center	Yes

Ordering data

Article No.

CP 1243-1 DNP3 communications processor

Communications processor for connecting SIMATIC S7-1200 to a control center via the DNP3 protocol

6GK7243-1JX30-0XE0

Accessories

Compact switch module CSM 1277

Unmanaged switch for connecting a SIMATIC S7-1200 and up to three further nodes to Industrial Ethernet with 10/100 Mbit/s; 4 x RJ45 ports; external 24 V DC power supply; diagnostics LEDs; S7-1200 module including electronic manual on CD-ROM

6GK7277-1AA10-0AA0

IE FC RJ45 plugs

RJ45 plug connector for Industrial Ethernet with a rugged metal enclosure and integrated insulation displacement contacts for connecting Industrial Ethernet FC installation cables

Article No.

IE FC RJ45 Plug 180

180° cable outlet; for network components and CPs/CPU with Industrial Ethernet interface

- 1 pack = 1 unit
- 1 pack = 10 units
- 1 pack = 50 units

6GK1901-1BB10-2AA0
6GK1901-1BB10-2AB0
6GK1901-1BB10-2AE0

IE FC TP Standard Cable GP 2 x 2 (Type A)

4-core, shielded TP installation cable for connection to IE FC outlet RJ45/IE F RJ45 Plug; PROFINET-compatible; with UL approval; sold by the meter; max. length 1 000 m, minimum order quantity 20 m

6XV1840-2AH10

IE FC Stripping Tool

Preadjusted tool for fast stripping of Industrial Ethernet FC cables

6GK1901-1GA00

Engineering Software STEP7 Basic (TIA Portal)

See catalog section 11

Overview

The CP 1243-1 IEC communications processor is used to connect a SIMATIC S7-1200 to a control center system via the IEC 60870 protocol and has the following characteristics:

- Support for the established communication standard in accordance with IEC 60870-5-104 for standardized linking of the SIMATIC S7-1200 to WinCC, PCS 7, or other commercially available control center systems
- Data transfer of measured values, control variables, or alarms optimized for telecontrol systems
- Automatic sending of alert emails
- Clearly laid out LED signaling for fast and easy diagnostics
- Compact industrial enclosure in S7-1200 design for mounting on a standard mounting rail
- Fast commissioning thanks to easy configuration using STEP 7
- Data buffering of up to 64,000 values ensures a secure database even with temporary connection failures

3**Technical specifications**

Article number	6GK7243-1PX30-0XE0
Product type designation	CP 1243-1 IEC
Transmission rate	
Transfer rate	
• at the 1st interface	10 ... 100 Mbit/s
Interfaces	
Number of interfaces acc. to Industrial Ethernet	1
Number of electrical connections	
• at the 1st interface acc. to Industrial Ethernet	1
• for power supply	0
Type of electrical connection	
• at the 1st interface acc. to Industrial Ethernet	RJ45 port
Supply voltage, current consumption, power loss	
Type of voltage of the supply voltage	DC
Supply voltage 1 from backplane bus	5 V
Consumed current	
• from backplane bus at DC at 5 V typical	0.25 A
Power loss [W]	1.25 W
Permitted ambient conditions	
Ambient temperature	
• for vertical installation during operation	-20 ... +60 °C
• for horizontally arranged busbars during operation	-20 ... +70 °C
• during storage	-40 ... +70 °C
• during transport	-40 ... +70 °C
Relative humidity at 25 °C without condensation during operation maximum	95%
Protection class IP	IP20

Article number	6GK7243-1PX30-0XE0
Product type designation	CP 1243-1 IEC
Design, dimensions and weight	
Module format	Compact module S7-1200 single width
Width	30 mm
Height	110 mm
Depth	75 mm
Net weight	0.122 kg
Mounting type	
• 35 mm DIN rail mounting	Yes
• wall mounting	Yes
Product properties, functions, components general	
Number of units	
• per CPU maximum	3
Performance data S7 communication	
Number of possible connections for S7 communication	
• Note	like CPU
Performance data IT functions	
Number of possible connections	
• as e-mail client maximum	1

SIMATIC S7-1200 Basic Controllers

I/O modules

Communication

CP 1243-1 IEC

Technical specifications (continued)

Article number	6GK7243-1PX30-0XE0
Product type designation	CP 1243-1 IEC
Performance data telecontrol	
Suitability for use	
• Node station	No
• substation	Yes
• TIM control center	No
Control center connection	control center with IEC 60870-5 function
• by means of a permanent connection	supported
• Note	Connection to SCADA system using IEC 60870-5
Protocol is supported	
• DNP3	No
• IEC 60870-5	Yes
Product function data buffering if connection is aborted	Yes; 64,000 values
Number of data points per station maximum	200

Article number	6GK7243-1PX30-0XE0
Product type designation	CP 1243-1 IEC
Performance data Teleservice	
Diagnostics function online diagnostics with SIMATIC STEP 7	Yes
Product function	
• program download with SIMATIC STEP 7	Yes
• Remote firmware update	Yes
Configuration software	
• required	STEP 7 Basic/Professional V13 (TIA Portal) or higher
Product functions Time	
Protocol is supported	
• NTP	No
time synchronization	
• from control center	Yes

Ordering data

Article No.

CP 1243-1 IEC communications processor

Communications processor for connecting SIMATIC S7-1200 to a control center via the IEC 60870-5-104 protocol

6GK7243-1PX30-0XE0

Accessories

CSM 1277 compact switch module

Unmanaged switch for connecting one SIMATIC S7-1200 and up to three further nodes to Industrial Ethernet with 10/100 Mbps; 4 x RJ45 ports; external 24 V DC power supply, diagnostics LEDs, S7-1200 module including electronic manual on CD-ROM

6GK7277-1AA10-0AA0

IE FC RJ45 plugs

RJ45 plug connector for Industrial Ethernet with a rugged metal housing and integrated insulation displacement contacts for connecting Industrial Ethernet FC installation cables; 180° cable outlet; for network components and CPs/CPUs with Industrial Ethernet interface

- 1 pack = 1 unit
- 1 pack = 10 units
- 1 pack = 50 units

6GK1901-1BB10-2AA0
6GK1901-1BB10-2AB0
6GK1901-1BB10-2AE0

Article No.

IE FC TP Standard Cable GP 2 x 2 (Type A)

4-core, shielded TP installation cable for connection to IE FC outlet RJ45/IE F RJ45 plug; PROFINET-compatible; with UL approval; sold by the meter; max. length 1 000 m, minimum order quantity 20 m

6XV1840-2AH10

IE FC Stripping Tool

Pre-adjusted stripping tool for fast stripping of the Industrial Ethernet FC cables

6GK1901-1GA00

STEP 7 Basic Engineering Software (TIA Portal)

See catalog section 11

Overview

The SIMATIC RF120C is a communication module for connecting the SIMATIC identification systems directly to the SIMATIC S7-1200. The readers of all RFID systems as well as the MV400 optical readers can be operated on the SIMATIC RF120C.

Integration into the TIA Portal and the uniform plug-in connection systems permit fast and simple commissioning.

3**Technical specifications**

Article number	6GT2002-0LA00
Product type designation	RF120C communication module
Suitability for operation	SIMATIC S7-1200 together with RF200/300/600, MV400, MOBY D/U
Transmission rate	
Transfer rate at the point-to-point connection serial maximum	115.2 kbit/s
Interfaces	
Design of the interface for point-to-point connection	RS 422
Number of readers connectable	1
Type of electrical connection	
• of the backplane bus	S7-1200 backplane bus
• for supply voltage	Screw terminals
Design of the interface to the reader for communication	D-sub, 9-pin, socket
Mechanical data	
Material	Xantar MX 1094
Color	Ti-grey 24L01
Tightening torque of the screw for securing the equipment maximum	0.45 N·m
Supply voltage, current consumption, power loss	
Supply voltage	
• at DC Rated value	24 V
• at DC	20 ... 30 V
Consumed current at DC at 24 V	
• without connected devices typical	0.03 A
• with connected devices maximum	1 A

Article number	6GT2002-0LA00
Product type designation	RF120C communication module
Permitted ambient conditions	
Ambient temperature	
• during operation	0 ... 55 °C
• during storage	-40 ... +70 °C
• during transport	-40 ... +70 °C
Protection class IP	IP20
Shock resistance	According to IEC 61131-2
Shock acceleration	300 m/s ²
Vibrational acceleration	100 m/s ²
Design, dimensions and weight	
Width	30 mm
Height	100 mm
Depth	75 mm
Net weight	0.15 kg
Mounting type	S7-1200 rack
Cable length for RS 422 interface maximum	1 000 m
Product properties, functions, components general	
Display version	4 LEDs for reader connection, 1 LED for device status
Product function transponder file handler can be addressed	No
Protocol is supported	
• S7 communication	Yes
Type of parameterization	HSP
Type of programming	Library with functions
Type of computer-mediated communication	acyclic communication
Standards, specifications, approvals	
Certificate of suitability	CE, FCC, cULus, KCC, C-Tick, FM, Ex: II 3G Ex nAA IIC T4 Gc
MTBF	196 y

SIMATIC S7-1200 Basic Controllers

I/O modules

Communication

SIMATIC RF120C

Ordering data	Article No.		Article No.
SIMATIC RF120C communication module	6GT2002-0LA00	Accessories for extended use	
Integrated in the S7-1200 controller for connection of a reader		Extension cable for all readers	
Accessories for all readers		PUR material, CMG approval, suitable for cable carriers	
Reader cable for SIMATIC RF200 / RF300 / RF600 / MV400		2 m, straight connector	6GT2891-4FH20
PUR material, CMG approval, suitable for cable carriers, straight reader connector		5 m, straight connector	6GT2891-4FH50
2 m	6GT2091-4LH20	10 m, straight connector	6GT2891-4FN10
5 m	6GT2091-4LH50	20 m, straight connector	6GT2891-4FN20
10 m	6GT2091-4LN10	50 m, straight connector	6GT2891-4FN50
		2 m, plug angled at reader	6GT2891-4JH20
		5 m, plug angled at reader	6GT2891-4JH50
		10 m, plug angled at reader	6GT2891-4JN10
		Reader adapter cable for MOBY D	6GT2691-4FH20
		Material PUR, CMG approval, suitable for cable carriers, 2 m. A cable of the type 6GT2091-4L... is also required.	
		DVD "RFID Systems Software & Documentation"	6GT2080-2AA20

3

Overview

- For fast, high-performance serial data exchange via point-to-point coupling
- Implemented protocols: ASCII, USS drive protocol, Modbus RTU
- Additional protocols can also be loaded
- Simple parameterization with STEP 7 Basic

Note:

SIPLUS extreme products are based on SIMATIC standard products. The contents listed here were taken from the respective standard products. SIPLUS extreme-specific information was added.

3**Technical specifications**

Article number	6AG1241-1AH32-4XB0	6AG1241-1AH32-2XB0	6AG1241-1CH32-4XB0	6AG1241-1CH32-2XB0
Based on	6ES7241-1AH32-0XB0 SIPLUS S7-1200 CM 1241 RS 232	6ES7241-1AH32-0XB0 SIPLUS S7-1200 CM1241 RS 232	6ES7241-1CH32-0XB0 SIPLUS S7-1200 CM 1241 RS 422/485	6ES7241-1CH32-0XB0 SIPLUS S7-1200 CM 1241 RS 422/485
Ambient conditions				
Free fall				
• Fall height, max.	0.3 m; five times, in product package	0.3 m; five times, in product package	0.3 m; five times, in product package	0.3 m; five times, in product package
Ambient temperature during operation				
• min.	-20 °C; = Tmin; Startup @ 0 °C	-40 °C; = Tmin; Startup @ -25 °C	-20 °C; = Tmin; Startup @ 0 °C	-40 °C; = Tmin; Startup @ -25 °C
• max.	60 °C	70 °C; = Tmax	60 °C; = Tmax	70 °C; Tmax > 60 °C, derating: Max. one module may be configured; this module must be the last module on the CM bus; minimum clearance on the left side of at least 45 mm

SIMATIC S7-1200 Basic Controllers

I/O modules

SIPLUS communication

SIPLUS CM 1241 communication modules

Technical specifications (continued)

Article number	6AG1241-1AH32-4XB0	6AG1241-1AH32-2XB0	6AG1241-1CH32-4XB0	6AG1241-1CH32-2XB0
Based on	6ES7241-1AH32-0XB0	6ES7241-1AH32-0XB0	6ES7241-1CH32-0XB0	6ES7241-1CH32-0XB0
	SIPLUS S7-1200 CM 1241 RS 232	SIPLUS S7-1200 CM1241 RS 232	SIPLUS S7-1200 CM 1241 RS 422/485	SIPLUS S7-1200 CM 1241 RS 422/485
Extended ambient conditions				
relative to ambient temperature-atmospheric pressure-installation altitude	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)
Relative humidity				
With condensation, tested in accordance with IEC 60068-2-38, max.	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)	100%; RH incl. condensation/frost (no commissioning under condensation conditions)
Resistance				
against biologically active substances / conformity with EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!
against chemically active substances / conformity with EN 60721-3-3	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!
against mechanically active substances / conformity with EN 60721-3-3	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!

Ordering data

SIPLUS CM 1241 communication module

(Extended temperature range and exposure to media)

Ambient temperature
-40 ... +70°C

Communication module for point-to-point connection, with one RS 232 interface

Communication module for point-to-point connection, with one RS 485 interface

Suitable for areas with extreme
medial exposure
(conformal coating)

Communication module for point-to-point connection, with one RS 232 interface

Communication module for point-to-point connection, with one RS 485 interface

Article No.

6AG1241-1AH32-2XB0

6AG1241-1CH32-2XB0

6AG1241-1AH32-4XB0

6AG1241-1CH32-4XB0

Article No.

Accessories

See SIMATIC S7-1200 CM 1241 communication modules, page 3/123

Overview

- For fast, high-performance serial data exchange via point-to-point connection
- Implemented protocols: ASCII, USS drive protocol, Modbus RTU
- Additional protocols can be loaded later
- Simple parameterization with STEP 7 Basic
- Can be plugged directly into the CPU

Note:

SIPLUS extreme products are based on SIMATIC standard products. The contents listed here were taken from the respective standard products. SIPLUS extreme-specific information was added.

Technical specifications

Article number	6AG1241-1CH30-5XB1
Based on	6ES7241-1CH30-1XB0 SIPLUS S7-1200 CB 1241 RS 485
General information	
Product type designation	CB 1241 RS 485
Input current	
from backplane bus 5 V DC, typ.	50 mA
Power loss	
Power loss, typ.	1.5 W
Interrupts/diagnostics/status information	
Diagnostic functions	Yes
Diagnostics indication LED	
• for status of the outputs	Yes
Degree and class of protection	
Degree of protection acc. to EN 60529	
• IP20	Yes
Standards, approvals, certificates	
CE mark	Yes
RCM (formerly C-TICK)	Yes
KC approval	Yes
Ambient conditions	
Free fall	
• Fall height, max.	0.3 m; five times, in product package
Ambient temperature during operation	
• min.	-40 °C; = Tmin; Startup @ -25 °C
• max.	55 °C

Article number	6AG1241-1CH30-5XB1
Based on	6ES7241-1CH30-1XB0 SIPLUS S7-1200 CB 1241 RS 485
Extended ambient conditions	
• relative to ambient temperature-atmospheric pressure-installation altitude	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m) // Tmin ... (Tmax - 10K) at 795 hPa ... 658 hPa (+2000 m ... +3500 m) // Tmin ... (Tmax - 20K) at 658 hPa ... 540 hPa (+3500 m ... +5000 m)
• At cold restart, min.	-25 °C
Relative humidity	
- With condensation, tested in accordance with IEC 60068-2-38, max.	100%; RH incl. condensation/frost (no commissioning under condensation conditions)
Pollutant concentrations	
- SO ₂ at RH < 60% without condensation	SO ₂ : < 0.5 ppm; H ₂ S: < 0.1 ppm; RH < 60 % condensation-free
Resistance	
- against biologically active substances / conformity with EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!
- against chemically active substances / conformity with EN 60721-3-3	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!
- against mechanically active substances / conformity with EN 60721-3-3	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!

Ordering data**Article No.****SIPLUS CB 1241 RS 485 communication board****6AG1241-1CH30-5XB1**

For point-to-point connection, with 1 RS 485 interface

Article No.**Accessories**

See SIMATIC CB 1241 RS 485 communication board, page 3/124

SIMATIC S7-1200 Basic Controllers

I/O modules

SIPLUS communication

SIPLUS CM 1242-5 communication modules

Overview



DP-M	DP-S	FMS	PG/OP	S7
	●			

The SIPLUS CM 1242-5 communication module is used to connect a SIMATIC S7-1200 to PROFIBUS as a DP slave and has the following characteristics:

- PROFIBUS DPV1 slave in accordance with IEC 61158
- Module replacement without PG supported
- Power is supplied via the backplane bus so that no extra cabling is required
- Support of all standard baud rates from 9.6 Kbit/s to 12 Mbit/s
- Compact industry-standard enclosure in S7-1200 design for mounting on a standard mounting rail
- Fast commissioning thanks to easy configuration using STEP 7 without additional programming overhead

The CM 1242-5 is intended for use in factory automation. Low-cost PROFIBUS-based automation solutions can be created on the basis of the S7-1200 for optimal production.

Note:

SIPLUS extreme products are based on SIMATIC standard products. The contents listed here were taken from the respective standard products. SIPLUS extreme specific information was added.

SIPLUS S7-1200 CM 1242-5

Article number	6AG1 242-5DX30-2XE0
Article number based on	6GK7 242-5DX30-0XE0
Ambient temperature range	-25 ... +55 °C
Ambient conditions	Suitable for exceptional exposure to media (e.g. sulfur chlorine atmosphere).
Technical data	The technical data of the standard product applies except for the ambient conditions.
Ambient conditions	
Relative humidity	100%, condensation/frost permitted. No commissioning in bedewed state.
Biologically active substances, compliance with EN 60721-3-3	Class 3B2 mold and fungal spores (excluding fauna). The supplied plug covers must remain in place over the unused interfaces during operation!
Chemically active substances, compliance with EN 60721-3-3	Class 3C4 incl. salt spray in accordance with EN60068-2-52 (degree of severity 3). The supplied plug covers must remain in place over the unused interfaces during operation!
Mechanically active substances, compliance with EN 60721-3-3	Class 3S4 incl. sand, dust. The supplied plug covers must remain in place over the unused interfaces during operation!
Air pressure (depending on the highest positive temperature range specified)	1 080 ... 795 hPa (-1 000 ... +2 000 m) see ambient temperature range 795 ... 658 hPa (+2 000 ... +3 500 m) derating 10 K 658 ... 540 hPa (+3 500 ... +5 000 m) derating 20 K

Technical documentation on SIPLUS can be found here:
<http://www.siemens.de/siplus-extreme>

Ordering data

Article No.

SIPLUS communication module CM 1242-5 (Extended temperature range and exposure to media) Communication module for electrical connection of SIMATIC S7-1200 to PROFIBUS as a DPV1 slave	6AG1242-5DX30-2XE0
Accessories	See SIMATIC S7-1200 CM 1242-5 communication module, page 3/126

Overview



DP-M	DP-S	FMS	PG/OP	S7
●			●	●

The CM 1243-5 communication module is used to connect a SIMATIC S7-1200 to PROFIBUS as a DP master and has the following characteristics:

- PROFIBUS DPV1 master in accordance with IEC 61158
- Support of up to 16 PROFIBUS DP slaves
- Communication with other S7 controllers based on S7 communication
- Allows the connection of programming devices and operator panels with a PROFIBUS interface to the S7-1200
- Module replacement without PG supported
- Support of all standard baud rates from 9.6 Kbit/s to 12 Mbit/s
- Compact industry-standard enclosure in S7-1200 design for mounting on a standard mounting rail
- Fast commissioning thanks to easy configuration using STEP 7 without additional programming overhead

The CM 1243-5 is intended for use in factory automation. Low-cost PROFIBUS-based automation solutions can be created on the basis of the S7-1200 for optimal production.

Note:

SIPLUS extreme products are based on SIMATIC standard products. The contents listed here were taken from the respective standard products. SIPLUS extreme specific information was added.

SIPLUS S7-1200 CM 1243-5

Article number	6AG1 243-5DX30-2XE0
Article number based on	6GK7 243-5DX30-0XE0

Ambient temperature range	-25 ... +70 °C
Ambient conditions	Suitable for exceptional exposure to media (e.g. sulfur chlorine atmosphere).
Technical data	The technical data of the standard product applies except for the ambient conditions.

Ambient conditions

Relative humidity	100%, condensation/frost permitted. No commissioning in bedewed state.
Biologically active substances, compliance with EN 60721-3-3	Class 3B2 mold and fungal spores (excluding fauna). The supplied plug covers must remain in place over the unused interfaces during operation!
Chemically active substances, compliance with EN 60721-3-3	Class 3C4 incl. salt spray in accordance with EN60068-2-52 (degree of severity 3). The supplied plug covers must remain in place over the unused interfaces during operation!
Mechanically active substances, compliance with EN 60721-3-3	Class 3S4 incl. sand, dust. The supplied plug covers must remain in place over the unused interfaces during operation!
Air pressure (depending on the highest positive temperature range specified)	1 080 ... 795 hPa (-1 000 ... +2 000 m) see ambient temperature range 795 ... 658 hPa (+2 000 ... +3 500 m) derating 10 K 658 ... 540 hPa (+3 500 ... +5 000 m) derating 20 K

Technical documentation on SIPLUS can be found here:
<http://www.siemens.com/siplus-extreme>

Ordering data**Article No.****SIPLUS CM 1243-5 communication module**

(Extended temperature range and exposure to media)

Communication module for electrical connection of SIMATIC S7-1200 to PROFIBUS as a DPV1 master

6AG1243-5DX30-2XE0

Accessories

See SIMATIC S7-1200 CM 1243-5 communication module, page 3/130

SIMATIC S7-1200 Basic Controllers

I/O modules

SIPLUS communication

SIPLUS NET CSM 1277**Overview**

- Unmanaged switch for connection of SIPLUS S7-1200 to an Industrial Ethernet network with a line, tree or star topology
- Multiplication of Ethernet interfaces on a SIPLUS S7-1200 for additional connection of up to three programming devices, operator controls, and further Ethernet nodes
- Simple, space-saving mounting on the SIPLUS S7-1200 rail
- Low-cost solution for implementing small, local Ethernet networks
- Problem-free connection using RJ45 plugs
- Simple and fast status display via LEDs on the device
- Integral autocrossover function permits use of uncrossed connecting cables

Note:

SIPLUS extreme products are based on SIMATIC standard products. The contents listed here were taken from the respective standard products. SIPLUS extreme-specific information was added.

SIPLUS NET CSM 1277	
Article number	6AG1 277-1AA10-4AA0
Article number based on	6GK7 277-1AA10-0AA0
Ambient temperature range	0 ... +60 °C

Ordering data**Article No.****SIPLUS NET CSM 1277 compact switch module**

(Extended temperature range and exposure to media)

Unmanaged switch for connection of SIPLUS S7-1200 and up to three further nodes to Industrial Ethernet with 10/100 Mbit/s; 4 x RJ45 ports; external 24 V DC power supply, LED diagnostics, S7-1200 module including electronic Manual on CD-ROM

6AG1 277-1AA10-4AA0**Accessories**

See CSM 1277 unmanaged, page 3/132

Overview



- Digital inputs as supplement to the integral I/O of the CPUs
- Integrated into the overall automation for the implementation of safety-related requirements
- With integrated safety functions
- Communication with fail-safe CPUs via PROFIsafe mechanisms
- For flexible adaptation of the controller to the corresponding task
- For subsequent expansion of the system with additional inputs
- Operable exclusively in the central system

3

Technical specifications

Article number	6ES7226-6BA32-0XB0 DIGITAL INPUT SM 1226, F-DI 16X 24VDC
General information	
Product type designation	SM 1226 F-DI 16x24VDC
Supply voltage	
Rated value (DC)	
• 24 V DC	Yes
Input current	
from backplane bus 5 V DC, max.	155 mA; Current consumption (SM Bus, 5 V DC): 155 mA
Digital inputs	
• from load voltage L+ (without load), max.	130 mA; 130 mA + 6 mA / input used + any Vs1/Vs2 current used
Power loss	
Power loss, typ.	7 W
Digital inputs	
Number of digital inputs	16; 16 (1oo1) or 8 (1oo2); Note: You can individually assign each pair of inputs "a.x" and "b.x" as a single (1oo2)-channel or as 2 separate (1oo1)-channels
horizontal installation	
- up to 50 °C, max.	16; 16 inputs at 55 °C horizontal
vertical installation	
- up to 40 °C, max.	16; 16 inputs at 45 °C vertical
Input voltage	
• Type of input voltage	DC
• for signal "0"	-30 V DC to +5 V DC
• for signal "1"	15 V DC to 30 V DC
Input current	
• for signal "0", max. (permissible quiescent current)	0.5 mA
Input delay (for rated value of input voltage) for standard inputs	
- parameterizable	Yes; 0.8 / 1.6 / 3.2 / 6.4 / 12.8 ms

Article number	6ES7226-6BA32-0XB0 DIGITAL INPUT SM 1226, F-DI 16X 24VDC
Cable length	
• shielded, max.	200 m; Unshielded with input filter time of 1.6 ms to 12.6 ms (With an input delay of 0.8 ms, shielded cables must be used for the digital inputs and the sensor supply)
• unshielded, max.	200 m; Shielded with input filter time of 0.8 ms to 12.6 s (With an input delay of 0.8 ms, shielded cables must be used for the digital inputs and the sensor supply)
Diagnostics indication LED	
• for status of the inputs	Yes
Standards, approvals, certificates	
CE mark	Yes
cULus	Yes
FM approval	Yes
Highest safety class achievable in safety mode	
• SIL acc. to IEC 61508	SIL 2 (single-channel), SIL 3 (two-channel)
Ambient conditions	
Free fall	
• Fall height, max.	0.3 m; five times, in product package
Ambient temperature during operation	
• min.	0 °C
• max.	55 °C
Mechanics/material	
Enclosure material (front)	
• Plastic	Yes
Dimensions	
Width	70 mm
Height	100 mm
Depth	75 mm
Weights	
Weight, approx.	250 g

SIMATIC S7-1200 Basic Controllers

I/O modules

Fail-safe digital modules

SM 1226 fail-safe digital input modules

Ordering data	Article No.	Article No.
SM 1226 fail-safe digital input signal module 16 inputs, 24 V DC (SIL 2/category 3/PL d) or 8 inputs 24 V DC (SIL 3/category 3 or category 4/PL e) or a combination of both	6ES7 226-6BA32-0XB0	
Accessories		
Terminal block (spare part) With 11 screws, tin-coated; 4 units	6ES7292-1AL30-0XA0	
Front flap set (spare part) For modules with a width of 70 mm	6ES7291-1BB30-0XA0	
STEP 7 Safety Advanced V14 SP1 Task: Engineering tool for configuring and programming fail-safe user programs for SIMATIC S7-1200 FC, S7-1500F, S7-1500F Software Controller, S7-300F, S7-400F, WinAC RTX F, ET 200SP F Controller and the fail-safe ET 200SP, ET 200S, ET 200M, ET 200iSP, ET 200pro and ET 200eco I/O Requirement: STEP 7 Professional V14 SP1 Floating license for 1 user, software and documentation on DVD; license key on USB flash drive Floating license for 1 user, software, documentation and license key for download ¹⁾ ; email address required for delivery	6ES7833-1FA14-0YA5 6ES7833-1FA14-0YH5	STEP 7 Safety Basic V14 SP1 Task: Engineering tool for configuring fail-safe user programs for SIMATIC S7-1200 FC Requirement: STEP 7 Basic V14 SP1 and higher Floating license for 1 user; software and documentation on DVD; license key on USB flash drive Floating license for 1 user; software, documentation and license key for download ¹⁾ ; email address required for delivery
		6ES7833-1FB14-0YA5 6ES7833-1FB14-0YH5

¹⁾ For up-to-date information and download availability, see:
<http://www.siemens.com/tia-online-software-delivery>

Overview



- Digital outputs as a supplement to the integral I/O of the CPUs
- Integrated into the overall automation for the implementation of safety-related requirements
- With integrated safety functions
- Communication with fail-safe CPUs via PROFIsafe mechanisms
- For flexible adaptation of the controller to the corresponding task
- For subsequent expansion of the system with additional outputs
- Operable exclusively in the central system

3

Technical specifications

Article number	6ES7226-6DA32-0XB0 DIGITAL OUTPUT SM 1226, F-DQ 4X 24VDC
General information	
Product type designation	SM 1226 F-DQ 4 x 24 V DC
Supply voltage	
Rated value (DC)	
• 24 V DC	Yes
Input current	
from backplane bus 5 V DC, max.	125 mA
Digital outputs	
• from load voltage L+, max.	170 mA
Power loss	
Power loss, typ.	8 W
Digital outputs	
Number of digital outputs	4
• in groups of	1
Short-circuit protection	Yes
Switching capacity of the outputs	
• with resistive load, max.	30 Hz
• on lamp load, max.	10 Hz
Output voltage	
• Rated value (DC)	24 V
Output current	
• for signal "1" rated value	2 A
• for signal "1" permissible range, max.	10 mA to 2.4 A
• for signal "0" residual current, max.	P-switch: 0.5 mA, maximum; M-switch: 0.5 mA, maximum
Cable length	
• shielded, max.	200 m
• unshielded, max.	200 m
Diagnostics indication LED	
• for status of the outputs	Yes

Article number	6ES7226-6DA32-0XB0 DIGITAL OUTPUT SM 1226, F-DQ 4X 24VDC
Standards, approvals, certificates	
CE mark	Yes
cULus	Yes
FM approval	Yes
Highest safety class achievable in safety mode	
• SIL acc. to IEC 61508	SIL 3
Ambient conditions	
Free fall	
• Fall height, max.	0.3 m; five times, in product package
Ambient temperature during operation	
• min.	0 °C
• max.	55 °C
Mechanics/material	
Enclosure material (front)	
• Plastic	Yes
Dimensions	
Width	70 mm
Height	100 mm
Depth	75 mm
Weights	
Weight, approx.	270 g

SIMATIC S7-1200 Basic Controllers

I/O modules

Fail-safe digital modules

SM 1226 fail-safe digital output modules

Ordering data	Article No.	Article No.
SM 1226 fail-safe digital output signal module 4 outputs; 24 V DC, current sourcing/sinking	6ES7226-6DA32-0XB0	
Accessories		
Terminal block (spare part) With 11 screws, tin-coated; 4 units	6ES7292-1AL30-0XA0	
Front flap set (spare part) For modules with a width of 70 mm	6ES7291-1BB30-0XA0	
STEP 7 Safety Advanced V14 SP1 Task: Engineering tool for configuring and programming fail-safe user programs for SIMATIC S7-1200 FC, S7-1500F, S7-1500F Software Controller, S7-300F, S7-400F, WinAC RTX F, ET 200SP F Controller and the fail-safe ET 200SP, ET 200S, ET 200M, ET 200iSP, ET 200pro and ET 200eco I/O Requirement: STEP 7 Professional V14 SP1 Floating license for 1 user, software and documentation on DVD; license key on USB flash drive Floating license for 1 user, software, documentation and license key for download ¹⁾ ; email address required for delivery	6ES7833-1FA14-0YA5 6ES7833-1FA14-0YH5	STEP 7 Safety Basic V14 SP1 Task: Engineering tool for configuring fail-safe user programs for SIMATIC S7-1200 FC Requirement: STEP 7 Basic V14 SP1 and higher Floating license for 1 user; software and documentation on DVD; license key on USB flash drive Floating license for 1 user; software, documentation and license key for download ¹⁾ ; email address required for delivery
		6ES7833-1FB14-0YA5 6ES7833-1FB14-0YH5

¹⁾ For up-to-date information and download availability, see:
<http://www.siemens.com/tia-online-software-delivery>

Overview



- Relay outputs as a supplement to the integral I/O of the CPUs
- Integrated into the overall automation for the implementation of safety-related requirements
- With integrated safety functions
- Communication with fail-safe CPUs via PROFIsafe mechanisms
- For flexible adaptation of the controller to the corresponding task
- For subsequent expansion of the system with additional outputs
- Operable exclusively in the central system

3

Technical specifications

Article number	6ES7226-6RA32-0XB0 DIGITAL OUTPUT SM 1226, F-DQ 2X RELAY
General information	
Product type designation	SM 1226 F-DQ 2 x Relay
Supply voltage	
Rated value (DC)	
• 24 V DC	Yes
Input current	
from backplane bus 5 V DC, max.	120 mA
Digital outputs	
• from load voltage L+, max.	300 mA
Power loss	
Power loss, typ.	10 W
Digital outputs	
Number of digital outputs	2
Short-circuit protection	No
Output voltage	
• Rated value (DC)	5 V DC to 30 V DC
• Rated value (AC)	5 V AC to 250 V AC
Output current	
• for signal "I" permissible range, max.	5 A maximum per circuit and 10 A maximum of all circuits per module
Relay outputs	
• Number of relay outputs	2; 2 circuits per output
Switching capacity of contacts	
- with inductive load, max.	0.1 Hz, accordance with IEC 60947-5-1, DC-13; 2 Hz, accordance with IEC 60947-5-1, AC-15
- with resistive load, max.	2 Hz
Cable length	
• shielded, max.	200 m
• unshielded, max.	200 m
Diagnostics indication LED	
• for status of the outputs	Yes

Article number	6ES7226-6RA32-0XB0 DIGITAL OUTPUT SM 1226, F-DQ 2X RELAY
Standards, approvals, certificates	
CE mark	Yes
cULus	Yes
FM approval	Yes
Highest safety class achievable in safety mode	
• SIL acc. to IEC 61508	SIL 3
Ambient conditions	
Free fall	
• Fall height, max.	0.3 m; five times, in product package
Ambient temperature during operation	
• min.	0 °C
• max.	55 °C
Mechanics/material	
Enclosure material (front)	
• Plastic	Yes
Dimensions	
Width	70 mm
Height	100 mm
Depth	75 mm
Weights	
Weight, approx.	300 g

SIMATIC S7-1200 Basic Controllers

I/O modules

Fail-safe digital modules

SM 1226 fail-safe relay output modules

Ordering data	Article No.	Article No.
SM 1226 fail-safe relay output signal module 2 relay outputs	6ES7226-6RA32-0XB0	STEP 7 Safety Basic V14 SP1 Task: Engineering tool for configuring fail-safe user programs for SIMATIC S7-1200 FC Requirement: STEP 7 Basic V14 SP1 and higher Floating license for 1 user; software and documentation on DVD; license key on USB flash drive Floating license for 1 user; software, documentation and license key for download ¹⁾ ; email address required for delivery
Accessories		
Terminal block (spare part) With 11 screws, tin-coated, coded; 4 units	6ES7292-1AL40-0XA0	
Front flap set (spare part) For modules with a width of 70 mm	6ES7291-1BB30-0XA0	
STEP 7 Safety Advanced V14 SP1 Task: Engineering tool for configuring and programming fail-safe user programs for SIMATIC S7-1200 FC, S7-1500F, S7-1500F Software Controller, S7-300F, S7-400F, WinAC RTX F, ET 200SP F Controller and the fail-safe ET 200SP, ET 200S, ET 200M, ET 200iSP, ET 200pro and ET 200eco I/O Requirement: STEP 7 Professional V14 SP1 Floating license for 1 user, software and documentation on DVD; license key on USB flash drive Floating license for 1 user, software, documentation and license key for download ¹⁾ ; email address required for delivery	6ES7833-1FA14-0YA5 6ES7833-1FA14-0YH5	
		6ES7833-1FB14-0YA5 6ES7833-1FB14-0YH5

¹⁾ For up-to-date information and download availability, see:
<http://www.siemens.com/tia-online-software-delivery>

Overview

- Digital inputs as supplement to the integral I/O of the CPUs
- Integrated into the overall automation for the implementation of safety-related requirements
- With integrated safety functions
- Communication with fail-safe CPUs via PROFIsafe mechanisms
- For flexible adaptation of the controller to the corresponding task
- For subsequent expansion of the system with additional inputs
- Operable exclusively in the central system

Note

SIPLUS extreme products are based on SIMATIC standard products. The contents listed here were taken from the corresponding standard products. SIPLUS extreme-specific information was added.

Technical specifications

Article number	6AG1226-6BA32-5XB0
Based on	6ES7226-6BA32-0XB0 SIPLUS S7-1200 SM 1226 F-DI 16x24VDC
Ambient conditions	
Free fall	
• Fall height, max.	0.3 m; five times, in product package
Ambient temperature during operation	
• min.	-25 °C; = Tmin
• max.	55 °C; = Tmax
Extended ambient conditions	
• relative to ambient temperature-atmospheric pressure-installation altitude	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m)
Relative humidity	
- With condensation, tested in accordance with IEC 60068-2-38, max.	100%; RH incl. condensation/frost (no commissioning under condensation conditions)
Resistance	
- against biologically active substances / conformity with EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!
- against chemically active substances / conformity with EN 60721-3-3	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!
- against mechanically active substances / conformity with EN 60721-3-3	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!

Ordering data**Article No.**

SIPLUS SM 1226 fail-safe digital input signal module (Extended temperature range and environmental stress) 16 inputs, 24 V DC (SIL 2/category 3/PL d) or 8 inputs 24 V DC (SIL 3/category 3 or category 4/PL e) or a combination of both	6AG1226-6BA32-5XB0
Accessories	See SIMATIC SM 1226 fail-safe digital input signal module, page 3/158

SIMATIC S7-1200 Basic Controllers

I/O modules

SIPLUS Fail-safe

SIPLUS SM 1226 fail-safe digital output modules**Overview**

- Digital outputs as a supplement to the integral I/O of the CPUs
- Integrated into the overall automation for the implementation of safety-related requirements
- With integrated safety functions
- Communication with fail-safe CPUs via PROFIsafe mechanisms
- For flexible adaptation of the controller to the corresponding task
- For subsequent expansion of the system with additional outputs
- Operable exclusively in the central system

Note

SIPLUS extreme products are based on SIMATIC standard products. The contents listed here were taken from the corresponding standard products. SIPLUS extreme-specific information was added.

Technical specifications

Article number	6AG1226-6DA32-5XB0
Based on	6ES7226-6DA32-0XB0 SIPLUS S7-1200 SM 1226 F-DQ 4x24VDC
Ambient conditions	
Free fall	
• Fall height, max.	0.3 m; five times, in product package
Ambient temperature during operation	
• min.	-25 °C; = Tmin
• max.	55 °C; = Tmax
Extended ambient conditions	
• relative to ambient temperature-atmospheric pressure-installation altitude	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m)
Relative humidity	
- With condensation, tested in accordance with IEC 60068-2-38, max.	100%; RH incl. condensation/frost (no commissioning under condensation conditions)
Resistance	
- against biologically active substances / conformity with EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!
- against chemically active substances / conformity with EN 60721-3-3	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!
- against mechanically active substances / conformity with EN 60721-3-3	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!

Ordering data**Article No.**

SIPLUS SM 1226 fail-safe digital output module	6AG1226-6DA32-5XB0
4 outputs; 24 V DC, current sourcing/sinking	
Accessories	See SIMATIC SM 1226 fail-safe digital output signal module, page 3/160

Overview

- Relay outputs as a supplement to the integral I/O of the CPUs
- Integrated into the overall automation for the implementation of safety-related requirements
- With integrated safety functions
- Communication with fail-safe CPUs via PROFIsafe mechanisms
- For flexible adaptation of the controller to the corresponding task
- For subsequent expansion of the system with additional outputs
- Operable exclusively in the central system

Note

SIPLUS extreme products are based on SIMATIC standard products. The contents listed here were taken from the corresponding standard products. SIPLUS extreme-specific information was added.

Technical specifications

Article number	6AG1226-6RA32-5XB0
Based on	6ES7226-6RA32-0XB0 SIPLUS S7-1200 SM 1226 F-DQ 2xRelay
Ambient conditions	
Free fall	
• Fall height, max.	0.3 m; five times, in product package
Ambient temperature during operation	
• min.	-25 °C; = Tmin
• max.	55 °C; = Tmax
Extended ambient conditions	
• relative to ambient temperature-atmospheric pressure-installation altitude	Tmin ... Tmax at 1080 hPa ... 795 hPa (-1000 m ... +2000 m)
Relative humidity	
- With condensation, tested in accordance with IEC 60068-2-38, max.	100%; RH incl. condensation/frost (no commissioning under condensation conditions)
Resistance	
- against biologically active substances / conformity with EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna). The supplied connector covers must remain on the unused interfaces during operation!
- against chemically active substances / conformity with EN 60721-3-3	Yes; Class 3C4 (RH < 75%) incl. salt spray according to EN 60068-2-52 (degree of severity 3). The supplied connector covers must remain on the unused interfaces during operation!
- against mechanically active substances / conformity with EN 60721-3-3	Yes; Class 3S4 incl. sand, dust. The supplied connector covers must remain on the unused interfaces during operation!

Ordering data**Article No.**

SIPLUS SM 1226 fail-safe relay output signal module 2 relay outputs	6AG1226-6RA32-5XB0
Accessories	See SIMATIC SM 1226 fail-safe relay output signal module, page 3/162

SIMATIC S7-1200 Basic Controllers

Power supplies

Single phase, 24 V DC (for S7-1200)

Overview



In terms of design and functionality, the SIMATIC PM 1207 single-phase load power supply (PM = power module) with automatic range selection of the input voltage is an optimal match to the SIMATIC S7-1200 PLC. It provides the supply to CPUs with 24 V input as well as to signal modules, and to 24 V loads connected to the modules. Comprehensive certifications, such as UL, ATEX and GL facilitate universal use.

Technical specifications

Article number	6EP1332-1SH71
Product	S7-1200 PM1207
Power supply, type	24 V/2.5 A
Input	
Input	1-phase AC
Supply voltage	
• 1 at AC Rated value	120 V
• 2 at AC Rated value	230 V
• Note	Automatic range selection
Input voltage	
• 1 at AC	85 ... 132 V
• 2 at AC	176 ... 264 V
Wide-range input	No
Overvoltage resistance	$2.3 \times V_{in \text{ rated}}$, 1.3 ms
Mains buffering at $I_{out \text{ rated}}$, min.	20 ms; at $V_{in} = 93/187 \text{ V}$
Rated line frequency 1	50 Hz
Rated line frequency 2	60 Hz
Rated line range	47 ... 63 Hz
Input current	
• at rated input voltage 120 V	1.2 A
• at rated input voltage 230 V	0.67 A
Switch-on current limiting (+25 °C), max.	13 A
Duration of inrush current limiting at 25 °C	
• maximum	3 ms
I^2t , max.	0.5 A ² ·s
Built-in incoming fuse	T 3, 15 A/250 V (not accessible)
Protection in the mains power input (IEC 898)	Recommended miniature circuit breaker: 16 A characteristic B or 10 A characteristic C

Article number	6EP1332-1SH71
Product	S7-1200 PM1207
Power supply, type	24 V/2.5 A
Output	
Output	Controlled, isolated DC voltage
Rated voltage $V_{out \text{ DC}}$	24 V
Total tolerance, static $\pm$	3%
Static mains compensation, approx.	0.1%
Static load balancing, approx.	0.2%
Residual ripple peak-peak, max.	150 mV
Spikes peak-peak, max. (bandwidth: 20 MHz)	240 mV
Product function Output voltage adjustable	No
Output voltage setting	-
Status display	Green LED for 24 V OK
On/off behavior	No overshoot of V_{out} (soft start)
Startup delay, max.	6 s; 2 s at 230 V, 6 s at 120 V
Voltage rise, typ.	10 ms
Rated current value $I_{out \text{ rated}}$	2.5 A
Current range	0 ... 2.5 A
Supplied active power typical	60 W
Short-term overload current	
• on short-circuiting during the start-up typical	6 A
• at short-circuit during operation typical	6 A
Duration of overloading capability for excess current	
• on short-circuiting during the start-up	100 ms
• at short-circuit during operation	100 ms
Parallel switching for enhanced performance	Yes
Numbers of parallel switchable units for enhanced performance	2

SIMATIC S7-1200 Basic Controllers

Power supplies

Single phase, 24 V DC (for S7-1200)

Technical specifications (continued)

Article number	6EP1332-1SH71
Product	S7-1200 PM1207
Power supply, type	24 V/2.5 A
Efficiency	
Efficiency at $V_{out rated}$, $I_{out rated}$, approx.	83%
Power loss at $V_{out rated}$, $I_{out rated}$, approx.	12 W
Closed-loop control	
Dynamic mains compensation ($V_{in rated} \pm 15\%$), max.	0.3%
Dynamic load smoothing (I_{out} : 50/100/50%), $U_{out} \pm$ typ.	3%
Load step setting time 50 to 100%, typ.	5 ms
Load step setting time 100 to 50%, typ.	5 ms
Setting time maximum	5 ms
Protection and monitoring	
Output overvoltage protection	< 33 V
Current limitation, typ.	2.65 A
Property of the output Short-circuit proof	Yes
Short-circuit protection	Constant current characteristic
Enduring short circuit current RMS value	
• typical	2.7 A
Overload/short-circuit indicator	-
Safety	
Primary/secondary isolation	Yes
Galvanic isolation	Safety extra-low output voltage U_{out} acc. to EN 60950-1 and EN 50178
Protection class	Class I
Leakage current	
• maximum	3.5 mA
CE mark	Yes
UL/cUL (CSA) approval	cULus-Listed (UL 508, CSA C22.2 No. 107.1), File E197259; cURus-Recognized (UL 60950-1, CSA C22.2 No. 60950-1) File E151273
Explosion protection	ATEX (EX) II 3G Ex nA II T4; cULus (ISA 12.12.01, CSA C22.2 No.213) Class I, Div. 2, Group ABCD, T4, File E330455
FM approval	Class I, Div. 2, Group ABCD, T4
CB approval	Yes
Marine approval	GL, ABS, BV, DNV, LRS, NK
Degree of protection (EN 60529)	IP20

Article number	6EP1332-1SH71
Product	S7-1200 PM1207
Power supply, type	24 V/2.5 A
EMC	
Emitted interference	EN 55022 Class B
Supply harmonics limitation	not applicable
Noise immunity	EN 61000-6-2
Operating data	
Ambient temperature	
• during operation	0 ... 60 °C
- Note	with natural convection
• during transport	-40 ... +85 °C
• during storage	-40 ... +85 °C
Humidity class according to EN 60721	Climate class 3K3, no condensation
Mechanics	
Connection technology	screw-type terminals
Connections	
• Supply input	L, N, PE: 1 screw terminal each for 0.5 ... 2.5 mm ²
• Output	L+, M: 2 screw terminals each for 0.5 ... 2.5 mm ²
• Auxiliary	-
Width of the enclosure	70 mm
Height of the enclosure	100 mm
Depth of the enclosure	75 mm
Required spacing	
• top	20 mm
• bottom	20 mm
• left	0 mm
• right	0 mm
Weight, approx.	0.3 kg
Product feature of the enclosure housing for side-by-side mounting	Yes
Installation	Snap onto DIN rail EN 60715 35x7.5/15, wall mounting
MTBF at 40 °C	1 492 537 h
Other information	Specifications at rated input voltage and ambient temperature +25 °C (unless otherwise specified)

Ordering data

Article No.

SIMATIC S7-1200 PM 1207

6EP1332-1SH71

Input: 120/230 V AC
Output: 24 V DC/2.5 A

SIMATIC S7-1200 Basic Controllers

SIPLUS power supplies

Single phase, 24 V DC (for SIPLUS S7-1200)

Overview



- Stabilized power supply for SIPLUS S7-1200
- In the S7-1200 design
- Input 120/230 V AC, output 24 V DC, 2.5 A (derating: 1.5 A above 60 °C)

Note:

SIPLUS extreme products are based on SIMATIC standard products. The contents listed here were adopted from the respective standard products. SIPLUS extreme specific information was added.

SIPLUS power supply PM 1207

Article number	6AG1332-1SH71-4AA0	6AG1332-1SH71-7AA0
Article number based on	6EP1332-1SH71	
Ambient temperature range	0 ... +60° C	-25 ... +70° C
Conformal coating	Coating of the printed circuit boards and the electronic components	
Technical specifications	The technical data of the standard product applies except for the ambient conditions.	
Ambient conditions		
Relative humidity	100%, condensation/frost permitted. No commissioning in bedewed state.	
Biologically active substances, compliance with EN 60721-3-3	Class 3B2 mold and fungal spores (excluding fauna). The supplied plug covers must remain in place over the unused interfaces during operation!	
Chemically active substances, compliance with EN 60721-3-3	Class 3C4 incl. salt spray in accordance with EN60068-2-52 (degree of severity 3). The supplied plug covers must remain in place over the unused interfaces during operation!	
Mechanically active substances, compliance with EN 60721-3-3	Class 3S4 incl. sand, dust. The supplied plug covers must remain in place over the unused interfaces during operation!	
Air pressure (depending on the highest positive temperature range specified)	1 080 ... 795 hPa (-1 000 ... +2 000 m) see ambient temperature range 795 ... 658 hPa (+2000 ... +3500 m) derating 10 K 658 ... 540 hPa (+3500 ... +5000 m) derating 20 K	

Technical documentation on SIPLUS can be found here:

<http://www.siemens.com/siplus-extreme>

SIMATIC S7-1200 Basic Controllers

SIPLUS power supplies

Single phase, 24 V DC (for SIPLUS S7-1200)

Technical specifications

Article number	SIPLUS PM 1207 6AG1332-1SH71-7AA0 6AG1332-1SH71-4AA0
Article number based on	6EP1332-1SH71
Input voltage, nominal value	120/230 V AC (auto-switching)
• Range	85...132 V / 176...264 V AC
Mains buffering	> 20 ms (at 93/187 V)
Line frequency, nominal	50/60 Hz
• Range	47 ... 63 Hz
Input current, nominal value	1.2/0.67 A
• Inrush current (25 °C)	<13 A
• Recommended circuit-breaker	16 A Charact. B, 10 A Charact. C
Output voltage, nominal value	24 V DC
• Tolerance	±3%
• Residual ripple	< 150 mVpp
• Adjustment	No
Output current, nominal value	2.5 A (derating: 1.5 A from 60 °C)
Efficiency at nominal values, approx.	83%
Parallel operation	Yes, 2 units
Electronic short-circuit protection	Yes, automatic restart
Radio interference suppression (EN 55022)	Class B
Operating display	Green LED for "24 V o.k."
Supply-harmonics limitation (EN 61000-3-2)	Not applicable
Degree of protection (EN 60529)	IP20
Protection class	Class 1
Electric isolation	SELV acc. to EN 60950 and EN 50178
Ambient temperature	0 ... +60 °C -25 ... 70 °C
Transport and storage temperature	-25 ... +85 °C
Installation	Standard rail EN 60715 35x7.5/15
Dimensions (W x H x D) in mm	70 x 100 x 75
Weight, approx.	0.3 kg
Certifications	CE

Ordering data

SIPLUS S7-1200 PM 1207 power supply

(Extended temperature range and exposure to media)

Input 120/230 V AC, output 24 V DC, 2.5 A; derating from + 55 °C to + 70 °C to 1.2 A output current

Ambient temperature
-25 ... +70 °C

Ambient temperature 0 ... +60 °C

Article No.

6AG1332-1SH71-7AA0**6AG1332-1SH71-4AA0**

SIRIUS

Halbleiterschütz

Semiconductor contactor

Contacteur statique

Contactor estático

Contattore a semiconduttore

Contator semiconductor

3RF24...BB..



DIN EN 60 947-4-2

Betriebsanleitung

Instructivo

Operating Instructions

Istruzioni Operative

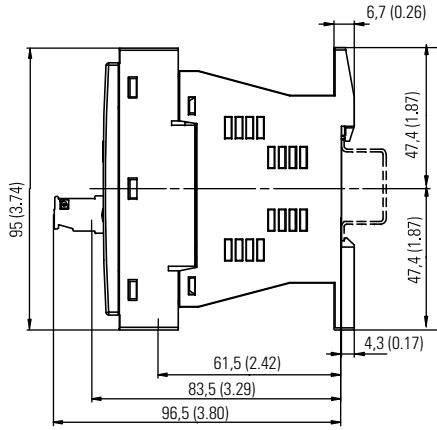
Instructions de Service

Instruções de Operação

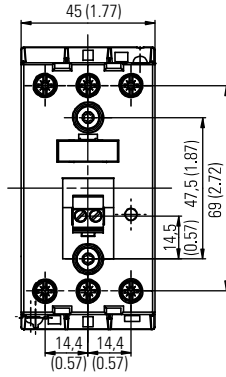
	Deutsch	English	Français
	Vor der Installation, dem Betrieb oder der Wartung des Geräts muss diese Anleitung gelesen und verstanden werden. Eine sichere Gerätefunktion ist nur mit zertifizierten Komponenten gewährleistet!	Read and understand these instructions before installing, operating, or maintaining the equipment. Reliable functioning of the equipment is only ensured with certified components.	Ne pas installer, utiliser ou intervenir sur cet équipement avant d'avoir lu et assimilé les présentes instructions et notamment les conseils de sécurité et mises en garde qui y figurent. La sécurité de fonctionnement de l'appareil n'est garantie qu'avec des composants certifiés.
	⚠ GEFAHR	⚠ DANGER	⚠ DANGER
	Gefährliche Spannung. Lebensgefahr oder schwere Verletzungsgefahr. Vor Beginn der Arbeiten Anlage und Gerät spannungsfrei schalten.	Hazardous voltage. Will cause death or serious injury. Turn off and lock out all power supplying this device before working on this device.	Tension électrique. Danger de mort ou risque de blessures graves. Mettre hors tension avant d'intervenir sur l'appareil.
	⚠ GEFAHR	⚠ DANGER	⚠ DANGER
	Gefährliche Spannung. Lebensgefahr oder schwere Verletzungsgefahr. Um elektrische Stromschläge oder Verbrennungen zu vermeiden, dürfen Sie die Klemmen des Halbleiterschützes nicht berühren, wenn an dessen Eingang Spannung liegt. Die Ausgangsklemme steht auch im AUS-Zustand des Halbleiterschützes unter Spannung (Leckstrom, Halbleiter durchlegiert).	Hazardous voltage. Will cause death or serious injury. To avoid electric shock and burns, never touch the terminals of the semiconductor contactor if voltage is present at its input. The output terminal remains live even in the OFF state of the semiconductor contactor (leakage current, semiconductor breakdown).	Tension dangereuse. Danger de mort ou risque de blessures graves. Ne pas toucher les bornes du relais statique lorsqu'une tension est appliquée à son entrée. Risque d'électrocution et de brûlure ! La borne de sortie est aussi sous tension à l'état BLOQUE du relais statique (courant de fuite, semi-conducteur claqué).
	Español	Italiano	Português
	Leer y comprender este instructivo antes de la instalación, operación o mantenimiento del equipo. El funcionamiento seguro del aparato sólo está garantizado con componentes certificados.	Leggere con attenzione queste istruzioni prima di installare, utilizzare o eseguire manutenzione su questa apparecchiatura. Il funzionamento sicuro dell'apparecchiatura è garantito soltanto con componenti certificati.	Ler e compreender estas instruções antes da instalação, operação ou manutenção do equipamento. O funcionamento seguro do aparelho apenas pode ser garantido se forem utilizados os componentes certificados.
	⚠ PELIGRO	⚠ PERICOLO	⚠ PERIGO
	Tensión peligrosa. Puede causar la muerte o lesiones graves. Desconectar la alimentación eléctrica antes de trabajar en el equipo.	Tensione pericolosa. Può provocare morte o lesioni gravi. Scollegare l'alimentazione prima di eseguire interventi sull'apparecchiatura.	Tensão perigosa. Perigo de morte ou ferimentos graves. Desligue a corrente antes de trabalhar no equipamento.
	⚠ PELIGRO	⚠ PERICOLO	⚠ PERIGO
	Tensión peligrosa. Puede causar la muerte o lesiones graves. Para evitar choques eléctricos o quemaduras no se deberán tocar los bornes del relé estático cuando su entrada esté bajo tensión. El borne de salida está también bajo tensión incluso si el relé estático se encuentra en estado DES (por causa de corriente de fuga o cortocircuito interno del semiconductor).	Tensione pericolosa. Può provocare morte o lesioni gravi. Per evitare shock elettrici e ustioni, non toccare i morsetti del relé statici quando il suo ingresso è sotto tensione. Il morsetto di uscita è sotto tensione anche quando il relé statici è disinserito (corrente di dispersione, semiconduttore in cortocircuito).	Tensão perigosa. Perigo de morte ou ferimentos graves. De modo a evitar choques elétricos ou queimaduras não pode tocar nos bornes do relé semicondutor se a entrada do mesmo estiver sob tensão. O borne de saída está sob tensão, mesmo quando o relé semicondutor estiver em estado DESLIGADO (corrente de fuga, semicondutor com liga).

DE	EN	FR	ES	IT	PT
Maßbilder	Dimension drawings	Encombrement	Croquis acotados	Disegni quotati	Desenhos dimensionais

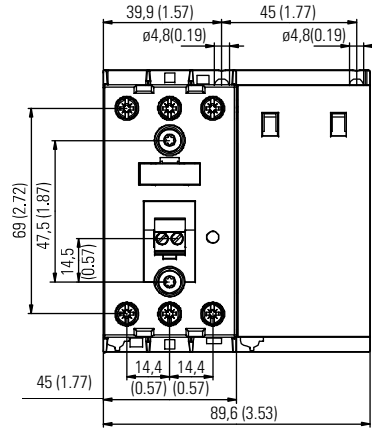
3RF24...-BB..



3RF2405...BB..

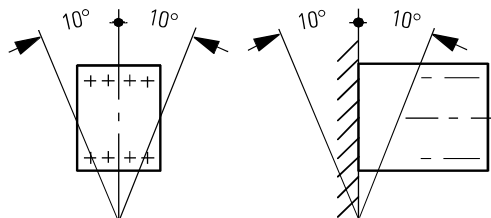
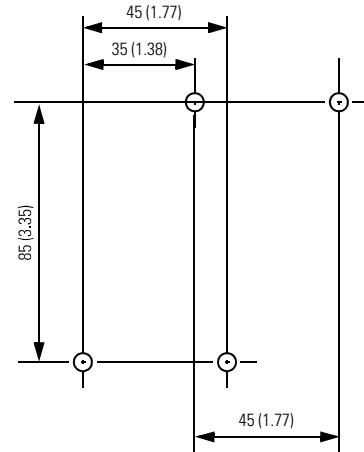
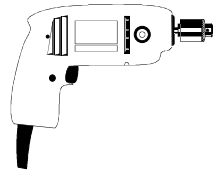
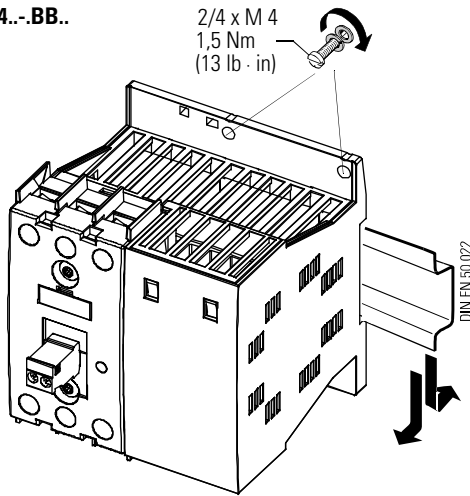


3RF2410...BB..
3RF2412...BB..
3RF2416...BB..



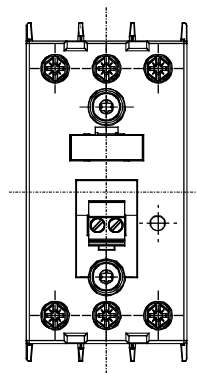
DE	EN	FR	ES	IT	PT
Montage	Assembly	Montage	Montaje	Montaggio	Montagem

3RF24...-BB..

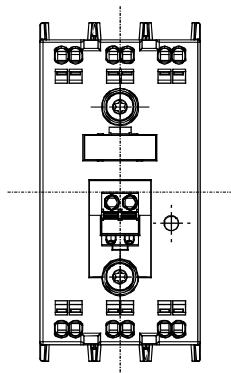


DE	EN	FR	ES	IT	PT
Anschluss	Connection	Connexion	Conexión	Connettore	Conexão

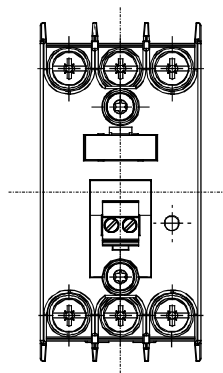
3RF24...-1



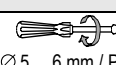
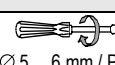
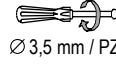
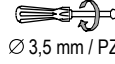
3RF24...-2



3RF24...-3

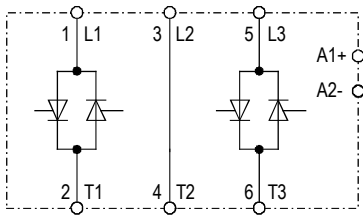


DE	EN	FR	ES	IT	PT
Anschlussquer- schnitte und Anzugs- drehmomente	Conductor cross- sections and tightening torques	Section des conducteurs de raccordement et couples de serrage	Secciones de cables de conexión y pares de apriete	Sezioni dei condut- tori di collegamento e coppie di serraggio	Seções transversais de conexão e torques de aperto

	3RF24...-1			3RF24...-2			3RF24...-3		
	L1 - T3	A1 - A2		L1 - T3	A1 - A2		L1 - T3	A1 - A2	
 Ø 5 ... 6 mm / PZ2	2 ... 2,5 Nm 18 ... 22 lbf · in	—	—	—	—	 Ø 5 ... 6 mm / PZ2	2 ... 2,5 Nm 18 ... 22 lbf · in	—	
	2 x 1,5 ... 2,5 mm² 2 x 2,5 ... 6 mm²			2 x 0,5 ... 2,5 mm²			DIN 46234 5 - 2,5 ... 5 - 16		
	2 x 1,5 ... 2,5 mm² 2 x 2,5 ... 6 mm² 1 x 10 mm²			2 x 0,5 ... 1,5 mm²					
				2 x 0,5 ... 2,5 mm²					
AWG	2 x 14 ... 10		AWG	2 x 18 ... 14			JIS C 2805 R 2 - 5 ... 14 - 5		
 Ø 3,5 mm / PZ1	—	0,5 ... 0,6 Nm 4.5 ... 5.3 lbf · in	—	—	—	 Ø 3,5 mm / PZ1	0,5 ... 0,6 Nm 4.5 ... 5.3 lbf · in	—	
		1 x 0,5 ... 2,5 mm² 2 x 0,5 ... 1 mm²			1 x 0,5 ... 2,5 mm²		1 x 0,5 ... 2,5 mm²		
		1 x 0,5 ... 2,5 mm² 2 x 0,5 ... 1 mm²			1 x 0,5 ... 2,5 mm²		2 x 0,5 ... 1 mm²		
AWG		1 x 20 ... 12	AWG		1 x 20 ... 12	AWG	1 x 20 ... 12		

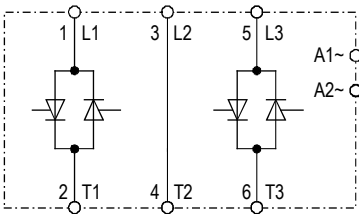
DE	EN	FR	ES	IT	PT
Schaltungsskizze	Circuit diagram	Schéma intérieur	Esquema eléctrico	Schema elettrico	Esquema do circuito

3RF24...-BB0.



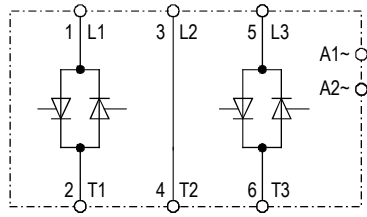
DC 24 V

3RF24...-BB2.



AC 110 ... 230 V

3RF24...-BB5.



AC 230 V

Technical Assistance:

Telephone: +49 (0) 911-895-5900 (8° - 17° CET)
E-mail: technical-assistance@siemens.com
Internet: www.siemens.de/lowvoltage/technical-assistance

Fax: +49 (0) 911-895-5907

Technical Support:

Telephone: +49 (0) 180 50 50 222

Technische Änderungen vorbehalten. Zum späteren Gebrauch aufbewahren.
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Bestell-Nr./Order No.: 3ZX1012-0RF24-3AA1
Printed in the Federal Republic of Germany



Betriebsanleitung/Operating Instructions

Bestell-Nr./Order No.: 3ZX1012-0RF00-1AW1

	⚠ WARNING Gefährliche elektrische Spannung! Kann zu elektrischem Schlag und Verbrennungen führen. Vor Beginn der Arbeiten Anlage und Gerät spannungsfrei schalten.	⚠ WARNING HAZARDOUS VOLTAGE. Can cause electrical shock and burns. Disconnect power before proceeding with any work on this equipment.	⚠ ATTENTION Tension dangereuse ! Risque d'électrocution et de brûlure. Isoler cet appareil du réseau avant d'y intervenir pour travaux.
	⚠ ADVERTENCIA ¡Tensión peligrosa! Puede causar choque eléctrico y quemaduras. Desconectar la alimentación antes de efectuar trabajo alguno en este equipo.	⚠ ATTENZIONE Tensione elettrica pericolosa! Rischio di shock elettrico e ustioni. Prima di eseguire qualsiasi tipo di lavoro, assicurarsi che l'apparecchio e l'impianto siano scollegati.	⚠ AVISO Tensão elétrica perigosa! Pode causar um choque elétrico e queimaduras. Antes de iniciar os trabalhos, desligue a tensão da unidade e do aparelho.

Eine sichere Gerätefunktion ist nur mit zertifizierten Komponenten gewährleistet.

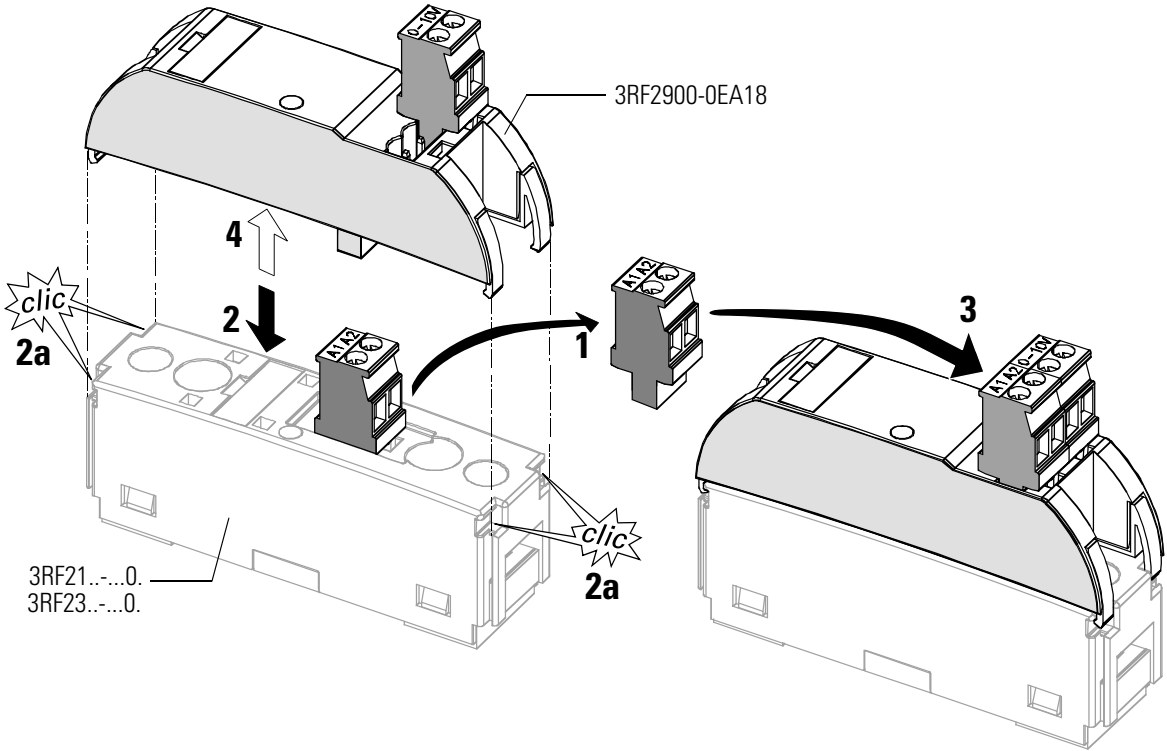
Reliable functioning of the equipment is only ensured with certified components.

Le fonctionnement sûr de l'appareil n'est garanti qu'avec des composants certifiés.

El funcionamiento seguro del aparato sólo está garantizado con componentes certificados.

Il funzionamento sicuro dell'apparecchiatura viene garantito soltanto con componenti certificati.

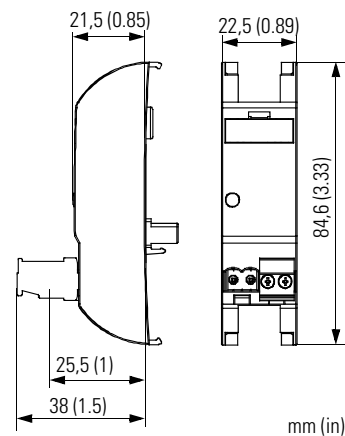
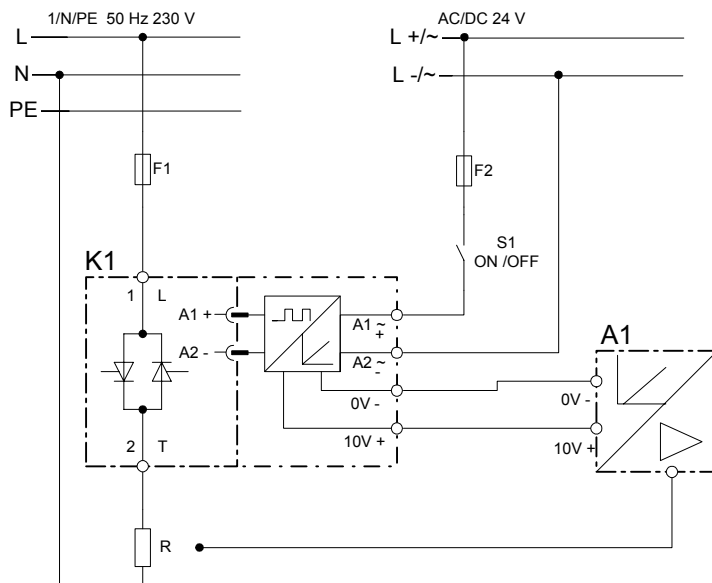
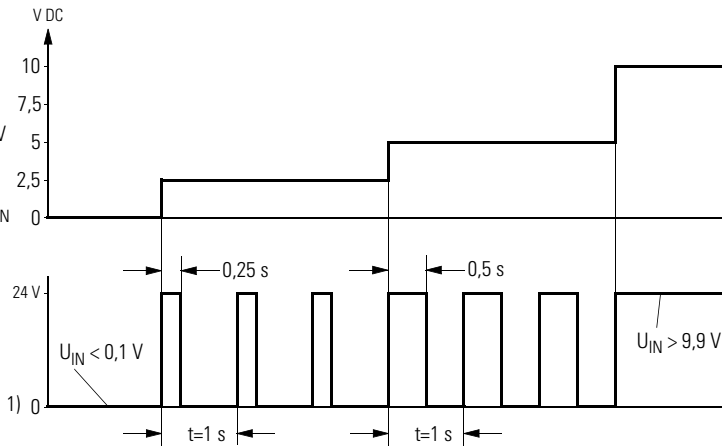
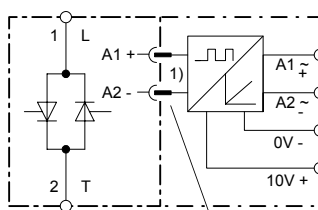
O funcionamento seguro do aparelho apenas pode ser garantido se forem utilizados os componentes certificados.

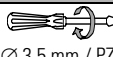


3RF21...-...0.

3RF23...-...0.

3RF2900-0EA18



	3RF2900-0EA18
 Ø 3,5 mm / PZ1	0,5 ... 0,6 Nm 4.5 to 5.3 lb · in
 7	1 x 0,5 ... 2,5 mm ² 2 x 0,5 ... 1 mm ²
 7	1 x 0,5 ... 2,5 mm ² 2 x 0,5 ... 1 mm ²
AWG	1 x 20 to 12

Technical Assistance: Telephone: +49 (0) 9131-7-43833 (8°° - 17°° CET)
E-mail: technical-assistance@siemens.com
Internet: www.siemens.de/lowvoltage/technical-assistance

Fax: +49 (0) 9131-7-42899

Technical Support: Telephone: +49 (0) 180 50 50 222

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I26076

Operator panels

Basic HMI

Basic Panels

Standard devices 2nd Generation

Overview



Basic Panels (2nd Generation)

SIMATIC HMI Basic Panels (2nd Generation) with their fully developed HMI basic functions are the ideal entry-level series for simple HMI applications.

The device family offers panels with 4", 7", 9" and 12" displays, as well as combined key and touch operation.

The innovative high-resolution widescreen displays with 64 000 colors are also suitable for upright installation, and they can be dimmed down to 100%. The innovative operator interface with improved usability opens up a diverse range of options thanks to new controls and graphics. The new USB interface enables the connection of keyboard, mouse or barcode scanner, and supports the simple archiving of data on a USB flash drive as well as the manual backup and restoring of the complete panel.

The integrated Ethernet or RS 485/422 interface (version-specific) enables simple connection to the controller.

<http://www.siemens.com/basic-panels>

Technical specifications

Article number	6AV2123-2DB03-0AX0 SIMATIC HMI KTP400 BASIC	6AV2123-2GB03-0AX0 SIMATIC HMI KTP700 BASIC	6AV2123-2GA03-0AX0 SIMATIC HMI KTP700 BASIC DP
General information			
Product type designation	KTP400 Basic color PN	KTP700 Basic color PN	KTP700 Basic color DP
Display			
Design of display	TFT widescreen display, LED backlighting	TFT widescreen display, LED backlighting	TFT widescreen display, LED backlighting
Screen diagonal	4.3 in	7 in	7 in
Number of colors	65 536	65 536	65 536
Resolution (pixels)			
• Horizontal image resolution	480 Pixel	800 Pixel	800 Pixel
• Vertical image resolution	272 Pixel	480 Pixel	480 Pixel
Backlighting			
• MTBF backlighting (at 25 °C)	20 000 h	20 000 h	20 000 h
• Backlight dimmable	Yes	Yes	Yes
Keyboard fonts			
• Function keys			
- Number of function keys	4	8	8
- Number of function keys with LEDs	0		
Touch operation			
• Design as touch screen	Yes	Yes	Yes
Installation type/mounting			
Mounting in portrait format possible	Yes	Yes	Yes
Supply voltage			
Type of supply voltage	DC	DC	DC
Rated value (DC)	24 V	24 V	24 V
Memory			
Memory available for user data	10 Mbyte	10 Mbyte	10 Mbyte
Type of output			
Acoustics			
• Buzzer	Yes	Yes	Yes
Time of day			
Clock			
• Software clock	Yes	Yes	Yes
• retentive	Yes; Back-up duration typically 6 weeks	Yes; Back-up duration typically 6 weeks	Yes; Back-up duration typically 6 weeks
• synchronizable	Yes	Yes	Yes

Technical specifications (continued)

Article number	6AV2123-2DB03-0AX0 SIMATIC HMI KTP400 BASIC	6AV2123-2GB03-0AX0 SIMATIC HMI KTP700 BASIC	6AV2123-2GA03-0AX0 SIMATIC HMI KTP700 BASIC DP
Interfaces			
Number of industrial Ethernet interfaces	1	1	0
Number of RS 485 interfaces	0	0	1
Number of USB interfaces	1; Up to 16 GB	1; Up to 16 GB	1; Up to 16 GB
Number of SD card slots	0	0	0
Protocols			
PROFINET	Yes	Yes	No
IRT	No	No	No
MRP	No	No	No
PROFIBUS	No	No	Yes
MPI	No	No	Yes
Degree and class of protection			
IP (at the front)	IP65	IP65	IP65
Enclosure Type 4x at the front	Yes	Yes	Yes
IP (rear)	IP20	IP20	IP20
Standards, approvals, certificates			
CE mark	Yes	Yes	Yes
cULus	Yes	Yes	Yes
RCM (formerly C-TICK)	Yes	Yes	Yes
KC approval	Yes	Yes	Yes
Marine approval			
• Germanischer Lloyd (GL)	Yes	Yes	Yes
• American Bureau of Shipping (ABS)	Yes	Yes	Yes
• Bureau Veritas (BV)	Yes	Yes	Yes
• Det Norske Veritas (DNV)	Yes	Yes	Yes
• Lloyds Register of Shipping (LRS)	Yes	Yes	Yes
• Nippon Kaiji Kyokai (Class NK)	Yes	Yes	Yes
• Polski Rejestr Statkow (PRS)	No	No	No
• Chinese Classification Society (CCS)	No	No	No
Use in hazardous areas			
• ATEX Zone 2	No	No	No
• ATEX Zone 22	No	No	No
• IECEx Zone 2	No	No	No
• IECEx Zone 22	No	No	No
• cULus Class I Zone 2, Division 2	No	No	No
• FM Class I Division 2	No	No	No
Ambient conditions			
Suited for indoor use	Yes		Yes
Suited for outdoor use	No		No
Ambient temperature during operation			
• Operation (vertical installation)			
- For vertical installation, min.	0 °C	0 °C	0 °C
- For vertical installation, max.	50 °C	50 °C	50 °C
Relative humidity			
• Operation, max.	90 %; no condensation	90 %; no condensation	90 %; no condensation
Configuration			
Configuration software			
• STEP 7 Basic (TIA Portal)	Yes; via integrated WinCC Basic (TIA Portal)	Yes; via integrated WinCC Basic (TIA Portal)	Yes; via integrated WinCC Basic (TIA Portal)
• WinCC flexible Compact	No	No	
• WinCC Basic (TIA Portal)	Yes	Yes	Yes

Operator panels

Basic HMI

Basic Panels

Standard devices 2nd Generation

Technical specifications (continued)

Article number	6AV2123-2DB03-0AX0 SIMATIC HMI KTP400 BASIC	6AV2123-2GB03-0AX0 SIMATIC HMI KTP700 BASIC	6AV2123-2GA03-0AX0 SIMATIC HMI KTP700 BASIC DP
Languages			
Online languages			
• Number of online/runtime languages	10	10	10
Functionality under WinCC (TIA Portal)			
Applications/options			
• Web browser	Yes	Yes	No
• SIMATIC WinCC Sm@rtServer	Yes; Available with WinCC (TIA Portal) V14 or higher	Yes; Available with WinCC (TIA Portal) V14 or higher	No
Number of Visual Basic Scripts	No	No	No
Task planner	Yes	Yes	Yes
• time-controlled	No	No	No
• task-controlled	Yes	Yes	Yes
Message system			
• Bit messages			
- Number of bit messages	1 000	1 000	1 000
• Analog messages			
- Number of analog messages	25	25	25
• Message buffer			
- Number of entries	256	256	256
- Circulating buffer	Yes	Yes	Yes
- retentive	Yes	Yes	Yes
Recipe management			
• Number of recipes	50	50	50
• Size of internal recipe memory	256 kbyte	256 kbyte	256 kbyte
• Recipe memory expandable	No	No	No
Variables			
• Number of variables per device	800	800	800
• Number of variables per screen	100	100	100
Images			
• Number of configurable images	250	250	250
Archiving			
• Number of archives per device	2; One message and one process value archive	2; One message and one process value archive	2; One message and one process value archive
• Number of entries per archive	10 000	10 000	10 000
• Message archive	Yes	Yes	Yes
• Process value archive	Yes	Yes	Yes
• Archiving methods			
- Sequential archive	Yes	Yes	Yes
- Short-term archive	Yes	Yes	Yes
• Memory location			
- Memory card	No	No	No
- USB memory	Yes	Yes	Yes
- Ethernet	No	No	No
• Data storage format			
- CSV	No	No	No
- TXT	Yes	Yes	Yes
- RDB	No	No	No

Technical specifications (continued)

Article number	6AV2123-2DB03-0AX0 SIMATIC HMI KTP400 BASIC	6AV2123-2GB03-0AX0 SIMATIC HMI KTP700 BASIC	6AV2123-2GA03-0AX0 SIMATIC HMI KTP700 BASIC DP
Functionality under WinCC (TIA Portal) (continued)			
Security			
• Number of user groups	50	50	50
• Number of users	50	50	50
• SIMATIC Logon	No	No	No
Transfer (upload/download)			
• MPI/PROFIBUS DP	No	No	Yes
• Ethernet	Yes	Yes	No
Process coupling			
• S7-1200	Yes	Yes	Yes
• S7-1500	Yes	Yes	Yes
• S7-200	Yes	Yes	Yes
• S7-300/400	Yes	Yes	Yes
• LOGO!	Yes	Yes	Yes
• WinAC	Yes	Yes	Yes
• SIMOTION	Yes	Yes	Yes
• Allen Bradley (EtherNet/IP)	Yes	Yes	No
• Allen Bradley (DF1)	No	No	Yes
• Mitsubishi (MC TCP/IP)	Yes	Yes	No
• Mitsubishi (FX)	No	No	Yes
• OMRON (FINS TCP)	No	No	No
• OMRON (LINK/Multilink)	No	No	Yes
• Modicon (Modbus TCP/IP)	Yes	Yes	No
• Modicon (Modbus)	No	No	Yes
Peripherals/Options			
Peripherals			
• Printer	No	No	No
• SIMATIC HMI MM memory card: Multi Media Card	No	No	No
• SIMATIC HMI SD memory card: Secure Digital memory card	No	No	No
• USB memory	Yes	Yes	Yes
Mechanics/material			
Enclosure material (front)			
• Plastic	Yes	Yes	Yes
• Aluminum	No	No	No
• Stainless steel	No	No	No
Dimensions			
Width of the housing front	141 mm	214 mm	214 mm
Height of housing front	116 mm	158 mm	158 mm
Mounting cutout, width	123 mm	197 mm	197 mm
Mounting cutout, height	99 mm	141 mm	141 mm
Overall depth	33 mm	39 mm	39 mm
Weights			
Weight without packaging	360 g	780 g	800 g

Operator panels

Basic HMI

Basic Panels

Standard devices 2nd Generation

Technical specifications (continued)

Article number	6AV2123-2JB03-0AX0 SIMATIC HMI KTP900 BASIC	6AV2123-2MB03-0AX0 SIMATIC HMI KTP1200 BASIC	6AV2123-2MA03-0AX0 SIMATIC HMI KTP1200 BASIC DP
General information			
Product type designation	KTP900 Basic color PN	KTP1200 Basic color PN	KTP1200 Basic color DP
Display			
Design of display	TFT widescreen display, LED backlighting	TFT widescreen display, LED backlighting	TFT widescreen display, LED backlighting
Screen diagonal	9 in	12 in	12 in
Number of colors	65 536	65 536	65 536
Resolution (pixels)			
• Horizontal image resolution	800 Pixel	1 280 Pixel	1 280 Pixel
• Vertical image resolution	480 Pixel	800 Pixel	800 Pixel
Backlighting			
• MTBF backlighting (at 25 °C)	20 000 h	20 000 h	20 000 h
• Backlight dimmable	Yes	Yes	Yes
Keyboard fonts			
• Function keys			
- Number of function keys	8	10	10
Touch operation			
• Design as touch screen	Yes	Yes	Yes
Installation type/mounting			
Mounting in portrait format possible	Yes	Yes	Yes
Supply voltage			
Type of supply voltage	DC	DC	DC
Rated value (DC)	24 V	24 V	24 V
Memory			
Memory available for user data	10 Mbyte	10 Mbyte	10 Mbyte
Type of output			
Acoustics			
• Buzzer	Yes	Yes	Yes
Time of day			
Clock			
• Software clock	Yes	Yes	Yes
• retentive	Yes; Back-up duration typically 6 weeks	Yes; Back-up duration typically 6 weeks	Yes; Back-up duration typically 6 weeks
• synchronizable	Yes	Yes	Yes
Interfaces			
Number of industrial Ethernet interfaces	1	1	0
Number of RS 485 interfaces	0	0	1
Number of USB interfaces	1; Up to 16 GB	1; Up to 16 GB	1; Up to 16 GB
Number of SD card slots	0	0	0
Protocols			
PROFINET	Yes	Yes	No
IRT	No	No	No
MRP	No	No	No
PROFIBUS	No	No	Yes
MPI	No	No	Yes
Degree and class of protection			
IP (at the front)	IP65	IP65	IP65
Enclosure Type 4x at the front	Yes	Yes	Yes
IP (rear)	IP20	IP20	IP20

Technical specifications (continued)

Article number	6AV2123-2JB03-0AX0 SIMATIC HMI KTP900 BASIC	6AV2123-2MB03-0AX0 SIMATIC HMI KTP1200 BASIC	6AV2123-2MA03-0AX0 SIMATIC HMI KTP1200 BASIC DP
Standards, approvals, certificates			
CE mark	Yes	Yes	Yes
cULus	Yes	Yes	Yes
RCM (formerly C-TICK)	Yes	Yes	Yes
KC approval	Yes	Yes	Yes
Marine approval			
• Germanischer Lloyd (GL)	Yes	Yes	Yes
• American Bureau of Shipping (ABS)	Yes	Yes	Yes
• Bureau Veritas (BV)	Yes	Yes	Yes
• Det Norske Veritas (DNV)	Yes	Yes	Yes
• Lloyds Register of Shipping (LRS)	Yes	Yes	Yes
• Nippon Kaiji Kyokai (Class NK)	Yes	Yes	Yes
• Polski Rejestr Statkow (PRS)	No	No	No
• Chinese Classification Society (CCS)	No	No	No
Use in hazardous areas			
• ATEX Zone 2	No	No	No
• ATEX Zone 22	No	No	No
• IECEx Zone 2	No	No	No
• IECEx Zone 22	No	No	No
• cULus Class I Zone 2, Division 2	No	No	No
• FM Class I Division 2	No	No	No
Ambient conditions			
Suited for indoor use			Yes
Suited for outdoor use			No
Ambient temperature during operation			
• Operation (vertical installation)			
- For vertical installation, min.	0 °C	0 °C	0 °C
- For vertical installation, max.	50 °C	50 °C	50 °C
Relative humidity			
• Operation, max.	90 %; no condensation	90 %; no condensation	90 %; no condensation
Configuration			
Configuration software			
• STEP 7 Basic (TIA Portal)	Yes; via integrated WinCC Basic (TIA Portal)	Yes; via integrated WinCC Basic (TIA Portal)	Yes; via integrated WinCC Basic (TIA Portal)
• WinCC flexible Compact	No	No	No
• WinCC Basic (TIA Portal)	Yes	Yes	Yes
Languages			
Online languages			
• Number of online/runtime languages	10	10	10

Operator panels

Basic HMI

Basic Panels

Standard devices 2nd Generation

Technical specifications (continued)

Article number	6AV2123-2JB03-0AX0 SIMATIC HMI KTP900 BASIC	6AV2123-2MB03-0AX0 SIMATIC HMI KTP1200 BASIC	6AV2123-2MA03-0AX0 SIMATIC HMI KTP1200 BASIC DP
Functionality under WinCC (TIA Portal)			
Applications/options			
• Web browser	Yes	Yes	No
• SIMATIC WinCC Sm@rtServer	Yes; Available with WinCC (TIA Portal) V14 or higher	Yes; Available with WinCC (TIA Portal) V14 or higher	No
Number of Visual Basic Scripts	No	No	No
Task planner	Yes	Yes	Yes
• time-controlled	No	No	No
• task-controlled	Yes	Yes	Yes
Message system			
• Bit messages			
- Number of bit messages	1 000	1 000	1 000
• Analog messages			
- Number of analog messages	25	25	25
• Message buffer			
- Number of entries	256	256	256
- Circulating buffer	Yes	Yes	Yes
- retentive	Yes	Yes	Yes
Recipe management			
• Number of recipes	50	50	50
• Size of internal recipe memory	256 kbyte	256 kbyte	256 kbyte
• Recipe memory expandable	No	No	No
Variables			
• Number of variables per device	800	800	800
• Number of variables per screen	100	100	100
Images			
• Number of configurable images	250	250	250
Archiving			
• Number of archives per device	2; One message and one process value archive	2; One message and one process value archive	2; One message and one process value archive
• Number of entries per archive	10 000	10 000	10 000
• Message archive	Yes	Yes	Yes
• Process value archive	Yes	Yes	Yes
• Archiving methods			
- Sequential archive	Yes	Yes	Yes
- Short-term archive	Yes	Yes	Yes
• Memory location			
- Memory card	No	No	No
- USB memory	Yes	Yes	Yes
- Ethernet	No	No	No
• Data storage format			
- CSV	No	No	No
- TXT	Yes	Yes	Yes
- RDB	No	No	No

Technical specifications (continued)

Article number	6AV2123-2JB03-0AX0 SIMATIC HMI KTP900 BASIC	6AV2123-2MB03-0AX0 SIMATIC HMI KTP1200 BASIC	6AV2123-2MA03-0AX0 SIMATIC HMI KTP1200 BASIC DP
Functionality under WinCC (TIA Portal) (continued)			
Security			
• Number of user groups	50	50	50
• Number of users	50	50	50
• SIMATIC Logon	No	No	No
Transfer (upload/download)			
• MPI/PROFIBUS DP	No	No	Yes
• Ethernet	Yes	Yes	No
Process coupling			
• S7-1200	Yes	Yes	Yes
• S7-1500	Yes	Yes	Yes
• S7-200	Yes	Yes	Yes
• S7-300/400	Yes	Yes	Yes
• LOGO!	Yes	Yes	Yes
• WinAC	Yes	Yes	Yes
• SIMOTION	Yes	Yes	Yes
• Allen Bradley (EtherNet/IP)	Yes	Yes	No
• Allen Bradley (DF1)	No	No	Yes
• Mitsubishi (MC TCP/IP)	Yes	Yes	No
• Mitsubishi (FX)	No	No	Yes
• OMRON (FINS TCP)	No	No	No
• OMRON (LINK/Multilink)	No	No	Yes
• Modicon (Modbus TCP/IP)	Yes	Yes	No
• Modicon (Modbus)	No	No	Yes
Peripherals/Options			
Peripherals			
• Printer	No	No	No
• SIMATIC HMI MM memory card: Multi Media Card	No	No	No
• SIMATIC HMI SD memory card: Secure Digital memory card	No	No	No
• USB memory	Yes	Yes	Yes
Mechanics/material			
Enclosure material (front)			
• Plastic	Yes	Yes	Yes
• Aluminum	No	No	No
• Stainless steel	No	No	No
Dimensions			
Width of the housing front	267 mm	330 mm	330 mm
Height of housing front	182 mm	245 mm	245 mm
Mounting cutout, width	251 mm	310 mm	310 mm
Mounting cutout, height	166 mm	221 mm	221 mm
Overall depth	55 mm	60 mm	60 mm
Weights			
Weight without packaging	1 130 g	1 710 g	1 710 g

Operator panels

Basic HMI

Basic Panels

Standard devices 2nd Generation

Ordering data	Article No.	Article No.
SIMATIC HMI Basic Panels (2nd Generation)		
Key and Touch devices		
SIMATIC HMI KTP400 Basic Key/touch-screen operation; 4" TFT widescreen display, 65 536 colors, PROFINET interface	6AV2123-2DB03-0AX0	Starter kits
SIMATIC HMI KTP700 Basic Key/touch-screen operation; 7" TFT display, 65 536 colors, PROFINET interface	6AV2123-2GB03-0AX0	Starter kit SIMATIC S7-1200 + KP300 Basic mono PN
SIMATIC HMI KTP700 Basic DP Key/touch-screen operation; 7" TFT display, 65 536 colors, PROFIBUS interface	6AV2123-2GA03-0AX0	Starter kit SIMATIC S7-1200 + KTP400 Basic
SIMATIC HMI KTP900 Basic Key/touch-screen operation; 9" TFT display, 65 536 colors, PROFINET interface	6AV2123-2JB03-0AX0	Starter kit SIMATIC S7-1200 + KTP700 Basic
SIMATIC HMI KTP1200 Basic Key/touch-screen operation; 12" TFT display, 65 536 colors, PROFINET interface	6AV2123-2MB03-0AX0	Starter kits with an S7-1200 consist of:
SIMATIC HMI KTP1200 Basic DP Key/touch-screen operation; 12" TFT display, 65 536 colors, PROFIBUS interface	6AV2123-2MA03-0AX0	- the respective SIMATIC HMI Basic Panel: SIMATIC HMI KP300 Basic mono PN, SIMATIC HMI KTP400 Basic, SIMATIC HMI KTP700 Basic - SIMATIC S7-1200 CPU 1212C AC/DC/Rly - SIMATIC S7-1200 Simulator Module SIM 12 - SIMATIC STEP 7 BASIC CD - SIMATIC S7-1200 HMI Manual Collection CD - Ethernet CAT5 cable, 2 m
		Starter kit LOGO! + KP300 Basic mono PN
		Starter kit LOGO! + KTP400 Basic
		Starter kit LOGO! + KTP700 Basic
		Starter kits with a LOGO! consist of:
		- the respective SIMATIC HMI Basic Panel: SIMATIC HMI KP300 Basic mono PN, SIMATIC HMI KTP400 Basic, SIMATIC HMI KTP700 Basic - LOGO! 12/24 RCE - LOGO! POWER 24 V 1.3 A - LOGO! SOFT COMFORT V7 - WINCC BASIC (TIA Portal) - Ethernet CAT5 cable, 2 m
		Documentation
		You can find the manual for the Basic Panels on the Internet at:
		http://support.automation.siemens.com
		Accessories
		See chapter 6

More information

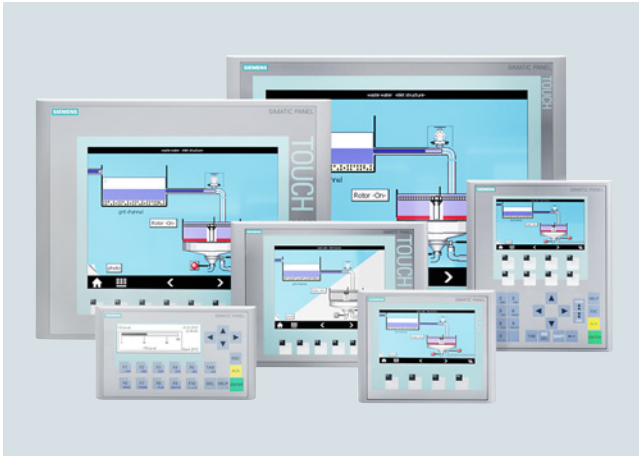
More information is available on the Internet at:

<http://www.siemens.com/basic-panels>

Note:

Do you need a specific modification or option for the products described here? Then look up "Customized products", where you will find information about additional sector-specific products that can be ordered as well as about options for customer-specific modification and adaptation.

Overview

**Basic Panels (1st Generation)**

- Ideal entry-level series from 3" to 15" for operator control and monitoring of compact machines and systems
- Clear process representation thanks to use of pixel-graphics displays
- Intuitive operation using touch and tactile function keys
- Equipped with all the necessary basic functions such as alarm logging, recipe management, plots, vector graphics, and language switching
- Simple connection to the controller via integral Ethernet interface or separate version with RS 485/422
- Faster commissioning thanks to integrated diagnostics viewer and IP setting for SIMATIC S7-1200 and S7-1500 PLCs

2

Technical specifications

Article number	6AV6647-0AH11-3AX0 SIMATIC HMI KP300 BASIC MONO PN	6AV6647-0AJ11-3AX0 SIMATIC HMI KP400 BASIC COLOR PN
General information		
Product type designation	KP300 Basic mono PN	KP400 Basic color PN
Display		
Design of display	FSTN	TFT
Screen diagonal	3.6 in	4.3 in
Number of colors	4; Backlit display only (white, red, green, yellow)	256
Resolution (pixels)		
• Horizontal image resolution	240 Pixel	480 Pixel
• Vertical image resolution	80 Pixel	272 Pixel
Backlighting		
• MTBF backlighting (at 25 °C)	50 000 h	50 000 h
• Backlight dimmable	No	No
Keyboard fonts		
• Function keys		
- Number of function keys	10	8
Touch operation		
• Design as touch screen	No	No
Installation type/mounting		
Mounting in portrait format possible	No	No
Supply voltage		
Type of supply voltage	DC	DC
Rated value (DC)	24 V	24 V

Article number	6AV6647-0AH11-3AX0 SIMATIC HMI KP300 BASIC MONO PN	6AV6647-0AJ11-3AX0 SIMATIC HMI KP400 BASIC COLOR PN
Memory		
Memory available for user data	1 Mbyte	1 Mbyte
Type of output		
Acoustics		
• Buzzer	No	No
Time of day		
Clock		
• Software clock	Yes	Yes
• retentive	No	No
• synchronizable	Yes	Yes
Interfaces		
Number of industrial Ethernet interfaces	1	1
Number of RS 485 interfaces	0	0
Number of USB interfaces	0	0
Number of SD card slots	0	0
Protocols		
PROFINET	Yes	Yes
IRT	No	No
PROFIBUS	No	No
MPI	No	No
Degree and class of protection		
IP (at the front)	IP65	IP65
Enclosure Type 4x at the front	Yes	Yes
IP (rear)	IP20	IP20

Operator panels

Basic HMI

Basic Panels

Standard devices 1st Generation

Technical specifications (continued)

Article number	6AV6647-0AH11-3AX0 SIMATIC HMI KP300 BASIC MONO PN	6AV6647-0AJ11-3AX0 SIMATIC HMI KP400 BASIC COLOR PN
Standards, approvals, certificates		
CE mark	Yes	Yes
cULus	Yes	Yes
RCM (formerly C-TICK)	Yes	Yes
KC approval	Yes	Yes
Marine approval		
• Germanischer Lloyd (GL)	Yes	No
• American Bureau of Shipping (ABS)	Yes	No
• Bureau Veritas (BV)	Yes	No
• Det Norske Veritas (DNV)	Yes	No
• Lloyds Register of Shipping (LRS)	Yes	No
• Nippon Kaiji Kyokai (Class NK)	Yes	No
• Polski Rejestr Statkow (PRS)	No	No
Use in hazardous areas		
• ATEX Zone 2	No	No
• ATEX Zone 22	No	No
• cULus Class I Zone 2, Division 2	Yes	No
• FM Class I Division 2	No	No
Ambient conditions		
Ambient temperature during operation		
• Operation (vertical installation)		
- For vertical installation, min.	0 °C	0 °C
- For vertical installation, max.	50 °C	50 °C
Relative humidity		
• Operation, max.	90 %; no condensation	90 %; no condensation
Configuration		
Configuration software		
• STEP 7 Basic (TIA Portal)	Yes; via integrated WinCC Basic (TIA Portal)	Yes; via integrated WinCC Basic (TIA Portal)
• WinCC flexible Compact	No	No
• WinCC Basic (TIA Portal)	Yes	Yes
Languages		
Online languages		
• Number of online/runtime languages	5	5
Functionality under WinCC (TIA Portal)		
Task planner		
• time-controlled	No	No
• task-controlled	Yes	Yes
Message system		
• Bit messages		
- Number of bit messages	200	200
• Analog messages		
- Number of analog messages	15	15
• Message buffer		
- Number of entries	256	256
- Circulating buffer	Yes	Yes
- retentive	Yes	Yes

Article number	6AV6647-0AH11-3AX0 SIMATIC HMI KP300 BASIC MONO PN	6AV6647-0AJ11-3AX0 SIMATIC HMI KP400 BASIC COLOR PN
Recipe management		
• Number of recipes	5	5
• Size of internal recipe memory	40 kbyte	40 kbyte
• Recipe memory expandable	No	No
Variables		
• Number of variables per device	250	500
• Number of variables per screen	30	30
Images		
• Number of configurable images	50	50
Archiving		
• Number of archives per device	0	0
Security		
• Number of user groups	50	50
• Number of users	50	50
• SIMATIC Logon	No	No
Transfer (upload/download)		
• MPI/PROFIBUS DP	No	No
• Ethernet	Yes	Yes
Process coupling		
• S7-1200	Yes	Yes
• S7-1500	Yes	Yes
• S7-200	Yes	Yes
• S7-300/400	Yes	Yes
• LOGO!	Yes	Yes
• WinAC	Yes	No
• SIMOTION	No	No
• Allen Bradley (EtherNet/IP)	Yes	Yes
• Allen Bradley (DF1)	No	No
• Mitsubishi (MC TCP/IP)	Yes	Yes
• Mitsubishi (FX)	No	No
• OMRON (FINS TCP)	No	No
• OMRON (LINK/Multilink)	No	No
• Modicon (Modbus TCP/IP)	Yes	Yes
• Modicon (Modbus)	No	No
Peripherals/Options		
Peripherals		
• Printer	No	No
• SIMATIC HMI MM memory card: Multi Media Card	No	No
• SIMATIC HMI SD memory card: Secure Digital memory card	No	No
• USB memory	No	No
Mechanics/material		
Enclosure material (front)		
• Plastic	Yes	Yes
• Aluminum	No	No
• Stainless steel	No	No
Dimensions		
Width of the housing front	165 mm	162 mm
Height of housing front	97 mm	189 mm
Mounting cutout, width	149 mm	135 mm
Mounting cutout, height	82 mm	171 mm
Overall depth	30 mm	33 mm
Weights		
Weight without packaging	0.25 kg	0.51 kg

Ordering data	Article No.
SIMATIC HMI Basic Panels (1st Generation)	
Key and Touch devices	
SIMATIC HMI KTP400 Basic mono PN Key/touch-screen operation; 4" STN display; 4 gray levels; PROFINET interface	6AV6647-0AA11-3AX0
SIMATIC HMI KTP400 Basic color PN Key/touch-screen operation; 4" widescreen TFT display, 256 colors, PROFINET interface	6AV6647-0AK11-3AX0
SIMATIC HMI KTP600 Basic mono PN Key/touch-screen operation; 6" STN display, 4 gray levels, PROFINET interface	6AV6647-0AB11-3AX0
SIMATIC HMI KTP600 Basic color DP Key/touch-screen operation; 6" TFT display, 256 colors, MPI/PROFIBUS DP interface	6AV6647-0AC11-3AX0
SIMATIC HMI KTP600 Basic color PN Key/touch-screen operation; 6" TFT display, 256 colors, PROFINET interface	6AV6647-0AD11-3AX0
SIMATIC HMI KTP1000 Basic color DP Key/touch-screen operation; 10" TFT display, 256 colors, MPI/PROFIBUS DP interface	6AV6647-0AE11-3AX0
SIMATIC HMI KTP1000 Basic color PN Key/touch-screen operation; 10" TFT display, 256 colors, PROFINET interface	6AV6647-0AF11-3AX0
Key devices	
SIMATIC HMI KP300 Basic mono PN Key operation; 3" FSTN LCD display, black and white, PROFINET interface	6AV6647-0AH11-3AX0
SIMATIC HMI KP400 Basic color PN Key operation; 4" widescreen TFT display, 256 colors, PROFINET interface	6AV6647-0AJ11-3AX0
Touch devices	
SIMATIC HMI TP1500 Basic color PN Touch-screen operation; 15" TFT display, 256 colors, PROFINET interface	6AV6647-0AG11-3AX0
Documentation	
You can find the manual for the Basic Panels on the Internet at:	http://support.automation.siemens.com
Accessories	See chapter 6

More informationNote:

Do you need a specific modification or option for the products described here? Then look up "Customized products", where you will find information about additional sector-specific products that can be ordered as well as about options for customer-specific modification and adaptation.

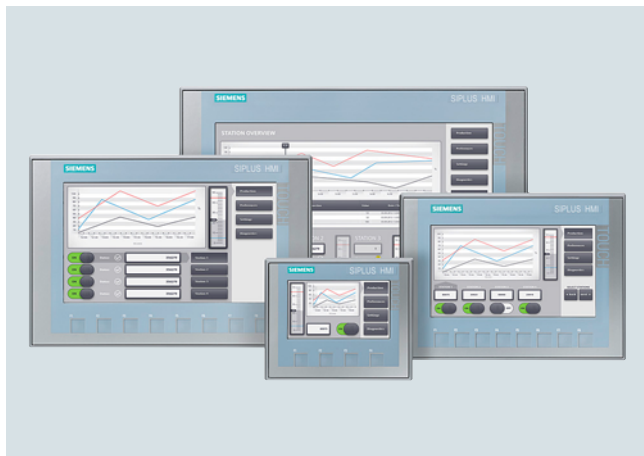
Operator panels

Basic HMI

Basic Panels

SIPLUS Basic Panels (2nd Generation)

Overview



Note:

SIPLUS extreme products are based on SIMATIC standard products. The contents listed here were taken from the respective standard products. SIPLUS extreme-specific information was added.

Technical documentation on SIPLUS can be found here:
<http://www.siemens.com/siplus-extreme>

Technical specifications

Article number	6AG1123-2DB03-2AX0 SIPLUS HMI KTP400 BASIC	6AG1123-2GB03-2AX0 SIPLUS HMI KTP700 BASIC	6AG1123-2GA03-2AX0 SIPLUS HMI KTP700 BASIC DP
Ambient conditions			
Suited for indoor use		Yes	Yes
Suited for outdoor use		No	No
Ambient temperature during operation			
• Operation (vertical installation)			
- For vertical installation, min.	-20 °C; = Tmin	-20 °C	-20 °C; = Tmin
- For vertical installation, max.	60 °C; = Tmax	50 °C	50 °C
• At cold restart, min.	0 °C	0 °C	0 °C
Altitude during operation relating to sea level			
• Installation altitude above sea level, max.	3 000 m	3 000 m	3 000 m
• Ambient air temperature-barometric pressure-altitude	Tmin ... Tmax at 1 140 hPa ... 795 hPa (-1 000 m ... +2 000 m) // Tmin ... Tmax -5K at 795 hPa ... 701 hPa (+2 000 m ... +3 000 m)		
Resistance			
Use in stationary industrial systems			
- to biologically active substances according to EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna); Class 3B3 on request		
- to chemically active substances according to EN 60721-3-3	Yes; Class 3C4 (RH < 75 %) incl. salt spray acc. to EN 60068-2-52 (severity degree 3); *		
- to mechanically active substances according to EN 60721-3-3	Yes; Class 3S4 incl. sand, dust, *	Yes; Class 3S4 incl. sand, dust, *	Yes; Class 3S4 incl. sand, dust, *
Use on ships/at sea			
- to biologically active substances according to EN 60721-3-6	Yes; Class 6B2 mold and fungal spores (excluding fauna); Class 6B3 on request		
- to chemically active substances according to EN 60721-3-6	Yes; Class 6C3 (RH < 75 %) incl. salt spray acc. to EN 60068-2-52 (severity degree 3); *		
- to mechanically active substances according to EN 60721-3-6	Yes; Class 6S3 incl. sand, dust; *	Yes; Class 6S3 incl. sand, dust; *	Yes; Class 6S3 incl. sand, dust; *
from supply voltage 1L+			
- Note regarding classification of environmental conditions acc. to EN 60721	* The supplied plug covers must remain in place over the unused interfaces during operation!		

Technical specifications (continued)

Article number	6AG1123-2JB03-2AX0 SIPLUS HMI KTP900 BASIC	6AG1123-2MB03-2AX0 SIPLUS HMI KTP1200 BASIC	6AG1123-2MA03-2AX0 SIPLUS HMI KTP1200 BASIC DP
Ambient conditions			
Suited for indoor/outdoor use	Yes/No	Yes/No	Yes/No
Ambient temperature during operation			
• Operation (vertical installation)			
- min./max.	-20 °C/50 °C	-10 °C; = Tmin/50 °C	-10 °C; = Tmin/50 °C
Altitude during operation relating to sea level			
• Ambient air temperature-barometric pressure-altitude	Tmin ... Tmax at 1 140 hPa ... 795 hPa (-1 000 m ... +2 000 m) // Tmin ... Tmax -5K) at 795 hPa ... 701 hPa (+2 000 m ... +3 000 m)		
Resistance			
Use in stationary industrial systems			
- to biologically active substances according to EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna); Class 3B3 on request		
- to chemically active substances according to EN 60721-3-3	Yes; Class 3C4 (RH < 75 %) incl. salt spray acc. to EN 60068-2-52 (severity degree 3); *		
- to mechanically active substances according to EN 60721-3-3	Yes; Class 3S4 incl. sand, dust, *	Yes; Class 3S4 incl. sand, dust, *	Yes; Class 3S4 incl. sand, dust, *
Use on ships/at sea			
- to biologically active substances according to EN 60721-3-6	Yes; Class 6B2 mold and fungal spores (excluding fauna); Class 6B3 on request		
- to chemically active substances according to EN 60721-3-6	Yes; Class 6C3 (RH < 75 %) incl. salt spray acc. to EN 60068-2-52 (severity degree 3); *		
- to mechanically active substances according to EN 60721-3-6	Yes; Class 6S3 incl. sand, dust; *	Yes; Class 6S3 incl. sand, dust; *	Yes; Class 6S3 incl. sand, dust; *
from supply voltage 1L+			
- Note regarding classification of environmental conditions acc. to EN 60721	* The supplied plug covers must remain in place over the unused interfaces during operation!		

Ordering data

Article No.	Article No.
SIPLUS HMI Basic Panels, Key and Touch	SIPLUS HMI KTP900 Basic
SIPLUS HMI KTP400 Basic	For areas with extreme exposure to environmental substances (conformal coating); ambient temperature -20 ... +50 °C
SIPLUS HMI KTP700 Basic	6AG1123-2JB03-2AX0
For areas with extreme exposure to environmental substances (conformal coating); ambient temperature -20 ... +60 °C	SIPLUS HMI KTP1200 Basic
SIPLUS HMI KTP700 Basic DP	For areas with extreme exposure to environmental substances (conformal coating); ambient temperature -10 ... +50 °C
For areas with extreme exposure to environmental substances (conformal coating); ambient temperature -20 ... +50 °C	6AG1123-2MB03-2AX0
SIPLUS HMI KTP700 Basic DP	SIPLUS HMI KTP1200 Basic DP
For areas with extreme exposure to environmental substances (conformal coating); ambient temperature -20 ... +50 °C	For areas with extreme exposure to environmental substances (conformal coating); ambient temperature -10 ... +50 °C
6AG1123-2GA03-2AX0	6AG1123-2MA03-2AX0
	Accessories
	See SIMATIC Basic Panels 2 nd Generation

More information

More information is available on the Internet at:

<http://www.siemens.com/basic-panels>

Note:

Do you need a specific modification or extension to the products described here? Then look up "Customized products", where you will find information about additional and generally available sector-specific products, as well as options for customer-specific modifications and adaptations.

Operator panels

Basic HMI

Basic Panels

SIPLUS Basic Panels (1st Generation)

Overview



Note:

SIPLUS extreme products are based on SIMATIC standard products. The contents listed here were taken from the respective standard products. SIPLUS extreme-specific information was added.

Technical documentation on SIPLUS can be found here:

<http://www.siemens.com/siplus-extreme>

Technical specifications

Article number	6AG1647-0AH11-2AX0 SIPLUS HMI KP300 BASIC MONO PN 3.6"	6AG1647-0AA11-2AX0 SIPLUS KTP400 BASIC MONO PN	6AG1647-0AD11-2AX0 SIPLUS KTP600 BASIC COLOR PN
Ambient conditions			
Suited for indoor use	Yes	Yes	Yes
Suited for outdoor use	No	No	No
Ambient temperature during operation			
• Operation (vertical installation)			
- For vertical installation, min.	-25 °C	-10 °C	-25 °C
- For vertical installation, max.	60 °C	60 °C	60 °C
Use in stationary industrial systems			
- to biologically active substances according to EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna); Class 3B3 on request		
- to chemically active substances according to EN 60721-3-3	Yes; Class 3C4 (RH < 75 %) incl. salt spray acc. to EN 60068-2-52 (severity degree 3); *		
- to mechanically active substances according to EN 60721-3-3	Yes; Class 3S4 incl. sand, dust, *	Yes; Class 3S4 incl. sand, dust, *	Yes; Class 3S4 incl. sand, dust, *
Use on ships/at sea			
- to biologically active substances according to EN 60721-3-6	Yes; Class 6B2 mold and fungal spores (excluding fauna); Class 6B3 on request		
- to chemically active substances according to EN 60721-3-6	Yes; Class 6C3 (RH < 75 %) incl. salt spray acc. to EN 60068-2-52 (severity degree 3); *		
- to mechanically active substances according to EN 60721-3-6	Yes; Class 6S3 incl. sand, dust; *	Yes; Class 6S3 incl. sand, dust; *	Yes; Class 6S3 incl. sand, dust; *
from supply voltage 1L+			
- Note regarding classification of environmental conditions acc. to EN 60721	* The supplied plug covers must remain in place over the unused interfaces during operation!		

Technical specifications (continued)

Article number	6AG1647-0AE11-4AX0 SIPLUS KTP1000 BASIC COLOR DP 10.4"	6AG1647-0AF11-4AX0 SIPLUS KTP1000 BASIC COLOR PN 10.4"	6AG1647-0AG11-4AX0 SIPLUS TP1500 BASIC COLOR PN 15" 10.4"
Ambient conditions			
Ambient temperature during operation			
• Operation (vertical installation)	0 to +50 °C	0 to +50 °C	0 to +50 °C
- For vertical installation, min.	0 °C	0 °C	0 °C
- For vertical installation, max.	50 °C	50 °C	50 °C
Use in stationary industrial systems			
- to biologically active substances according to EN 60721-3-3	Yes; Class 3B2 mold, fungus and dry rot spores (with the exception of fauna); Class 3B3 on request		
- to chemically active substances according to EN 60721-3-3	Yes; Class 3C4 (RH < 75 %) incl. salt spray acc. to EN 60068-2-52 (severity degree 3); *		
- to mechanically active substances according to EN 60721-3-3	Yes; Class 3S4 incl. sand, dust, *	Yes; Class 3S4 incl. sand, dust, *	Yes; Class 3S4 incl. sand, dust, *
Use on ships/at sea			
- to biologically active substances according to EN 60721-3-6	Yes; Class 6B2 mold and fungal spores (excluding fauna); Class 6B3 on request		
- to chemically active substances according to EN 60721-3-6	Yes; Class 6C3 (RH < 75 %) incl. salt spray acc. to EN 60068-2-52 (severity degree 3); *		
- to mechanically active substances according to EN 60721-3-6	Yes; Class 6S3 incl. sand, dust; *	Yes; Class 6S3 incl. sand, dust; *	Yes; Class 6S3 incl. sand, dust; *
from supply voltage 1L+			
- Note regarding classification of environmental conditions acc. to EN 60721	* The supplied plug covers must remain in place over the unused interfaces during operation!		

Ordering data

Article No.	Article No.
SIPLUS HMI KTP300 Basic mono PN For areas with extreme exposure to media (conformal coating); ambient temperature -25 ... +60 °C	6AG1647-0AH11-2AX0
SIPLUS HMI KTP400 Basic mono PN For areas with extreme exposure to media (conformal coating); ambient temperature -10 ... +60 °C	6AG1647-0AA11-2AX0
SIPLUS HMI KTP600 Basic Color PN For areas with extreme exposure to media (conformal coating); ambient temperature -25 ... +60 °C	6AG1647-0AD11-2AX0
SIPLUS HMI KTP1000 Basic color DP For areas with extreme exposure to media (conformal coating); ambient temperature 0 ... +50 °C	6AG1647-0AE11-4AX0
SIPLUS HMI KTP1000 Basic color PN For areas with extreme exposure to media (conformal coating); ambient temperature 0 ... +50 °C	6AG1647-0AF11-4AX0
SIPLUS HMI TP1500 Basic color PN For areas with extreme exposure to media (conformal coating); ambient temperature 0 ... +50 °C	6AG1647-0AG11-4AX0
Accessories	See SIMATIC Basic Panels

More informationNote:

Do you need a specific modification or option for the products described here? Then look up "Customized products", where you will find information about additional sector-specific products that can be ordered as well as about options for customer-specific modification and adaptation.

The tolerances shown in the table below apply to new, essentially homogeneous thermocouple wire in the size range of 30 AWG to 8 AWG. These tolerances only apply to thermocouples used at temperatures not exceeding the recommended limits. If thermocouples are used at temperatures above the recommended limits, or in detrimental environments, the below stated tolerances may not apply.

Tolerances on Initial Values of EMF vs Temperature for Thermocouples

Reference Junction 0 °C [32 °F]. Published in ASTM E230

TYPE	TEMPERATURE RANGE for STANDARD TOLERANCES	STANDARD TOLERANCES	TEMPERATURE RANGE for SPECIAL TOLERANCES	SPECIAL TOLERANCES
J	(0 to 293) °C [32 to 559] °F (293 to 760) °C [559 to 1400] °F	± 2.2 °C [± 4 °F] ± 0.75%	(0 to 275) °C [32 to 527] °F (275 to 760) °C [527 to 1400] °F	± 1.1 °C [± 2 °F] ± 0.4%
K	(-200 to -110) °C [-328 to -166] °F (-110 to 0) °C [-166 to 32] °F (0 to 293) °C [32 to 559] °F (293 to 1260) °C [559 to 2300] °F	± 2% ^[1] ± 2.2 °C [± 4 °F] ^[1] ± 2.2 °C [± 4 °F] ± 0.75%	(0 to 275) °C [32 to 527] °F (275 to 1260) °C [527 to 2300] °F	^[2] ^[2] ± 1.1 °C [± 2 °F] ± 0.4%
N	(0 to 293) °C [32 to 559] °F (293 to 1260) °C [559 to 2300] °F	± 2.2 °C [± 4 °F] ^[1] ± 0.75%	(0 to 275) °C [32 to 527] °F (275 to 1260) °C [527 to 2300] °F	± 1.1 °C [± 2 °F] ± 0.4%
T	(-200 to -67) °C [-328 to -89] °F (-67 to 0) °C [-89 to 32] °F (0 to 133) °C [32 to 271] °F (133 to 370) °C [271 to 700] °F	± 1.5% ^[1] ± 1 °C [± 1.8 °F] ^[1] ± 1 °C [± 1.8 °F] ± 0.75%	(0 to 125) °C [32 to 257] °F (125 to 370) °C [257 to 700] °F	^[2] ^[2] ± 0.5 °C [± 0.9 °F] ± 0.4%
E	(0 to 870) °C [32 to 1600] °F	± 1.7 °C [± 3.06 °F] ^[3] or ± 0.5%	(0 to 870) °C [32 to 1600] °F	± 1.0 °C [± 1.8 °F] ^[3] or ± 0.4%
R	(0 to 600) °C [32 to 1112] °F (600 to 1480) °C [1112 to 2642] °F	± 1.5 °C [± 2.7 °F] ± 0.25%	(0 to 600) °C [32 to 1112] °F (600 to 1480) °C [1112 to 2700] °F	± 0.6 °C [± 1.1 °F] ± 0.1%
S	(0 to 600) °C [32 to 1112] °F (600 to 1480) °C [1112 to 2700] °F	± 1.5 °C [± 2.7 °F] ± 0.25%	(0 to 600) °C [32 to 1112] °F (600 to 1480) °C [1112 to 2642] °F	± 0.6 °C [± 1.1 °F] ± 0.1%
B	(870 to 1700) °C [1600 to 3100] °F	± 0.5%	(870 to 1700) °C [1600 to 3100] °F	± 0.25%
C	(0 to 400) °C [32 to 752] °F (400 to 2315) °C [752 to 4200] °F	± 4.4 °C [± 8 °F] ± 1.0%	Not Available	

[1] Thermocouples and thermocouple materials are supplied to meet the tolerance specified for temperatures above 0 °C. A thermocouple material may not conform to the published sub-zero limits of error for that material when purchased, unless conformance is agreed upon by customer and Pyromation when ordering.

[2] Special tolerances for sub-zero temperatures have not yet been established. The following limits for calibrations of types E and T are useful to start discussion between customer and Pyromation.

(-200 to 0) °C Type T ± 0.5 °C or ± 0.8%, whichever is greater

[3] The standard tolerances shown do not apply to Type E mineral-insulated, metal-sheathed (MIMS) thermocouples and thermocouple cables. The standard tolerances for MIMS Type E constructions are the greater of ± 2.2 °C or ± 1.75 % from 0 to 870 °C and the greater of ± 2.2 °C or ± 2 % from -200 to 0 °C.

Initial values of tolerance for Type J and special tolerance for Type K thermocouples below 0 °C are not given due to the characteristics of the materials.

Tolerances on Initial Values of EMF vs Temperature for Thermocouples

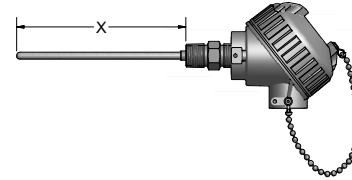
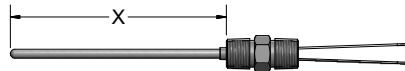
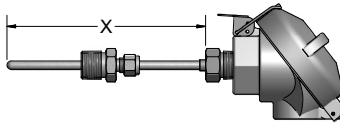
CODE	MATERIAL	TEMPERATURE RANGE	TOLERANCE
M	Ni18Mo/Ni	(-50 to 1410) °C [-58 to 2570] °F	± 0.75%
P	Platinel® II	(0 to 1395) °C [32 to 4200] °F	± 0.10 mV

Platinel® is a registered trademark of BASF Catalysts.

SENSORS WITH CONNECTION HEADS

Configuration Code GP02 Fixed-Sheath Thermocouple Assemblies with General-Purpose Connection Heads

Fixed-Sheath Thermocouple Assemblies with General-Purpose Connection Heads have head mounting fittings that are welded or brazed to the sheath for direct immersion into a process. To order an assembly with an optional 4 to 20 mA transmitter, select the assembly below and the transmitter from the back of this section. The MgO-insulated thermocouple assemblies are offered in a variety of calibrations, sheath diameters, and sheath materials.



ORDER CODES

Example Order Number:

1-0 1-1 1-2 1-3 2-0 3 4-0 4-1 4-2
J 3 8 U - 012 - 01A - 9HP 49, T Select Type and Range from back of section

1-0 Thermocouple Types

CODE	
SINGLE	DUPLEX
E	EE
J	JJ
K	KK
T	TT

1-1 Sheath Diameters

CODE	DIAMETER (inches)
2	1/8
3	3/16
4	1/4
6	3/8

1-2 Sheath Materials

CODE	MATERIAL	STANDARD AVAILABLE TYPES
3	Alloy 600	K
4	310 SS	K
5	446 SS	K ^[1]
8	316 SS	E, J, K, T

[1] All sensors with 446SS sheaths must have an ungrounded measuring junction.

1-3 Measuring Junctions

CODE	DESCRIPTION
G	Grounded junction
U	Ungrounded junction
E	Exposed junction

2-0 'X' Dimension

Insert three digit "X" length in inches

Sheath lengths over 72" will be shipped in a coiled configuration unless otherwise specified.

3-0 No Fitting

CODE	
00	

3-1 One-Time Adjustable Fittings

CODE	TYPE	NPT SIZE (inches)	PRESSURE-RATED	AVAILABLE SHEATH DIAMETERS (inches)
01A	303 SS	1/8	NO	1/8, 3/16, 1/4
05A	316 SS	1/8	YES	1/8, 3/16, 1/4
05B	316 SS	1/4	YES	1/8, 3/16, 1/4, 3/8
05C	316 SS	1/2	YES	1/8, 1/4, 3/8
15A	Brass	1/8	NO	1/8, 3/16, 1/4
15B	Brass	1/4	NO	3/16, 1/4, 3/8
15C	Brass	1/2	NO	1/4, 3/8
14	Brass/Steel	Flange	NO	1/8, 3/16, 1/4, 3/8

3-2 Re-Adjustable Compression Fittings

CODE	TYPE	NPT SIZE (inches)	AVAILABLE SHEATH DIAMETERS (inches)
10A	303 SS	1/8	1/8, 3/16
10B	303 SS	1/4	1/4, 3/8
10C	303 SS	1/2	1/4, 3/8
12A	316 SS	1/8	1/8, 3/16, 1/4
12B	316 SS	1/4	1/8, 3/16, 1/4, 3/8
12C	316 SS	1/2	1/8, 1/4, 3/8
11A	Brass	1/8	1/8, 3/16, 1/4
11B	Brass	1/4	1/8, 3/16, 1/4, 3/8
11C	Brass	1/2	1/4, 3/8
19C	Spring-loaded SS well fitting	1/2	3/16, 1/4

FEP gland standard 204 °C [400 °F] max.

3-3 Fixed Bushings^[1]

CODE	MOUNTING THREAD NPT (inches)	AVAILABLE SHEATH DIAMETERS (inches)
316 SS		
8A __ ^[2]	1/8	1/8, 3/16, 1/4
8B __ ^[2]	1/4	1/8, 3/16, 1/4, 3/8
8C __ ^[2]	1/2	1/8, 3/16, 1/4, 3/8
8D __ ^[2]	3/4	1/8, 3/16, 1/4, 3/8

[1] Requires Table 4, Option 9HP Selection

[2] When ordering fixed bushings, specify order code above plus insert length "U", as measured from hot tip to bottom of threaded bushing. EXAMPLE: order code 8A06 is 1/8" NPT, 316 SS bushing located 6" from hot tip.

4-0 Head Mounting Fittings

CODE	DESCRIPTION
6HN	1/2" x 1/2" NPT steel hex nipple 1" "E" length
8HN	1/2" x 1/2" NPT stainless steel hex nipple 1" "E" length
9HP	1/2" NPT stainless steel bushing (no process threads)
8RND	3/4" x 1/2" NPT stainless steel hex nipple

4-1 Head and Sheath Terminations

CODE	DESCRIPTION
22	3" Individual fluoropolymer leads with terminal pins
31	Aluminum screw-cover head
34	Cast iron screw-cover head
35T-642A	(4 to 20) mA HART® Field Transmitter with aluminum general-purpose housing
36T82-D10	(4 to 20) mA dual input HART® transmitter with digital display and general-purpose aluminum housing with glass lid
49	Flip-top aluminum head
63	White polypropylene screw-cover head
91	316 L stainless steel screw-cover head

4-2 Options

W ^[1]	Epoxy Coating
GS	Ground screw
I	Stainless tag
NB	1/2" NPT nylon conduit reducer bushing
SB	1/2" NPT conduit reducer bushing
T-441	(4 to 20) mA isolated head-mounted transmitter
T-442	(4 to 20) mA isolated Hart® head-mounted transmitter
T82-00	(4 to 20) mA dual input HART® head-mounted transmitter

See transmitter ordering information in back of section.

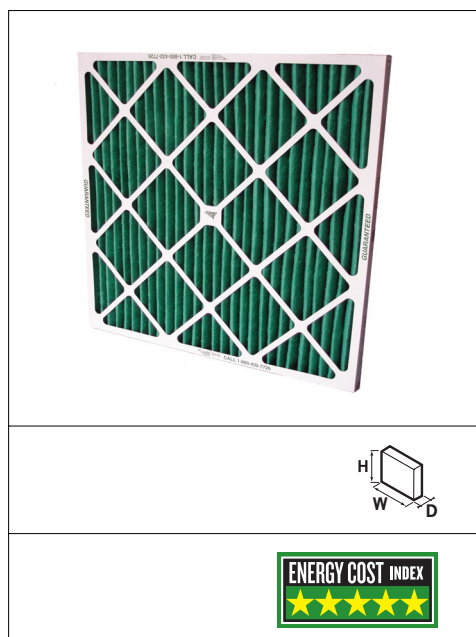
[1] Available with option 31 only.

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30/30®



Advantages

- Guaranteed to last longer
- True MERV 8 maintained efficiency
- Meets all published hospital filter efficiency guidelines

Description: High capacity pleated panel filter available in 1", 2" or 4" depths.

Typical applications: Prefilters to higher efficiency filters, stand-alone filter for rooftops, split systems, free-standing units and package systems and air handlers.

Efficiency: Value of 8 for MERV and MERV-A when evaluated under ASHRAE Testing Standard 52.2.

Media: Proprietary blend of cotton and synthetic fibers in a uniform lofted media blanket.

Recommended final pressure drop: 1.0" w.g. when operated at 500 fpm. System design may dictate alternative changeout point.

Temperature: Maximum continuous operating temperature of 200° F (93° C).

Ratings: ECI value of five stars, listed as UL 900.

See Literature 1002 for more details.



Part Number	Nominal Size H x W x D (inches)	Height (inches)	Width (inches)	Depth (inches)	Airflow Capacity (cfm)	Media Area (sq. ft.)	Resistance @ Capacity (inches w.g.)	Pleats per Linear Foot
054862-001	20 x 16 x 1	19.50	15.50	0.88	780	5.4	0.23	16
054862-002	20 x 20 x 1	19.50	19.50	0.88	970	6.6	0.23	16
054862-003	25 x 20 x 1	24.50	19.50	0.88	1215	8.3	0.23	16
054862-004	25 x 16 x 1	24.50	15.50	0.88	970	6.7	0.23	16
054862-005	24 x 24 x 1	23.50	23.50	0.88	1400	9.8	0.23	16
054862-006	20 x 14 x 1	19.50	13.50	0.88	680	4.6	0.23	16
054862-007	25 x 14 x 1	24.50	13.50	0.88	850	5.7	0.23	16
054862-008	20 x 15 x 1	19.50	14.50	0.88	730	5.1	0.23	16
054862-009	20 x 7 x 1	19.50	6.50	0.88	340	2.4	0.23	16
054862-010	24 x 12 x 1	23.50	11.50	0.88	700	4.9	0.23	16
054862-011	24 x 20 x 1	23.50	19.50	0.88	1170	8.0	0.23	16
054862-012	16 x 16 x 1	15.50	15.50	0.88	620	4.3	0.23	16
054862-013	25 x 15 x 1	24.50	14.50	0.88	910	6.4	0.23	16
054862-014	25 x 25 x 1	24.50	24.50	0.88	1520	10.5	0.23	16
054862-015	24 x 16 x 1	23.50	15.50	0.88	930	6.5	0.23	16
054862-016	20 x 10 x 1	19.50	9.50	0.88	490	3.3	0.23	16
054862-017	25 x 18 x 1	24.50	17.50	0.88	1100	7.6	0.23	16
054862-019	20 x 12 x 1	19.50	11.50	0.88	580	4.1	0.23	16
054862-020	20 x 18 x 1	19.50	17.50	0.88	880	6.1	0.23	16
054862-021	22 x 22 x 1	21.50	21.50	0.88	1180	8.2	0.23	16
054862-022	24 x 10 x 1	23.50	9.50	0.88	580	4.0	0.23	16
054862-023	25 x 10 x 1	24.50	9.50	0.88	610	4.1	0.23	16
054862-024	25 x 12 x 1	24.50	11.50	0.88	730	5.2	0.23	16
054862-025	12 x 12 x 1	11.50	11.50	0.88	350	2.5	0.23	16

Pleated Panel Filters

Part Number	Nominal Size H x W x D (inches)	Height (inches)	Width (inches)	Depth (inches)	Airflow Capacity (cfm)	Media Area (sq. ft.)	Resistance @ Capacity (inches w.g.)	Pleats per Linear Foot
054862-028	24 x 18 x 1	23.50	17.50	0.88	1050	7.3	0.23	16
049880-001	20 x 16 x 2	19.50	15.50	1.75	1110	9.9	0.31	15
049880-002	20 x 20 x 2	19.50	19.50	1.75	1390	11.9	0.31	15
049880-003	25 x 20 x 2	24.50	19.50	1.75	1740	14.9	0.31	15
049880-004	25 x 16 x 2	24.50	15.50	1.75	1390	12.4	0.31	15
049880-005	24 x 24 x 2	23.38	23.38	1.75	2000	17.3	0.31	15
049880-006	24 x 12 x 2	23.38	23.38	1.75	1000	8.4	0.31	15
049880-007	20 x 12 x 2	19.50	19.50	1.75	835	7.4	0.31	15
049880-008	20 x 10 x 2	19.50	19.50	1.75	700	6.0	0.31	15
049880-009	20 x 14 x 2	19.50	19.50	1.75	975	8.3	0.31	15
049880-010	25 x 14 x 2	24.50	24.50	1.75	1220	10.4	0.31	15
049880-011	20 x 15 x 2	19.50	19.50	1.75	1045	9.3	0.31	15
049880-012	24 x 20 x 2	23.50	23.50	1.75	1670	14.3	0.31	15
049880-013	20 x 18 x 2	19.50	19.50	1.75	1250	10.8	0.31	15
049880-014	25 x 18 x 2	24.50	24.50	1.75	1565	13.5	0.31	15
049880-015	24 x 18 x 2	23.50	23.50	1.75	1500	13.0	0.31	15
049880-016	24 x 16 x 2	23.50	23.50	1.75	1335	11.8	0.31	15
049880-018	25 x 25 x 2	24.50	24.50	1.75	2170	19.0	0.31	15
049880-019	16 x 16 x 2	15.50	15.50	1.75	890	7.8	0.31	15
049880-020	25 x 15 x 2	24.50	15.50	1.75	1300	11.6	0.31	15
059413-001	24 x 24 x 4	23.38	23.38	3.75	2000	27.7	0.27	11
059413-002	24 x 12 x 4	23.38	11.38	3.75	1000	13.9	0.27	11
059413-003	20 x 20 x 4	19.38	19.38	3.75	1390	18.9	0.27	11
059413-004	20 x 16 x 4	19.38	15.38	3.75	1110	15.7	0.27	11
059413-005	25 x 16 x 4	24.38	15.38	3.75	1390	19.7	0.27	11
059413-006	25 x 20 x 4	24.38	19.38	3.75	1740	23.6	0.27	11
059413-007	25 x 29 x 4	24.38	28.38	3.75	2520	35.4	0.27	11
059413-008	24 x 20 x 4	23.38	19.38	3.75	1670	22.7	0.27	11
059413-009	24 x 18 x 4	23.38	17.38	3.75	1500	20.2	0.27	11
059413-010	25 x 25 x 4	24.38	24.38	3.75	2170	30.1	0.27	11
059413-011	24 x 16 x 4	23.38	15.38	3.75	1330	18.7	0.27	11

As part of our program for continuous improvement, Camfil reserves the right to change specifications without notice. 2015-11-08

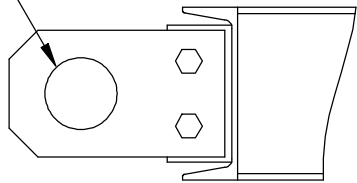


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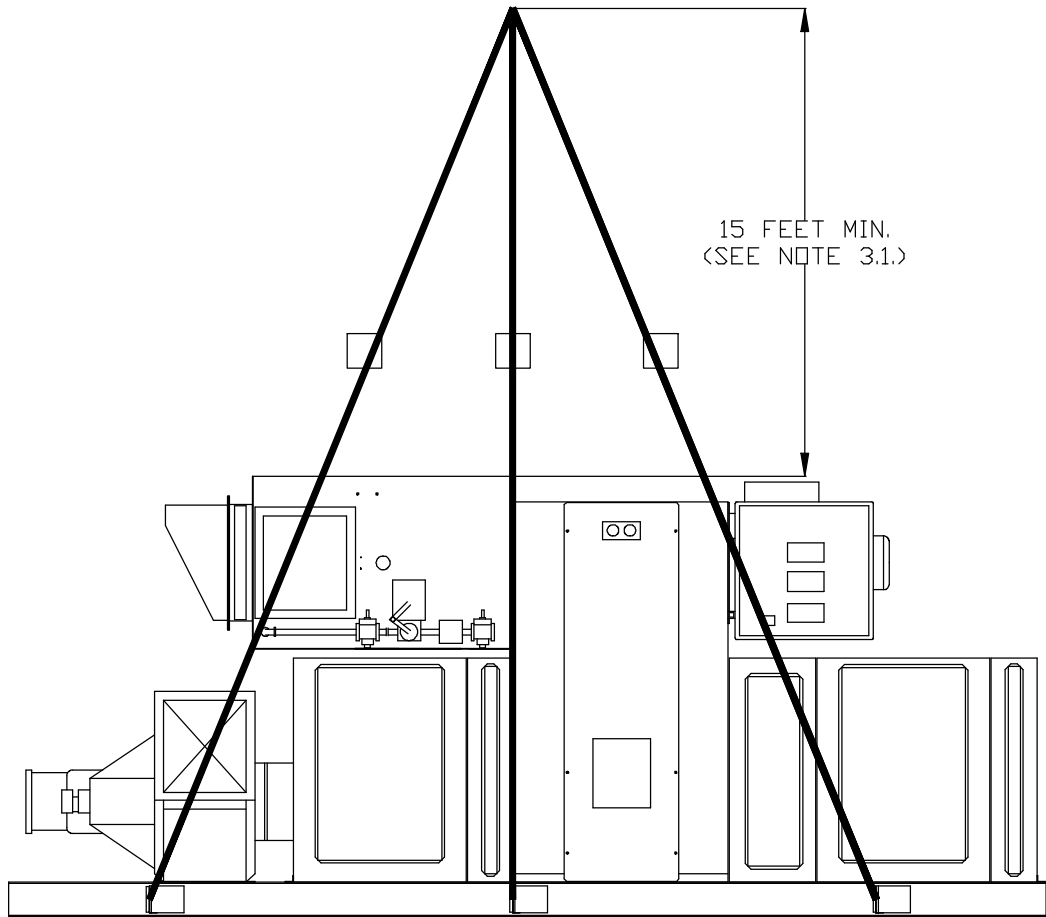
ALWAYS ORDER BY STOCK NUMBER		Stock Size (Nominal)	Stock No.	Performance		
E Z KLEEN[®] AIR FILTERS UL CLASS 2 FILTER RATED VELOCITY 150-650 FPM (0.76-3.31 m/s) OPTIMUM VELOCITY 350-520 FPM (1.78-2.64 m/s)		H x W x D		1" EZ Kleen Filters		
		20" x 25" x 1"	9801	1" (25 mm) & 3/32" (2.4 mm)	At 350 fpm	At 520 fpm
		20" x 20" x 1"	9802		(1.78 m/s)	(2.64 m/s)
		16" x 25" x 1"	9803	Average Arrestance	54%	48%
		16" x 20" x 1"	9804	Clean Resistance	.045" w.c.	.09" w.c.
		15" x 20" x 1"	9805		0.011 kPa	0.023 kPa
		10" x 20" x 1"	9806	Dust-Holding Capacity	95 gms/ft² (1023 gms/m²)	67 gms/ft² (721 gms/m²)
MV E Z KLEEN[®] AIR FILTERS UL CLASS 2 FILTER RATED VELOCITY 150-650 FPM (0.76-4.58 m/s) OPTIMUM VELOCITY 350-520 FPM (1.78-2.64 m/s)		H x W x D		2" MV EZ Kleen Filters		
		20" x 25" x 2"	9151		At 350 fpm	At 520 fpm
		20" x 20" x 2"	9152		(1.78 m/s)	(2.64 m/s)
		16" x 25" x 2"	9153	Average Arrestance	72%	68%
		16" x 20" x 2"	9154	Clean Resistance	.07" w.c.	.14" w.c.
		15" x 20" x 2"	9155		0.018 kPa	0.035 kPa
		10" x 20" x 2"	9156	Dust-Holding Capacity	130 gms/ft²	91 gms/ft²
24" x 24" x 2"	9157	(1399 gms/m²)	(980 gms/m²)			
INDUSTRIAL E Z KLEEN[®] AIR FILTERS UL CLASS 2 FILTER RATED VELOCITY 150-900 FPM (0.76-4.58 m/s) OPTIMUM VELOCITY 350-520 FPM (1.78-2.64 m/s)		H x W x D		2" Industrial Filters		
		20" x 25" x 2"	901		At 350 fpm	At 520 fpm
		20" x 20" x 2"	902		(1.78 m/s)	(2.64 m/s)
		16" x 25" x 2"	903	Average Arrestance	67%	64%
		16" x 20" x 2"	904	Clean Resistance	.05" w.c.	.10" w.c.
		15" x 20" x 2"	905		0.013 kPa	0.025 kPa
		10" x 20" x 2"	906	Dust-Holding Capacity	143 gms/ft² (1539 gms/m²)	117 gms/ft² (1259 gms/m²)
TRIM-TO-SIZE[®] E Z KLEEN AIR FILTERS UL CLASS 2 FILTER		H x W x D		1/4" Trim-To-Size Filters	5/8" Trim-To-Size Filters*	
		15" x 24" x 1/4"	9200		At 350 fpm (1.78 m/s)	At 350 fpm (1.78 m/s)
		40" x 64" x 1/4"	9250	Average Arrestance	61%	54%
		40" x 64" x 5/8"	9251	Clean Resistance	.07" w.c.	.045" w.c.
		40" x 50" x 1/4"	9260		0.018 kPa	0.011 kPa
		40" x 50" x 5/8"	9261	Dust-Holding Capacity	100 gms/ft² (1075 gms/m²)	95 gms/ft² (1023 gms/m²)
				* Note: See 1" EZ Kleen performance for 520 fpm		
Made-To-Order E Z KLEEN[®] AIR FILTERS Minimum quantities may apply.				EZ Kleen Filters: 3/32", 3/8", 1/2"		
				3/32", 3/8" & 1/2" Nominal Frame Thickness	At 350 fpm (1.78 m/s)	
				Average Arrestance	64%	
				Clean Resistance	.06" w.c. (.016 kPa)	
				Dust-Holding Capacity	44 gms/ft² (474 gms/m²)	

Revisions					
Rev	Description	Date	Dft	Chk	App
1	CONVERTED TO CAD FROM PDF, MODIFIED NOTES	12/11/09	MRB	EJL	EJL

ø2 1/2" HOLE

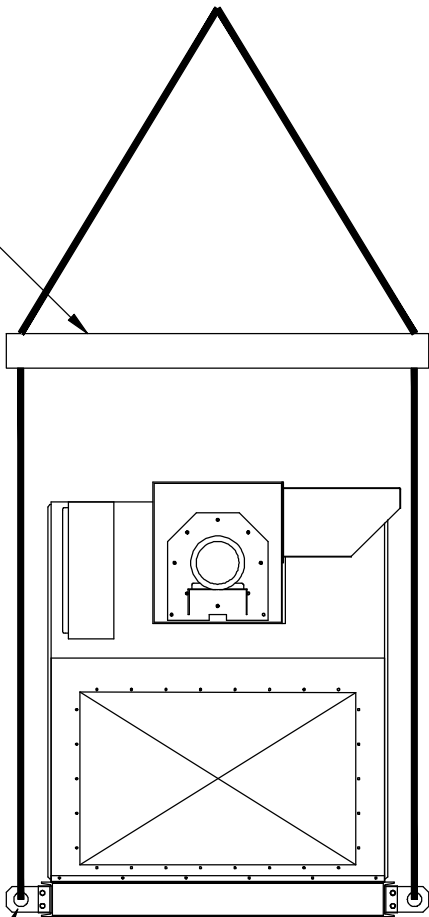


DETAIL 'A'



FRONT ELEVATION

SPREADER BAR



REMOVABLE LIFTING LUGS
SEE DETAIL "A"

END ELEVATION

- NOTES:
- RECEIVING: WHEN EQUIPMENT IS RECEIVED ALL ITEMS SHOULD BE CAREFULLY CHECKED AGAINST THE BILL OF LADING TO BE SURE THAT THE TOTAL NUMBER OF PIECES HAVE BEEN RECEIVED, THE SYSTEM WAS THOROUGHLY INSPECTED BEFORE LEAVING THE FACTORY AND THE CARRIER HAS ACCEPTED AND SIGNED FOR IT. ANY DAMAGE OR IRREGULARITIES SHOULD BE NOTED TO THE CARRIER.
 - HANDLING & RIGGING;
 - WHEN LIFTING AN HCD SYSTEM FROM A FLAT BED TRUCK OR A FLAT CAR WITH A HOOK AT A SINGLE POINT, A LIFTING/SPREADING DEVICE SIMILAR TO THE ONE SHOWN IS SUGGESTED.
 - THE LIFTING/SPREADING DEVICE AND LIFTING BARS MUST BE FURNISHED BY THE RIGGING CONTRACTOR AND MUNTERS ASSUMES NO RESPONSIBILITY OR LIABILITY FOR METHODS OR MATERIALS USED. LIFTING/SPREADING DEVICE MUST BE CAPABLE OF LIFTING THE TOTAL SYSTEM WEIGHT (REFER TO GENERAL ARRANGEMENT FOR APPROXIMATE WEIGHT).
 - A LIFTING/SPREADING DEVICE IS ESSENTIAL TO PREVENT CRUSHING SYSTEM SIDES AND DAMAGE TO SENSITIVE CONTROL COMPONENTS.
 - RIGGING MUST BE DESIGNED SO THAT LOAD REMAINS LEVEL AT ALL TIMES.
 - LIFTING:
 - RECOMMENDED LIFTING/SPREADING DEVICE SHOULD HAVE A LENGTH SUFFICIENT TO PREVENT THE CABLES FROM DAMAGING THE EQUIPMENT. THE WIDTH SHALL NOT EXCEED 18 INCHES WIDER THAN THE WIDTH OF THE LIFTING LUGS HOLE CENTER TO CENTER DISTANCE AS VIEWED FROM THE END ELEVATION. ALL LIFTING LUGS ARE TO BE UTILIZED. CABLES MUST BE LONG ENOUGH TO PERMIT A MINIMUM DISTANCE OF 15 FEET BETWEEN CABLE CENTER RING AND HIGHEST POINT OF THE SYSTEM.
 - SAFETY LINES SHOULD BE ATTACHED TO SKID AT NOT LESS THAN 2 POINTS TO PREVENT SWINGING OR TWISTING DURING LIFTING.
 - PRIOR TO LIFTING, CHECK THAT ALL ACCESS PANELS ARE IN PLACE, THAT CORNERS ALONG TOP SURFACES ARE PROTECTED FROM POSSIBLE DAMAGE BY SLINGS AND THAT ALL EXPOSED COMPONENTS ARE FULLY PROTECTED.
 - LIFTING SHOULD BE ACCOMPLISHED USING SAFEST SPEED OF LIFT BASED ON CONDITIONS AT TIME OF OPERATION.
 - WHEN A SYSTEM IS SPLIT INTO TWO OR MORE SECTIONS, EACH SECTION IS TO BE LIFTED INDIVIDUALLY AS SHOWN. DO NOT LIFT A "SPLIT SYSTEM" AS A COMPLETE UNIT.
 - PLACEMENT:
 - THE HCD SYSTEM IS DESIGNED TO BE PLACED AS A SINGLE STRUCTURE ON A CONCRETE PAD OR OTHER TYPE OF SUPPORTING BASE DESIGNED FOR SUPPORTING THE WEIGHT OF THE SYSTEM.
 - THE SYSTEM SHOULD BE PLACED AS TO ALLOW ACCESS TO ALL COMPONENTS AND EQUIPMENT. CARE SHOULD BE TAKEN SO THAT AIR INLETS AND OUTLETS ARE NOT BLOCKED BY OTHER STRUCTURES OR PIECES OF EQUIPMENT THAT DO NO INTERFACE WITH THE SYSTEM.
 - THE SYSTEM MUST BE LOWERED GENTLY ONTO THE BASE OR SUPPORT STRUCTURE WITHOUT BOUNCING OR DROPPING.
 - CARE MUST BE TAKEN DURING PLACEMENT OF UNIT WITH REGARD TO AVAILABILITY OF UTILITIES.

CONFIDENTIALITY STATEMENT
THE TECHNICAL INFORMATION AND DESIGN DATA DISCLOSED HEREIN CONSTITUTE PROPRIETARY INFORMATION OF MUNTERS CORPORATION - CARGOCAIRE DIVISION AND ARE TO BE MAINTAINED IN STRICT CONFIDENCE. THIS INFORMATION IS FOR THE SOLE USE OF OUR CUSTOMERS AND END USERS OF OUR EQUIPMENT.

Job No.	STANDARD	
Customer		
Dr.By.	D. REILLY	4/23/97
Chk.By.	RG	4/97
Appr.By.	RG	4/97



Munters Corporation
79 Monroe St. Amesbury, Ma 01913
TEL 978-241-1100 FAX 978-241-1214

RIGGING INSTRUCTIONS
HCD SYSTEMS

SCALE	SHEET	SIZE	REV
NTS	1 of 1	B	1

27137



Spare Parts List

CCN: US60
BCR Type: CUR
Item: 22367602-01
Description: HCD-2250-EA-SFS

QTY	UM	COMPONENT	DESCRIPTION	BALLOON
2	EA	10461	CONE INLET #12-1/4 3-4-2-3 AL	0021/0027
2	EA	15563	MINIHELIC 0-3 IN 2-5003	0007/0008
1	EA	30145-27	WHEEL ASSY 30" TIGEL 400mm	0012
1	EA	43125-04	SEAL KIT HC-2250	0011
1	EA	90476-01	SWITCH LIMIT BA-2RV2-A2	0003
2	EA	90916-02	WHEEL BLOWER 12-1/4 CW 8-5-3-1	
2	EA	90992-06	COVER, FUSE, 1 POLE, CLASS CC/MIDGET, MERSEN DFC-7	TRF-1K
2	EA	91410-04	FUSE MIDGET TIME DEL 600V, 1 AMP	FUS30
3	EA	91816-05	FUSE CLASS J 15 AMP JKS-15	FUS03
6	EA	91816-08	FUSE CLASS J 30 AMP JKS-30	FUS01/FUS02
12	EA	91816-40	COVER,FUSE,INDICATING MERSEN P/N DFC-10I	FUS01/FUS02/FUS03/FUS10
1	EA	91832-01	THERMOCOUPLE,4","J",STRAIGHT	TE02
1	EA	91862-02	THERMOCOUPLE,7.5","J",RT.ANGLE	TE01
3	EA	91981-21	FUSE TIME DELAY CLASS J 15A	FUS10
2	EA	92011-10	MOTOR 3HP 3450RPM TRI VOLT 00336ET3E182T	0016/0023
4	EA	92022-03	WHEEL ROLLER GREASELESS 3-1/4"	
1	EA	92247-01	MOTOR DRIVE 1/100HP 115V	
1	EA	92283-05	BELT TIMING TYPE L, 3/8P, 1"W, 102"L, 272Tooth	
1	EA	92466-01	FUSE,3AG,TD,250V,2.0AMP	FUS31
2	EA	93962-02	FUSE, 1/4X1-1/4, CERAMIC, 0.5A	FUS26/FUS38
1	EA	93962-04	FUSE,1A FAST ACTING, ABC-1	FUS18
1	EA	94763-30	HEATER OPEN ELEMENT, 460V, 16KW	
1	EA	94763-35	HEATER OPEN ELEMENT, 460V, 8KW	
23	FT	95001-01	SEAL BULB,SILICONE,CLIP ON,OMEGA,(11' LONG)	0001/0003
1	EA	95007-05	METAL MESH FILTER, 12 X 12 X 2	0010
4	EA	K95002-38	TERMINAL,FUSE,500V,SPRING CLAMP,1616460000	FUS18/FUS26/FUS31/FUS38
4	EA	K95002-39	TERMINAL, FUSE, END COVER,1610840000	FUS18/FUS26/FUS31/FUS38
1	EA	K95002-59	TERMINAL,5MM,4 CONN,GROUND,SPRING CLAMP,1608660000	TB0
1	EA	K95050-102	KIT,SEL SW, 3 POS,ILL WHT,120VAC,4 NO, CP, SIEMENS	SS01
2	EA	K95050-29	PUSHBUTTON, ILLUM RED, 120VAC, 1 NO/1 NC, SIEMENS	LT02/LT22
2	EA	K95050-30	YELLOW PB KIT, 120V, 1NO/1NC, PUSH TO TEST	LT03/LT23
1	EA	K95050-31	PUSHBUTTON, ILLUM GREEN, 120VAC, 1 NO/1 NC,SIEMENS	LT24
1	EA	K95309-03	TRANSFORMER, 250VA, 230/400/460/575-95/115/125	TRF-1K
2	EA	K95552-04	SS RELAY, 70 AMP, 24VDC CNTRL,600V OPER,RESIS LOAD	SSR1
2	EA	K95552-06	CONVERTER MODULE,0-10V TO PULSE, 24VAC/DC, PLUG-IN	SSR1
1	EA	K95935-03	S7-1200, CPU VA, AC/DC/RLY, 14DI DC, 10DO RLY, 2AI	PLC1
1	EA	K95935-13	S7-1200, ANALOG INPUT MOD, 4 INPUTS, TC	PLC1
1	EA	K95935-24	S7-1200, SIGNAL BOARD, ANALOG OUTPUT, 1 OUTPUT	PLC1
1	EA	K95936-03	2O1P3E3R1 IN5T8E8R-0F1C,4" TOUCH,COLOR,SIEMENS,KTP	KTP400
3	EA	K95941-01	CONTACTOR, 120V COIL 3HP@460VAC 20A(R) ,1NO EATON	MS01/MS02/MS03

QTY	UM	COMPONENT	DESCRIPTION	BALLOON
6	EA	K95941-05	CONTACTOR, 120V COIL 10HP 460VAC 40A(R),1NO EATON	CON01/CON02/CON03/CON51/CON52/CON53
1	EA	K95945-04	MSP, 230-575VAC, ROT. ON/OFF, FRAME B .4-.63 AMP	MSP1
2	EA	K95945-08	MSP, 230-575VAC, ROT. ON/OFF, FRAME B 2.5-4 AMP	MSP2/MSP3
1	EA	K95946-01	AUX. CONTACT, 16A, FRAME B-C, 2POLE, 2NO	MS02
3	EA	K95946-14	LINE SIDE ADAPTER, FRAME B	MSP1/MSP2/MSP3
3	EA	K95946-51	AUXILIARY CONTACTS FOR ALL XTPB, XTPR AND XTPM MSR	MSP1/MSP2/MSP3